The Holographic Alaya-Field Framework

The formation of accessible world structure

Complete Collected Volume

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Introduction

What makes a physical relation into part of a world, rather than a description imposed on it? HAFF follows the formation of persistent distinctions, their readable relations, temporal order and geometric response. The question is generative: which structures are produced by interaction, and which still enter through a physical dictionary or an initial boundary?

A and B begin with process-selected blocks and coupling-dependent coherence-decay relations. E asks how those distinctions become actual records. F shows how a strengthened recording protocol can identify an unknown order of events, while ordinary copying can remain order-blind. C connects acquired history to intervention. D uses a common conserved source to link modular constraints and physical response; G distinguishes internal execution from complete explanation.

These routes have different scopes. The decay-rate relation is not automatically space; transition responses require a separately identified transfer law. Recovered event order is not a generated time parameter, and the conditional linearized gravitational law still needs its field and entropy–area inputs. Finite return supplies a genuine common realization, not a substitute for those formation questions. The postscript brings the results back to cosmic origin and closure through an explicit formation-and-return conjecture.

Reading map

HAFF-A Physical coarse-graining selects persistent distinctions, with a coupling-dependent coherence-decay relation. Transfer responses and a spatial dictionary require further inputs; finite geometry certificates test this generative route.

HAFF-B Interaction-selected blocks, protected content and finite lifetimes establish which relations survive. Stable sectors are not automatically tensor factors or spatial sites.

HAFF-C Acquired history changes the interventions available to a body. The relation between functional agency and experience remains a separate explanatory question.

HAFF-D One conserved physical source enters modular and geometric response constraints. Conditional field-law synthesis, finite physical return and stronger thermal conjectures have distinct assumptions.

HAFF-E Actual instruments connect permissions to stable and public records. Public agreement requires more than redundancy to certify correctness.

HAFF-F Endpoint records identify an unknown exchange-event order under specified gates. Ordinary copying can fail this test; acquisition, renewal and time parametrization remain distinct.

HAFF-G Internal programs alter permissions and sources. Protocol-relative indistinguishability constrains complete explanation without creating a universal final boundary.

HAFF-PS Cosmic origin and closure return as a formation-and-renewal conjecture, alongside conditional finite-body participation bounds.

PHYSICAL COARSE-GRAINING AND THE PROBLEM OF EMERGENT GEOMETRY: THE HOLOGRAPHIC ALAYA–FIELD FRAMEWORK

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ABSTRACT

How can a physical process select the structures from which geometry is reconstructed? We study this question through implemented coarse-graining. For finite systems with commuting Hermitian noise, the dynamics determines a direct-sum decomposition through the centre of its fixed algebra. Cross-sector coherence decay determines a Euclidean rate geometry, and a finite-window certificate identifies persistent blocks from separated rates. Coherent probes can estimate these relations; sector-label measurements alone cannot. Identical system channels can also leave different readable environmental records when a purification is inaccessible. These results distinguish process-selected structure, physical permissions and geometric interpretation. Two quantitative tests then constrain reconstruction: an exact qubit criterion exposes reductions that entropy cannot distinguish, while a local interaction and recording protocol certifies non-isometric correlation metrics from a fixed GHZ source. The latter uses a declared site representation and distance rule. Neither its tensor sites nor spatial transport follow from the noise sectors. We formulate the missing bridge as a same-process locality conjecture: interaction histories must select persistent organization and produce intervention responses admitting a common spatial interpretation.

1 Introduction

A reconstruction of space already has parts to relate. Decoherence distinguishes a system, apparatus and environment; entanglement-based reconstructions assign observables to regions or tensor factors [1, 4, 3]. A Hilbert space alone selects none of these divisions. Accessible observables can induce subsystem structure [5, 6, 7]; conversely, tailoring the observables can give a fixed pure state any prescribed Schmidt decomposition [8]. The emergence problem therefore includes the physical selection of the parts, as well as the geometric reading of their relations.

We approach selection through the implemented process. A coupling determines a reduced channel and can protect a direct-sum organization of its carrier. In a commuting-noise class, this organization is the centre decomposition of the channel's fixed algebra; a supplied spatial factorization is unnecessary for that calculation. The rates at which its sectors lose coherence also determine an operational relation. Sections 3 and 4 make both statements precise and identify the measurements needed to test them. They use standard dephasing structure [28, 25], while separating it from the availability of instruments and from spatial transport.

The distinction changes what a geometric claim requires. The same observable algebra can arise from reductions with different resolution. Different channels can also have identical entropy functions. Section 5 characterizes this obstruction for qubits. Section 6 carries a fixed correlation reconstruction through noise interactions, environmental records and finite measurements. Its non-isometric metrics are physical consequences of that protocol, with the site representation and distance rule stated explicitly. Section 7 returns to the remaining question: when can process-selected relations acquire a spatial interpretation through actual propagation or intervention?

We use *Alaya-Field*, borrowed from Yogācāra's storehouse consciousness, as a technical name for an unfactorized source $(\mathcal{H}_U, \mathcal{A}_U, \Omega)$, with Ω cyclic and separating. The name specifies no additional ontology. A Type III algebra has no intrinsic density operator or trace entropy [9]. All entropy results here instead concern finite-dimensional carriers; a faithful finite state has a cyclic and separating vector in its standard representation, whereas a pure carrier vector does not separate the full matrix algebra. No Type III entropy formula or Lorentzian metric is inferred from the finite calculations.

2 Coarse-graining structures, algebras and channels

2.1 Definition

An observable algebra does not by itself reinstate a tensor factorization: operators can act irreducibly on $\mathcal{H}_{\mathrm{total}}$, and in algebraic quantum field theory local algebras are assigned to regions without any global tensor product [9]. A coarse-graining here is meant to be realized by a physical interaction, not chosen by an agent.

Definition 2.1 (Coarse-graining structure)**.** *A coarse-graining structure is a pair* $\mathbf{c} = (\mathcal{A}_{\mathrm{sel},\mathbf{c}}, \Phi_{\mathbf{c}})$*, where*

1. $\mathcal{A}_{\mathrm{sel},\mathbf{c}} \subset \mathcal{B}(\mathcal{H}_U)$ *is a selected unital* $*$*-subalgebra used to define the effective representation. It is not identified with Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set [26]; a physical model must state any map between them.*
2. *In the finite-dimensional results,* $\Phi_{\mathbf{c}}$ *is a completely positive trace-preserving (CPTP) map on states that implements an operational reduction; its Heisenberg dual* $\Phi_{\mathbf{c}}^*$ *is unital and completely positive. Partial trace over a specified factor and channels induced by a conditional expectation are examples. In a general von Neumann setting the corresponding object is a normal channel with a normal unital completely positive dual, and no entropy result below is claimed for that extension.*

No theorem guarantees a unique maximal observable algebra for a given physical system without additional structure. Several admissible algebras can coexist, and for a fixed algebra several channels can implement different losses of information, such as tracing out inaccessible degrees of freedom or effective decoherence. The two entries of $\mathbf{c}$ must therefore be related explicitly.

2.2 The algebra generated by a channel

For a channel $\Phi : \mathcal{B}(\mathcal{H})_* \to \mathcal{B}(\mathcal{K})_*$ put

$$\mathcal{S}_\Phi = \operatorname{ran} \Phi^*, \qquad \mathcal{A}_\Phi = C^*(\mathcal{S}_\Phi).$$

$\mathcal{S}_\Phi$ is a unital self-adjoint subspace, not necessarily closed under products. The effects that the channel actually reads out are $\Phi^*([0, I])$, which is in general smaller than the set of positive contractions in $\mathcal{S}_\Phi$. We call $\mathbf{c}$ weakly compatible if $\mathcal{S}_{\Phi_\mathbf{c}} \subseteq \mathcal{A}_{\mathrm{sel},\mathbf{c}}$ and minimally compatible if $\mathcal{A}_{\mathrm{sel},\mathbf{c}} = \mathcal{A}_{\Phi_\mathbf{c}}$.

Proposition 2.2 (Algebras and post-processing)**.**

1. *If two states agree on* $\mathcal{S}_\Phi$*, in particular if they agree on* $\mathcal{A}_\Phi$*, then their outputs under* Φ *coincide.*
2. *If* $\Phi_2 = \Lambda \circ \Phi_1$ *for a channel* Λ*, then* $\mathcal{S}_{\Phi_2} \subseteq \mathcal{S}_{\Phi_1}$ *and* $\mathcal{A}_{\Phi_2} \subseteq \mathcal{A}_{\Phi_1}$*.*
3. *The converse of (2) fails. For the depolarizing channels* $D_p(\rho) = p\rho + (1-p)I/2$ *on a qubit with* $0 < p < q \leq 1$*, both* $\mathcal{A}_{D_p}$ *and* $\mathcal{A}_{D_q}$ *equal the full matrix algebra,* $D_p = D_{p/q} \circ D_q$*, and* D_q *is not a post-processing of* D_p*.*

Proof. (1) For any output operator Y, $\mathrm{Tr}[\Phi(\rho)Y] - \mathrm{Tr}[\Phi(\sigma)Y] = \mathrm{Tr}[(\rho - \sigma)\Phi^*(Y)]$, and $\Phi^*(Y) \in \mathcal{S}_\Phi$. (2) $\Phi_2^* = \Phi_1^*\Lambda^*$, so the range of Φ_2^* lies in the range of Φ_1^*, and generated algebras inherit the inclusion. (3) Both maps are invertible on the operator space, so their dual ranges are everything. The identity $D_{p/q}(D_q(\rho)) = p\rho + (1-p)I/2$ is direct. Two orthogonal pure inputs have outputs at trace distance p under D_p and q under D_q; a channel cannot increase trace distance, so no Λ satisfies $\Lambda \circ D_p = D_q$. □

Two reductions with the same algebra can thus have different operational resolution. An algebra label is not a substitute for the channel and its allowed post-processing. If a conditional expectation is chosen as the implementation, that is an additional structure, not a synonym for every CPTP reduction.

3 Structures selected by a physical process

Consider a finite carrier $\mathcal{H}$ with commuting Hermitian couplings A_k at rates $\gamma_k > 0$. In a specified interaction picture the Heisenberg generator is

$$\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k, [A_k, X]], \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

This generator is self-adjoint for the Hilbert–Schmidt inner product, so the same formula acts on states. A Hamiltonian contribution must either be included in a different analysis or removed by a justified interaction picture. The carrier, system–environment division, couplings and time scale are physical inputs.

Proposition 3.1 (Intrinsic sectors of a dephasing process)**.** *Let $\mathcal{B} = \mathrm{Fix}(\Phi) = \ker \mathcal{L}$, and let P_a, $a \in V$, be the minimal nonzero projections of $Z(\mathcal{B})$. Then*

$$\mathcal{H} = \bigoplus_{a\in V} P_a\mathcal{H}, \qquad \mathcal{B} = \bigoplus_{a\in V} \mathcal{B}(P_a\mathcal{H}).$$

The projections are determined by the semigroup, up to relabelling. For every $X \in P_a\mathcal{B}(\mathcal{H})P_b$,

$$\mathcal{L}(X) = -D_{ab}X, \qquad D_{ab} = \frac{1}{2}\sum_k \gamma_k(a_k(a) - a_k(b))^2, \quad A_kP_a = a_k(a)P_a. \tag{2}$$

These rates are intrinsic to the semigroup: $D_{aa} = 0$ and $D_{ab} > 0$ for distinct sectors. A common unitary conjugation rotates the P_a and preserves D.

Proof. The identity

$$-\langle X, \mathcal{L}(X)\rangle_{\mathrm{HS}} = \frac{1}{2}\sum_k \gamma_k \|[A_k, X]\|^2_{\mathrm{HS}}$$

gives $\mathcal{B} = \{A_k\}'$. Simultaneous diagonalization yields full joint eigenspaces $P_a\mathcal{H}$, on which every coupling is scalar. Their full matrix algebras form the commutant, and their projections are precisely its central atoms. No basis inside a degenerate eigenspace enters this description. Applying the double commutator to a cross-corner gives (2). Its scalar is fixed by $\mathcal{L}$, independently of the coupling representation; different atoms have different joint eigenvalue tuples. Conjugating the entire generator conjugates its corners and leaves their scalar eigenvalues unchanged. □

The fixed-algebra construction and its dephasing rates are known [28, 25]. Their role here is to remove a separate choice of effective sectors after the physical process has been supplied. A direct sum is not a tensor product. In particular, the P_a are not the sites of the correlation reconstruction below.

For finite t, the channel multiplies each corner by $e^{-tD_{ab}} > 0$, so it has full linear operator range and $\mathcal{A}_{\Phi_t} = \mathcal{B}(\mathcal{H})$. Its long-time limit is the pinching $\mathcal{E}_\infty(X) = \sum_a P_aXP_a$, whose range is $\mathcal{B}$. Thus $(\mathcal{B}, \mathcal{E}_\infty)$ is minimally compatible in Definition 2.1; replacing Φ_t by this projection at finite time requires an error estimate.

3.1 Selection on a finite window

Exact fixed algebras can jump under arbitrarily weak symmetry breaking. A finite capability instead asks how well an entire block survives over a stated time. Following Paper B, call a unital algebra ε-stable on $[0, T]$ if

$$\sup_{0\le t\le T} \sup_{\substack{X\in\mathcal{B}\\ \|X\|\le 1}} \|\Phi_t(X) - X\| \le \varepsilon.$$

Put $u_a = (\sqrt{\gamma_k/2}\, a_k(a))_k$. For a partition π of the intrinsic sectors into clusters C, define

$$Q_C = \sum_{a\in C} P_a, \qquad \mathcal{B}_\pi = \bigoplus_C \mathcal{B}(Q_C\mathcal{H}), \qquad r_C = \inf_c \max_{a\in C} \|u_a - c\|,$$

and $\kappa_{\mathrm{in}} = 4\max_C r_C^2$, $\Delta_{\mathrm{out}} = \min_{a,b \text{ in different } C} D_{ab}$. An empty minimum is $+\infty$.

Proposition 3.2 (A representation-independent window certificate)**.** *If $0 < \varepsilon < 1$, $T > 0$ and*

$$T\kappa_{\mathrm{in}} \le \varepsilon < 1 - e^{-T\Delta_{\mathrm{out}}}, \tag{3}$$

then $\mathcal{B}_\pi$ is ε-stable. No strictly containing unital algebra is ε-stable. At most one cluster partition satisfies this certificate; its algebra contains every stable block algebra in a common joint eigenbasis of the couplings.

The quantities in this certificate depend only on the rate matrix, since $D_{ab} = \|u_a - u_b\|^2$ determines its Euclidean configuration. The radius bound replaces Paper B's coordinate-box bound. Centring the noise coordinates on each cluster gives a generator norm at most $4r_C^2$; the separated cross-cluster matrix units exclude stable extensions. Appendix A gives the proof. The maximality statement does not compare all rotated, incomparable subalgebras.

Stability of individual coherences is insufficient. Three equally spaced points can have $TD_{01} = TD_{12} = .06$ but $TD_{02} = .24$: the first two matrix units lose less than .1 of their amplitude, while their product loses more than .1. Generating an algebra from all slow matrix units can therefore introduce unstable observables. The whole-block condition in (3) supplies a sufficient selection rule; absence of a gap does not determine a unique partition.

3.2 What a stable algebra makes available

A fixed observable need not have an available reader or gate. For example, a pointer-controlled collision can implement

$$\Gamma(\rho) = \sum_{z=0}^{1} \mathrm{Tr}(P_z\rho)|e_z\rangle\langle e_z|, \qquad \Gamma^*(F) = \sum_z \langle e_z|F|e_z\rangle P_z.$$

When $0 < |\langle e_0|e_1\rangle| < 1$, the dual range spans the pointer algebra, but its implementable effects exclude the sharp projections [26]. Source dephasing and the record channel arise from the same collision and have different outputs. Foundation I makes the same distinction between an ideal seed algebra and a finite instrument realizing a task [29].

For a faithful stationary state of (1), the density matrix is block diagonal and commutes with all P_a. Their modular orbit is therefore constant. Foundation I's stable hull of central seeds S is exactly $W^*(S)$; it is abelian and has no proper half-sided modular inclusion. The selected labels thus do not acquire a spacetime Cut by taking their hull. Noncentral seeds or an independently supplied infinite source require their own interface.

4 Which relations can the process identify?

4.1 A rate geometry and its measurements

Equation (2) gives a metric

$$d_D(a,b) = \sqrt{D_{ab}} = \|u_a - u_b\|. \tag{4}$$

If joint eigenvectors were labelled separately, this would be a pseudometric; equal-rate-zero labels must first be merged into the central sectors. If all couplings are scalar there is one sector. For $r = |V|$ and $J = I - \mathbf{1}\mathbf{1}^T/r$, the centred Gram matrix is $G = -JDJ/2$. Its positive eigenspaces reconstruct the minimal Euclidean configuration, up to isometry. Its dimension is a rank of the noise responses, not a spatial dimension. Moreover, d_D has units $\mathrm{time}^{-1/2}$; $\sqrt{t_0 D}$ is dimensionless for a supplied reference time t_0. Neither quantity is a physical length.

Proposition 4.1 (Visibility identifies the rate relation)**.** *Suppose a reader can choose unit vectors $x \in P_a\mathcal{H}$, $y \in P_b\mathcal{H}$, prepare $(x+y)/\sqrt{2}$, and measure $X_{ab} = |x\rangle\langle y| + |y\rangle\langle x|$ after known time $t > 0$. The mean is $v_{ab}(t) = e^{-tD_{ab}}$, hence $D_{ab} = -t^{-1}\log v_{ab}(t)$. If the true and estimated visibilities both belong to $[v_{\min}, 1]$ with $v_{\min} > 0$ and differ by at most ζ, then*

$$|\widehat{D}_{ab} - D_{ab}| \le \frac{\zeta}{t v_{\min}}, \qquad |\sqrt{\widehat{D}_{ab}} - \sqrt{D_{ab}}| \le \sqrt{\frac{\zeta}{t v_{\min}}}.$$

For M tested pairs with n independent preparations each, the visibility error bound holds simultaneously with failure probability at most $2Me^{-n\zeta^2/2}$. The sample mean may be clipped to $[v_{\min}, 1]$ when the true visibility is known to lie there.

Proof. The two off-diagonal entries of the prepared state are multiplied by $e^{-tD_{ab}}$, while its populations are constant. Their sum is the stated expectation. The logarithm has derivative bounded by $1/v_{\min}$ on the relevant interval, and $|\sqrt{x} - \sqrt{y}| \le \sqrt{|x-y|}$ for $x, y \ge 0$. Each measurement has outcomes ± 1 on the prepared subspace; Hoeffding's inequality and a union bound give the final statement. Clipping to an interval containing the true mean cannot increase the error. □

The proposition assumes the stated coherent preparations and readers, rather than obtaining them from the fixed algebra. It identifies a relation of the physical dynamics. It cannot certify exact zero rates with finite data, and says nothing about the stability of the number of exact sectors.

Proposition 4.2 (Sector-only access cannot identify the rates)**.** *Fix projections P_a. If all allowed preparations, instruments and outputs act within the block-diagonal algebra $\bigoplus_a \mathcal{B}(P_a\mathcal{H})$, every adaptive record law is independent of the cross-sector rates of a dephasing semigroup with these projections.*

Proof. Every such semigroup restricts to the identity on this algebra. Each permitted step, conditional on the previous record, therefore has the same output in every model. Induction over the steps gives the same complete record law. □

Thus mathematical knowledge of a channel and its identification by a reader with finite permissions are different resources.

4.2 A conditional record interpretation

With a pure initial environment and access to its complete output, a dilation of this dephasing channel has

$$V_t\psi_a = \psi_a \otimes e_a(t), \qquad \langle e_b(t)|e_a(t)\rangle = e^{-tD_{ab}}.$$

Indeed, preserving every pure state in a sector forces its environment vector to be independent of the state within that sector; the cross-corner multiplier fixes the overlaps. Their Gram matrix is positive because it is the Schur kernel of the channel. The isometry has a unitary extension at each fixed time. Complete-environment binary discrimination then has trace distance

$$q_{ab}(t) = \sqrt{1 - e^{-2tD_{ab}}}. \tag{5}$$

This is another metric, supplied by an explicit discrimination task, and its inversion requires the appropriate environmental measurements. A restricted fragment can carry less information.

Example 4.3 (Same system channel, different readable records)**.** *Use the collision $U = P_0 \otimes I + P_1 \otimes e^{-i\theta Z_E}$ with $0 < \theta < \pi/2$. A memory initially in $|+\rangle$ has conditional states $|+\rangle$ and $e^{-i\theta Z}|+\rangle$, overlap $\cos\theta$ and trace distance $\sin\theta$. A memory initially in $I/2$ gives the identical system off-diagonal multiplier $\mathrm{Tr}[(I/2)e^{i\theta Z}] = \cos\theta$, but both its conditional states are $I/2$. No measurement on that memory distinguishes the labels. If its purifier is also accessible, the joint conditional pure states again have overlap $\cos\theta$ and trace distance $\sin\theta$.*

Taking $\cos\theta = e^{-tD}$ matches any positive two-sector rate at a fixed time. This example concerns a reduced channel, rather than claiming equivalence of complete multitime processes. It locates the missing information precisely: the specified environment port excludes a purification. Decoherence alone therefore does not fix readable records, in agreement with Paper B's leakage relation [25].

The metrics (4) and (5) describe response to noise and environmental discrimination. The dynamics has no sector population transport: $\mathrm{Tr}\, P_a\Phi_t(\rho) = \mathrm{Tr}\, P_a\rho$. Neither metric is consequently a propagation distance in this model. Establishing a spatial interpretation needs another observable relation, addressed in Section 7.

5 Entropy as a reconstruction test

A reconstruction may use entropies rather than coherent response data. Those entropies can distinguish channels only to a limited extent. The exact outputs of distinct channels agree on a measure-zero set of pure inputs, but entropy separation needs an additional nondegeneracy condition (Theorem B.4 in Appendix B). For qubits the condition can be decided exactly. The following results constrain any entropy-based dictionary, including those that retain the same generated algebra.

5.1 Qubit reductions: the criterion

A qubit channel acts on Bloch vectors as an affine map

$$\Phi_i:\ r \mapsto M_i r + t_i, \qquad i = 1, 2,$$

with M_i a real 3×3 matrix and $t_i \in \mathbb{R}^3$, subject to the complete-positivity conditions [15]. Put

$$A = M_1^T M_1 - M_2^T M_2, \qquad b = M_1^T t_1 - M_2^T t_2, \qquad c = \|t_1\|^2 - \|t_2\|^2.$$

The difference of the squared output Bloch lengths is

$$q(r) = \|M_1 r + t_1\|^2 - \|M_2 r + t_2\|^2 = r^T A r + 2b^T r + c. \tag{6}$$

The entropy of a qubit state is a strictly decreasing function of its Bloch length: $S = h\big((1 + \|s\|)/2\big)$ with h the binary entropy in nats.

Theorem 5.1 (Equal entropy on all pure inputs)**.** *Φ_1 and Φ_2 give equal output entropy on every pure input if and only if there is a real number α with*

$$A = \alpha I, \qquad b = 0, \qquad c = -\alpha. \tag{7}$$

Proof. Equal entropy on pure inputs means $q(r) = 0$ on the unit sphere. Comparing $q(r)$ and $q(-r)$ gives $b^T r = 0$ for all unit r, so $b = 0$. A real symmetric quadratic form that is constant on the unit sphere is a multiple of the identity, so $A = \alpha I$ and then $c = -\alpha$. Conversely, (7) gives $q(r) = \alpha(\|r\|^2 - 1)$, which vanishes on the sphere. No rank condition on the outputs is needed, and pure outputs on the boundary are included. □

Together with positivity of the Choi matrices, (7) decides membership for any given pair. It is a decision rule, not yet a classification of all solutions into physically distinct normal forms.

5.2 The two branches

Proposition 5.2 (Zero branch)**.** *Let $X_i = [M_i \mid t_i]$ be the 3×4 matrix with t_i appended. If $\alpha = 0$, then $X_2 = OX_1$ for an orthogonal $O \in O(3)$. If $\operatorname{rank} X_1 \leq 2$, O can be chosen with $\det O = +1$, so Φ_2 is Φ_1 followed by a unitary conjugation of the output. If $\operatorname{rank} X_1 = 3$, O is unique; $\det O = +1$ again gives a unitary conjugation, and $\det O = -1$ gives a unitary conjugation composed with transposition.*

Proof. For $\alpha = 0$, (7) says exactly $X_1^T X_1 = X_2^T X_2$. Equal Gram matrices define an isometry between the column spaces of X_1 and X_2, which extends to an orthogonal map of $\mathbb{R}^3$. If the column space has dimension at most two, the extension can be composed with a reflection in an unused direction to make the determinant positive. Rotations of the Bloch ball are unitary conjugations, and the reflection $\operatorname{diag}(1, -1, 1)$ is transposition in the standard basis. □

Transposition is not completely positive, but both endpoints of a transposed pair can still be channels.

Example 5.3 (Equal entropies everywhere, no simulation either way)**.** *Take $M_1 = pI$, $M_2 = p\operatorname{diag}(1, -1, 1)$ and $t_1 = t_2 = 0$. The first map is a depolarizing channel for $0 \leq p \leq 1$; by the conditions for unital qubit channels [16], the second is completely positive exactly when $0 \leq p \leq 1/3$. They give equal output entropy on every input, pure or mixed. For $0 < p \leq 1/3$ neither is a post-processing of the other. Indeed both are invertible on the Bloch ball, so a post-processing Λ with $\Lambda \circ \Phi_1 = \Phi_2$ would have to act as $r \mapsto \operatorname{diag}(1, -1, 1)r$ on an open set and hence everywhere; that map is transposition, whose normalized Choi matrix has smallest eigenvalue $-1/2$. The same argument excludes the reverse direction.*

Identical entropy functions on the whole input space therefore do not imply operational equivalence, and the two channels are incomparable in the order of Proposition B.2.

Solutions with $\alpha \neq 0$ are not rare. Every pair of complementary qubit channels satisfies (7): if $V : \mathcal{H} \to \mathcal{K} \otimes \mathcal{E}$ is an isometry with $\mathcal{K}$ and $\mathcal{E}$ two-dimensional, $\Phi(\rho) = \operatorname{Tr}_{\mathcal{E}} V\rho V^\dagger$ and $\Phi^c(\rho) = \operatorname{Tr}_{\mathcal{K}} V\rho V^\dagger$, then for a pure input the joint output is pure and its two marginals have the same entropy. For amplitude damping with decay probability γ and its complementary channel, $\alpha = 1 - 2\gamma$, which is nonzero unless $\gamma = 1/2$. A family with full Choi rank is

$$\Phi_{a,b} : r \mapsto ar + b\hat{z}, \qquad \Phi_{b,a} : r \mapsto br + a\hat{z}, \qquad a, b > 0,\ a + b < 1.$$

$\Phi_{a,b}$ is a convex combination of the identity and the replacer with Bloch vector $b\hat{z}/(1-a)$, so it is CPTP, and likewise $\Phi_{b,a}$. Both give squared pure-output length $a^2 + b^2 + 2abr_z$, and $\alpha = a^2 - b^2$. For $a = .3$, $b = .2$ both normalized Choi matrices have smallest eigenvalue .125.

5.3 Mixed inputs expose the nonzero branch

Proposition 5.4 (Entropy gap on mixed inputs)**.** *For any two qubit channels and any input,*

$$|S_1(r) - S_2(r)| \geq \tfrac{1}{2}|q(r)|. \tag{8}$$

If (7) holds, then $q(r) = \alpha(\|r\|^2 - 1)$ on the whole Bloch ball, so

$$|S_1(r) - S_2(r)| \geq \frac{|\alpha|}{2}\big(1 - \|r\|^2\big), \tag{9}$$

and $\operatorname{sgn}(S_1 - S_2) = \operatorname{sgn}\alpha$ for every strictly mixed input.

Proof. Write the entropy as a function of the squared Bloch length, $f(x) = h\big((1+\sqrt{x})/2\big)$. Then $-f'(x) = \operatorname{artanh}(\sqrt{x})/(2\sqrt{x}) \geq \frac{1}{2}$, with the continuous extension at $x = 0$ and the integral limit at $x = 1$. The mean value theorem gives (8), and substituting (7) into (6) gives (9). A larger Bloch length means smaller entropy, which fixes the sign. □

When all pure inputs agree, a single strictly mixed input with equal entropies therefore forces $\alpha = 0$ and the orthogonal relation of Proposition 5.2. The argument uses only the one-to-one relation between qubit entropy and Bloch length, not a polynomial form of the entropy.

Finite data can test a violation of this criterion through purity. Corollary B.5 and Proposition B.7 give a nine-probe finite-copy test with explicit confidence bounds; Appendix B supplies their proofs and sample assumptions. The test does not establish exact equality from finitely many outcomes.

6 A physical correlation geometry

Here the vertices are supplied tensor sites, not the central sectors of Section 3. The construction tests a correlation-to-distance rule within that representation.

6.1 Mutual information as connectivity

To compute correlations, declare a subsystem or region representation of the channel output and its relation to the selected source algebra. The channel and state alone do not determine that representation. For output factors $A, B, \ldots$ in this representation, set

$$I_{\mathbf{c}}(A:B) = S_{\mathbf{c}}(A) + S_{\mathbf{c}}(B) - S_{\mathbf{c}}(AB),$$

with entropies computed after coarse-graining.

Definition 6.1 (Connectivity)**.** *For disjoint output factors A, B in the declared representation, the connectivity is $\mu_{\mathbf{c}}(A,B) := I_{\mathbf{c}}(A:B)$. It defines a weighted graph but is not a distance: large mutual information denotes strong correlation rather than large separation, and the metric axioms generally fail.*

A state-parameter metric, such as the SLD Fisher metric in Appendix E, is a different construction. Neither it nor the graph weights alone supply a Lorentzian metric.

6.2 A finite reconstruction rule

For a declared family of finite-dimensional subsystems, a positive connected correlation graph supports a shortest-path metric once an edge-length rule has been chosen. We use $\ell_e = 1/I_e$ on edges with $I_e > 0$. An edge with $I_e = 0$ is absent; distinct connected components then have infinite separation under this rule. Mutual information itself is not a distance. The rule assigns longer distances to weaker correlations, but its physical interpretation depends on the model.

A positive lower bound on the used mutual informations makes this rule stable. No assumption of exponential decay is needed for this finite statement. Exponential decay relative to a pre-existing locality structure can help construct a spatial interpretation [24, 23]; it would supply an additional input rather than derive that locality from the graph.

Proposition 6.2 (Stability of a graph metric)**.** *On the complete graph with v vertices, give the edge e the length $1/I_e$ and use the shortest-path metric d. If every $I_e \geq m > 0$ and the perturbed weights satisfy $|I'_e - I_e| \leq \varepsilon < m$, then*

$$\max_{i,j} |d'_{ij} - d_{ij}| \leq (v-1)\frac{\varepsilon}{m(m-\varepsilon)}.$$

Proof. For each edge, $|1/I'_e - 1/I_e| = |I_e - I'_e|/(I_e I'_e) \leq \varepsilon/(m(m-\varepsilon))$. A shortest path with positive weights uses at most $v-1$ edges. Comparing the new shortest path with the old lengths, and the old shortest path with the new lengths, bounds the difference in both directions. □

6.3 One state, one operation family, one rule

Proposition 6.3 (Non-isometric finite geometries)**.** *Fix the three-qubit GHZ state. Allow only local bit-flip channels*

$$\Phi_{\mathbf{p}} = \bigotimes_{i=1}^{3} \big[(1-p_i)\,\mathrm{id} + p_i\,\mathrm{Ad}_{X_i}\big], \qquad 0 < p_i < \tfrac{1}{2},$$

which are generated by local bit-flip semigroups with $p_i = (1-e^{-2\gamma_i t})/2$. Fix the reconstruction rule: on the complete graph of the three qubits, the edge $\{i, j\}$ has length $1/I_{ij}$, and distances are shortest-path lengths. Then

$$I_{ij} = \ln 2 - h\Big(\frac{1+c_{ij}}{2}\Big) > 0, \qquad c_{ij} = (1-2p_i)(1-2p_j),$$

and different flip probabilities can give metric spaces that are not isometric under any relabelling of the vertices:

flip probabilities	*mutual informations of the three edges*	*diameter*
$(.1, .1, .1)$	$(.221753694,\ .221753694,\ .221753694)$	4.509507747
$(.05, .1, .2)$	$(.288183695,\ .153870839,\ .120090263)$	8.327069751
$(.05, .1, .15)$	$(.288183695,\ .214256042,\ .166239219)$	6.015427679

The first and third rows have the same expected number of applied flips, $\sum_i p_i = .3$, when each channel is implemented by applying X_i with probability p_i.

Proof. Every two-qubit marginal of the GHZ state is an equal mixture of $|00\rangle$ and $|11\rangle$. Local flips leave both bits uniform and make them agree with probability $(1+c_{ij})/2$, so $I_{ij} = 2\ln 2 - \big(\ln 2 + h((1+c_{ij})/2)\big)$. All edge lengths are finite and positive, so the rule defines a metric. The diameters differ, so no relabelling makes the spaces isometric. The numbers were also computed from the full eight-dimensional density matrix with the channels applied and partial traces taken; the analytic formula agrees to better than 3×10^{-16}. □

All three rows use the same source, operation family, sites and reconstruction rule. The finite-time flip channels have full operator range, so their algebra label is identical even when their metrics differ. Rows one and three locate the difference in the allocation of the same expected flip count. That count is a resource in this implementation, rather than a hardware-independent energy price. Proposition 6.2 bounds the distance error by $2\varepsilon/(m(m-\varepsilon))$ when the mutual informations vary by at most $\varepsilon < m$ and m is their smallest true value.

6.4 The reduction is a local interaction

For each source qubit prepare a noise ancilla N_i in $|0\rangle$ and apply

$$U_i(\theta_i) = e^{-i\theta_i X_i \otimes Y_{N_i}}, \qquad U_i(\theta_i)|\psi\rangle|0\rangle = \cos\theta_i|\psi\rangle|0\rangle + \sin\theta_i X_i|\psi\rangle|1\rangle.$$

Discarding N_i realizes exactly the local flip channel with $p_i = \sin^2\theta_i$. One possible interaction Hamiltonian is $\hbar g_i X_i \otimes Y_{N_i}$ applied for θ_i/g_i. The angles, tensor factors and timing are supplied by the apparatus. They do not encode a target metric.

Next attach, at each site, a fragment of k_i fresh memory qubits. Use Paper E's collisions $P_{0,i} \otimes I + P_{1,i} \otimes R_{i,a}$, with real conditional overlaps $c_{i,a}$, so the fragment overlap is $r_i = \prod_{a=1}^{k_i} c_{i,a}$. These collisions dephase the source in the Z basis. Every two-site marginal of the noisy GHZ state is already diagonal, so its probabilities and mutual information are unchanged by recording. The post-recording source therefore has the same graph metric as before, although its three-site coherence may have changed. This is preservation of the data used by this rule, not preservation of the entire source state.

The noise gate ends before recording begins. This sequencing and its switching resources matter: the flip couples through X, while the recorder uses Z. Their common fixed algebra is only the scalars. Indeed the composed single-site source channel sends a Bloch vector to $(r_i x, r_i(1-2p_i)y, (1-2p_i)z)$; for $0 < p_i < 1/2$ and $0 < r_i < 1$ it has no fixed nonidentity direction, although its linear range is still the full matrix algebra. The record channel nevertheless reads the final pointer probabilities with nonzero resolution. This supplies an actual finite-access example beyond an exact stable-algebra label.

Tracing out the source after the collisions gives the exact joint fragment state

$$\rho_{\mathrm{rec}} = \sum_{\mathbf{z}} P_{\mathbf{p}}(\mathbf{z}) \bigotimes_{i=1}^{3} |e_{z_i}^{(i)}\rangle\langle e_{z_i}^{(i)}|, \tag{10}$$

where $z_i = \pm 1$ labels the local pointer and $P_{\mathbf{p}}$ is the actual diagonal distribution of the noisy GHZ state. Thus conditional product structure follows from the unitary couplings and fresh product memories; it is not an extra assumption about independent answers. Each site has equal pointer priors.

For $r_i < 1$, the equal-prior Helstrom measurement on fragment i has a symmetric binary error probability $\epsilon_i = (1 - \sqrt{1 - r_i^2})/2$ [19, 26]. To see the symmetry directly, choose real vectors $e_0 = (1, 0)$ and $e_1 = (r, \sqrt{1 - r^2})$. The sign of $|e_0\rangle\langle e_0| - |e_1\rangle\langle e_1|$ has expectations $+\sqrt{1 - r^2}$ and $-\sqrt{1 - r^2}$ in the two states. Writing $Y_i = \pm 1$ for the actual outcome and $v_i = 1 - 2\epsilon_i = \sqrt{1 - r_i^2}$ therefore gives

$$\mathbb{E}(Y_i \mid \mathbf{z}) = v_i z_i, \qquad \mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}. \tag{11}$$

Local measurement on (10) gives these identities even though the records are correlated before conditioning.

6.5 Finite records certify the geometric difference

Let $f(c) = \ln 2 - h((1 + c)/2)$, $-1 \le c \le 1$. Consider K noise configurations, with independently prepared source states and fresh ancillas in each run. For every configuration collect n triples (Y_1, Y_2, Y_3) from the local measurements above. Calibration supplies estimates $\widehat{v}_i \in [0, 1]$ with $\widehat{v}_i \ge \nu > 0$ and $|\widehat{v}_i - v_i| \le \xi$. The estimate

$$\widehat{c}_{ij} = \mathrm{clip}_{[-1,1]} \frac{n^{-1} \sum_{s=1}^{n} Y_i^{(s)} Y_j^{(s)}}{\widehat{v}_i \widehat{v}_j}$$

uses the measurement outcomes, not hidden pointer values.

Theorem 6.4 (Record-to-metric confidence certificate)**.** *Suppose the calibration event has failure probability at most δ_{cal}, and the record law is* (10) *with the stipulated local measurements. For every $\tau > 0$, with probability at least $1 - \delta_{\mathrm{cal}} - 6Ke^{-n\tau^2/2}$, all $3K$ correlations satisfy*

$$|\widehat{c}_{ij} - c_{ij}| \le e_c := \frac{\tau + 2\xi}{\nu^2}. \tag{12}$$

Intersect each interval $[\widehat{c}_{ij} - e_c, \widehat{c}_{ij} + e_c]$ with $[-1, 1]$ and minimize and maximize f on it, obtaining I_{ij}^-, I_{ij}^+. If every $I_{ij}^- > 0$, form shortest-path metrics d^- and d^+ from edge lengths $1/I_{ij}^+$ and $1/I_{ij}^-$. The true metric obeys $d^- \le d \le d^+$ entrywise, and its diameter lies in $[\operatorname{diam} d^-, \operatorname{diam} d^+]$. Disjoint diameter intervals for two configurations certify non-isometry on the same confidence event.

For a sample-size guarantee, if true and estimated correlations obey $|c_{ij}|, |\widehat{c}_{ij}| \le b < 1$, set $e_I = \operatorname{artanh}(b) e_c$. When $I_{ij} \ge \mu > e_I$,

$$\|\widehat{d} - d\|_\infty \le R := \frac{2e_I}{\mu(\mu - e_I)}. \tag{13}$$

For two configurations with true diameter gap $\Delta > 4R$, the intervals $[\operatorname{diam} \widehat{d} - R, \operatorname{diam} \widehat{d} + R]$ are disjoint on that event.

The proof and full calibration budgets are in Appendix C. For example, the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$ with fragment overlap .2 can be separated at confidence .95 using 1 217 920 runs per configuration when their readout visibilities are specified. Unknown visibilities require additional reference preparations, included in the appendix. Equal expected flip count does not imply equal total energy or a free reset.

6.6 Limits of record inversion

At $r_i = 1$ the record states coincide and $v_i = 0$: algebraic inversion of their span supplies no source information. Even for $v_i > 0$ the error grows as the inverse product visibility. A narrow effect set can span the pointer algebra while requiring too many trials for the available budget.

Calibration of individual error rates is also insufficient when the readers share an unmodelled error. Let a common random sign B multiply all otherwise ideal outcomes, with $\mathbb{E}B = v$. Each local reference test then reports visibility v, while the pair products are unchanged, since $B^2 = 1$. Dividing by v^2 overestimates the true correlations. A common inversion has $v = -1$ and is invisible to agreement. Joint reference tests, conditional error modelling or a verified local apparatus are needed to exclude such contamination. The confidence certificate is conditional on the joint record law, rather than an inference of that law from single-site calibration.

A spacetime reconstruction must solve a further problem: finite measurements determine a profile within an error region, not an exact profile. The relevant dictionary must be stable on that region and must quotient its coordinate freedom. Entropy equality from Section 5 supplies a genuine obstruction, including operationally different channels with identical entropy profiles. More accurate readout cannot repair a dictionary's failure to distinguish such profiles.

A coordinate change preserves the reconstructed geometry up to its coordinate equivalence. Changing a physical reduction can instead change its measured correlations. Algebraic difference alone does not prove this distinction: it needs a separating dictionary or an explicit invariant such as the diameter in Proposition 6.3.

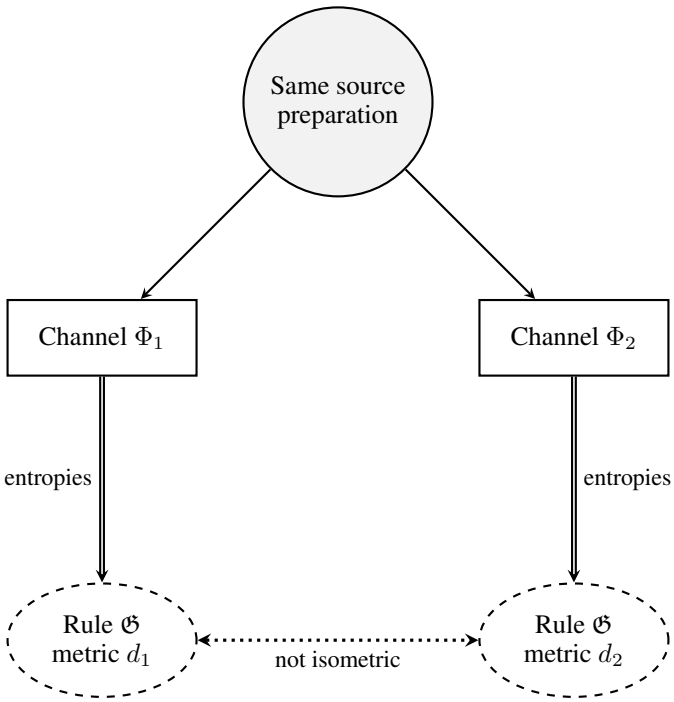


Figure 1: One global state is reduced by two channels from the same allowed family. The same reconstruction rule turns their entropy data into metrics. The tensor sites and the rule are fixed inputs. Proposition 6.3 exhibits different diameters; the process-selected sector geometry is a separate construction.

7 From relations to spatial structure

7.1 What the different geometries compare

The process-selected sectors and the GHZ sites answer different questions. The first are central atoms of a persistent algebra; the second are tensor factors supplied to a correlation protocol. There is no canonical identification between these two sets. In the rate geometry the channel determines a relation independently of the input state. In the correlation geometry the state, its reduction, the site representation and the reconstruction rule all matter.

A spatial interpretation needs evidence linking these relations to propagation or intervention. Pure dephasing supplies none of the required sector transport. A weak Hamiltonian connecting different sectors is a possible extension, but degenerate blocks generally retain quantum states that prevent a closed scalar equation for their populations. Paper B's protected Bell-pair interaction acts inside a joint noise sector and does not supply transitions between central labels [25]. Deriving a transfer law would require its own approximation, error bound and actual intervention ports.

If a reversible transfer process were identified through those ports, its stationary currents and responses could fix a conductance form and operational resistance distances. This would give a distance rule through a stated physical task, as (5) does for discrimination. An arbitrary support of the added Hamiltonian can still encode an arbitrary interaction graph. Inferring spatial locality, rather than inserting it into that support, remains the stronger problem.

For a precisely specified response candidate, suppose calibrated experiments identify a connected reversible jump generator with rates q_{ab} from a to b and stationary weights $\pi_a > 0$. The response fixes

$$\mathcal{E}(f,f) = \frac{1}{2}\sum_{a,b} \pi_a q_{ab}(f_b - f_a)^2, \qquad \Gamma(f)(a) = \frac{1}{2}\sum_{b} q_{ab}(f_b - f_a)^2.$$

A possible task convention fixes a reference rate $r_0 > 0$, with units time^{-1}, and bounds local quadratic variation:

$$d_{\Gamma,r_0}(a,b) = \sup\{f_a - f_b : \Gamma(f)(c) \le r_0 \text{ for all } c\}, \qquad f \in \mathbb{R}^V. \tag{14}$$

On a finite connected graph this is a finite separating metric: edgewise gradient bounds make every difference bounded, small multiples of separating functions are feasible, and the supremum gives the triangle inequality. For dimensionless

f, this distance is dimensionless; its conversion to length requires a separate calibration. The reference rate is part of the task convention. It is defined here by this task convention; it is not identified with the effective resistance or asserted to be the sole intrinsic metric of a jump process. For diffusion forms the energy-measure analogue defines a standard intrinsic pseudometric, whose nondegeneracy and agreement with the topology need extra conditions [30].

7.2 A same-process locality conjecture

The world-structure proposal is that the process selecting persistent organization can also constrain the relations that become spatial. Selection, record access and propagation should be tested within one history, rather than supplied by incompatible models.

Conjecture 7.1 (Locality from process-selected relations)**.** *There are families of finite carriers, specified by noise and Hamiltonian operators without assigning their matrix entries to spatially local gates, for which interaction histories produce refining persistent block descriptions and calibrated transfer responses on those blocks. Under refinement these responses admit a common continuum limit described by a nondegenerate strongly local Dirichlet form. After calibration, the distances* (14) *converge with these descriptions to the nondegenerate intrinsic distance of that limit. This distance is unchanged by alternative descriptions preserving the response, stationary weights and reference rate.*

The conjecture asks for an existence mechanism in one microscopic family, not a conclusion for every dephasing channel. Its substantial requirement is the joint formation of stable organization and a local response limit without a spatial dictionary supplied to the reconstruction. The finite selection result supplies a persistent-block ingredient; Paper B's interactions forming protected relations show that preservation and formation can be distinguished experimentally. Neither establishes the proposed limit.

A proposed realization must supply the refining blocks, allowed preparations and readers, measured transfer kernels, limiting form and its intrinsic distance. It would fail if block dynamics did not close at the claimed resolution, if persistent long-range shortcuts prevented a local limit, or if response-preserving descriptions produced inequivalent limiting distances after common calibration. Its strongest competing explanation is that any observed locality was already encoded in the supplied coupling architecture. A useful realization must show which locality constraints follow from the history and which remain in that architecture.

7.3 Reconstruction and the series

In holography, extremal-surface areas provide a specific entropy dictionary [2, 4]; tensor-network distances use another [22]. To infer distinct spacetimes from separated entropy profiles, a dictionary $\mathfrak{G}$ must identify its equivalence relation and satisfy, for example,

$$\mathfrak{G}[S_1] \simeq_{\mathrm{diff}} \mathfrak{G}[S_2] \implies S_1 \simeq_{\mathfrak{G}} S_2.$$

This separation property is an input about that dictionary. Finite readings also require its stability over a confidence region. The finite model in Section 6 proves separation under its declared rule rather than assuming a spacetime dictionary. Imprint's cross-scale analysis asks the related question of which readable relations survive a change of resolution and at what cost [27].

Conjecture 7.2 (Finite permissions approaching geometric access)**.** *In a source family with an independently justified local reconstruction dictionary, histories that enlarge the available record fragments and calibrate their readout can make the inferred geometry converge while the generated observable algebra remains constant. The required growth of visibility, preparation count and dictionary stability bounds must jointly dominate sampling and model error.*

The finite certificate supplies the mechanism and the costs of this candidate. For E edges with visibility at least ν, error in the correlations scales as $\nu^{-2}\sqrt{\ln(2E/\delta)/n}$ before calibration error. If reconstructed edge mutual informations have lower bound μ, distance error carries another factor of order μ^{-2} times the number of edges in a shortest path. Consequently, a growing graph or an approaching zero-information edge can defeat convergence even while all fixed-algebra labels are unchanged. The conjecture would fail in a proposed model if its dictionary is unstable, if common error survives calibration, or if its available budget cannot overcome these factors. This is a route from finite permissions to geometric access, not an assumption that a continuum or its dynamics is already generated.

The spatial/eigenmode example in Appendix E illustrates dependence of connectivity on factorization. Unlike Proposition 6.3, its empty eigenmode graph has no finite reconstructed paths.

8 Conclusion

A physical process can select an effective organization before spatial sites have been assigned. In the commuting-noise class, the organization is a direct sum fixed by the dynamics; its central sectors and cross-sector coherence rates are intrinsic to the channel. A finite stability window can select persistent blocks, and allowed coherent probes identify their rate relation.

What is selected must still be distinguished from what can be read. Equal system channels can leave different readable records at a restricted environmental port. Equal entropy functions can hide operationally incomparable reductions. The qubit criterion and finite-copy tests make those obstructions quantitative.

The GHZ protocol carries a separate correlation reconstruction through actual interactions and finite records. Changing the allocation of a fixed expected flip count changes its certified metric. Its tensor sites and distance rule remain specified inputs. Turning process-selected organization into spatial geometry requires a common account of persistence and actual transport. Conjecture 7.1 states that remaining mechanism as a testable target within the framework.

A Proof of the intrinsic window certificate

On a cluster C, choose a centre c of its smallest enclosing ball and put $B_k = \sum_{a\in C}(u_{a,k} - c_k)P_a$. Scalar shifts do not alter double commutators. The restricted generator is

$$\mathcal{L}_C(X) = 2\Psi(X) - \{R, X\}, \qquad \Psi(X) = \sum_k B_k X B_k, \qquad R = \sum_k B_k^2.$$

Here Ψ is completely positive and $\|\Psi\| = \|\Psi(I)\| = \|R\| \le r_C^2$. Thus $\|\mathcal{L}_C\| \le 4r_C^2$. The semigroup is a contraction on each invariant block, so Duhamel's formula gives $\|\Phi_t(X) - X\| \le t\kappa_{\mathrm{in}}\|X\|$ on $\mathcal{B}_\pi$.

A strictly containing unital algebra has a nonzero entry between two different Q_C blocks. Multiplication on both sides by matrix units already in $\mathcal{B}_\pi$ isolates a cross-cluster matrix unit E_{xy}. Its loss at time T is $1-e^{-TD_{ab}} \ge 1-e^{-T\Delta_{\mathrm{out}}} > \varepsilon$. Such a containing algebra cannot be stable.

Conversely, any stable block algebra in a common joint eigenbasis can put x and y together only if $1 - e^{-TD_{ab}} \le \varepsilon$. The separation in (3) forbids this across two clusters, so every stable block of that form lies in one Q_C block. Two certified partitions therefore refine each other and coincide.

The radii are determined by the Euclidean configuration of D. A computable sufficient upper bound uses the equal-weight centroid:

$$\|u_a - \bar{u}_C\|^2 = \frac{1}{|C|}\sum_{b\in C} D_{ab} - \frac{1}{2|C|^2}\sum_{b,c\in C} D_{bc}.$$

Replacing r_C^2 by its largest value over $a \in C$ retains the certificate. It need not be optimal.

B Channel order and entropy tests

B.1 Refinement

Definition B.1 (Refinement)**.** *For coarse-graining structures $\mathbf{c}_1, \mathbf{c}_2$ write $\mathbf{c}_1 \succeq \mathbf{c}_2$ if there is a CPTP map Λ from states on the output space of $\Phi_{\mathbf{c}_1}$ to states on the output space of $\Phi_{\mathbf{c}_2}$ with $\Phi_{\mathbf{c}_2} = \Lambda \circ \Phi_{\mathbf{c}_1}$. Then $\mathbf{c}_2$ is a further coarse-graining of $\mathbf{c}_1$.*

Proposition B.2 (Order on classes)**.** *The relation $\succeq$ is reflexive and transitive but not antisymmetric. Writing $\Phi \sim \Psi$ when each is a post-processing of the other, $\succeq$ induces a partial order on equivalence classes.*

Proof. Reflexivity uses $\Lambda = \mathrm{id}$, transitivity uses composition of channels. Two different replacer channels, which output fixed states $\omega_1 \neq \omega_2$, simulate each other by a further replacement, so antisymmetry fails. On classes the relation is well defined because composing with a simulating channel does not change membership, and antisymmetry holds by construction. □

If the allowed post-processings are restricted, for example to local, energy-bounded or specified control operations, the restricted class must be closed under composition before the same quotient can be formed. The comparison of channels by approximate post-processing has an operational characterization through success probabilities of discrimination tasks [14]; we use only exact post-processing.

B.2 Mutual information under refinement

Proposition B.3 (Local and global post-processing)**.** *For channels Λ_A, Λ_B acting on A and B separately,*

$$I(A' : B')_{(\Lambda_A \otimes \Lambda_B)(\rho)} \leq I(A : B)_\rho.$$

A global CPTP post-processing can increase mutual information: the channel that replaces every two-qubit input by a Bell state takes $|00\rangle$, with $I = 0$, to a state with $I = 2 \ln 2$.

Proof. $I(A : B)_\rho = D(\rho_{AB} \| \rho_A \otimes \rho_B)$, and $\Lambda_A \otimes \Lambda_B$ maps $\rho_A \otimes \rho_B$ to the product of the new marginals, so data processing for relative entropy gives the inequality. The Bell example is a direct computation. □

A requirement that mutual information decrease under refinement is therefore a theorem for local refinements and false for global ones. A geometric reading of mutual information must say which refinements it allows.

B.3 Generic separation

Theorem B.4 (Generic separation for a fixed pair of channels)**.** *Let $\mathcal{H}$ and $\mathcal{K} = \mathcal{K}_A \otimes \mathcal{K}_{A^c}$ be finite dimensional, and let $\Phi_1 \neq \Phi_2$ be CPTP maps from states on $\mathcal{H}$ to states on $\mathcal{K}$. The pure inputs on which Φ_1 and Φ_2 agree form a proper real-algebraic subset of Haar measure zero. If, in addition, the real-analytic function*

$$f_A(\Psi) := S(\mathrm{Tr}_{A^c}\, \Phi_1(|\Psi\rangle\langle\Psi|)) - S(\mathrm{Tr}_{A^c}\, \Phi_2(|\Psi\rangle\langle\Psi|))$$

is not identically zero on each connected component under consideration of the open set where both reduced outputs have full rank, then $f_A(\Psi) \neq 0$ for generic inputs in those components.

Proof. Pure-state projectors span the Hermitian operators, so $\Phi_1 - \Phi_2$ cannot vanish on all of them. The equality condition is polynomial in the real and imaginary parts of $|\Psi\rangle$, so its solution set is a proper real-algebraic subset and has Haar measure zero. On the full-rank domain the von Neumann entropy, and hence f_A, is real analytic, and a nonzero real-analytic function has a zero set of measure zero. □

The nondegeneracy hypothesis cannot be dropped: channels that differ by a unitary acting on A alone can give different outputs with identical entropies on every input. Channel inequivalence alone therefore does not yield an entropic or geometric difference. Section 5 gives the exact qubit criterion; the tests below turn its violation into a finite measurement problem.

B.4 Probes and random inputs

Corollary B.5 (Nine pure probes and one mixed probe)**.** *Let the probe set consist of the nine Bloch vectors*

$$\pm e_1,\ \pm e_2,\ \pm e_3,\ (e_1 + e_2)/\sqrt{2},\ (e_1 + e_3)/\sqrt{2},\ (e_2 + e_3)/\sqrt{2}.$$

If the two channels give equal output entropies on these nine inputs, they give equal output entropies on every pure input. If they also agree at $r = 0$, they agree on every input.

Proof. The six axial probes give $b_i = 0$ and $A_{ii} + c = 0$; the three diagonal probes then give $A_{ij} = 0$ for $i \neq j$. This is (7). At $r = 0$, $q = c = -\alpha$, so agreement there gives $\alpha = 0$. Then q vanishes on the entire Bloch ball, hence the output lengths and entropies agree everywhere. □

The corollary concerns exact entropy values on nine inputs, not nine measurements. On the unit sphere $q(r) = r^T A_0 r + 2b^T r + d$ with $\alpha = \frac{1}{3}\,\mathrm{tr}\, A$, $A_0 = A - \alpha I$ and $d = c + \alpha$, which is linear in nine real parameters.

Proposition B.6 (Random pure inputs)**.** *For Haar-random pure inputs,*

$$Q := \mathbb{E}\, q(r)^2 = \frac{2}{15}\|A_0\|_F^2 + \frac{4}{3}\|b\|^2 + d^2. \tag{15}$$

$Q = 0$ exactly when (7) holds. If $Q > 0$, then $\mathbb{E}(S_1 - S_2)^2 \geq Q/4$, and for $0 < \tau < 1$

$$\Pr\Big\{|S_1 - S_2| > \tfrac{1}{2}\sqrt{\tau Q}\Big\} \geq \Pr\{q^2 > \tau Q\} \geq \frac{(1-\tau)Q}{1-\tau Q}. \tag{16}$$

Proof. Use $\mathbb{E} r_i r_j = \delta_{ij}/3$ and $\mathbb{E} r_i r_j r_k r_l = (\delta_{ij}\delta_{kl} + \delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk})/15$; odd moments and the cross terms with the traceless A_0 vanish. The second claim follows from (8). Both outputs are states, so $0 \le q^2 \le 1$; splitting $\mathbb{E} q^2 = Q$ at $q^2 = \tau Q$ gives $Q \le \tau Q(1-P) + P$ with $P = \Pr\{q^2 > \tau Q\}$, which is (16). □

With n independent Haar inputs, the probability of missing the difference at the level $\frac{1}{2}\sqrt{\tau Q}$ is at most $(1-p_\tau)^n$, where p_τ is the right side of (16). This assumes exact entropies; the next proposition removes that assumption. The average change of purity over pure inputs is also the basis of the unitarity measure of noise [17]; Eq. (15) concerns the difference between two channels rather than the coherence of one.

B.5 Finitely many copies

For a qubit, purity and entropy are both monotone functions of the Bloch length, so equal entropy is equivalent to equal purity, and purity can be measured directly. A SWAP test on two copies of a state σ returns the outcome 0 with probability $\frac{1}{2}\big(1 + \operatorname{Tr}\sigma^2\big)$ [18]. For a qubit with Bloch vector s this is $\frac{3}{4} + \frac{1}{4}\|s\|^2$. We assume that every channel use acts on a freshly prepared, known probe state, that the channels do not drift between uses, and that two output copies can be measured jointly.

Finite data cannot establish that the two entropy functions are exactly equal. They can separate equality from a violation of prescribed size, and they can bound how large a violation can be. Put

$$\kappa_9 = \frac{75 - 3\sqrt{465}}{8} = 1.28855\ldots.$$

Proposition B.7 (Finite-copy test and confidence bound)**.** *For each of the nine probes r_k of Corollary B.5, send $2m$ copies of the input through Φ_1, apply a SWAP test to each of the m output pairs, and do the same for Φ_2. Let D_k be the difference of the two observed frequencies of the outcome 0.*

1. *Given $\gamma > 0$, declare the channels different on pure inputs if $\max_k |D_k| > \gamma/8$. If (7) holds, a difference is declared with probability at most $18\, e^{-m\gamma^2/64}$. If $\max_k |q(r_k)| \ge \gamma$, the difference is missed with probability at most $2\, e^{-m\gamma^2/64}$.*
2. *$\sum_k q(r_k)^2 \ge \kappa_9\, Q$, and κ_9 is the largest constant with this property. Hence $Q \ge Q_0 > 0$ forces $\max_k |q(r_k)| \ge (\kappa_9 Q_0/9)^{1/2}$, and*
$$m \ge \frac{576\, \ln(18/\delta)}{\kappa_9\, Q_0}$$
output pairs per probe and channel keep both error probabilities below δ for the hypotheses $Q = 0$ and $Q \ge Q_0$.
3. *For every $t > 0$, with probability at least $1 - 18\, e^{-mt^2}$,*
$$Q \le \frac{16}{\kappa_9} \sum_{k=1}^{9} \big(|D_k| + t\big)^2.$$

Proof. The expected value of D_k is $q(r_k)/4$. D_k is a sum of $2m$ independent Bernoulli variables with coefficients $\pm 1/m$, so Hoeffding's inequality [20] gives $\Pr\{|D_k - \mathbb{E} D_k| \ge t\} \le 2e^{-mt^2}$. Under (7) every $\mathbb{E} D_k$ vanishes, and $t = \gamma/8$ with a union bound over the nine probes gives the first error bound. If $|q(r_k)| \ge \gamma$ for some k, then $|\mathbb{E} D_k| \ge \gamma/4$, and missing it requires $|D_k - \mathbb{E} D_k| \ge \gamma/8$, which gives the second.

The smallest generalized eigenvalue of the two quadratic forms $\sum_k q(r_k)^2$ and Q is κ_9, as computed in Appendix B.6. The largest of nine terms is at least their mean; setting $\gamma^2 = \kappa_9 Q_0/9$ and requiring $18e^{-m\gamma^2/64} \le \delta$ gives item 2. For item 3, on the event that all nine deviations are at most t, which fails with probability at most $18e^{-mt^2}$, we have $|q(r_k)| = 4|\mathbb{E} D_k| \le 4(|D_k| + t)$, and $Q \le \kappa_9^{-1} \sum_k q(r_k)^2$. □

If the pure-input criterion is known to hold, the same argument at the mixed probe $r = 0$ tests the size of α: the expected difference there is $-\alpha/4$, and $m \ge 64 \ln(2/\delta)/\alpha_0^2$ pairs per channel separate $\alpha = 0$ from $|\alpha| \ge \alpha_0$ with error at most δ in each direction. The pure test uses $18m$ channel uses per channel, $36m$ in all. The constants are conservative. Beyond qubits, purity and entropy are different functionals and this reduction no longer applies.

B.6 The nine-probe constant

On the unit sphere, $\sum_k q(r_k)^2$ and Q are quadratic forms in the nine parameters (A_0, b, d), and both are positive definite; the first because the nine probe values determine the parameters, as in the proof of Corollary B.5. Their generalized eigenvalues do not depend on the basis chosen for the parameters, and a direct computation gives the characteristic polynomial, up to a constant factor,

$$(\lambda - 6)\,(4\lambda^2 - 75\lambda + 90)\,(8\lambda^3 - 225\lambda^2 + 1620\lambda - 2700)^2.$$

Its roots are 1.2886, 2.3962, 6, 7.9001, 17.4614 and 17.8287, the three roots of the cubic each counted twice. The smallest is the smaller root of the quadratic factor, κ_9, and it is attained.

C Record certificate: proof and calibration budgets

C.1 Proof of Theorem 6.4

Proof. Each $Y_iY_j \in [-1, 1]$ has expectation $v_iv_jc_{ij}$. Hoeffding's inequality gives a two-sided tail $2e^{-n\tau^2/2}$; a union bound covers the $3K$ edge estimates. The three edges within a run need not be independent. On the calibration event, $|v_iv_j - \widehat{v}_i\widehat{v}_j| \le 2\xi$, by telescoping and the bounds $v_i, \widehat{v}_i \le 1$. Dividing by $\widehat{v}_i\widehat{v}_j \ge \nu^2$, using $|c_{ij}| \le 1$, and clipping cannot increase the error, proving (12). The interval construction contains the true mutual informations. Shortest-path lengths and their diameter are monotone in positive edge lengths, proving the metric enclosure. Finally $f'(c) = \operatorname{artanh} c$, so $|\widehat{I} - I| \le e_I$. Proposition 6.2, with three vertices, gives (13). Each estimated diameter differs from its true value by at most R, so the estimated gap is at least $\Delta - 2R$; two radius-R intervals are disjoint when $\Delta > 4R$. □

C.2 Specified and empirically calibrated readers

For the equal-flip-count configurations $(.1, .1, .1)$ and $(.05, .1, .15)$, take one recording qubit per site with overlap $r_i = .2$. Then $v_i = \sqrt{.96}$ and the correlations lie between $.56$ and $.72$. With exactly specified v_i, use $\nu = \sqrt{.96}$, $\xi = 0$, $\tau = .003$, $b = .8$, $\mu = .16$ and $\delta = .05$. The choice

$$n = \left\lceil \frac{2\ln(12/\delta)}{\tau^2} \right\rceil = 1\,217\,920$$

runs per configuration gives $e_c = .003125$, $e_I = .0034332$ and $R < .275$. The true diameter gap is $1.505919931 > 4R$. Thus a finite record-only experiment separates the two geometries with confidence at least $.95$, under the stated source and apparatus model. The sample bound is conservative and does not claim optimal tomography.

A known unitary specification is not free empirical calibration. If the v_i are unknown, prepare reference sources in $Z = +1$ and apply the same record couplings and readers. Each reference outcome has mean v_i. For N independent reference runs per site, clipping the empirical mean to $[0, 1]$ yields

$$\Pr\{\max_i |\widehat{v}_i - v_i| > \xi\} \le 6e^{-N\xi^2/2}$$

for three shared readers; separately calibrated readers for K configurations replace 6 by $6K$. Thus $N \ge \lceil 2\ln(6/\delta_{\mathrm{cal}})/\xi^2 \rceil$ per site suffices. Reject calibrations below ν. For the same two geometries, take $\xi = .001$, $\delta_{\mathrm{cal}} = .025$, $\nu = .978$ and $\tau = .002$. Then $N = 10\,961\,278$ reference runs per site and $n = 3\,086\,894$ main runs per configuration give total failure at most $.05$, $e_c < .004182$, $R < .36955$ and $4R < 1.4782 < \Delta$. On the calibration event the true $v_i = \sqrt{.96}$ ensures $\widehat{v}_i > \nu$; the correlation bound also stays inside $b = .8$. The reference budget is therefore $3N$ additional preparations and readouts, not part of the $2n$ main runs. Equation (12) includes their residual error. Prepared reference pointers, their basis labels and the stability of the reader between calibration and use are additional physical resources.

The comparison uses $2n$ preparations of the same GHZ source, $6n$ fresh noise ancillas, $6n$ fresh recording qubits, $6n$ local binary readouts and communication of the outcomes to a comparator, plus any calibration runs. The expected noise-flip count is $.3$ per source in both configurations; their recording and readout schedule is identical. Preparation, switching work, storage, communication and renewal do not vanish because the counts agree. No heat or total-energy equality follows from the flip count, and this finite protocol supplies no reservoir reset.

D No canonical tensor factorization

For a Hilbert space $\mathcal{H}_{\text{total}}$ of dimension $d_A d_B$ with $d_A, d_B > 1$, there is no unique way to write $\mathcal{H}_{\text{total}} = \mathcal{H}_A \otimes \mathcal{H}_B$ without further physical input. Given any pure state $|\Psi_U\rangle$, choose a unitary U with $U(|0\rangle_A \otimes |0\rangle_B) = |\Psi_U\rangle$ and pull the reference factors through U; in the resulting factorization $|\Psi_U\rangle$ is a product. More generally, observables can be chosen so that a given pure state has any prescribed Schmidt decomposition [8].

What survives is typicality for a family of factorizations fixed in advance.

Theorem D.1 (Entanglement typicality for a specified family)**.** *Let $\mathcal{H}_{\text{total}}$ be finite dimensional and let $\mathfrak{F} = \{\mathbf{f}_1, \ldots, \mathbf{f}_m\}$ be a finite family of nontrivial tensor factorizations selected by a physical rule that does not depend on the sampled state. Write $\mathcal{H}_{\text{total}} = \mathcal{H}_{A_j} \otimes \mathcal{H}_{B_j}$ with $d_{A_j} \le d_{B_j}$. For $\delta > 0$ let*

$$p_j(\delta) := \Pr_{\Psi \sim \text{Haar}} \left[S(\rho_{A_j}) < \log d_{A_j} - \delta \right].$$

Then

$$\Pr\left[S(\rho_{A_j}) \ge \log d_{A_j} - \delta \text{ for every } j \right] \ge 1 - \sum_{j=1}^{m} p_j(\delta),$$

and standard concentration estimates make each $p_j(\delta)$ small at large dimension under the usual conditions on the dimension ratio.

Proof. For each fixed $\mathbf{f}_j$, Page's formula gives [10]

$$\mathbb{E}[S(\rho_{A_j})] = \sum_{k=d_{B_j}+1}^{d_{A_j} d_{B_j}} \frac{1}{k} - \frac{d_{A_j} - 1}{2 d_{B_j}}, \tag{17}$$

and concentration of measure bounds the lower tail [11]; this tail is $p_j(\delta)$. The union bound over the finite, preselected family proves the inequality. □

The size of the statement matters. For a balanced split $d_A = d_B$, Eq. (17) gives a mean entropy of $\log d_A - \frac{1}{2} + o(1)$, so typical states are close to maximally entangled but keep a gap of about half a nat. Drawing the factorization at random with the state fixed is equivalent to drawing the state at random with the factorization fixed. Hence for a fixed state, product factorizations form a set of measure zero, while factorizations with small but nonzero entanglement have small positive measure.

Theorem D.1 says nothing about all factorizations and does not pick out any member of $\mathfrak{F}$. Which family is admissible, and why, is a question of quantum mereology [12, 13]. Earlier work showed that subsystem structure is induced by the accessible observables rather than given by the Hilbert space [5, 6]. The channel-level results in the main text retain the accessible-observable view while distinguishing its algebra from its resolution.

E Other connectivity constructions

When a metric on a parameter space of coarse-grained states is wanted, the finite-dimensional construction is the quantum Fisher metric. For a smooth faithful family $\lambda \mapsto \rho_{\mathbf{c}}(\lambda)$, define the symmetric logarithmic derivatives by $\partial_a \rho_{\mathbf{c}} = \frac{1}{2}(L_a \rho_{\mathbf{c}} + \rho_{\mathbf{c}} L_a)$ and

$$g^{\text{QF}}_{ab}(\lambda) := \frac{1}{2} \operatorname{Tr}[\rho_{\mathbf{c}}(\lambda)\{L_a, L_b\}]. \tag{18}$$

This is a positive-semidefinite Riemannian metric on identifiable parameter directions [21]. It is not a Lorentzian spacetime metric, and the pairwise numbers $\mu_{\mathbf{c}}(A, B)$ do not determine it without an additional map from graph or state parameters to geometry.

E.1 Spatial and eigenmode descriptions of a free-fermion chain

Example E.1 (A dimerized free-fermion chain)**.** *Take a periodic chain of $2L$ sites with alternating hopping amplitudes $t_1 = 1$ and $t_2 = 1/2$,*

$$H = -\sum_{c=0}^{L-1} \left(t_1 \, c^\dagger_{2c} c_{2c+1} + t_2 \, c^\dagger_{2c+1} c_{2c+2} + \text{h.c.} \right),$$

at half filling. The single-particle spectrum has two bands separated by the gap $2|t_1 - t_2| = 1$*, and the ground state fills the lower band,* $|\Psi_0\rangle = \prod_k a^\dagger_{k,-}|0\rangle$*, where* $a_{k,-}$ *are the lower-band Bloch modes. In the description by occupation numbers of the band-resolved Bloch modes the ground state is a product of filled and empty modes: every set of modes has a pure reduced state, disjoint sets have zero mutual information, and the connectivity graph is empty. In the spatial description blocks of sites are correlated. For* $L = 60$ *and blocks of four sites, the block entropy saturates at* $.3554$ *as the block grows, as the area law for gapped one-dimensional systems requires [24, 23]. The mutual information between two blocks falls from* $.333$ *for adjacent blocks to* 9.0×10^{-5} *at a separation of ten sites and* 5.2×10^{-8} *at twenty, close to* $e^{-r/1.32}$*. The connectivity graph is dominated by edges between neighbouring blocks.*

The second description has to be chosen with care. Plain Fourier modes are not enough when the Hamiltonian is not translation invariant: for two disconnected dimers on four sites, $H = -(c_0^\dagger c_1 + c_2^\dagger c_3 + \text{h.c.})$ at half filling, the gap at fixed particle number is 2, yet the Fourier modes $k = \pi/2$ and $3\pi/2$ share one particle in superposition and have mutual information $2\ln 2$. A translation-invariant single-band chain with particle conservation is gapless unless its band is full or empty, so a gapped example whose ground state is a product of mode occupations needs band-resolved modes, as above, or in general the single-particle eigenmodes of the Hamiltonian.

The example shows that connectivity depends on the factorization: the spatial description has short-range edges and the eigenmode description has none. It does not exhibit two reconstructed geometries, since its zero-connectivity graph supplies no finite paths under the chosen rule; Proposition 6.3 is the example in which both descriptions produce geometries.

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COUPLING, RECORDS, AND STABLE RELATIONS: FORMATION, PROTECTED CONTENT AND THE LIFETIME OF A GEOMETRY

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ABSTRACT

A coupling can preserve an inherited relation without having formed it. We ask how interactions select stable structure and how records make that structure usable. Commuting Hermitian noise selects a direct-sum algebra on a given carrier; its central sectors do not require an independently chosen spatial basis. A separated range of decay rates further selects a unique block partition over a finite window within the stated block class. We distinguish this selection from the formation of correlations and from learning the operations that use them. A collective recorder creates a flagged protected Bell branch from a product input, while a physical link creates a deterministic protected Bell pair across noise-disconnected components. Both retain the resources that produce the relation. Block-constant records preserve internal quantum content, whereas independent leakage imposes a sharp $R^{-1/2}$ symmetry threshold for R records. An exact entropy identity separates sector and within-sector contributions to a supplied correlation geometry. The GHZ example then exhibits a selected geometry, its finite plateau and its loss under symmetry breaking. Finite experimental capabilities remain continuous under perturbations even when the exact fixed algebra changes discontinuously. These results support a process-selected subsystem programme: an effective component must acquire reproducible operations as well as stable correlations. The carrier, available interactions and spatial interpretation remain distinct inputs to be tested rather than inferred from a noise graph alone.

1 Introduction

A stable distinction is a physical achievement. It must survive the interactions that would erase it, be readable through an available instrument, and last long enough to matter. An algebra of invariant observables describes one part of that achievement; it does not specify a reader, a preparation history or a set of available gates.

Paper A [1] shows why the distinction matters for geometry. Channels generating the same observable algebra can resolve different states and produce non-isometric finite correlation geometries. Its local bit-flip family has a Stinespring realization and a finite fragment-readout certificate. Here we ask a complementary question: what structure a coupling protects, how actual records and interactions form usable relations within that structure, and when those relations lose their support.

Commuting Hermitian noise gives an exact starting point. The fixed observables are the commutant of the coupling operators, familiar from einselection and noiseless codes [9, 10, 12, 13]. The infinite-time channel pinches onto this algebra. At finite times its operator range is full, so a protected block algebra is additional stability data rather than the range of the actual channel. A window and a tolerance give a unique block partition when intra-block and inter-block rates are separated (Theorem 4.2). The maximality statement concerns block algebras in the specified joint eigenbasis, not every rotated algebra or every useful finite task.

Recording adds a different restriction. A coupling constant on each block can record its label without learning its quantum content. Independent fragments that resolve two states within a block destroy their coherence according to a multiplicative visibility budget (Theorem 5.2). The centre and the blocks thus perform different jobs within one stable structure. They are not disjoint sources of geometry: classical sector weights can sustain mutual information even when every block is one-dimensional. Section 6 gives the exact decomposition needed to distinguish the contributions.

Selection alone also says too little about formation. Two disconnected collective-noise components can preserve a logical Bell pair shared at the start. Conversely, local independent processes cannot create cross-component correlations from independent inputs. Section 7 supplies the missing process distinction and two explicit formation mechanisms. One retains a collective measurement flag and all its branches; the other uses an actual cross-component Hamiltonian link. Both generate relations from product inputs, and both expose the interaction resources responsible for them. Foundation III [6] supplies a further mechanism in which acquired records select an encoder and decoder for the same pre-existing quantum content. Storage, formation and learned use then have separate, testable meanings.

The GHZ calculation in Section 8 shows how a coupling changes a finite correlation geometry and how weak symmetry breaking ends its plateau. For noncommuting perturbations, Section 4.1 bounds complete adaptive record laws and task risks without requiring a stable algebra. A transverse qubit exhibits a diverging record lifetime despite a scalar exact fixed algebra.

The Hilbert space and coupling permissions are supplied. The formation and GHZ experiments additionally specify site factors; the exact sector-selection result does not need a prior spatial factorization. Consciousness I [4] selects a Virasoro fixed net inside a given current-net representation; Foundation's hulls concern closure under declared operations. Neither result derives the ambient carrier. Our formation results address how relations and usable protected sectors arise on that carrier. Section 2 states the effective-subsystem question before the constructions. Section 10 examines its extension beyond commuting noise.

2 What has to form

The generation question has three physically different outputs. A coupling can select a direct-sum sector without supplying a tensor factorization. An interaction can create a relation between supplied components without teaching a reader how to use it. Acquiring a component also requires repeatable operations whose errors stay controlled over the intended task. These distinctions set the target of the constructions below.

Conjecture 2.1 (Process-selected effective subsystems)**.** *On an initially specified carrier, robust effective subsystems form through a history of interactions that both builds relations and makes a noncommuting set of internal operations reproducible across a finite task window. The relevant connectivity is that of the complete physical process, including recording ports and Hamiltonian controls, not the collective-noise hypergraph alone. Within a separated-rate regime, a learned encoding can turn the process's indistinguishable sectors into stable operational components; their boundaries change when the reproducible operations or their lifetime change.*

The proposal is tested by changing the preparation history, recording port or coupling while keeping the task fixed. A stable Bell pair present initially demonstrates preservation. A product preparation followed by the actual collective port or Hamiltonian of Section 7 demonstrates formation of a relation. A diagnostic record that selects a working encoder demonstrates learned use. None of these tests alone establishes a complete subsystem gate algebra. This is why the complete interaction history, rather than just the noise graph or the final state, is the candidate source of effective organization.

The mathematical starting point needs only a carrier and its specified couplings. Site factors are introduced later for correlation and formation experiments. The general tensor-factorization question is therefore kept separate from the exact selection of stable sectors.

3 Reductions selected by a coupling

3.1 Setting

Let $\mathcal{H}$ be a finite-dimensional system Hilbert space. The environment couples to the system through Hermitian operators $A_1, \ldots, A_K$ with $[A_k, A_l] = 0$, at rates $\gamma_k > 0$. In the Markovian limit the reduced dynamics is generated by

$$\mathcal{L}(\rho) = \sum_{k=1}^{K} \gamma_k \Big(A_k \rho A_k - \tfrac{1}{2}\{A_k^2, \rho\} \Big), \qquad \Phi_t = e^{t\mathcal{L}}. \tag{1}$$

A system Hamiltonian that commutes with every A_k only rotates states inside the blocks defined below; we work in the corresponding interaction picture and return to Hamiltonians that do not commute in Remark 3.3. Section 5 treats the same couplings in an exact unitary model with environmental fragments.

Choose an orthonormal basis $\{|x\rangle : x \in \mathcal{X}\}$ of joint eigenvectors, $A_k|x\rangle = a_k(x)|x\rangle$, and put

$$D(x,y) = \tfrac{1}{2}\sum_k \gamma_k\big(a_k(x) - a_k(y)\big)^2 = \|u(x) - u(y)\|^2, \qquad u(x) = \big(\sqrt{\gamma_k/2}\, a_k(x)\big)_{k=1}^{K} \in \mathbb{R}^K. \tag{2}$$

We call $u(\mathcal{X}) \subset \mathbb{R}^K$ the rate image of the basis; decoherence rates are squared Euclidean distances in it. Write $x \approx y$ when $a_k(x) = a_k(y)$ for every k, let π_∞ be the partition of $\mathcal{X}$ into $\approx$-classes a, and let P_a be the projection onto $\mathcal{H}_a = \operatorname{span}\{|x\rangle : x \in a\}$. For any partition π of $\mathcal{X}$ into classes C, put

$$\mathfrak{B}_\pi = \bigoplus_{C\in\pi} \mathcal{B}(\mathcal{H}_C), \qquad \mathbb{E}_\pi(\rho) = \sum_{C\in\pi} P_C \rho P_C.$$

$\mathfrak{B}_\pi$ is a unital $*$-algebra, $\mathbb{E}_\pi$ is a channel (the pinching), and $\mathbb{E}_\pi$ is its own dual, the conditional expectation onto $\mathfrak{B}_\pi$.

Proposition 3.1 (The reduction selected by a commuting coupling)**.** *1. Φ_t acts as a Schur multiplier in the eigenbasis: $\Phi_t(\rho)_{xy} = e^{-tD(x,y)}\rho_{xy}$, and $\Phi_t^* = \Phi_t$.*

2. *The observables fixed by every Φ_t^* form the algebra $\mathfrak{B}_{\pi_\infty} = \{A_1, \dots, A_K\}'$.*
3. *$\Phi_t \to \mathbb{E}_{\pi_\infty}$ as $t \to \infty$, and the algebra generated by the dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$. In the terminology of Paper A, $(\mathfrak{B}_{\pi_\infty}, \mathbb{E}_{\pi_\infty})$ is a minimally compatible coarse-graining structure.*
4. *For every finite t, Φ_t is invertible on the operator space, so the algebra generated by its dual range is $\mathcal{B}(\mathcal{H})$.*

Proof. (1) In the eigenbasis, $(A\rho A - \frac{1}{2}\{A^2, \rho\})_{xy} = (a(x)a(y) - \frac{1}{2}a(x)^2 - \frac{1}{2}a(y)^2)\rho_{xy} = -\frac{1}{2}(a(x) - a(y))^2\rho_{xy}$, so $\mathcal{L}$ multiplies the (x, y) entry by $-D(x, y)$. The multiplier is real and symmetric, hence $\Phi_t^* = \Phi_t$. (2) X is fixed for all t exactly when $X_{xy} = 0$ whenever $D(x, y) > 0$, that is, whenever $x \not\approx y$. An operator commutes with every diagonal A_k exactly when it has no entries between basis vectors with different eigenvalue tuples. (3) Entries with $D(x, y) > 0$ tend to zero and the others are constant. The dual range of $\mathbb{E}_{\pi_\infty}$ is $\mathfrak{B}_{\pi_\infty}$, which is already an algebra. (4) All multipliers $e^{-tD(x,y)}$ are nonzero. □

Part (2) is the elementary commuting case of the known structure of fixed points of quantum dynamical semigroups [14, 15]. For this series the consequence matters more than the statement: the given coupling determines both its channel and its invariant algebra. Part (4) is Paper A's result that an algebra label does not record resolution, now in dynamical form. Before the long-time limit the channel generates everything, and the structure that a coupling makes accessible has to be identified by how well observables survive over time.

Corollary 3.2 (The selected sectors are intrinsic)**.** *In the interaction picture of* (1)*, the minimal central projections of* $\operatorname{Fix}(\Phi)$ *are precisely the complete joint-eigenvalue projections P_a. They are determined by the semigroup up to relabelling. A common unitary conjugation of the carrier and all couplings conjugates these projections and preserves the rates between them.*

Proof. Proposition 3.1 gives $\operatorname{Fix}(\Phi) = \bigoplus_a \mathcal{B}(P_a\mathcal{H})$. The centre of this algebra is $\operatorname{span}\{P_a\}$; its minimal nonzero projections are unique. On $P_a\mathcal{B}(\mathcal{H})P_b$, the generator is multiplication by $-\frac{1}{2}\sum_k \gamma_k(a_k(a) - a_k(b))^2$. Conjugating the entire model preserves these scalars. □

The eigenbasis used in the proof is not an extra choice of spatial sites. What has been selected is a direct sum, including quantum content within its blocks. It neither supplies the instruments for that content nor identifies the blocks with spatial tensor factors.

The objects of Paper E [3] can be located in this class. Its environmental fixed set $\operatorname{Fix}(\mathcal{E})$ is $\mathfrak{B}_{\pi_\infty}$. In Paper E's collision model, with a qubit pointer, the pointer algebra equals the fixed algebra of the induced dephasing and contains every effect a fragment can report. The centre of $\mathfrak{B}_{\pi_\infty}$ is generated by the block projections P_a; it is commutative, and Section 5 shows that it is what the environment records, so it serves as Paper E's pointer or record algebra. The inclusion of the record algebra in the fixed set is thus a conclusion of the model, as Paper E requires of any such overlap. The fixed set is strictly larger than its centre as soon as some block has dimension greater than one; a nontrivial two-sector qubit pointer has only one-dimensional blocks. Such blocks are decoherence-free subspaces [12, 13]. They retain quantum content; sector weights can retain classical correlations as well.

Remark 3.3 (Hamiltonians that do not commute with the couplings)**.** *If the system Hamiltonian H has a part H_{off} connecting different blocks, $\mathfrak{B}_{\pi_\infty}$ need not remain exactly invariant. Strong dephasing can then confine the dynamics to the blocks, with leakage on a time scale of order $\Gamma_{\mathrm{out}}/\|H_{\mathrm{off}}\|^2$, where Γ_{out} is the smallest between-block rate; this is the mechanism of quantum Zeno subspaces [16]. Section 4.1 gives an exact two-level lifetime and a finite-time stability bound without a commutativity assumption. Paper D's chain with a transverse field requires its own many-body estimate.*

4 Selection over a finite window

Definition 4.1 (Stable algebra on a window)**.** *A unital $*$-subalgebra $\mathfrak{B} \subseteq \mathcal{B}(\mathcal{H})$ is ε-stable on $[0,T]$ if*

$$\sup_{0\le t\le T}\ \sup_{X\in\mathfrak{B},\ \|X\|\le 1} \|\Phi_t^*(X) - X\| \le \varepsilon.$$

The diagnostic tests the whole unit ball of an algebra, rather than selected states or a list of slow observables. For a partition π define the width of a class C in the direction k, $w_k(C) = \max_{x\in C} a_k(x) - \min_{x\in C} a_k(x)$, and the two rates

$$\Gamma_{\mathrm{in}}(\pi) = \max_{C\in\pi}\ \tfrac{1}{2}\sum_k \gamma_k w_k(C)^2, \qquad \Gamma_{\mathrm{out}}(\pi) = \min\{D(x,y) :\ x,y \text{ in different classes of } \pi\}. \tag{3}$$

The minimum over an empty set is $+\infty$, so a single-block partition has $\Gamma_{\mathrm{out}} = +\infty$. If every coupling is scalar, the full algebra is fixed and selected for every window. In the rate image, $\Gamma_{\mathrm{in}}(\pi)$ is the largest squared diagonal of the axis-parallel boxes enclosing the classes, and $\Gamma_{\mathrm{out}}(\pi)$ is the smallest squared distance between points of different classes.

Theorem 4.2 (Selection by a window)**.** *Let $0 < \varepsilon < 1$ and $T > 0$.*

1. *For every $X \in \mathfrak{B}_\pi$ and $t \ge 0$, $\|\Phi_t^*(X) - X\| \le t\,\Gamma_{\mathrm{in}}(\pi)\|X\|$. For every state $\rho \in \mathfrak{B}_\pi$, $\|\Phi_t(\rho) - \rho\|_1 \le t\,\Gamma_{\mathrm{in}}(\pi)$. In particular $\mathfrak{B}_\pi$ is ε-stable on $[0,T]$ when $T\Gamma_{\mathrm{in}}(\pi) \le \varepsilon$.*
2. *If $1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$, no unital $*$-algebra that contains $\mathfrak{B}_\pi$ properly is ε-stable on $[0,T]$.*
3. *Call π selected by (T,ε) if*
$$T\,\Gamma_{\mathrm{in}}(\pi) \le \varepsilon < 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)}. \tag{4}$$
At most one partition is selected by (T,ε). If π is selected by (T,ε) and π' by (T',ε) with $T \le T'$, then π' refines π, so $\mathfrak{B}_{\pi'} \subseteq \mathfrak{B}_\pi$.
4. *For $T > -\ln(1-\varepsilon)/\Gamma_{\mathrm{out}}(\pi_\infty)$ the selected partition exists and is π_∞, so the selected algebra is the fixed algebra of Proposition 3.1.*

Proof. (1) $\mathfrak{B}_\pi$ is invariant under $\mathcal{L}$, and on it $\mathcal{L}(X) = -\frac{1}{2}\sum_k \gamma_k [A_k,[A_k,X]]$. On a class C, replacing A_k by $A_k - m_k$ with m_k the midpoint of its eigenvalues on C does not change the commutator and gives an operator of norm $w_k(C)/2$; hence $\|[A_k,X_C]\| \le w_k(C)\|X_C\|$ and $\|[A_k,[A_k,X_C]]\| \le w_k(C)^2\|X_C\|$. Since $\|X\| = \max_C \|X_C\|$, $\|\mathcal{L}(X)\| \le \Gamma_{\mathrm{in}}(\pi)\|X\|$. Then $\Phi_t^*(X) - X = \int_0^t \Phi_s^*(\mathcal{L}(X))\,ds$ and Φ_s^* is a contraction in operator norm. For a state $\rho \in \mathfrak{B}_\pi$ and $\|Y\| \le 1$, $\mathrm{Tr}[(\Phi_t(\rho) - \rho)Y] = \mathrm{Tr}[\rho(\Phi_t^* - \mathrm{id})(\mathbb{E}_\pi Y)]$, because $\rho = \mathbb{E}_\pi\rho$, $\mathbb{E}_\pi$ is self-dual and commutes with the Schur multiplier Φ_t^*; since $\mathbb{E}_\pi Y \in \mathfrak{B}_\pi$ and $\|\mathbb{E}_\pi Y\| \le 1$, the first statement bounds this by $t\Gamma_{\mathrm{in}}(\pi)$.

(2) Let $\mathfrak{B} \supsetneq \mathfrak{B}_\pi$ and $X \in \mathfrak{B} \setminus \mathfrak{B}_\pi$. Some entry $X_{x'y'}$ with $x' \in C$, $y' \in C'$, $C \ne C'$, is nonzero. For any $x \in C$, $y \in C'$ the matrix units $E_{xx'}$ and $E_{y'y}$ lie in $\mathfrak{B}_\pi \subseteq \mathfrak{B}$, so $E_{xx'}XE_{y'y} = X_{x'y'}E_{xy} \in \mathfrak{B}$. Then $\|\Phi_T^*(E_{xy}) - E_{xy}\| = 1 - e^{-TD(x,y)} \ge 1 - e^{-T\Gamma_{\mathrm{out}}(\pi)} > \varepsilon$.

(3) Let π and π' be selected by (T,ε) and (T',ε), $T \le T'$. If x,y lie in one class of π', then $D(x,y) \le \Gamma_{\mathrm{in}}(\pi') \le \varepsilon/T' \le \varepsilon/T$. If x,y lay in different classes of π, then $D(x,y) \ge \Gamma_{\mathrm{out}}(\pi) > -\ln(1-\varepsilon)/T \ge \varepsilon/T$. Hence every class of π' lies in a class of π. For $T = T'$ the same argument with the roles exchanged gives $\pi = \pi'$.

(4) $\Gamma_{\mathrm{in}}(\pi_\infty) = 0$ and $\Gamma_{\mathrm{out}}(\pi_\infty) > 0$, so π_∞ satisfies (4) for such T; uniqueness is part (3). □

A window and a tolerance therefore pick out at most one algebra of the form $\mathfrak{B}_\pi$, and they pick one out when the rate image of the basis splits into clusters that are small compared with ε/T and far apart compared with $-\ln(1-\varepsilon)/T$. This is a separation of time scales, the mechanism behind metastability in open quantum systems [17]; for commuting couplings it can be stated exactly. The selected algebra is moreover the largest ε-stable algebra of block form: if $\mathfrak{B}_\sigma$

is ε-stable on $[0,T]$, every matrix unit E_{xy} with x,y in one class of σ satisfies $1-e^{-TD(x,y)} \le \varepsilon$, so x and y lie in one class of the selected partition π, and $\mathfrak{B}_\sigma \subseteq \mathfrak{B}_\pi$. As the window lengthens, the selected algebra shrinks in discrete steps located at the gaps of the rate spectrum and ends at the fixed algebra. In the order of Paper A this is a chain of further coarse-grainings, since $\mathbb{E}_{\pi'} = \mathbb{E}_{\pi'} \circ \mathbb{E}_\pi$ when π' refines π.

Without a gap there is no selected algebra at that window. Selection uses the sufficient bound of part (1), so between that bound and the necessary bound of part (2) a unique maximal stable block algebra can still exist; but the window may also fail to determine one, and two incomparable maximal stable block algebras can occur. The set of ε-stable matrix units can then be an operator system whose generated algebra contains unstable elements: if $D(x,y)$ and $D(y,z)$ are small, $D(x,z)$ can still be as large as $(\sqrt{D(x,y)}+\sqrt{D(y,z)})^2$, and a chain of small steps can connect distant points of the rate image. Theorem 4.2 is a sufficient selection certificate. Failure of its gap condition does not by itself decide existence or uniqueness of a maximal stable algebra. Nor does stability of a block algebra supply the instruments needed to use every element as a gate.

4.1 Continuous capabilities when the fixed algebra jumps

A weak coupling can change the exact commutant discontinuously. It cannot change every finite experiment discontinuously. The distinction matters when a selected algebra is used to describe a finite observer.

Proposition 4.3 (Finite-time operational stability)**.** *Let $\mathcal{L}_0, \mathcal{L}_1$ be finite-dimensional Lindblad generators, and put $v = \|\mathcal{L}_1 - \mathcal{L}_0\|_\diamond$. Then*

$$\|e^{t\mathcal{L}_1} - e^{t\mathcal{L}_0}\|_\diamond \le \min\{2, tv\}. \tag{5}$$

Consider any experiment with the same initial states, retained ancillas and permitted instruments in both models. Between its instruments the system evolves for prescribed intervals $t_1,\dots,t_m$, with total $T=\sum_i t_i$. Instruments may depend on earlier outcomes. The total variation distance between its complete record laws is at most $\min\{1, Tv/2\}$. Consequently, for a common decision problem with loss in $[0,L]$, the two optimal risks differ by at most $L\min\{1,Tv/2\}$. In particular, a Hamiltonian perturbation $-i\epsilon[K,\cdot]$ gives $v \le 2|\epsilon|\|K\|$.

Proof. Duhamel's formula writes the channel difference as $\int_0^t e^{(t-s)\mathcal{L}_1}(\mathcal{L}_1-\mathcal{L}_0)e^{s\mathcal{L}_0}\,ds$. CPTP maps have diamond norm one, giving tv; two channels differ by at most two. A hybrid argument replaces the evolution intervals one at a time, keeping outcome-controlled instruments as CPTP maps on the system and classical record. Contractivity and the triangle inequality give a trace-norm difference at most Tv. The bounded-loss estimate holds uniformly over each common policy, so it also holds for the infima. Finally, $\|[K,A]\|_1 \le 2\|K\|\|A\|_1$ holds with every retained ancilla. □

The estimate needs neither a gap nor a selected algebra. It bounds distinguishing, storing and acting through the permitted experiment, not just the state of an isolated system. The available instruments and the total time remain the same on both sides. A discontinuous change of permissions, or an unbounded observation time, is a different comparison.

For an exact noncommuting example, take

$$\mathcal{L}_\epsilon(\rho) = -i\epsilon[X,\rho] + \kappa(Z\rho Z - \rho), \qquad \kappa > 0. \tag{6}$$

The fixed algebra is $\operatorname{span}\{I,Z\}$ at $\epsilon = 0$ and only $\mathbb{C}I$ at every $\epsilon \neq 0$. To see the latter, the Hilbert–Schmidt dissipative form of a fixed observable forces $[Z,O]=0$; the remaining Hamiltonian equation then forces $[X,O]=0$. Nevertheless, the two initially perfect records $(I \pm Z)/2$ remain readable for a long time. If $0 < 2|\epsilon| < \kappa$, put $\omega = \sqrt{\kappa^2 - 4\epsilon^2}$. Their Bloch vectors at time t are $\pm(0, y(t), z(t))$, where

$$z(t) = e^{-\kappa t}\left(\cosh\omega t + \frac{\kappa}{\omega}\sinh\omega t\right), \qquad y(t) = -\frac{2\epsilon}{\omega}e^{-\kappa t}\sinh\omega t. \tag{7}$$

These expressions solve $\dot y = -2\kappa y - 2\epsilon z$, $\dot z = 2\epsilon y$, with $(y,z) = (0,1)$ initially. Reading only Z has error $(1-z(t))/2$; allowing the optimal system measurement gives error $(1-\sqrt{y(t)^2+z(t)^2})/2$. The slow decay rate is exactly

$$r = \kappa - \sqrt{\kappa^2 - 4\epsilon^2} = \frac{2\epsilon^2}{\kappa} + O(\epsilon^4/\kappa^3). \tag{8}$$

Thus arbitrarily weak transverse coupling erases the infinite-time record, while its finite-time lifetime diverges as $\kappa/(2\epsilon^2)$. Exact hulls and fixed algebras describe the ideal endpoint; finite-time risks describe what remains available on the way to it. This is also the distinction needed by Foundation's hulls and the maintenance tasks of T-DOME and Brain Anchor. Neither the general stability estimate nor this example asserts that every slow operator space is an algebra or a physical projection.

5 Records live on the centre

5.1 A recording model

To see what the environment learns we replace the Markovian generator by an exact unitary model. Environmental fragments $E_1, \dots, E_K$ start in states ω_k, and the interaction is

$$U = \sum_{x \in \mathcal{X}} |x\rangle\langle x| \otimes V_1(x) \otimes \cdots \otimes V_K(x), \tag{9}$$

with unitaries $V_k(x)$ that depend on the basis state through the coupling, for example $V_k(x) = \exp[-i\theta_k a_k(x) G_k]$ for a fragment generator G_k. The joint state after the interaction is

$$\sigma = \sum_{x,y} \rho_{xy} |x\rangle\langle y| \otimes \bigotimes_k V_k(x)\, \omega_k\, V_k(y)^\dagger,$$

so the system undergoes the Schur multiplier

$$\rho_{xy} \mapsto c(x,y)\, \rho_{xy}, \qquad c(x,y) = \prod_k c_k(x,y), \qquad c_k(x,y) = \mathrm{Tr}\big[\omega_k V_k(y)^\dagger V_k(x)\big]. \tag{10}$$

The generator (1) arises from such models in weak-coupling and repeated-collision limits. Here the collision statements are exact for the specified unitary and an initial product $\rho \otimes \bigotimes_k \omega_k$. Fresh independent fragments are an input: recollisions or shared initial fragment correlations do not obey the product formula automatically.

Proposition 5.1 (Records carry only the block label)**.** *Let π be a partition of $\mathcal{X}$ such that every $V_k(x)$ depends on x only through its class $a(x) \in \pi$. Then:*

1. *$c(x,y) = 1$ whenever x and y lie in one class, so $\mathbb{E}_\pi(\rho)$, and with it the state on every block, is unchanged.*
2. *For every $O \in \mathfrak{B}_\pi$ and every operator F on any collection $\mathcal{F}$ of fragments,*

$$\mathrm{Tr}[\sigma(O \otimes F)] = \sum_{a \in \pi} \mathrm{Tr}[P_a \rho P_a\, O]\ \mathrm{Tr}[\omega_{\mathcal{F}}(a) F], \qquad \omega_{\mathcal{F}}(a) = \bigotimes_{k \in \mathcal{F}} V_k(a)\, \omega_k\, V_k(a)^\dagger.$$

The state of the fragments is $\sum_a p(a)\, \omega_{\mathcal{F}}(a)$ with $p(a) = \mathrm{Tr}\, P_a \rho$; it depends on ρ only through the distribution of the label, and given the label, block observables and fragments are uncorrelated.

Proof. (1) $V_k(x) = V_k(y)$ when x, y lie in one class, so $c_k(x,y) = 1$. (2) $O_{yx} \neq 0$ only when x and y lie in one class a, and then the fragment factor is $\omega_{\mathcal{F}}(a)$ for both. Summing $\rho_{xy} O_{yx}$ over $x, y \in a$ gives $\mathrm{Tr}[P_a \rho P_a O]$. □

Whatever the environment learns under such couplings is the block label, which generates the centre of $\mathfrak{B}_\pi$. Many fragments can hold readable copies of that label while leaving the blocks alone. Block dependence alone does not guarantee distinguishability: the conditional states $\omega_k(a)$ must also be different, and a permitted measurement must resolve them. Records and the correlations inside the blocks are thus carried by one selected algebra, the first by its centre and the second by its blocks. That the redundantly available information is classical, and is information about one measurement, holds far more generally [18]; a two-qubit system with a decoherence-free subspace that proliferates part of its information while the remaining subspace keeps non-classical features was studied in [19]. What the commuting class adds is the exact location of both kinds of information in one algebra and, next, the price of imperfect symmetry.

5.2 Leakage costs coherence, and the costs multiply

Theorem 5.2 (Leakage and coherence)**.** *For basis states x, y and a fragment k, let $D_k(x,y) = \frac{1}{2}\big\| V_k(x)\omega_k V_k(x)^\dagger - V_k(y)\omega_k V_k(y)^\dagger \big\|_1$ be the fragment's distinguishability of x and y, so that an optimal measurement on the fragment identifies one of the two equally likely states with probability $\frac{1}{2}(1 + D_k)$ [20]. Then*

$$|c_k(x,y)|^2 + D_k(x,y)^2 \leq 1, \tag{11}$$

with equality when ω_k is pure. Consequently

$$|c(x,y)| \leq \prod_k \big(1 - D_k(x,y)^2\big)^{1/2} \leq \exp\Big(-\tfrac{1}{2} \sum_k D_k(x,y)^2\Big), \tag{12}$$

and if the coherence between x and y must stay above $1 - \varepsilon$, then

$$\sum_k D_k(x,y)^2 \leq 2 \ln \frac{1}{1-\varepsilon}. \tag{13}$$

Proof. Purify $\omega_k = \mathrm{Tr}_R\,|\Omega\rangle\langle\Omega|$. Then $c_k(x,y) = \langle\Omega|(V_k(y)^\dagger V_k(x)\otimes I)|\Omega\rangle$ is the overlap of the pure states $(V_k(x)\otimes I)|\Omega\rangle$ and $(V_k(y)\otimes I)|\Omega\rangle$, whose trace distance is $(1-|c_k|^2)^{1/2}$. Partial trace over R does not increase trace distance, which gives (11), with equality when there is nothing to trace out. Inequality (12) follows from (10) and $\ln(1-u)\le -u$, and (13) by taking logarithms. □

Inequality (11) is Englert's relation between visibility and which-way distinguishability [21], applied to one fragment. It bounds what a fragment learns by the coherence it destroys, not the reverse: a fragment in a mixed state can destroy coherence while learning nothing, as in Paper E's maximally mixed memory example. Purity is one useful preparation resource; mixed fragments can also record when their conditional states differ. For many fragments it becomes the budget (13): the sum of their squared pairwise distinguishabilities is at most $2\ln(1/(1-\varepsilon))$ for a protected pair of basis states. This is not a bound on their total Shannon information about an arbitrary block.

Corollary 5.3 (Redundancy against symmetry)**.** *Suppose that R fragments each distinguish two states x, y of one block with $D_k(x,y)\ge\beta$. Then $|c(x,y)|\le(1-\beta^2)^{R/2}$, and keeping the coherence above $1-\varepsilon$ requires*

$$\beta \le \Big(\frac{2\ln(1/(1-\varepsilon))}{R}\Big)^{1/2}. \tag{14}$$

In a sequence of models with $R\to\infty$ and $D_k(x,y)=\beta_R=\lambda R^{-\alpha}$, $\alpha>0$, for the R fragments that couple to the block, the coherence tends to zero if $\alpha<\frac12$. If the fragment states are pure and no other fragment distinguishes x from y, it tends to 1 for $\alpha>\frac12$ and to $e^{-\lambda^2/2}$ for $\alpha=\frac12$.

Proof. The bound and (14) are (12) and (13). For pure fragments $|c(x,y)| = (1-\beta_R^2)^{R/2}$ by the equality case, and $\frac{R}{2}\ln(1-\lambda^2R^{-2\alpha})\to 0,\ -\lambda^2/2,\ -\infty$ for $\alpha>\frac12, =\frac12, <\frac12$. □

A coupling of the form (9) is block-constant only by symmetry: the fragments must respond identically to all states of a block. Real couplings break such a symmetry slightly, and each recording fragment then leaks a little. Corollary 5.3 says that objectivity by redundancy and protection of the blocks can coexist, but only if the symmetry error per fragment falls as $R^{-1/2}$. In the limit of many records this is a sharp threshold in the exponent α. It is the closest analogue of a transition in this class. It is a statement about how a symmetry error scales with the number of records, and it is not a phase transition in the thermodynamic sense.

6 Sector records and protected correlations

A mutual-information geometry does not require entanglement. Even local dephasing preserves $\rho = \frac12|00\rangle\langle 00| + \frac12|11\rangle\langle 11|$, for which $I(1:2)=\ln 2$. With local Z couplings every selected block is one-dimensional and every sector state is a product. Its entire edge comes from sector weights.

For a selected state $\bar\rho = \mathbb{E}_\pi(\rho) = \sum_k p_k\rho_k$, with $\rho_k = P_k\rho P_k/p_k$, retain the label in a classical register:

$$\Omega_{SK} = \sum_k p_k\rho_k\otimes|k\rangle\langle k|_K. \tag{15}$$

This extension is physically realized by a perfectly resolving block-constant recorder followed by measurement or dephasing of its label register. It is not the coherent source–memory state before that readout. If the actual fragment states overlap, K is still a useful mathematical label, but it is not a freely readable record. Zero-weight sectors are omitted.

Proposition 6.1 (The two contributions to correlation)**.** *For any supplied disjoint subsystems A, B of S,*

$$I(A:B)_{\bar\rho} = \sum_k p_k I(A:B)_{\rho_k} + I(A:K)_\Omega + I(B:K)_\Omega - I(AB:K)_\Omega. \tag{16}$$

The last three terms need not have a nonnegative sum.

Proof. For $X = A, B, AB$, the classical–quantum entropy formula gives $I(X:K) = S(\bar\rho_X) - \sum_k p_k S((\rho_k)_X)$. Substitution in $S(A)+S(B)-S(AB)$ proves the identity. □

The first term locates correlations protected within sectors; it can include classical as well as quantum correlations. The correction describes what mixing sector-conditioned states does to those correlations. In the locally dephased example above it equals $\ln 2$. In the next recording example it is negative. Thus neither a noncommutative block nor a classical centre alone determines the geometry. The sector states, their weights, the site marginals and the dictionary all enter.

7 How protected relations are formed

A noise commutant specifies what can be stored. It does not determine whether that content was inherited or created. The distinction already invalidates a noise-only mereology rule.

7.1 Disconnected noise can preserve a shared Bell pair

Take four qubits with only $A_{12} = Z_1 + Z_2$ and $A_{34} = Z_3 + Z_4$. Each pair has a zero-magnetization sector with logical basis $|0_L\rangle = |01\rangle$, $|1_L\rangle = |10\rangle$. The state

$$|\Psi_L\rangle = \frac{|0_L\rangle_{12}|0_L\rangle_{34} + |1_L\rangle_{12}|1_L\rangle_{34}}{\sqrt{2}} \tag{17}$$

is annihilated by both noise operators and is fixed exactly by (1). Yet $I(12:34) = 2\ln 2$. The two noise-hypergraph components are disconnected. Applying a Hadamard to each qubit gives the same counterexample for the X couplings of Section 8. Classical and quantum correlations can both cross disconnected noise components when they are present initially.

7.2 A collective recorder forms a flagged protected relation

Start two qubits in the physical product $|+_X\rangle|+_X\rangle$ and a fresh three-level memory in $|0\rangle$. Let $M = Z_1 + Z_2$ and let P_+, P_0, P_- project onto its eigenvalues $2, 0, -2$. For the shift $S_3|k\rangle = |k+1 \bmod 3\rangle$, use

$$U_{\rm rec} = P_+ \otimes I + P_0 \otimes S_3 + P_- \otimes S_3^2. \tag{18}$$

This is a block-constant recording interaction, not a product of independent local readers. It has a finite Hamiltonian implementation: choose a Hermitian logarithm G with $e^{-iG} = S_3$ and $\|G\| = 2\pi/3$, put $a(M) = (2I - M)/2$, and apply $H_{\rm rec} = (\hbar/t_g)a(M) \otimes G$ for a time $t_g > 0$. The same memory couples to both sites. Its preparation, shared physical port, interaction time, control and subsequent readout are resources.

The complete output is

$$|\Omega\rangle = \tfrac{1}{2}|00\rangle|0\rangle + \tfrac{1}{\sqrt{2}}|\Psi^+\rangle|1\rangle + \tfrac{1}{2}|11\rangle|2\rangle, \qquad |\Psi^+\rangle = (|01\rangle + |10\rangle)/\sqrt{2}. \tag{19}$$

A computational-basis measurement of the memory produces the classical–quantum state with probabilities $(1/4, 1/2, 1/4)$ and conditional source states $(|00\rangle\langle 00|, |\Psi^+\rangle\langle\Psi^+|, |11\rangle\langle 11|)$. All branches remain in the protocol. Flag 1, of probability $1/2$, certifies a Bell pair in the $M = 0$ protected sector; flags $0, 2$ certify product states.

If the flag is discarded, the source is instead

$$\rho_* = \tfrac{1}{4}|00\rangle\langle 00| + \tfrac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \tfrac{1}{4}|11\rangle\langle 11|, \qquad I(1:2)_{\rho_*} = \tfrac{1}{2}\ln 2. \tag{20}$$

It is separable. Indeed it is the uniform collective Z-rotation twirl of $|+_X, +_X\rangle\langle +_X, +_X|$, a mixture of product rotations. Its partial-transpose eigenvalues are $0, 1/4, 1/4, 1/2$. The unconditional source therefore gains a mutual-information edge without entanglement; separability does not imply that every correlation measure is classical. The retained flagged ensemble permits a conditional protected Bell use, with herald probability and every unsuccessful branch charged. The pure source–memory state (19), the measured flagged state and the unconditional source are three different outputs.

For this flagged ensemble, $\sum_k p_k I(1:2)_{\rho_k} = \ln 2$, $I(1:K) = I(2:K) = \frac{1}{2}\ln 2$ and $I(12:K) = H(K) = \frac{3}{2}\ln 2$. Equation (16) recovers the unconditional $\frac{1}{2}\ln 2$ with correction $-\frac{1}{2}\ln 2$. Collective dephasing of the product input alone reaches the same ρ_* but supplies no accessible flag. A flag requires the specified recording interaction and actual readout.

7.3 A real link forms a deterministic protected Bell pair

The distinction is sharper in the four-qubit example. Define

$$L^x_{ab} = \tfrac{1}{2}(X_aX_b + Y_aY_b), \qquad H_{\rm link} = \hbar J L^x_{12}L^x_{34}, \quad J > 0.$$

L^x_{ab} exchanges $|01\rangle$ and $|10\rangle$ and vanishes on $|00\rangle, |11\rangle$. Consequently $[H_{\rm link}, A_{12}] = [H_{\rm link}, A_{34}] = 0$. Starting from the physical product $|01\rangle_{12}|01\rangle_{34}$, the full noisy evolution remains in the joint zero sector and is exactly

$$|\psi(t)\rangle = \cos(Jt)|0_L0_L\rangle - i\sin(Jt)|1_L1_L\rangle. \tag{21}$$

At $t = \pi/(4J)$ it has $I(12:34) = 2\ln 2$ and one ebit across the pair partition. The noise term vanishes throughout the trajectory.

This is new protected correlation from an initially uncorrelated state. It uses an actual cross-pair Hamiltonian link, here a four-body control term. The noise graph remains disconnected but the complete interaction graph does not. The factorization, prepared input, gate permissions, coupling strength and duration are supplied; there is no inferred free implementation or heat price for the link.

7.4 The process constraint and learned use

Proposition 7.1 (Correlation provenance across independent components)**.** *Let a supplied partition of a carrier have initially independent states and auxiliary resources. Suppose its full channel, including record registers, factors across these components. Then it creates no cross-component mutual information. If the input is correlated, a product channel cannot increase the mutual information across any bipartition of the components, but it can preserve initially shared classical or quantum correlations.*

Proof. A product channel sends a product input to a product output, including its local records. For a correlated input, $I(A:B) = D(\rho_{AB}\|\rho_A \otimes \rho_B)$, so the second statement is data processing under the product channel. □

An implementation satisfies the factor premise when its Hamiltonian and noise generators act separately on the components, its fresh ports are independent, and every control and instrument is component-local. Shared random controls, communication or common memories must be added to the process description: they can correlate unconditional outputs. Conditioning on a joint event can also change correlations and is not a product unconditional channel. The proposition is a diagnostic, not a claim that every process with a disconnected noise graph factors.

The constructions above do more than add initial-state qualifications to a conjecture. They exhibit how the missing relations can form: a collective port records an indistinguishability sector, while an interaction inside protected sectors builds usable quantum content. The other indispensable step is learning which encoding an unknown physical plant protects. Foundation III [6] implements that step for $A_s = n_1 + sn_2$ with unknown $s = \pm 1$. A diagnostic sharing the plant's actual sign writes a record; that record, not s, selects odd or even encoding and the matching decoder. The same logical content and its old reference survive from before diagnosis. The record-controlled frame changes the later contact, rather than merely changing a description of the fixed commutant.

That result assumes a physical relation between diagnostic and storage couplings. A diagnostic carrying an unrelated sign would not learn the protected frame. Its retained-history scalar/JT implementation also contains real body forces, a source and a second acquisition; it is not identified with the full quantum material model. The complementary contribution here is the separation of formation, protection and use: each can fail while the others hold. A noiseless block can be unreadable, a readable label can outlive its quantum content, and a learned encoder can protect content without supplying every logical gate.

8 A geometry selected by the coupling

We now apply these results to the state and the reconstruction rule of Paper A. The rule acts on the complete graph of the sites: the edge $\{i,j\}$ has length $1/I_{ij}$, where I_{ij} is the mutual information of sites i and j, and distances are shortest-path lengths. We extend it to $I_{ij} = 0$ by giving that edge infinite length, so that a site whose edges all have zero mutual information is disconnected. The state is the three-qubit GHZ state $(|000\rangle + |111\rangle)/\sqrt{2}$. In the eigenbasis $|\pm\rangle$ of the Pauli operator X it reads

$$|\mathrm{GHZ}\rangle = \tfrac{1}{2}\big(|{+}{+}{+}\rangle + |{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle\big).$$

All couplings below are sums of X_i, hence diagonal in this basis, and Sections 3–5 apply. A local coupling $A = X_i$ with rate γ_i is the bit-flip noise of Paper A, with flip probability $p_i = (1 - e^{-2\gamma_i t})/2$. A collective coupling $A = X_1 + X_2 + X_3$ dephases between different values of the X-magnetization $M = \sum_i x_i$ and leaves states of equal M coherent.

Proposition 8.1 (Three couplings, three geometries)**.** *Let* $\Psi^+ = (|{+}{-}\rangle + |{-}{+}\rangle)/\sqrt{2}$ *and* $\tilde{W} = (|{+}{-}{-}\rangle + |{-}{+}{-}\rangle + |{-}{-}{+}\rangle)/\sqrt{3}$*. The long-time selected states* $\mathbb{E}_{\pi_\infty}(|\mathrm{GHZ}\rangle\langle\mathrm{GHZ}|)$ *and their geometries are:*

1. *Collective coupling* $X_1 + X_2 + X_3$*:* $\frac{1}{4}|{+}{+}{+}\rangle\langle{+}{+}{+}| + \frac{3}{4}|\tilde{W}\rangle\langle\tilde{W}|$*. Each pair has* $I_{ij} = \frac{1}{2}\ln 2$*; the geometry is an equilateral triangle with side* $2/\ln 2 = 2.8854$*.*

2. *Collective coupling $X_1 + X_2$ and local coupling X_3:* $\frac{1}{4}|++\rangle\langle+++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+|_{12} \otimes |-\rangle\langle-|_3 + \frac{1}{4}|--+\rangle\langle--+|$. *Here $I_{12} = \frac{1}{2}\ln 2$ and $I_{13} = I_{23} = 0$; the geometry is a single edge, and site 3 is disconnected.*

3. *Local couplings X_1, X_2, X_3: the uniform mixture of the four even strings $|+++\rangle, |+--\rangle, |-+-\rangle, |--+\rangle$. Every $I_{ij} = 0$; no edge survives.*

Proof. The pinching keeps the amplitudes inside each class of equal eigenvalue tuples. (1) The classes met by the GHZ state are $M = 3$ and $M = -1$. The marginal of sites $1, 2$ is $\frac{1}{4}|++\rangle\langle++| + \frac{1}{2}|\Psi^+\rangle\langle\Psi^+| + \frac{1}{4}|--\rangle\langle--|$, with eigenvalues $\frac{1}{4}, \frac{1}{2}, \frac{1}{4}$ and entropy $\frac{3}{2}\ln 2$; each single-site marginal is $I/2$; so $I_{12} = 2\ln 2 - \frac{3}{2}\ln 2$, and the other pairs follow by symmetry. (2) The classes are labelled by $(x_1 + x_2, x_3)$; the strings $|+--\rangle$ and $|-+-\rangle$ share the label $(0, -)$. The marginal of sites $1, 2$ is as in (1). The marginal of sites $1, 3$ is uniform on the four strings $|++\rangle, |+-\rangle, |--\rangle, |-+\rangle$, and so is that of sites $2, 3$. (3) Every pair marginal of the uniform mixture of even strings is uniform. Direct evaluation of the full eight-dimensional density matrices gives the same values. □

The state and the rule are the same in all three cases; only the coupling differs. At zero time each pair has $I_{ij} = \ln 2$, Paper A's GHZ geometry at zero noise. Under local noise Paper A's triangles relax toward the empty graph; a collective coupling stops the relaxation at an equilateral triangle, and a coupling that is collective on one pair keeps exactly that edge. The surviving correlations come from blocks of dimension greater than one: the three-dimensional block $M = -1$, which contains the entangled state $\tilde{W}$, in case (1), and the two-dimensional block spanned by $|+--\rangle$ and $|-+-\rangle$ in case (2). In case (3) the selected algebra is commutative with product minimal projections, every selected state is classically correlated, and for this state the classical pair correlations vanish as well. By Proposition 3.1(4) the channels at any finite time generate the full matrix algebra in all three cases, so an algebra label cannot tell them apart.

8.1 The lifetime of the selected geometry

Real couplings are not exactly collective. Add weak local couplings X_i with rates γ_i to the collective coupling with rate γ_c. The exact fixed algebra is then commutative and the long-time geometry is empty, but on a window the magnetization blocks are still selected. With π the partition by M, the rates (3) are $\Gamma_{\text{in}}(\pi) = 2(\gamma_1 + \gamma_2 + \gamma_3)$ and $\Gamma_{\text{out}}(\pi) = 2\gamma_c + 2\min_i \gamma_i$, while the coherences of the GHZ state between its two blocks decay at the rates $8\gamma_c + 2(\gamma_i + \gamma_j)$.

Lemma 8.2 (Plateau bound)**.** *For a state ρ and a partition π, let $\Gamma_{\text{out}}(\rho, \pi)$ be the smallest $D(x, y)$ over pairs in different classes with $\rho_{xy} \neq 0$, and let r be the number of basis vectors with $\rho_{xx} \neq 0$. Then for all $t \geq 0$*

$$\|\Phi_t(\rho) - \mathbb{E}_\pi(\rho)\|_1 \leq \sqrt{r}\, e^{-t\Gamma_{\text{out}}(\rho,\pi)}\, \|\rho - \mathbb{E}_\pi(\rho)\|_2 + t\,\Gamma_{\text{in}}(\pi). \tag{22}$$

Proof. Split $\Phi_t(\rho) - \mathbb{E}_\pi\rho = \Phi_t(\rho - \mathbb{E}_\pi\rho) + (\Phi_t(\mathbb{E}_\pi\rho) - \mathbb{E}_\pi\rho)$. The second term is bounded by Theorem 4.2(1). The first is the Schur product of $\rho - \mathbb{E}_\pi\rho$ with multipliers of modulus at most $e^{-t\Gamma_{\text{out}}(\rho,\pi)}$; its rank is at most r, because a row of a positive matrix with zero diagonal entry vanishes, and $\|Z\|_1 \leq \sqrt{\operatorname{rank} Z}\, \|Z\|_2$. □

The bound turns into a certified geometry through continuity of entropy. If two states of a qubit pair are at trace distance $\delta = \frac{1}{2}\|\rho - \sigma\|_1 \leq \frac{1}{2}$, then by the Fannes–Audenaert inequality [22] and Winter's bound for conditional entropy [23] their mutual informations differ by at most

$$f(\delta) = h(\delta) + 2\delta\ln 2 + (1 + \delta)\, h\Big(\frac{\delta}{1 + \delta}\Big), \tag{23}$$

with h the binary entropy in nats. Partial traces do not increase trace distance, and Paper A's stability bound for graph metrics converts a uniform change $f < m$ of mutual informations bounded below by m into a change of at most $2f/(m(m - f))$ of every distance among three sites.

Proposition 8.3 (A certified plateau)**.** *For $\gamma_c = 1$ and $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$, the bounds* (22) *and* (23) *certify that every distance of the GHZ geometry lies within 25% of the plateau value $2/\ln 2$ for $0.716 \leq t \leq 3.58$. Numerical root finding of the exact evolution gives the approximate 25% window $0.053 \lesssim t \lesssim 53.6$ and 5% window $0.163 \lesssim t \lesssim 8.86$. If the local rates are scaled to $s(1, 2, 4)$ with $s \leq 10^{-3}$, the late edges scale to leading order as $1/s$: approximately $5.36 \times 10^{-3}/s$ for the 25% window, $8.86 \times 10^{-4}/s$ for the 5% window and, when $s \leq 10^{-4}$ so that it exists, $3.58 \times 10^{-4}/s$ for the certified 25% window. The windows begin at times of order $1/\gamma_c$.*

Proof. For this state $r = 4$, $\|\rho - \mathbb{E}_\pi\rho\|_2 = \sqrt{6}/4$, $\Gamma_{\text{out}}(\rho, \pi) = 8.0006$ and $\Gamma_{\text{in}}(\pi) = 1.4 \times 10^{-3}$. The right side of (22) is the convex function $b(t) = (\sqrt{6}/2)e^{-8.0006t} + 0.0014t$. On $[0.716, 3.58]$ its maximum is at an endpoint and is

t	$\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1$	bound (22)	distances (d_{12}, d_{13}, d_{23})
0	0.8660	1.2247	(1.4427, 1.4427, 1.4427)
1	0.0010	0.0018	(2.8968, 2.9034, 2.9066)
10	0.0094	0.0140	(2.9728, 3.0221, 3.0458)
100	0.0897	0.1400	(3.5620, 3.9855, 4.2034)
300	0.2442	0.4200	(4.8943, 6.5032, 7.4508)
1000	0.5969	1.4000	(12.559, 28.867, 41.426)

Table 1: GHZ state under a collective coupling with $\gamma_c = 1$ and local couplings with $(\gamma_1, \gamma_2, \gamma_3) = 10^{-4}(1, 2, 4)$. The plateau value of every distance is $2/\ln 2 = 2.8854$.

less than 0.0050121. The trace distance used in (23) is $\delta(t) = \frac{1}{2}\|\Phi_t(\rho) - \mathbb{E}_\pi\rho\|_1 \le b(t)/2$, not $b(t)$. The increasing entropy and graph bounds therefore give a distance error less than $0.72130 < \frac{1}{4} \cdot 2/\ln 2$ throughout that interval. The exact-evolution window edges were located by root finding of the Schur multipliers. Once the coherences between blocks have decayed, the in-block multipliers depend on t only through st, which gives the late-edge scaling. The root values are numerical diagnostics, distinct from the conservative analytic interval. □

Table 1 shows the three stages. The GHZ geometry, with side $1/\ln 2$, relaxes on the time scale $1/\gamma_c$ to the selected triangle with side $2/\ln 2$; the triangle persists while $t\Gamma_{\text{in}}(\pi)$ is small; then the unequal local rates deform it into scalene triangles, which are not isometric to the plateau, before every edge diverges. This is Paper A's non-isometry produced by one time-ordered dynamics rather than by a comparison of separately chosen channels. Proposition 8.3 gives a state-specific stability certificate for Paper A's declared graph dictionary on an explicit window. The plateau exists when its end lies after its beginning, that is, when the ratio of in-block to between-block rates is small, as Theorem 4.2 requires.

The constants in Proposition 8.3 are small. At the end of the exact 25% window the three in-block coherences have fallen by only 3.2, 5.2 and 6.2 per cent. The selected pair marginal has a zero eigenvalue, and in-block decoherence populates it, so pair entropies respond to a coherence loss δ like $\delta\ln(1/\delta)$. Geometries read from entropies are therefore far more fragile than the coherences that carry them, and the gap between the certified and the exact window comes mostly from this logarithmic response together with worst-case continuity bounds.

8.2 Imperfect redundant records detach a vertex

Finally we let fragments record the magnetization with a small asymmetry. Each of K fragment qubits starts in $|0\rangle$ and interacts through $V_k(x) = \exp[-i\theta(M(x) + \eta x_1)Y]$ with $\theta = \pi/8$, where Y is the Pauli operator of the fragment. For $\eta = 0$ the coupling is block-constant. The two magnetizations present in the GHZ state, $M = 3$ and $M = -1$, then produce orthogonal fragment states, so every fragment holds a perfect copy of the label, and by Proposition 5.1 the equilateral triangle survives for any number of fragments. For $\eta > 0$ each fragment distinguishes strings of the block $M = -1$ that differ at site 1, with $D_k = |\sin 2\theta\eta|$.

η	K	coherence of $\|+--\rangle, \|-+-\rangle$	$I_{12} = I_{13}$	I_{23}	label error
0.02	1	0.99988	0.34631	0.34657	6.2×10^{-5}
0.02	100	0.98774	0.32790	0.34657	6.2×10^{-5}
0.02	1000	0.88393	0.23581	0.34657	6.2×10^{-5}
0.05	1	0.99923	0.34525	0.34657	3.9×10^{-4}
0.05	100	0.92577	0.26724	0.34657	3.9×10^{-4}
0.05	1000	0.46243	0.05555	0.34657	3.9×10^{-4}

Table 2: Recording the magnetization with a site-1 asymmetry η. The last column is the minimal error probability with which a measurement on one fragment identifies the label M, whose prior is $(\frac{1}{4}, \frac{3}{4})$. Coherences are relative to their initial values; the coherence between $|-+-\rangle$ and $|--+\rangle$ stays equal to 1.

Table 2 shows the effect. Every fragment is a nearly perfect and readable record of the label for both asymmetries, so the loss of geometry is not a price paid for better records. It is the small leakage about site 1, multiplied over fragments as in Theorem 5.2. The coherences between strings that differ at site 1 decay as $(1 - D_k^2)^{K/2}$, the edges to site 1 weaken, and as $K \to \infty$ the state approaches $\frac{1}{4}|+++\rangle\langle+++| + \frac{1}{4}|+--\rangle\langle+--| + \frac{1}{2}|-\rangle\langle-|_1 \otimes |\Psi^+\rangle\langle\Psi^+|_{23}$. Then

$I_{12} = I_{13} = 0$ and $I_{23} = \frac{1}{2}\ln 2$: site 1 is detached and the geometry becomes the single edge of Proposition 8.1(2), now on sites $2, 3$. With $\eta_K = 0.5K^{-\alpha}$ the coherence after $K = 10^6$ fragments is 0 for $\alpha = \frac{1}{4}$, $0.9258 = e^{-(\pi/8)^2/2}$ for $\alpha = \frac{1}{2}$, and 0.9999 for $\alpha = \frac{3}{4}$, as Corollary 5.3 predicts with $\lambda = 2\theta \cdot 0.5$ to leading order.

9 What selection settles

The selected partition is fixed by the supplied dynamics, window and tolerance, and is robust to perturbations preserving the rate gap. This differs in emphasis from QBism's state assignments and relational quantum mechanics' reference-system dependence [26, 25], without deciding their wider interpretations. Public agreement further needs actual fragments and calibrated readers; Paper A provides one finite realization. A stable algebra alone does not grant those permissions.

The GHZ comparison changes a state under one graph dictionary, not the coordinates of an unchanged metric. Sector mixing can sustain edges after quantum content is lost; the leaking GHZ block instead supports the specific edges destroyed in Table 2. Formation requires the collective port or Hamiltonian link of Section 7, not noise selection alone. Removing either resource supplies its corresponding opposing control.

AQFT and Consciousness I start with their local or current-net carriers [24, 4]. Observable-induced factorization and quantum mereology address when retained relations support an effective subsystem description [28, 29, 30]. Decoherence and Everettian accounts share the underlying dynamics [10, 27]; the issue here is whether its resulting structure is readable, operable and persistent for a finite task.

10 Beyond the commuting class

Conjecture 2.1 separates the existence of a stable block from its acquisition as an operational component. We now examine the further conditions required outside the exactly commuting class.

The conjecture concerns effective factorization, not the existence of elementary sites. It predicts more than correlation alone: successful formation should yield an encoder, decoder and repeatable operations on the formed component, with calibration records and finite error bounds. The exact formation examples give protected relational content; Foundation III adds record-controlled encoding of unknown content. Together they support a route from coupling through history to changed capability. They do not yet supply a general autonomous acquisition of a complete internal gate set. A proposed component that has only a state-dependent correlation but no reproducible internal operations would fail this criterion. The disconnected Bell counterexample rules out noise-only connectivity as a necessary condition, while Proposition 7.1 identifies the real no-creation obstruction for product processes.

For general Lindblad dynamics, a separated spectrum defines slow states and an effective metastable evolution [17]. A finite-window extension should select the operational structure of this manifold rather than assume that slow observables close under multiplication. Even complete positivity and idempotence do not ensure closure in the original multiplication. For example,

$$P(X) = \operatorname{diag}\big(x_{11}, x_{22}, (x_{11} + x_{22})/2\big) \tag{24}$$

is a unital completely positive idempotent on M_3. It is completely positive because it is a sum of two positive evaluation maps into positive diagonal matrices. Its range contains $E = \operatorname{diag}(1, 0, 1/2)$ but not $E^2 = \operatorname{diag}(1, 0, 1/4)$. A completely positive projection gives its range a Choi–Effros product $x \circ y = P(xy)$ [7, 8]. This abstract product does not make xy a physically permitted observable or control in the original representation. An ordinary conditional expectation onto a represented subalgebra needs the additional algebraic closure. A metastable spectral projection may not even be completely positive. Thus a general selection theory must provide actual instruments or recovery maps, their errors and their operational closure, rather than infer them from a slow spectral space.

The physical price of that capability is the next link to Q-RAIF and the gravitational line. A passive GHZ geometry has lifetime of order $1/s$ under symmetry breaking, while independent redundant fragments require an $R^{-1/2}$ protecting symmetry. Q-RAIF B realizes correction as field coupling with work and stress [5]; HAFF D distinguishes entropy removal, usable side information and supplied purity, and gives specified source–response constructions [2]. These results suggest a maintained-structure mechanism: actual records learn a protected frame, operations install it, maintenance changes the physical source, and the response changes subsequent records. Foundation III realizes a finite retained-history scalar/JT return, whereas the full quantum material source requires its own support and preparation account. Connecting those implementations and renewing their resources is a substantive requirement, not an irreducible heat flux inferred from algebra inclusion.

Many-site and continuum selection require further work when rate gaps close. Consciousness I's ideal fixed-net closure and finite-time noise estimates are not already a continuum uniqueness theorem for window-selected blocks. Likewise, the graph dictionary is supplied here. A general emergence of spacetime needs a physical family of readers and a reconstruction law that remains controlled as the operational structure grows.

11 Conclusion

A commuting coupling determines its fixed block algebra and its long-time pinching. On a separated finite window it selects a unique block partition in the supplied joint eigenbasis; the selected algebra contains every stable block algebra of that form. The actual finite-time channel has full operator range. Block-constant record couplings report sector weights without resolving protected content. Independent leakage multiplies the coherence loss and imposes a sharp redundancy–symmetry threshold.

The relation to geometry has two parts. Sector mixing and within-sector content contribute through the exact identity (16); neither alone determines the mutual information graph. For the GHZ source, collective, pair-collective and local couplings yield a triangle, one edge and no edge under one rule. Weak asymmetry gives a finite plateau and then destroys its supporting edges. Noncommuting perturbations can erase the exact fixed algebra without abruptly erasing a finite record or task capability.

Formation requires a history. An initially shared logical Bell pair can survive between noise-disconnected protected components, so a noise graph cannot delimit all surviving quantum relations. A collective recorder forms a flagged protected Bell branch from a product input, with all branches retained; an actual Hamiltonian link forms a deterministic Bell pair protected throughout its generation. These constructions reduce the missing formation step to identifiable physical operations without hiding their costs. Record-controlled encoding supplies the further route from learned relations to changed capability. The remaining general problem is to derive reproducible subsystem operations and their sustained maintenance from such histories, rather than merely rename correlated states as subsystems.

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Causation, Agency, and Existence: Historical Control and Maintained Intervention

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Abstract

A cause changes what an allowed intervention can bring about; a record alone need not identify that change. We define interventional influence relative to a process, an intervention set and a readout. It contracts under post-processing and along channel chains. A direct cause and a common cause can have identical declared record laws and influences 1 and 0.

Agency requires more than a persistent register. In a binary tracking task, the memoryless failure probability is bounded by $\frac{1}{2}[1-(1-2q)\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}]$. Three retained readings beat every present-reading controller in the specified sensor menu, despite a smaller influence of the present world on the action. We then give the same agent a damageable body: an actual mixed-port diagnosis controls repair, and the repaired body determines the next sensing channel. The finite reversible protocol maintains tracking error at most 0.15820 from a healthy start, against a long-run lower bound 0.22172 for damage-independent maintenance calendars at the same repair rate. All spent ports and old records remain; fresh blank supply is not a thermal reset. This is a mechanism for maintaining intervention capability, not a derivation of the initial body or its couplings.

These results support relational existence: structures can be objective within their coupling, history and observation window without being intrinsic to a state alone. The essay develops implications for will, self-maintenance and structural realism, then compares traces and dependence with Yogācāra and Mādhyamaka. These comparisons have their own arguments and do not establish phenomenal consciousness.

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1 Introduction

A register can remain unchanged while doing nothing. An agent can change almost every internal value while continuing to sense, act and preserve the conditions for doing so again. The distinction matters when a quantum state does not by itself select a division into things, causes and agents.

The earlier papers provide different parts of the physical account. Paper A [1] separates process-selected sectors, identification through actual responses and reconstruction of spatial geometry. Its finite correlation geometries keep their source and dictionary explicit; its continuum target requires more than relabelling those examples. Paper B [2] derives stable block content in a commuting coupling class and distinguishes pre-existing relations from relations formed by actual recording or Hamiltonian links. Block-constant couplings record central labels; arbitrary environmental records need not have that form. Paper E [3] derives measurement permissions, partly mixed records and calibrated public readout. Paper F [4] distinguishes event-order identification from record retention and updating; Paper G [5] asks how much of the selection can be performed from inside the process.

The present question is how these structures acquire causal and agentive roles. Section 2 separates observation from intervention. Section 3 bounds viable control with and without history, then closes a missing step: records act on the same body whose condition controls later sensing. Section 4 develops relational objectivity rather than identifying reality with a reader's answer. Section 5 compares the argument with Buddhist accounts of traces and dependence. The initial partition and gate permissions are stated inputs to the finite mechanism; their primitive formation remains a broader question.

2 Part I: Causation, records and interventions

2.1 Observational asymmetry and its limits

Standard quantum mechanics assigns no joint density operator to non-commuting observables at two times without a process formalism, so a cross-time mutual information is defined only relative to declared instruments.

Definition 2.1 (Instrument-relative directional asymmetry)**.** *Fix a two-time process $\mathcal{N}_{t_1\to t_2}$, an initial preparation and instruments $\mathcal{I}_A, \mathcal{I}_B$ with classical outcomes $X_A^{t_i}, X_B^{t_i}$, measured in separate forward and reverse arrangements. Put*

$$\Delta^{\mathcal{I}}_{A\to B} = I(X_A^{t_1} : X_B^{t_2}) - I(X_B^{t_1} : X_A^{t_2}),$$

with Shannon mutual informations of the outcome distributions.

$\Delta^{\mathcal{I}}$ is a well-defined observational diagnostic. Its sign depends on the instruments, the preparation and the process, and it can arise from a common cause or from asymmetric noise. Other conventions for quantum two-time objects exist [9, 10]; none turns a correlation into a cause without an account of intervention, which we now give.

2.2 Interventional influence

Definition 2.2 (Interventional influence)**.** *Fix a process in which a declared set $\mathcal{J}_A$ of interventions can be applied at A, each replacing the mechanism that ordinarily determines A while everything else is held fixed, and fix a readout at B. Let ρ_B^J be the state, or the outcome distribution, at B under the intervention J. The interventional influence of A on B is*

$$\kappa_{\mathcal{J}}(A \to B) = \sup_{J,J' \in \mathcal{J}_A} D\big(\rho_B^J, \rho_B^{J'}\big),$$

where D is the trace distance, or the total-variation distance for classical outcomes.

For several steps, with memory in the environment and controls at several times, the natural setting is a quantum comb or process tensor [11, 12]; the definition applies to any slot of such an object.

Proposition 2.3 (Properties of interventional influence)**.** *1. $0 \le \kappa \le 1$, and $\kappa = 0$ exactly when no allowed intervention changes the state at B.*

2. Post-processing the readout at B by any channel, or restricting $\mathcal{J}_A$, cannot increase κ.

3. If the interventions at A affect B only through M, in the sense that $\rho_B^J = \Lambda(\rho_M^J)$ for a channel Λ that does not depend on J, then

$$\kappa(A \to B) \le \eta(\Lambda)\, \kappa(A \to M), \qquad \eta(\Lambda) = \sup_{\rho \neq \sigma} \frac{D(\Lambda\rho, \Lambda\sigma)}{D(\rho, \sigma)}.$$

For a classical channel with kernel $\Lambda(\cdot \mid m)$, $\eta(\Lambda) = \max_{m,m'} D(\Lambda(\cdot \mid m), \Lambda(\cdot \mid m'))$, Dobrushin's coefficient [13], which is the influence of M on B under all point interventions on M.

Proof. (1) is immediate. (2) Trace and total-variation distances do not increase under channels, and a supremum over a smaller set is smaller. (3) $D(\Lambda\rho_M^J, \Lambda\rho_M^{J'}) \le \eta(\Lambda) D(\rho_M^J, \rho_M^{J'})$ by the definition of η. For classical kernels the supremum is attained on pairs of point masses, which is Dobrushin's formula. □

Part (3) is the causal counterpart of the data-processing inequality: influence passed along a chain is at most the product of the influences of its links. Section 3 uses it to show that an agent needs both sensing and control.

2.3 Records do not reveal causes

Proposition 2.4 (Same observations, different causes)**.** *Consider two processes on bits A, B observed in the computational basis at times $t_1 < t_2$.*

1. Direct cause: A is a uniform bit at t_1, $B = 0$ at t_1, and a controlled NOT from A to B acts between t_1 and t_2.

2. Common cause: a hidden uniform bit C is copied into A before t_1 and into B between t_1 and t_2; $B = 0$ at t_1.

The joint distributions of $(X_A^{t_1}, X_B^{t_1}, X_A^{t_2}, X_B^{t_2})$ *coincide, both equal to* $\frac{1}{2}$ *on* $(0,0,0,0)$ *and on* $(1,0,1,1)$*, and both give* $\Delta^{\mathcal{I}}_{A\to B} = \ln 2$*. For the interventions "set* A *to* 0*" and "set* A *to* 1*" at* t_1*, the influence on* B *at* t_2 *is* 1 *in the first process and* 0 *in the second.*

Proof. In both processes A keeps its value, B starts at 0 and ends equal to the uniform bit carried by A. Under the intervention, B at t_2 equals the set value in the first process and the hidden bit in the second. □

Every statistic of the declared observed variables is the same in the two processes, so accumulating more of those records cannot settle which structure produced them. For quantum variables the situation is partly different: some causal structures can be inferred from observations alone when entanglement or coherence is available [14]. That is an advantage of particular quantum protocols, not a general replacement for interventions.

2.4 Records and interventions in one process

A record carries a recoverable difference from an earlier process. An intervention changes a declared mechanism or operation. They can occur within the same interaction without being the same role. Paper E shows that a memory need not be pure to record: the relevant budget is its purity deficit $\ln d - S(\eta)$, and partly mixed pointer ports have nonzero readable information. Records of a commuting pointer label are a specific acquisition model, not a classification of every record.

For an already detached port, a later trace-preserving operation on the source leaves its marginal unchanged, by no-signalling. A renewed interaction with the port, or a direct operation on it, can change its state. Reversible relabelling, repair, transfer into another memory and reset have different resource balances. Paper F treats them through the joint state; they cannot all be priced as erasure of a marginal entropy. The Imprint Chain [6] makes a complementary distinction: a passive record and a feedback can carry the same information about an interface while only the feedback changes the world state.

In Proposition 2.4, the recorded correlation between A and B does not make A a cause of B. Conversely, an influence can change a state without a record being retained. Memory consists here of physical records that remain available to later decisions. Their availability may itself depend on repair, storage and permission.

The influence comparisons here already have a time order. Paper F also asks whether an unknown contact order can be recovered from endpoint relations. Ordinary pointer-copying gates fail that test even with complete endpoint access; its retained-label record–swap protocol passes within a known one-pass architecture. That theorem does not make the observational process in Proposition 2.4 causally identifiable. The permitted contacts, interventions and preparation are different. Recontact can restore coherence and reduce local entropy; neither recovery of a permutation nor record growth produces an external time parameter.

3 Part II: Historical control and maintained capability

3.1 What a boundary criterion must allow

A tempting boundary criterion asks every internal observable to stay nearly unchanged: $\sup_t \sup_{\|O\|\le 1} \|\mathcal{E}_t^*(O) - O\| \le \epsilon$. It admits an isolated frozen register and excludes a changing tracker. In the example below, the internal $Z_r = (-1)^r$ has deviation $1+(1-2\nu)(1-2q) = 1.768$, although the tracker succeeds. What must persist is a capability and the physical organization that supports it, not every stored value.

A stable algebra of the kind derived in Paper B can hold part of that organization. It does not by itself provide the sensor, actuator or maintenance procedure. We first specify the task and then construct the maintenance mechanism.

Definition 3.1 (Viable agent on a window)**.** *Fix a declared partition into a world variable W, an agent register R and an actuated variable b, declared interventions on W and on R, a viability condition $v(W, b)$ and a window. The register is a viable agent on the window, with margin $\eta < \frac{1}{2}$, if*

1. *the viability condition fails with probability at most η at every step of the window;*
2. *the sensing influence $\kappa_{\mathrm{s}} = \kappa(W \to R)$ is positive;*
3. *the control influence $\kappa_{\mathrm{c}} = \kappa(R \to b)$ is positive.*

The agent is memoryless if its action at a step depends on the world only through its reading at that step.

3.2 Sensing and control multiply

Consider a world bit W_t that flips with probability $q < \frac{1}{2}$ at each step. The agent reads it into R, acts on b, and is viable when $b_{t+1} = W_{t+1}$.

Proposition 3.2 (Viability needs sensing and control)**.** *Let a uniform world bit be read through a fixed sensor channel with independent fresh randomness. The actuator receives the world only through that reading, uses independent randomness, and the next world flip is independent of both. Thus the sensor's ordinary conditional output law equals its output under interventions setting W_t, with no initial world–controller common cause or retained world-correlated side input. For this memoryless menu, with κ_{s} computed for interventions setting W_t to 0 or 1 and κ_{c} for all point interventions on R,*

$$\Pr(b_{t+1} \neq W_{t+1}) \;\geq\; \tfrac{1}{2}\big(1 - (1 - 2q)\,\kappa_{\mathrm{s}}\kappa_{\mathrm{c}}\big). \tag{1}$$

For a sensor that errs with probability ν and an actuator that errs with probability μ, both symmetric, equality holds and the misalignment is $\frac{1}{2}(1 - (1 - 2\nu)(1 - 2\mu)(1 - 2q))$.

Proof. Let K be the channel from W_t to b_{t+1}. With $W_{t+1} = W_t \oplus f$, $\Pr(f = 1) = q$, the success probability is $q + \frac{1}{2}(1 - 2q)(K(0|0) + K(1|1))$, and $K(0|0) + K(1|1) \leq 1 + D(K(\cdot|0), K(\cdot|1)) = 1 + \kappa(W_t \to b_{t+1})$. By Proposition 2.3(3), $\kappa(W_t \to b_{t+1}) \leq \kappa_{\mathrm{s}}\kappa_{\mathrm{c}}$. For symmetric channels $\kappa_{\mathrm{s}} = 1 - 2\nu$, $\kappa_{\mathrm{c}} = 1 - 2\mu$, and the misalignment is the probability that an odd number of the three independent events sensor error, actuator error and flip occurs. □

Corollary 3.3 (Records without control, control without records)**.** *In this fresh-sensor task, a passive recorder with $\kappa_{\mathrm{c}} = 0$, an open-loop actuator with $\kappa_{\mathrm{s}} = 0$, and an initially uncorrelated frozen register fail with probability $\frac{1}{2}$.*

The channel condition is necessary. If a hidden fair bit is copied into W_t and a frozen R before the slot, then $b_{t+1} = R$ has ordinary error q, although intervening on current W_t does not change R: $\kappa_{\mathrm{s}} = 0$. The common-cause record is useful without a new sensing influence. It is not a fresh sensor channel from the intervened world. Treating observed $\Pr(R \mid W_t)$ as that channel would erase the distinction proved in Proposition 2.4. Retained world-correlated inputs must instead be included as memory, as in the next subsection.

The magnitudes $\kappa_{\mathrm{s}}, \kappa_{\mathrm{c}}$ alone do not certify the labels needed for the directional task: a reversed sensor has the same influence as a correctly oriented one. T-DOME III shows that acting cannot reveal parameters moved by a symmetry of the complete record law. Certifying this directional task therefore needs a symmetry-breaking labelled operation or retained standard [31]. It may be physically implemented inside the agent; the requirement concerns the information it supplies, not its location outside the loop.

With $\nu = 0.1$, $\mu = 0.05$ and $q = 0.02$ the memoryless agent fails with probability 0.1544, while its register changes value in one step with probability 0.193. It is viable in Definition 3.1 and fails the pointwise-invariance criterion; an initially uncorrelated frozen register passes that criterion and has no sensing route.

3.3 Memory changes what an agent can do

Let the agent instead act on the majority of its last three readings. Exact enumeration gives a misalignment of 0.1082. The influence of the present world state on the action falls at the same time: an intervention on W_t changes only the newest reading, which decides the majority only when the two older readings disagree, so the influence is $(1-2\nu)(1-2\mu)\Pr(\text{older readings disagree}) = 0.139$, against 0.72 for the memoryless agent.

Corollary 3.4 (Memory is necessary for this level of viability)**.** *In the specified drifting-bit menu with one fresh sensor reading per slot, every memoryless agent fails with probability at least* $\frac{1}{2}(1-(1-2q)(1-2\nu))$*, which is* 0.116 *for* $\nu = 0.1$*,* $q = 0.02$*. The majority-of-three agent fails with probability* 0.1082 *even with an imperfect actuator,* $\mu = 0.05$*. Within this menu, a failure probability below* 0.116 *therefore requires dependence on past readings.*

Proof. A memoryless agent's register depends on W_t only through the reading, so $\kappa_{\mathrm{s}} \le 1-2\nu$ by Proposition 2.3(2); Proposition 3.2 with $\kappa_{\mathrm{c}} \le 1$ gives the bound; the majority value follows from enumerating the 2^8 values of the initial world bit, three flips, three reading errors and actuator error. The exact value is 0.10822784; Appendix A gives the verification method. □

The memory agent tracks the world better while the present world influences it less. Its success rests on records of past world states, which are correlated with the present one because the world persists; an intervention on the present world cannot reach those records. Agency therefore involves two different relations to the world: causal influence through current sensing and control, and correlation carried by records. Proposition 3.2 bounds what the first can achieve alone. This is the task-level form of the memory resource studied in T-DOME I [8], and it is the step at which the self-reading of Imprint W [7] and the internal selection rules of Paper G become relevant: an agent that also keeps a model of its own sensor and actuator, and revises it from its records, can choose which readings to take.

3.4 Records maintain the body that reads next

The tracker so far has a supplied sensor. Give it a body bit D_t: 0 is functional and 1 damaged. The same physical body remains throughout the run; its condition modulates the world-sensing coupling, and it can be repaired before sensing. Damage is independent of the world, arrives after each task slot with probability $0 < d < 1$, and is absorbing until repair. This is a specified failure mechanism, not a biological model.

Each slot begins with diagnosis. A fresh port in $\eta_{\mathrm{d}} = [I + (1-2e)Z]/2$, $0 < e < 1/2$, interacts by controlled X with D_t. Its Z record is $Y_t = D_t \oplus N_t$, $\Pr(N_t = 1) = e$, from the mixed-port instrument of Paper E. Exactly when $Y_t = 1$, the controller swaps a fresh $|0\rangle$ into D_t; the old body state goes into that port and remains in the spent bank. The post-repair body is $D'_t = D_t(1-Y_t)$. The decision uses Y_t, not hidden D_t.

The world sensor then gives

$$R_t = W_t \oplus N_t^{\mathrm{s}} \oplus D'_t Q_t, \qquad \Pr(N_t^{\mathrm{s}} = 1) = \nu, \quad \Pr(Q_t = 1) = \tfrac{1}{2}. \tag{2}$$

For a healthy body it is the original sensor; for a damaged body its reading is fair. Two gates realize this law: controlled X from W_t to a port in $[I+(1-2\nu)Z]/2$, followed by a flip controlled by D'_t and a fresh fair bit Q_t. The spent Q_t is retained but outside the controller's readout menu. Joint access to it would change the sensing permission.

The reading drives the same actuated variable b. A fresh blank is swapped into b, retaining its old value; copying R_t and adding independent actuator error gives $b_{t+1} = R_t \oplus N_t^{\mathrm{a}}$, $\Pr(N_t^{\mathrm{a}} = 1) = \mu$. The world then flips with probability q, and finally the body suffers damage: $D_{t+1} = D'_t \vee J_t$, $\Pr(J_t = 1) = d$. Fresh ports are independent of the retained body, world and history. Diagnostic orientation and error are known or separately calibrated; a common unknown sign is not eliminated by a self-certifying reference.

Proposition 3.5 (Maintenance of a sensing route)**.** *Let* $x_t = \Pr(D_t = 1)$*,* $z_t = \Pr(D'_t = 1)$*, and* $r_t = \Pr(Y_t = 1)$*. Then*

$$z_t = ex_t, \qquad r_t = e + (1-2e)x_t, \qquad x_{t+1} = d + (1-d)ex_t, \tag{3}$$

$$x_* = \frac{d}{1-(1-d)e}, \qquad z_* = ex_*, \qquad r_* = e + (1-2e)x_*. \tag{4}$$

With $\rho = (1-d)e$*,* $x_t = x_* + \rho^t(x_0 - x_*)$*. For a uniform world independent of the initial body, the raw sensing influence and tracking failure are*

$$\kappa_{\mathrm{s},t} = (1-z_t)(1-2\nu), \tag{5}$$

$$\mathcal{R}_t = \mathcal{R}_0 + (\tfrac{1}{2} - \mathcal{R}_0)z_t, \qquad \mathcal{R}_0 = \tfrac{1}{2}[1-(1-2q)(1-2\nu)(1-2\mu)]. \tag{6}$$

A healthy start gives $z_t \le z_*$ *and* $\mathcal{R}_t \le \mathcal{R}_0 + (\frac{1}{2} - \mathcal{R}_0)z_*$ *at every slot. Without repair, sensing influence tends to zero and tracking failure tends to* $1/2$*.*

Proof. Only a damaged body missed by diagnosis remains damaged, with probability ex_t. The diagnostic is positive on a healthy body with probability e and on a damaged body with probability $1-e$. Independent damage gives $x_{t+1} = z_t + d(1-z_t)$. Solving the recurrence gives the fixed point and convergence. Repair and damage do not depend on the world, so the body remains independent of W_t. The sensor mixes the original channel, with weight $1-z_t$, and a fair channel, with weight z_t, giving its influence and risk. Without repair, $x_t = 1-(1-x_0)(1-d)^t$. □

This is intervention rather than relabelling. Replacing diagnosis by *never repair* or *always repair*, with the pre-slot distribution fixed, changes post-repair damage from x_t to zero and risk by $(\frac{1}{2} - \mathcal{R}_0)x_t$. Setting the body to healthy or damaged after repair changes the available sensing channel. The record maintains a physical condition of later intervention, not merely confidence in a previous answer.

Proposition 3.6 (A matched calendar comparison)**.** *Consider repair calendars independent of damage and of all information about it, including diagnostic or world-sensor histories. They may use private randomness and have positive long-run repair frequency* r*, with empirical mean gap* $1/r$ *and vanishing end effects in the occupancy average. In the same present-reading control menu, their long-run mean post-repair damage is at least*

$$z_{\mathrm{cal}}(r) = r[(1-\omega)G(L) + \omega G(L+1)], \quad L = \lfloor r^{-1} \rfloor, \quad \omega = r^{-1} - L, \tag{7}$$

where

$$G(\ell) = \sum_{j=0}^{\ell-1}[1-(1-d)^j] = \ell - \frac{1-(1-d)^\ell}{d}.$$

Gaps $L, L+1$ *in these proportions attain the bound. Frequency at most* r *gives the same lower bound.*

Proof. A gap of length ℓ, starting with repair, has expected damaged occupancy $1-(1-d)^j$ at age j, with sum $G(\ell)$. Conditioning on an independent calendar preserves the damage law. The differences $G(\ell+1) - G(\ell) = 1-(1-d)^\ell$ increase. The convex envelope on the integers minimizes average gap occupancy at fixed mean gap $1/r$ by balancing adjacent lengths. Multiplication by gaps per slot gives (7). Adding repairs cannot increase occupancy, so the allowance is optimally used in full. □

For $d = e = 0.1$ and $q = 0.02, \nu = 0.1, \mu = 0.05$,

$$x_* = \frac{10}{91}, \quad z_* = \frac{1}{91}, \quad r_* = 0.18791209, \quad \mathcal{R}_* = 0.15819780.$$

The optimal independent calendar at this repair frequency has $z_{\mathrm{cal}} = 0.19479225$ and mean error 0.22172020. Both receive the same diagnostic and world-reading ports and the same repair and actuator menu. One unit is charged per actual repair, with other per-slot inputs equal. A controller inferring damage from sensor history is another record-guided strategy, outside the calendar class. Equal long-run rate does not mean equal repair counts on every finite trajectory.

3.5 Finite stock and a maintained boundary

For n slots, the realization has three continuing bits D, W, b and nine fresh ports per slot: diagnostic, repair blank, world sensor, fair sensor-noise, actuator-noise, world-flip, damage coin, damage-dilation blank and actuator blank. Maximum repair count is n; a healthy start gives expected count at most nr_*. Supplying the bank, storing spent ports and routing records belong to the finite-window budget. Mean activation count is not stock capacity.

Each of the three blank ports per slot supplies initial purity deficit $\ln 2$. Diagnostic and sensor ports supply $\ln 2 - h(e)$ and $\ln 2 - h(\nu)$; actuator, world-flip and damage coins supply $\ln 2 - h(\mu), \ln 2 - h(q), \ln 2 - h(d)$; fair noise supplies none. Here h is binary entropy in nats. Unused repair blanks may remain pure; supplied purity is not necessarily exhausted on every slot. Old states and correlations remain; the reversible circuit in Appendix A conserves joint entropy. No bath is reset and no work or heat price follows from counting swaps. For the numerical parameters, the nine-port input purity is $8\ln 2 - 3h(0.1) - h(0.05) - h(0.02) = 4.27337417$ nats per slot. For degenerate port Hamiltonians, $k_B T$ times this quantity is the free-energy difference from a maximally mixed preparation at temperature T, not heat dissipated or net work consumed by the run. The initialized healthy body and blank actuator in the explicit check add two initial pure bits; the world starts fair. Gate strengths, preparation, duration, clock, storage and later bank renewal remain physical inputs.

The body's condition regulates whether a reading is informative, and a record of that condition regulates the operation preserving the route. Organization survives while its values change. Continued intervention capability is produced by record-conditioned action, rather than merely imposed as a viability condition. Repair uses the current diagnosis only; earlier diagnoses remain stored but are not read by this policy. The majority tracker establishes the value of longer history, whereas this circuit establishes the maintenance of later permissions. Neither result alone constructs a long-term self-model.

Remark 3.7 (Maintained organization)**.** *A broader agency hypothesis is that an effective agent boundary is the organization of sensing and action routes whose continued availability is causally regulated by retained history. Disrupting use of a bodily diagnostic should impair later intervention capability even when current discrimination is spared. At a matched budget, an independent calendar should fail where a condition-dependent policy succeeds.*

The binary body establishes one instance. Formation must also explain how routes and the repair menu arise from earlier interactions and how histories coordinate several demands. The hypothesis explains no additional boundary if a fixed external controller accounts for every apparent self-maintenance effect, or if manipulating history never changes later permissions. BRAIN supplies learned protection and routing for a different body model [32]; Imprint W and Imprint SH price reliability-based action [7, 33]. Their dynamics and budgets must match before they form a larger construction; they are not assumed to be this circuit.

3.6 Which aspects of a body are being formed?

The models connected to this account change different physical relations. In the present circuit, D'_t gates the next sensor channel and Y_t controls actual replacement of the damaged body state. The repair menu and its known diagnostic orientation are supplied. T-DOME III instead asks what a retained model must know to choose a priced repair: histories with the same sign

confidence can give different values of future access, because gain, physical body state and their correlations differ [31]. BRAIN learns relations among several ports and uses their connected components to route and maintain access [32]. These are distinct mechanisms, not repeated observations of one neural system.

There is a common intervention question. Hold current world discrimination and available repair operations fixed, then change whether the policy can use its own diagnostic history. A difference in later sensing capability tests the path from history through maintenance to future access. In this paper the matched calendar comparison isolates that path within a declared repair budget. A loss of performance alone does not identify it: destroying a port, changing the world or reducing the available repair stock can produce the same loss. The comparisons therefore match those inputs and use the intermediate body state and sensor law as separate readouts.

A formation claim requires a further change: history must select some relation used by later sensing or control, rather than merely operate a preselected repair schedule. The learned route in BRAIN and the retained foreground of T-DOME address that step in their own task classes. Combining them with this circuit would require a common transition law, diagnostic permission and resource ledger. Their conceptual fit is a reason to seek that construction, not evidence that it has already been made.

3.7 Will, degrees of agency and free will

On this account, "will" names the control route from an agent's records to the world: the influence κ_c, together with the dependence of the action on the agent's records. A thermostat has a sensing route, a control route and a viability condition, and it is a viable agent in the weak sense of Definition 3.1. Organisms differ from it in degree along several axes: the length of the window over which viability is maintained, the margin, the strength of sensing and control, the depth of memory, and whether the agent can revise its own selection rules. Agency is graded along these axes and is not an all-or-nothing property.

The traditional free-will question presupposes well-defined agents and causal histories. The present account supplies both relative to a declared partition and intervention set, and it turns part of the question into questions that can be answered: how strongly the agent's records, rather than the present state of the world, determine what it does, and whether its selection rules are themselves revised from its records. It does not settle the metaphysical question and is not offered as doing so.

The account says nothing about what it is like to act. The explanatory gap between functional organization and experience [15, 16] remains. One empirical route connects these notions to neuroscience, where habitual and flexible control have been associated with distinct cortico-basal ganglia loops [17, 18] and the sense of agency with motor prediction [19]; the relevant times there are classical control times, not quantum coherence times [20]. Whether the quantities defined here are useful for that research is open.

4 Part III: Existence without substance

4.1 Relational existence

Parts I and II treat causes and agents as real within declared contexts and dependent on them. This raises the question of their ontological status. Two extremes are available and both are mistaken: treating context-dependent structures as fundamental, and denying them objective reality because they depend on context. We propose structural realism about relational patterns [21, 22].

Definition 4.1 (Relational existence)**.** *An entity X exists relative to a context $\mathcal{S}$ if X is well defined and stable within $\mathcal{S}$, participates in objective relational patterns there, and may be absent or differently constituted in another context $\mathcal{S}'$.*

Definition 4.2 (Emptiness, technical sense)**.** *An entity is empty in the technical sense if it lacks being independent of every context while exhibiting stable patterns within specified contexts. Emptiness in this sense entails neither non-existence nor illusion.*

Paper B supplies physical instances of a context. In its commuting-coupling class, the stated stability and observability criteria select a unique largest admissible block algebra on a window. Smaller admissible algebras also exist. Selected content depends on coupling and window, yet the same inputs define the same criterion for all readers. Readable redundant labels additionally require actual recording couplings, fragment preparation and calibrated readout, as in Paper E; stability alone does not make them public.

The maintained body adds history and budget to the context. The sensor's operational distinction survives under record-guided repair and disappears under unmaintained damage. That difference is real in the task outcomes even for a reader unaware of the controller's policy. Relational objectivity concerns these reproducible consequences, not agreement on a chosen label.

4.2 Examples

Temperature depends on which degrees of freedom are treated as macroscopic, yet given that choice it is measurable and predictive. Quasiparticles such as phonons and magnons are collective excitations with no counterpart among the microscopic constituents; operationally they are stable peaks in spectral functions, dispersion relations, lifetimes and scattering amplitudes, all of which depend on temperature, doping and resolution. Subsystems are the case closest to this framework. A state alone does not privilege one of its many factorizations, and Paper A shows that entropies, mutual informations and reconstructed geometries depend on the channel through which the state is read; Paper B shows how couplings select one structure over a window.

4.3 What is invariant

Objectivity does not require independence from every context. It requires that patterns be stable and predictive within a context and that the transformations between contexts be stated. What stays invariant depends on that class of transformations. Under a change of factorization, implemented by a unitary relabelling of the Hilbert space, the spectrum of the global state is invariant, while subsystem entropies, mutual informations and the geometries built from them are not. For readers applying the same selection criterion with couplings and window fixed, the selected algebra is unchanged. Access to its labels requires the declared record protocol; readers with different measurement permissions need not obtain the same information. A mere unitary change of coordinates is also different from physically changing a coupling.

5 Part IV: Interpretive bridges to Buddhist philosophy

5.1 Method

The comparison below is between formal patterns. It does not claim historical influence in either direction, endorsement of Buddhist doctrine, or that physics confirms any religious view; "Buddhism" is not treated as one doctrine, and the discussion is limited to some Yogācāra and Mādhyamaka texts. One limitation must be stated plainly. The framework was named after the ālaya-vijñāna, so a resemblance between the two was partly built in by the naming and cannot count as independent evidence for either. The mathematical results of Parts I–III do not depend on the comparison; the comparison depends on them.

Kind	Characterization	Example
Intrinsic existence	independent of every context	classical particles, if fundamental
Relational existence	dependent on context, objective within it	temperature, selected subsystems, records
Conventional existence	dependent on context and on agreement among agents	money, legal rights
Non-existence	absent from every relevant context	phlogiston, luminiferous ether

Table 1: Kinds of existence claim. The causal influences, agents and geometric features of this series belong to the second kind.

5.2 Yogācāra: traces that condition later appearance

Yogācāra texts describe the ālaya-vijñāna, the store consciousness, as the stream in which the impressions (*vāsanā*) of past actions are kept as seeds (*bīja*) that condition later experience [23, 24, 26, 27]. Vasubandhu's *Triṃśikā* says that it proceeds like a torrent; it is a changing stream, not a fixed container. The following correspondences are formal analogies.

Yogācāra notion	Counterpart in this series (analogy only)
Ālaya-vijñāna, a stream that keeps traces	the global state together with the records its history has formed
Bīja and vāsanā, seeds and impressions	records that change what later readouts can distinguish (Papers E, F; the Imprint papers)
Pravṛtti-vijñāna, the active consciousnesses	selected readouts and the effective subsystems they define (Papers A, B)
Āśraya-parāvṛtti, transformation of the basis	a change of couplings that changes which structure is selected, as opposed to a change of description (Paper B)

Table 2: Formal analogies. Yogācāra concerns consciousness and liberation; HAFF concerns physical structure, and its global state carries no phenomenal properties.

The analogy concerns the role of traces. In both pictures what can appear later depends on what earlier processes have left behind, and the traces are not inert stores but conditions on later access. The differences are as important. The Yogācāra substrate is described as consciousness; the HAFF state is physical and the framework is neutral on mind and matter. The analogy does not establish panpsychism, idealism or the claim that quantum information is conscious.

5.3 Mādhyamaka: dependence without a foundation

For Nāgārjuna, whatever arises dependently is empty of intrinsic nature (*svabhāva*), and that emptiness is itself a dependent designation, the middle way (*Mūlamadhyamakakārikā* 24:18) [25, 28, 29]. The technical emptiness of Definition 4.2 has the same form: denial of intrinsic being together with affirmation of functional reality.

Dependence also applies to the apparatus of selection. Selected content depends on couplings and observation windows (Paper B). Paper G constructs choices inside a fixed executor and separates that execution closure from an explanation of the executor's own organization. It does not prove that every possible model has an unrevisable last selector. The supplied dynamics of a finite construction is an explanatory boundary of that construction, not an independently existing ground. A larger model may explain some of it. Reading this revisability as dependence without an ultimate base is a philosophical proposal, not a consequence of a finite-register count.

The comparison with Mādhyamaka concerns this form of dependence; the tradition's use of it to loosen attachment to views has a different practical purpose.

5.4 Karma: actions, traces and later access

Buddhist accounts of karma hold that intentional actions leave traces that condition future experience [30, 34]. In the stated fresh-port process, an intervention on the source changes later records while detached earlier ports retain their marginals (Paper F). Records then condition later reading and action. In Section 3.4, they condition whether later sensing remains informative. The analogy is formal and stops at normativity. Karma classifies actions as wholesome or unwholesome; interventional influence and records carry no ethical valuation.

5.5 Where the traditions and the framework differ

The differences are substantive. Buddhist philosophy starts from first-person experience and aims at liberation from suffering; the physical results used here begin with third-person structure. Rebirth, karmic continuity across lives, cosmological realms and meditative attainment have no counterpart here, and the comparison neither supports nor undermines them. Yogācāra readings that give consciousness priority differ from the framework's neutrality. And the methods differ: contemplative inquiry, textual interpretation and dialectic on one side, mathematics and experiment on the other. The comparison is worth making because both traditions and the framework deny intrinsic subsystem structure while affirming stable relational patterns, and because precise statements on one side can sharpen questions on the other.

6 Conclusion

Causal influence is a property of a declared intervention process, not of correlation alone. It contracts through links and distinguishes processes whose observed records coincide. In the tracking task, sensing and control jointly limit memoryless viability; history can surpass that limit without raising the influence of the current world on the action.

The same finite task now includes a damageable body. An actual diagnostic controls repair of that body, whose condition determines the next sensor channel. The recurrence and matched calendar comparison show how a record maintains later intervention capability. Fresh blanks and spent history are explicit resources; this is a finite-stock mechanism, not a perpetual thermal cycle or the primitive origin of the initial body.

Agency so understood is a maintained relational organization. Partition, loss and permissions specify the task; success and failure are physical consequences, not matters of agreement. A fuller formation account must generate and coordinate the routes rather than supply all of them. The Buddhist comparison clarifies a related philosophical possibility: traces and dependent structures can have functional reality without an independently existing substrate. Its experiential and ethical questions remain independent of the physical mechanism.

A Reversible realization and checks

The body protocol is an explicit basis-permutation unitary on twelve bits. Besides continuing D, W, b, use fresh $(C, B, R, Q, A, F, J, G, V)$ with initial one-probabilities

$$(e, 0, \nu, \tfrac{1}{2}, \mu, q, d, 0, 0).$$

Here C, B are the diagnostic and repair ports; R, Q the sensor and fair-noise ports; A, F, J actuator, world and damage randomness; G, V blank damage and actuator ports. On diagonal states perform, in order:

1. $C \leftarrow C \oplus D$; swap D, B conditional on $C = 1$.

2. $R \leftarrow R \oplus W$; $R \leftarrow R \oplus DQ$.

3. Swap b, V; $b \leftarrow b \oplus R \oplus A$.

4. $W \leftarrow W \oplus F$.

5. $G \leftarrow G \oplus D$; $D \leftarrow D \oplus [(1 - G)J]$.

Each gate is reversible on all basis inputs, not just the initialized subspace. Since G starts at zero, the last step stores the post-repair body in G and gives $D_{\text{next}} = D' \vee J$. Repair uses $C = Y$ alone. The physical control of noise by the body does not grant a hidden diagnostic read. All ports remain in the spent bank, and the next slot uses fresh ones. Z reading of C, R is compatible with the diagonal protocol and does not change its joint law. Realizing these gates and their known orientations is a charged hardware permission.

The check enumerates all 4096 input strings and verifies that the composite is a permutation. For diagonal product inputs with $x = \Pr(D = 1)$ equal to $0, 1/5, 10/91, 1$, exact rational probabilities reproduce every term of (3) and (6). Old b does not affect the output: it remains in V. Joint entropy is unchanged. This is a circuit distribution check for the stated classical task, not unknown quantum-source or bath certification.

The calendar check verifies discrete convexity of G and the optimal $5, 6$ gap mixture at the stated rate. Independently, enumeration of the original three-reading task gives 0.10822784. The probability that the two older readings disagree is $\frac{1}{2}[1 - (1 - 2q)(1 - 2\nu)^2] = 0.1928$, giving present-world influence 0.138816. Analytic contraction and calendar proofs, rather than numerical sampling, establish the general statements.

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Selection, Maintenance and Gravitational Response: Modular Constraints and a Physical Return

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Abstract

Can the physical selection of observable algebras account for gravitational dynamics, rather than merely change what an observer can reconstruct? We formulate this question as an adiabatic evolution of selected localization structures and a common-action consistency requirement. Modular stability alone does not select the algebra. A supplied half-sided modular family and holographic dictionary give a conditional linearized gravity route. In the same-spacetime conformal scalar model of Consciousness II, the improved stress and its Wald surface coefficient connect accessible preparations to an all-ball test of the Einstein response. The test still requires modular consistency; changing a stress-containing algebra at fixed matter history does not change its first-order sourced Einstein tensor.

The source side is constructive. Calibrated correction, selected projections and learned protection change stress through actual interactions. In a specified Jackiw–Teitelboim theory, an acquired quantum record drives a material force and radiation, whose response changes later operations. The finite mean-action return retains old references, clock energy, support work and recoil. Separate spatial and full-mode bulk constructions extend the source account without identifying their distinct models.

Maintenance supplies a possible selection pressure. For complete entropy removal at rate $\dot{s}$ through a thermal chiral exhaust of fixed capacity c_{ex}, the heat rate obeys $\dot{Q} \geq k_B T \dot{s} + 3\hbar \dot{s}^2/(\pi c_{\mathrm{ex}})$. Pointer alignment lowers the complete renewal debit, and matched horizon transport reads its price. Sustained renewal and joint sharp exhaust with persistent causal return remain open. The proposed gravitational mechanism must connect formed access, a conserved total source and geometric consistency in one dynamics; neither a visibility filter nor a finite response in an assigned gravity theory supplies that connection.

Technical appendices and their original numbering are retained in HAFF-D technical supplement (`HAFF_D_supplement.pdf`). The certificate guide at the end of this paper identifies their claims.

1 The problem: what makes selection gravitational?

An apparatus changes which distinctions it can preserve and which operations it can execute. This can reorganize a reconstructed geometry. It can also change matter through

the selecting interaction. A theory of gravity based on observable-algebra selection must connect these two effects: the localization structure selected by physical histories must constrain a geometric law, and the same histories must supply the source and respond to it.

The central proposal is that gravitational dynamics describes a slow evolution of the selected geometric algebra, not just motion of matter within a fixed localization structure. This is a mechanism conjecture, developed in Section 4, rather than an identification of every algebra change with curvature. Its strongest obstruction is already visible in a conformal model: two selected algebras that retain the same stress readings can give different relative entropies while fixing the same matter-sourced Einstein tensor. A physical change of localization must therefore be distinguished from a change of description.

HAFF-A, B and E relate channels, stable content and actual records [19, 20]. Foundation I constructs a stable hull relative to a readable seed [54]. These results make parts of algebra selection precise; they do not determine a spacetime metric or a universal gravitational dynamics. Here we ask what further relation would make that selection gravitational.

Sections 2–5 state the algebraic question, its central conjecture and the modular constraints. The holographic route needs an explicit region family and dictionary. Section 6 gives a different conditional bridge in the same spacetime: one conformal scalar source, its surface charge, and its all-ball geometric test. That bridge identifies which law is compatible with the supplied consistency condition, while leaving the physical origin of that condition to the common-action conjecture.

Sections 8 and 9 then test the source and return edges through actual interactions in a specified two-dimensional JT theory. The decisive finite mechanism is an acquired record that changes a material source, not a command label attached to a separate emitter. Section 10 prices the full maintained state. Section 11.2 asks whether the source, renewal and informative return can persist together. Long controller, clock, support and source proofs are in the appendices.

Table 1 separates these constructions. Its rows are not pieces of one already realized apparatus. In particular, a supplied JT response cannot establish the higher-dimensional consistency law, and a modular constraint cannot manufacture a controller.

2 The Factorization Problem

2.1 No canonical tensor factorization

A Hilbert space of dimension $d = d_1 d_2$ admits a tensor factorization, but its Hilbert structure alone does not select one. Commuting observable algebras can induce the subsystem structure [17, 18]. Which of those algebras an apparatus can use is a physical question about its couplings and resources.

2.2 Coarse-graining and effective descriptions

Paper A [19] gives the following conditional separation.

Proposition 2.1 (Conditional coarse-graining separation)**.** *Let $\Phi_1 \neq \Phi_2$ be finite-dimensional coarse-graining channels with a common output space. Their outputs differ for generic pure inputs. Entropy profiles differ only under the additional nondegeneracy*

Object and supplied inputs	Result	Remaining interface
Modular family, dictionary and ball laws	Conditional translations and linearized geometric consistency (Sections 2–5)	Formation of the candidate family and its geometric dictionary.
Conformal scalar, source reference, collar prescription and MC	One prepared source, matched Wald coefficient and all-ball Einstein test (Section 6)	Physical origin of MC; apparatus source and finite-resource readings.
Calibrated field operations and a JT action	Derived outgoing source, retarded response and finite echo/contact readout (Section 8; Appendix G)	Total apparatus stress and physical references; the action is an input.
Quantum core with classical mechanics and mean-action source	Finite retained-history return and strong-domain perturbation bounds (Section 9; Foundation III)	Fixed-window mean-action result; protected content, old references and initial work remain charged. No quantum stress tensor for the scalar fields or support is supplied.
Spatial quadratic fields and real-quartet bulk	Work-corrected full-mode stress matching, a finite spatial return, and a separately constrained quartet interior	Joining their receivers, moving endpoints, preparation and thermal ports in one material model.
Complete retained entropy and paid finite carrier	Long-window heat floor and neutral-sector sorting of a resettable source (Section 10; Proposition 11.2)	Actual adaptive source concentration, protected-content compatibility and local delayed readout.
One reused body over growing runs	Conjecture C.1 jointly requires complete-output sharpness, normalized window contrast and causal transition gain	Sustained inlet, support, sensing and resource account in that same dynamics; none is supplied by sorting alone.

Table 1: Objects, results and interfaces. Finite constructions retain their stated inputs; the joint growing-window target is a conjecture.

hypothesis stated in Paper A, and metric inequivalence follows only when a supplied reconstruction dictionary separates those profiles modulo diffeomorphisms.

Channel inequivalence therefore establishes a difference of effective descriptions, not by itself a difference of spacetime metrics. Paper A also decides, for qubit reductions, exactly when two channels give the same output entropy on every pure input, and it constructs one global state, one family of local noise channels and one fixed reconstruction rule that yield non-isometric finite metrics. Metric inequivalence of spacetime geometries remains a conditional model claim throughout this paper.

2.3 Geometry from entanglement

In a holographic model the Ryu–Takayanagi formula [27] relates an extremal area to boundary entropy,

$$S_A = \frac{\mathrm{Area}(\gamma_A)}{4G_N\hbar}, \tag{1}$$

and motivates an entanglement account of connectivity [28]. Such a reconstruction requires its dictionary. Outside a specified model, the accessible data determine operator relations rather than a spacetime metric.

3 A Structural Distinction: Gauge Forces and Gravity

Gauge interactions evolve matter and connection fields within a supplied localization structure. The algebraic proposal asks whether gravity instead describes how physical correlations, accessible operations and their stability select that structure. With a geometric dictionary, changing those relations can change effective distances and causal structure. The source constructions below test a necessary interface: changed access reaches geometry through a specified matter process.

If the proposal holds, quantum gravity must describe selection of localization as well as fields within it. This is not an explanation of why quantization fails. Background independence [29, 30] asks for covariance under changes of selected geometric structure, not equality of the metrics assigned to operationally different algebras. Selection is constrained by interactions and stability, rather than subjective choice [20]; changed knowledge affects stress only through an operation.

A common reconstruction dictionary gives every matter representation in one selected geometric sector the same metric. This conditional universality is not a derivation of free fall or equality of inertial and gravitational mass. Degrees outside the sector cannot be declared geometrically irrelevant while changing its metric: a dark-sector application would need an enlarged source algebra and explicit backreaction map. No dark-matter or dark-energy conclusion follows here.

4 Algebra selection as a gravitational mechanism

4.1 Setup

We separate the operator-algebraic and finite-dimensional regimes result by result. Modular statements use a Type III_1 ambient factor with a faithful normal state and explicit cyclic and separating hypotheses. Density matrices, traces, the quantum Fisher information metric and the qubit chain are finite-dimensional statements. The Rindler example is an algebraic quantum field theory check. Theorem 5.6 uses the holographic regime; Section 6 uses a distinct same-spacetime conformal model. The JT model of Section 8 is a supplied two-dimensional gravity theory.

Definition 4.1 (Selected geometric algebra)**.** *A* ***selected geometric algebra*** $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ *is a von Neumann subalgebra chosen as input to a specified algebra-to-geometry reconstruction model. It is not Paper E's probe-reachable effect system, pointer or record algebra, or environmental fixed set. A coarse-graining model must declare the channel or inclusion relating any of those objects to* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$*.*

No canonical tensor factorization is assumed. State restrictions and relative-entropy data on $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ can be associated, as in holography and tensor networks, with an effective distance structure. That structure depends on the choice of $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$, on the stability of correlations under coarse-grained dynamics and on the redundancy of the encoding; it is an effective description valid in the context the algebra defines.

4.2 Central conjecture

Conjecture 4.2 (Gravity as adiabatic algebra evolution)**.** *Gravitational dynamics and the* ***adiabatic flow*** *of selected geometric algebras* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t)$ *arise from one physical selection process:*

1. *the global state evolves unitarily,* $|\Psi_U(t)\rangle = U(t)|\Psi_U(0)\rangle$*;*
2. *a model-dependent rule selects a geometric candidate* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \subset \mathcal{M}_U$ *at each time;*
3. *as the state evolves, the selected candidate may shift,* $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(t) \to \mathcal{A}_{\mathrm{geo},\mathbf{c}}(t+dt)$*;*
4. *the interaction that changes the selected algebra also determines the complete physical source and the compatible localization data. A change of algebra alone, with the matter history and source readings held fixed, supplies no additional first-order sourced Einstein tensor in the common-source regime below. A changed sourced response must be carried by the changed total source under that regime's field-law conditions.*

The proposed distinction from ordinary matter feedback is that the same dynamics selects the localization structure in which the source and response are represented, rather than merely evolving matter on a permanently supplied stage. Evolution within a fixed selected localization can describe gauge dynamics, but neither unitarity nor a moving algebra by itself identifies an interaction as gauge or gravitational. The selection claim must be established through the common-action conditions below; finite sourced return alone does not establish it.

The proposed flow needs a physical selector. A readable history changes which controls and correlations remain stable; those controls determine a selected hull and the matter interactions that sustain it. The changing hull must then admit compatible geometric region data, rather than merely receive an arbitrary distance assignment. In an adiabatic sector, the relaxation time of the selected structure must be short compared with its change time. This separation is a condition to establish in a model, not a consequence of writing $\mathcal{A}(t)$.

Two explanations compete here. In the reconstruction explanation, the ambient geometry and complete source stay fixed, while smaller algebras resolve less of them. In the dynamical explanation, history changes an actual coupling or preparation, changes the complete source, and changes later physical readings. The latter has a finite realization below; the stronger claim that the localization law itself is selected requires Section 7. Algebra variation alone distinguishes neither explanation.

A useful test compares histories under one action and one source calibration. Erasing command access before execution should remove the record-mediated part of the source; erasing a log after execution should not. If two selected structures contain the same total stress and surface data, a first-order difference of their sourced Einstein tensors would contradict the conformal bridge in Section 6. A difference only in relative entropy supports a reconstruction claim unless a corresponding physical intervention is demonstrated. Conversely, a history-dependent source that fails the same all-ball response test does not realize the proposed gravitational flow, even if it produces a detectable JT signal. Rapid algebra transitions lie outside the adiabatic claim; whether they admit a different geometric regime is open.

4.3 Comparison with quantum information geometry

For comparison consider the quantum Fisher information metric (QFIM) on a smooth finite-dimensional family of faithful effective states $\rho(\lambda)$ [36, 37],

$$g_{\mu\nu}(\lambda) = \frac{1}{2}\mathrm{Tr}\left(\rho(\lambda)\{L_\mu, L_\nu\}\right), \qquad \partial_\mu\rho = \frac{1}{2}(\rho L_\mu + L_\mu\rho). \tag{2}$$

It is coordinate independent, reduces to the classical Fisher metric in commuting limits, and measures how hard nearby states are to distinguish. It is also positive semidefinite, whereas a Lorentzian spacetime metric is indefinite and carries causal structure. Nothing in Eq. (2) supplies the negative direction, so the QFIM is a metric on model space and an analogy, not the spacetime metric. A change $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \to \mathcal{A}_{\mathrm{geo},\mathbf{c}} + \delta\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ may change this state-space metric; calling the change a spacetime perturbation requires an additional Lorentzian reconstruction map. In AdS/CFT, relative entropy and the entanglement first law constrain bulk perturbations [38, 39]; those results use the holographic dictionary and do not make an arbitrary boundary QFIM the bulk metric.

5 Algebraic Foundation: Modular-Stable Candidates

This section places one candidate class in Tomita–Takesaki modular theory. It does not translate measurement accessibility or environmental redundancy into modular conditions, and it does not select a unique algebra.

Standard modular definitions and their two checks are in Appendix E.

5.1 Modular definition of candidate algebras

Let $\mathcal{M}_U$ be a Type III_1 factor in standard form on $\mathcal{H}_U$, let $|\Psi_U\rangle$ be cyclic and separating, and write $\omega(\cdot) = \langle\Psi_U| \cdot |\Psi_U\rangle$. Then ω is faithful and normal and its ambient modular group $\sigma_t^{\omega,U}$ exists. Let $\{\mathcal{E}_t\}_{t\geq 0}$ be a normal unital completely positive (UCP) semigroup acting in the Heisenberg picture on $\mathcal{M}_U$.

Definition 5.1 (Modular-stable geometric candidate)**.** *A von Neumann subalgebra $\mathcal{A}_{\mathrm{geo},\mathbf{c}} \subset \mathcal{M}_U$ is a* ***modular-stable geometric candidate*** *if:*

(U1) ***Faithful restriction.*** *$\omega|_{\mathcal{A}_{\mathrm{geo},\mathbf{c}}}$ is faithful and normal. In the faithful ambient setting this holds for every unital subalgebra, so it is a regularity condition, not a selector.*

(U2) ***Ambient modular stability.*** *$\sigma_t^{\omega,U}(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) = \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all real t.*

(U3) ***Semigroup stability.*** *$\mathcal{E}_t(\mathcal{A}_{\mathrm{geo},\mathbf{c}}) \subseteq \mathcal{A}_{\mathrm{geo},\mathbf{c}}$ for all $t \geq 0$.*

Remark 5.2 (Typed relation to the measurement objects)**.** *Conditions (U1)–(U3) do not derive Paper E's implementable effects* Eff*, their operator system* $\mathrm{Eff}_{\mathrm{span}}$ *or the pointer algebra $\mathcal{P}_{\mathrm{env}}$, and setwise semigroup stability is not the exact fixed set* $\mathrm{Fix}(\mathcal{E})$*. A downstream model may specify a channel or inclusion from those objects into $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$; without it the relation is motivational. Environmental redundancy does not imply faithfulness of a state restriction. A causal restriction map and a Lindblad evolution are different types, and each model must supply its own $\mathcal{E}_t$ and verify invariance.*

5.2 Selection is not settled by these conditions

Remark 5.3 (No uniqueness from (U1)–(U3))**.** *Conditions (U1)–(U3), even with only scalars fixed by both dynamics, do not determine a unique candidate. There are four independent obstructions.*

1. *Uniqueness of the modular flow for a fixed pair $(\mathcal{A}, \omega)$ does not select the algebra $\mathcal{A}$.*

2. *A normal completely positive map may preserve two subalgebras separately without preserving their join, because it is not multiplicative. Join preservation holds for modular automorphisms but not for a general $\mathcal{E}_t$; it does hold for subalgebras of the fixed algebra of a state-preserving dynamics, on which $\mathcal{E}_t$ acts as the identity [54].*

3. *Pointwise ergodicity constrains fixed elements, while (U2)–(U3) are setwise conditions; no implication to a unique maximal element of the lattice was proved.*

4. *With a faithful ambient state, $\mathcal{M}_U$ itself satisfies (U1)–(U3), so a bare maximality rule selects the ambient algebra unless an independent seed, properness, read-out or minimal-hull condition is supplied.*

Replacing $\mathcal{B}(\mathcal{H}_U)$ by a faithful Type III factor repairs the existence of the ambient modular flow but none of these selection failures. The class of candidates may contain several members that are not unitarily equivalent, and classifying it requires additional structure. A nontrivial selection principle could start from a foreground seed and construct a minimal invariant hull, or impose an explicit read-out constraint. Foundation I carries out the first

route for a supplied seed: under state-preserving dynamics the hull of a seed in the fixed algebra lies between that seed and the fixed algebra. For a minimal compact symmetry, it equals the seed-stabilizer fixed algebra when that algebra is preserved by the hull generators and the hull contains the group-fixed algebra [54]. These hypotheses establish uniqueness relative to the seed; the seed's physical origin is a further question.

5.3 Connection to linearized gravity

Theorem 5.4 (Entanglement first law in a density-matrix setting)**.** *Let ρ_0 be a faithful finite-dimensional density matrix and $\rho_\varepsilon = \rho_0 + \varepsilon\,\delta\rho + O(\varepsilon^2)$ a normalized perturbation, with $K_0 = -\log\rho_0$. Then $\delta S = \delta\langle K_0\rangle$. An infinite-dimensional extension requires entropy differentiability and control of the modular-energy variation; faithfulness alone is insufficient. For a Type III local algebra the statement must instead use Araki relative entropy.*

Proof sketch. $S(\rho_\varepsilon\|\rho_0) = -S(\rho_\varepsilon) + \langle K_0\rangle_{\rho_\varepsilon}$ plus a constant is nonnegative and stationary at $\varepsilon = 0$. □

Conjecture 4.2 posits a family $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(\lambda)$. Modular theory treats two kinds of perturbation differently. For a state perturbation on a fixed algebra the Connes cocycle $u_t = (D\omega' : D\omega)_t$ relates the automorphism groups by $\sigma_t^{\omega'}(A) = u_t\sigma_t^{\omega}(A)u_t^*$ [50, 40]. This is not a left-multiplication formula for the two full modular operators. If the algebra family is generated by the positive translation of an HSMI, $\mathcal{A}_{\mathrm{geo},\mathbf{c}}(a) = e^{iaP}\mathcal{A}_{\mathrm{geo},\mathbf{c}}(0)e^{-iaP}$, and the vacuum is fixed by this translation, its full modular generator obeys $K(a) = e^{iaP}K(0)e^{-iaP}$. On a common smooth domain,

$$\frac{dK(a)}{da}\Big|_{a=0} = i[P, K(0)]. \tag{3}$$

Equation (3) concerns an HSMI-generated family and a derivative of matrix elements on that domain, not arbitrary subalgebra changes or an operator-norm Taylor expansion. The one-sided CHM charge used for a ball below is distinct from the full modular generator; their relevant local-unitary expectation differences agree, but the operators themselves do not.

Theorem 5.5 (Linearized gravity from entanglement, [39])**.** *In a holographic CFT dual to Einstein gravity in asymptotically AdS spacetime, the entanglement first law $\delta S = \delta\langle K_{\mathrm{mod}}\rangle$ for all boundary balls is equivalent to the linearized bulk Einstein equations $G^{(1)}_{\mu\nu} + \Lambda g^{(1)}_{\mu\nu} = 8\pi G_N\, T^{(1)}_{\mu\nu}$.*

The derivation uses the ball modular Hamiltonian [47]

$$K^B_{\mathrm{mod}} = 2\pi \int_B d^{d-1}x\, \frac{R^2 - |\mathbf{x} - \mathbf{x}_0|^2}{2R}\, T_{00}(x), \tag{4}$$

together with the Ryu–Takayanagi formula and the relation between bulk and boundary modular Hamiltonians [48]; complementary arguments are due to Jacobson [49].

Theorem 5.6 (Conditional holographic matching for an HSMI-generated family)**.** *Let $\mathcal{A}_{\mathrm{geo},\mathbf{c}}$ be a modular-stable candidate in a holographic CFT vacuum representation, and assume:*

(H1) the AdS/CFT correspondence holds;

(H2) the Ryu–Takayanagi formula and its quantum corrections hold;

(H3) the JLMS relation between bulk and boundary modular Hamiltonians holds;

(H4) the compared algebra family is an explicitly supplied HSMI/ball-region family whose generator P implements the boundary-region variation used in the holographic first-law argument;

(H5) the regulated entropy variation, or its relative-entropy replacement, satisfies the hypotheses of the Faulkner et al. theorem for all required balls.

Then Eq. (3) *describes the modular-generator variation within that family, and the holographic first law matches the linearized Einstein equations by the imported result:*

$$\begin{aligned} \textit{HSMI/ball family (H4)} &\Longrightarrow \delta K_{\mathrm{mod}} \xrightarrow{(H5)} \delta S = \delta\langle K_{\mathrm{mod}}\rangle, \\ \textit{(H1)–(H3)} &\Longrightarrow \delta(G_{\mu\nu} + \Lambda g_{\mu\nu}) = 8\pi G_N T^{(1)}_{\mu\nu}. \end{aligned} \tag{5}$$

The theorem does not show that a generic stability-selected algebra variation is an HSMI translation, nor that HAFF algebra flow is gravitational dynamics.

5.4 The circularity issue

Jacobson's thermodynamic derivation [34] presupposes a background geometry, while HAFF wants geometry to emerge from algebra. Three strategies are discussed in the literature. (A) In the algebraic programme, wedges are defined through modular inclusions without a background metric, and Theorem E.5 recovers a positive translation from an HSMI; constructing a sufficient net and showing that the candidates of this paper form such inclusions remain open tasks. (B) A bootstrap starts from a seed geometry, derives linearized equations from the first law, updates the geometry and iterates. (C) In holography the boundary theory supplies the algebra and the bulk geometry is derived from boundary data. Theorem 5.6 uses strategy (C) and assumes the HSMI/ball interface. A geometry-free reconstruction from the present candidate class remains open.

6 One source and a same-spacetime geometric test

The holographic matching above is not the only conditional route. Consciousness II [24] uses a free massless conformally coupled complex scalar in $D = d + 1 \geq 3$ dimensions. Its ball algebras, physical translation generator and vacuum are supplied in Minkowski spacetime. This matters: they are not the finite pointer algebra of a JT reader. What can be combined in this model is the prepared source, its modular charge and a geometric consistency test in that same spacetime.

In units $\hbar = c = 1$, the scalar curvature coupling is $-\xi\mathcal{R}\Phi^2/2$, with $\Phi^2 = \phi_1^2 + \phi_2^2$ and $\xi = (d-1)/(4d)$. Metric variation gives the conformally improved stress. For a ball $B = B(x_0, R, u)$, write

$$\begin{aligned} K_B &= 2\pi \int_B w_R T_{uu}\, d^d x, \qquad w_R = \frac{R^2 - |x - x_0|^2}{2R}, \\ H_B &= K_B + Q_B, \qquad Q_B = 2\pi\xi \int_{\partial B} {:}\Phi^2{:}\; dA. \end{aligned} \tag{6}$$

These are vacuum-subtracted expectation data with a fixed scalar boundary prescription. Surface readings mean controlled collar limits. H_B is the physical one-sided energy, K_B its bulk charge; neither is the full modular generator $-\log\Delta_B$. Keeping the same curvature coupling with Einstein coefficient $1/(16\pi G)$ gives Wald density $1/(4G) - 2\pi\xi\Phi^2$. The surface coefficient is therefore fixed by the same matter action, not chosen separately for every reader.

Proposition 6.1 (A common-source conditional bridge)**.** *Use the conformal scalar representation and surface prescription above. Let δT and h be smooth source and metric perturbations on an open domain. For the intrinsic geodesic ball of fixed proper radius set*

$$\delta F_B[h] = \delta\left(A(\partial B) - \frac{d-1}{R}V(B)\right), \qquad \eta_A > 0.$$

Then the following statements hold.

1. *The bulk condition*
$$\eta_A \delta F_B[h] + \delta\langle H_B\rangle - \delta\langle Q_B\rangle = 0 \tag{7}$$
for all centres, radii and rest frames is equivalent to $G^{(1)}[h] = (2\pi/\eta_A)\delta T$. It suffices to test the leading small-ball term at every centre and an open set of rest frames.

2. *At $\eta_A = 1/(4G)$ the same condition is $\mathcal{B}_B + \delta\langle H_B\rangle = 0$, where*
$$\mathcal{B}_B = \delta S_{\mathrm{W}}[\partial B] - \eta_A\frac{d-1}{R}\delta V(B) = \eta_A\delta F_B - \delta\langle Q_B\rangle$$
at linear order about the vacuum reference. The source normalization is $8\pi G$ in both forms.

3. *If selected ball subalgebras have vacuum-preserving normal conditional expectations and contain the bounded stress readings and the required surface readings, restricting one fixed physical state to them leaves Eq.* (7) *and its sourced Einstein tensor unchanged. A metric additionally requires common gravitational initial or boundary data and gauge identification.*

4. *Suppose a preparation has a smooth source with integrable energy density of strictly positive total energy, and that the source and response Fourier transforms belong to the tempered tensor-valued Radon measure class of Consciousness II. If consistency is tested at all centres and an open frame set, its nonzero-frequency data identify the canonical intrinsic response up to one normalization at each tested radius, within that paper's intrinsic, gauge-invariant, isotropic, compactly supported linear response class.*

Proof. The exact flat ball identity is $\delta F_B = -\int_B w_R G^{(1)}_{uu}$ [24, Theorem on the exact identity]. Subtracting Q_B from H_B in Eq. (6) gives the other integrand $2\pi\delta T_{uu}$. The leading coefficient of their sum is a nonzero multiple of $R^{d+1}(-\eta_A G^{(1)} + 2\pi\delta T)_{uu}$. An open frame set recovers the symmetric tensor; the field equation conversely makes every ball residual zero. The curvature derivative of $(1/(16\pi G) - \xi\Phi^2/2)\mathcal{R}$ gives the displayed Wald density, so its scalar part is exactly $-Q_B$ at this order. Conditional expectations preserve the vacuum modular restriction and the common stress readings, proving the third statement. For the fourth, a Radon Fourier measure supported only at zero would

give a constant source. Integrability forces its energy density to vanish, contrary to the positive energy. The nonzero-frequency identification theorem of Consciousness II now applies with its stated response hypotheses. These are consequences of that model's results, not an independent derivation of modular consistency. □

The preparation hypotheses are nonempty. In the same scalar theory, a local Weyl pulse gives a smooth improved stress with positive finite energy and a rapidly decreasing mass-shell convolution measure. A neutral pulse $U_\theta = \exp[i\theta(\phi_1(f)^2 + \phi_2(f)^2)/2]$ belongs to a proper symmetry-fixed stable hull when its bounded seed is permitted. It too has a smooth positive-energy nonzero-frequency source [24, Sections on prepared sources and selected algebras]. The permission to execute that pulse is an input; containing stress observables alone need not contain its generator.

There is a direct first-order comparison using this same preparation. Fix $\theta \neq 0$ and take the one-sided family $\psi_\epsilon = (1-\epsilon)\omega + \epsilon\omega_{U_\theta}$, $0 \leq \epsilon \leq 1$. Its source and collar tangents are exactly the neutral pulse's finite-amplitude differences. Choosing the constraint-compatible linear metric response to that conserved source makes Eq. (7) hold for every ball, with the same Wald coefficient. The mixture can be prepared by a recorded random choice of the permitted pulse or identity; that choice and the apparatus are charged. The prepared matter source is not the whole apparatus source when controls occupy the tested domain. This common-model comparison uses the Einstein coupling and gravitational constraint data as inputs, not outputs of the pulse.

A bounded non-scalar accessible observable with a permitted finite translation interval can instead be time-smoothed to give a bounded pulse of strictly positive finite energy. Local stress regularity and the Radon condition still require separate proof in that general route. Exact positive-energy translation permission on a nontrivial separated finite-dimensional algebra is impossible in the unique-vacuum setting; a finite clock is not that translation generator.

The fixed-source statement also survives finite adaptive reading within closed permissions. If $E : \mathcal{A} \to \mathcal{B}$ is a normal conditional expectation and every history-conditioned Heisenberg instrument preserves $\mathcal{B}$, initial states ρ and $\rho \circ E$ give the same record probabilities and, on positive-probability histories, conditional bounded source readings in $\mathcal{B}$ [24]. For each history, the pulled-back record effect and source observable stay in $\mathcal{B}$, where E is identity. Unbounded readings need absolute expectation convergence of spectral truncations. This permits real feedback that changes a common source; it forbids that same restricted feedback from recovering a discarded initial distinction. Enlarged controls or side information change the premise.

The bridge constrains the law rather than merely measuring one response: within its intrinsic response class, a nonzero spectral witness removes the higher-spatial-derivative ambiguity left by a homogeneous source. It does not select an entire metric from matter data; homogeneous gravitational solutions and common boundary data remain. Raw scalar H_B cannot replace the subtracted charge on all small balls: its surface term is $O(R^{d-1})$, while smooth bulk stress and geometry begin at $O(R^{d+1})$. Nor is the Wald coefficient match a modular first law for an arbitrary curved diamond. The area coefficient, region family, source reference, preparation permissions and Eq. (7) are still supplied.

6.1 Conservation before geometric interpretation

A source eligibility rule cannot remove unread stress at will. Q-RAIF NV [26, Section on physical selection and reaction] proves a limited obstruction: on a connected flat domain

of $D \geq 2$, a filter $T \mapsto a(x)T + b(x)\,\mathrm{tr}(T)g$ that conserves every smooth symmetric conserved T has constant a and $b = 0$. If it also kills every constant vacuum source, it is zero. This is a statement about that local algebraic filter family, not a no-go theorem for additional-field or nonlocal gravity models.

The same paper exhibits the appropriate alternative at the classical level. For an actual coupling field χ and canonical matter $\mathcal{L}_m = -\frac{1}{2}(\nabla\psi)^2 - U(\psi)$, let

$$S = \int \sqrt{-g}\left[-\tfrac{1}{2}(\nabla\chi)^2 - V(\chi) + f(\chi)\mathcal{L}_m\right], \qquad f > 0.$$

On both field equations,

$$\nabla^\mu(fT^m_{\mu\nu}) = \mathcal{L}_m\partial_\nu f = -\nabla^\mu T^\chi_{\mu\nu}.$$

The changed material coupling carries a compensating physical stress. This standard action-and-reaction identity is not a quantum total-source construction or a history-derived selector. It also does not cancel a vacuum term: shifting $\mathcal{L}_m$ by $-\rho$ changes both the total stress and the χ equation.

There is a concrete incompatibility to resolve before combining this mechanism with Proposition 6.1. A varying $f(\chi)$ changes the matter theory, so its ball modular Hamiltonian is not automatically the free conformal one. One may compare the prepared source in a causal domain where that interaction has ended and the specified conformal dynamics applies, with selector stress absent there or explicitly accounted for; or one must derive the interacting modular response anew. A separately prepared switching field plus the free CHM formula is not a common-model proof.

6.2 The second-order test

At the next order the source must follow the changed geometry. Consciousness II writes $g_\epsilon = \eta + \epsilon h + \epsilon^2 j + \cdots$ and $T_\epsilon = \epsilon t_1 + \epsilon^2 t_2 + \cdots$. Its corrected small-ball test yields

$$G^{(1)}[j] + G^{(2)}[h,h] = (2\pi/\eta_A)t_2$$

only after adding the slice term $-\frac{V_d}{2(d+2)}R^{d+1}[(\mathrm{tr}\,K)^2 - K_{ij}K^{ij}]$ to the area–volume functional, or choosing a geodesic slice with $K(p) = 0$. The source coefficient obeys the expanded covariant Ward identity, including its metric variation and changed propagation. A flat coherent pulse kept unchanged in the perturbed metric generally fails this identity. The same nonminimal action supplies the scalar propagation correction, the $\xi G_{\mu\nu}\Phi^2$ stress and the Wald coefficient.

This is a conditional leading two-jet stress law, not an exact finite-radius curved modular identity. An arbitrary bent slice already gives a false source without its extrinsic correction. The full quantum reference stress and preparation apparatus must also enter the same renormalized source. Thus a common action supplies necessary source and coefficient relations, while the actual curved modular response remains an additional problem.

7 The mechanism to be realized

A physical history must do two jobs at once: form and maintain the accessible structure, and supply the total source to which the geometric law responds. The finite return

of Section 9 establishes a source-to-operation edge inside an assigned gravity theory. Proposition 6.1 establishes what an all-ball consistency law requires in a different, higher-dimensional matter theory. Their relation is the following strong target.

Conjecture 7.1 (Common-action selection and geometric consistency)**.** *There is a local matter–controller theory in which readable histories form and maintain a nontrivial selected geometric structure, and whose causal effective action uses the same prepared fields, support and clock in its source and response. In a low-energy sector with a common cone, its selected ball data satisfy the geometric consistency condition of Consciousness II. Its first-order response is the corresponding Einstein response with one source normalization; its higher-order response includes the interaction, contact and clock work that produced the selection. Selection changes this response through physical couplings and sources, while a change of description at fixed matter history leaves the source unchanged. The sustainable family of Conjecture C.1 is a finite operating sector of this theory, with a response-to-operation gain that survives the growing-window limit.*

The conjecture has tests beyond assigning gravity to a massive apparatus. Removing the history's access to a command must remove the record-mediated component of the source, while holding the commanded interaction fixed and merely discarding a log must preserve it. Source identification must survive switched physical references; logical noise records alone fail this test in Proposition F.1. The same calibrated source must then pass the all-ball consistency test with fixed initial or boundary data. A law that reproduces only the JT dilaton or one binary return has not passed the higher-dimensional test. At second order the source, geometry and changed preparation must be varied together; entropy Hessians do not become SLD or spacetime metrics by name.

A model satisfying the conjecture must output a nontrivial selected sector from a family of retained histories; a prescribed sequence of couplings is not enough. With a supplied noise and readable seed, Foundation I gives a relative stable-hull construction. The next demand is that the same history dynamics prepares those inputs and that the resulting region data satisfy Eq. (7), with the total source and one area coefficient. The scalar surface match removes an independent matter boundary coefficient, but does not remove MC. This is the precise gap between common-action source formation and a selected gravitational law.

The hardest competing explanation is ordinary matter feedback on an unchanged gravitational stage. It explains every finite JT return here. To go beyond it, the selected localization family must predict a common geometric response without refitting the source reference, area coefficient or region prescription for each history. A permanent mismatch of that test would refute this common-action realization, even if its controller remained useful.

8 From a Selected Interface to a Physical Source

A calibrated interface acts through a field coupling whose outgoing stress sources the supplied JT theory. This gives a physical selection–source map; the origin of the gravitational law in Conjecture 4.2 is a separate question.

8.1 What records can and cannot do to a source

Let a classical record $R \sim \mathrm{Bernoulli}(p)$ control whether a physical operation prepares the matter state ψ or leaves the reference ω. Without reading R, the matter state is $\rho_p = (1-p)\omega + p\psi$ and its stress is the mixture of the two stresses. Deleting or ignoring R after the preparation does not act on the matter, so ρ_p and its source are unchanged. Changing the preparation probability p changes the source. The information side has a matching statement. For the joint state $\rho_{RM} = (1-p)|0\rangle\langle 0| \otimes \omega + p|1\rangle\langle 1| \otimes \psi$ and the reference $\pi_R \otimes \omega$, with π_R the marginal of R, additivity of relative entropy on direct sums and data processing give

$$D(\rho_{RM} \| \pi_R \otimes \omega) = p\, D(\psi\|\omega), \qquad D(\rho_p\|\omega) \le p\, D(\psi\|\omega) \tag{8}$$

[51]. Losing the record lowers the distinguishability available to a reader; it does not remove the matter source. A gravitational consequence of selection must therefore pass through a preparation or interaction, and the conditional geometry of a branch, the average of branch geometries and the geometry of the averaged source are three different objects.

8.2 The source of a calibrated feedback actuator

The companion paper [21] realizes a corrective flip of a degenerate bit as a field coupling. A right-moving free massless scalar with null coordinate u, current $j = \partial_u \phi$ and stress $T_{uu} =: j^2 :$ is coupled through a real smooth switch g. We require a finite nonzero one-particle norm $\lambda^2 = \int_0^\infty \omega |\hat{g}(\omega)|^2 d\omega/(4\pi)$ and $\int (1+u^2)|g'(u)|^2 du < \infty$, where $\hat{g}(\omega) = \int g(u) e^{-i\omega u} du$ in the L^2 sense when needed. For example, $g \in H^1(\mathbb{R})$ together with the weighted derivative condition is sufficient. Smooth compact switches and the full-line packet below both belong to this class. The smeared current defines a Weyl unitary,

$$U = \exp\Big(-iX \int g(u) j(u)\, du\Big) = \exp\big[-i\lambda X (b + b^\dagger)\big], \tag{9}$$

where b is a normalized field mode with mean frequency $\bar{\omega}$ fixed by g, and $\lambda^2 \bar{\omega} = \frac{1}{4}\int g'^2 du$. With the drive mode in the coherent state $|\sqrt{N}\rangle$ and $2\lambda\sqrt{N} = \pi/2$, the coupling flips the bit with success probability $s_N = (1 + e^{-\pi^2/(8N)})/2$. From the current commutator $[j(u), j(v)] = \frac{i}{2}\delta'(u-v)$ one has $U^\dagger j(u) U = j(u) + \frac{1}{2} X g'(u)$ and

$$U^\dagger T_{uu}(u) U = T_{uu}(u) + g'(u) X j(u) + \tfrac{1}{4} g'(u)^2. \tag{10}$$

The cross term has zero expectation if the bit and the fresh field input are initially independent and $\mathrm{tr}(\rho X) = 0$, or if their joint state has no coherence in the bit's Z basis; a Z-diagonal bit marginal alone does not suffice [21]. Writing $j = \psi(u) b + \psi^*(u) b^\dagger + j_\perp$, the incoming coherent beam has stress $T_{\mathrm{in}} = 4N[\mathrm{Re}\,\psi(u)]^2$ and each activation adds

$$F_g(u) = \tfrac{1}{4} g'(u)^2 = 4\lambda^2 [\mathrm{Im}\,\psi(u)]^2. \tag{11}$$

This added stress is a property of the coupling. Its integral is the switching work $\lambda^2 \bar{\omega}$ supplied by the external switching agent.

The feedback task is that of the Imprint Chain [22] with the flip realized by this coupling. A hidden instrument label is estimated from calibration records with mismatch

probability q; each task step reads the bit with reliability ν, accepts feedback with probability h, switches g on when accepted and the reading is negative, and lets a bath with contraction c act. Each opportunity receives its own scheduled incoming beam, independent of the bit and earlier records. In the companion paper the stationary loss $L_\infty(q)$ and the stationary coupling rate satisfy [21]

$$p(q) = h\nu L_\infty(q) + \tfrac{h(1-\nu)}{2}. \tag{12}$$

Proposition 8.1 (Source–loss relation)**.** *At fixed drive, switch and task parameters, the stationary mean source per opportunity outside the apparatus is* $\langle T_{uu}\rangle_q = T_{\rm in} + p(q)F_g$. *For two calibrations,*

$$\langle T_{uu}(u)\rangle_q - \langle T_{uu}(u)\rangle_{q'} = h\nu\,[L_\infty(q) - L_\infty(q')]\,F_g(u). \tag{13}$$

Proof. The incoming beam is present at every opportunity. A coupling occurs with stationary probability $p(q)$ and, under the joint initial condition, adds F_g in expectation by Eq. (10). Equation (12) gives the difference. □

Better calibration lowers the task loss and, through it, the source. The coefficient comes from the same interaction that performs the correction; no separate rule maps a calibration error to an emitted pulse. The relation does not assign stress to knowledge. Deleting the calibration record after the commands have been executed leaves the outgoing source unchanged, in line with Section 8.1. The relation is a stationary average per opportunity; a single trajectory, with its fixed hidden match label, does not follow the population mean step by step.

8.3 A source set by learned quantum protection

The activation-rate mechanism is not the only way that retained records can change a source. Brain Anchor [57] constructs two degenerate qubits with a programmable coupling

$$A_a = Z_1 + \alpha a Z_2, \qquad U_a = e^{-iA_aQ_f}, \quad Q_f = \int f(u)j(u)du = \sqrt{k}(b + b^\dagger). \tag{14}$$

The real smooth compact switch f and positive gain interval are supplied. The gain α is unknown, and each encounter receives a fresh independent vacuum field. On $\mathrm{span}\{|01\rangle, |10\rangle\}$ the coupling is $(1 - \alpha a)Z_L$. Calibration records from the commands $\pm h$ have means $r_\pm = e^{-2k(1\mp\alpha h)^2}$, so $\log(r_+/r_-) = 8k\alpha h$. The finite-record certificate in that paper bounds $|1 - \alpha a|$, including command rounding and known drift. It does not promise an exact fixed algebra after finite training.

Proposition 8.2 (One residual controls protection and source)**.** *Fix the acquired command* a, *write* $m = 1 - \alpha a$ *and* $F_f = f'^2/4$, *and keep the initial field independent of the logical body and retained ancillas. For every logical input, one encounter has added mean stress* m^2F_f *and switching work* m^2E_f, *with* $E_f = \int F_f du$. *After* M *independent identical encounters, with the output ports discarded and no interleaved gates or active recovery, its passive protection error is*

$$\epsilon_M = \tfrac{1}{2}(1 - e^{-2Mkm^2}). \tag{15}$$

For a single pulse with stress moments $E_j = \int u^j F_f du$ and the initial data of Proposition 8.4, the late invariant is

$$\mathcal{C}(m) = H_0^2 + 2\kappa H_0 m^2 E_1 + \kappa^2 m^4 (E_1^2 - E_0 E_2). \tag{16}$$

Thus retained history can change the physical protection and the gravitational response at fixed pulse shape and activation schedule.

Proof. The current shift is $j \mapsto j + A_a f'/2$. The centered independent field input makes the stress cross term vanish for every logical state, and $A_a^2 = m^2 I_L$ on the code. Its characteristic function multiplies the logical off-diagonal by e^{-2km^2} per encounter; a dephasing channel's half-diamond error is its phase-flip probability. Time ordering may give a factor $e^{i\vartheta_f A_a^2}$, which is scalar on the code but not on the whole two-qubit register. Finally substitute $m^2 F_f$ in the fixed-initial-data source solution below. □

This link does not require a rule that emits a pulse only after a wrong answer. The same interaction produces both the residual error and its source, even when the logical body is entangled with retained memory. At $m = 0$ the body decouples in the code; a noiseless stored qubit need not continuously export entropy. The finite-rate entropy-export bound of Section 10 applies to actual renewal, not to every protected state. Calibration itself adds field energy $2N(1 + \alpha^2 h^2)E_f$ for N independent tests at each sign. Preparations, measurement, the controller and switching mechanism add further charges. The source suppression during operation alone is not a total-energy saving.

Brain Anchor also treats an unknown centered Gaussian variance K_ω. Logical records at $0, \pm h$ give $K_\omega = -\log r(0)/2$ and $\alpha = [\log r(h) - \log r(-h)]/(8hK_\omega)$; two settings suffice when the signs of $1 \mp \alpha h$ are known. Its finite-record mismatch certificate needs no supplied temperature label. The coherence is now $e^{-2K_\omega m^2}$, but the coupling-added source remains $m^2 F_f$ on centered independent input.

Incoming stress must still be added. A u-clock chiral KMS input has background $\pi/(12\beta^2)$ in natural units. Within a supplied bounded one-temperature family, strictly decreasing K_β permits a finite-error background estimate; a general Gaussian variance does not determine incoming energy. At common preparations and initial data, the linear dilaton equation compares coupling-source differences. Equation (16) cannot be used for the full source after dropping its thermal background, nor is u thermality the rapidity-clock reference of Section 10.

The same command can also operate on one reused vacuum field. For nonnegative compact switches f_j on disjoint intervals, with no noncommuting logical gates, the smeared currents commute but have negative correlations. Brain Anchor gives exact coherence $e^{-2m^2 k_M}$ with $k_M = \langle (\sum_j Q_{f_j})^2 \rangle \le Mk$, and added source $m^2 \sum_j f_j'^2/4$ of energy $Mm^2 E_f$. Its JT response uses the actual pulse-train moments and initial coefficients in a valid operating domain. No independent-port approximation, thermal input or infinite-lifetime worldline is inferred.

There is a finite causal pulse in this construction. For example, set $f(u) = A \exp[-1/(1 - (u-3)^2)]$ on $2 < u < 4$, zero outside, and normalize its one-particle norm to $k = .2$. Numerical quadrature gives $(E_0, E_1, E_2) = (.4568522, 1.3705565, 4.3215400)$. For $H_0 = 1$, $\kappa = .05$ and $\alpha = 1.1$, the supplied command $a = 48/55$ has residual $m = .04$ and gives $\mathcal{C} \simeq 1.00021929$; the comparison $a = 1$ has $m = -.1$ and $\mathcal{C} \simeq 1.00137053$. These are deterministic evaluations of the displayed pulse, with code and outputs in `supplemental/pulse_debit`, not statistics of a calibration pilot. For a random learned

command, the invariant of the mean source uses $(\mathbb{E}m^2)^2$, whereas the mean branch invariant uses $\mathbb{E}m^4$ in the quadratic term. The compact source ends before a later boundary echo is scheduled, but the calibrated wall and timing instrument remain the inputs of Section G.1.

Stress readback also has an exact blind direction. At fixed $a_0 > 0$, the gains $\alpha_\pm = (1 \pm d)/a_0$ give residuals $\mp d$ and therefore the same protection channel and source. Field parity conjugates their full code interactions while leaving the vacuum and stress invariant. No parity-covariant stress instrument with invariant outcome labels can distinguish them, nor can their mean JT responses at common initial data. Their correcting commands $1/\alpha_\pm$ differ. Probes at $a_0 \pm h$ break this ambiguity through the opposite signs of $8k\alpha h(1-\alpha a_0)$ in their log contrast. A gravitational response is therefore not a complete calibration reference for the interface that produced it. Off-frame coupling probes, or a signed field quadrature, are additional access, with their own sources and costs. Full output-field parity and a conditional logical pulse can instead restore the stored state without identifying the gain, as shown in Ref. [57]. This recovery leaves the emitted source unchanged. Active state correction and learning a match that suppresses the actual source are therefore distinct operations.

The gain is a physical actuator parameter, not a freely changed description. The selected finite code is not the geometric algebra of Section 5, and the supplied JT action still supplies the gravitational response. A link to continuous modular selection requires that further interface; it is not obtained by identifying a protected M_2 sector with a half-sided modular inclusion.

8.4 A physical seed changes the source

The rate-controlled model changes how often a fixed interaction occurs. The coupled observable can itself change, giving a direct relation between an exactly represented algebra and a source. Keep the incoming field, switch and activation schedule fixed. On a qubit put $P_\theta = (I + \cos\theta\, Z + \sin\theta\, X)/2$ and use $U_\theta = \exp\big[-i\lambda P_\theta(b + b^\dagger)\big]$. The projection selects a pointer algebra and its source population under this fixed schedule.

Proposition 8.3 (Selected projection and outgoing stress)**.** *For $\lambda \neq 0$, an initially independent qubit state ρ and normal field state σ, the exactly represented algebra of the induced qubit channel is $\{P_\theta\}'$. Let $r_\theta = \mathrm{tr}(\rho P_\theta)$. The stress formula below requires its displayed current moments to exist. The outgoing field is*

$$(1 - r_\theta)\sigma + r_\theta e^{-i\lambda Q}\sigma e^{i\lambda Q}, \qquad Q = b + b^\dagger,$$

and its stress is

$$T_\theta(u) = T_{\mathrm{in}}(u) + r_\theta \left[g'(u)\langle j(u)\rangle_\sigma + \tfrac{1}{4}g'(u)^2\right]. \tag{17}$$

Thus two selected projections can change the source at fixed global input state and fixed activation schedule.

Proof. The quadrature has non-atomic spectral distribution in a normal state, so the multiplicative-domain theorem of Q-RAIF C gives $\{P_\theta\}'$ [14]. Expanding U_θ in its two projection blocks and tracing the qubit gives the field mixture; off-diagonal blocks have zero trace even when ρ does not commute with P_θ. In the current identity replace the Pauli X by P_θ and use $P_\theta^2 = P_\theta$. Initial independence gives (17). □

For $\rho = (I + 0.8Z)/2$, the seeds P_0 and $P_{\pi/2}$ select different pointer algebras, with $r_0 = 0.9$ and $r_{\pi/2} = 0.5$. In the worked coherent packet, the cross profile is $8\lambda\sqrt{N}\,\mathrm{Re}\,\psi\,\mathrm{Im}\,\psi$. Its first three moments vanish, so the invariant increment is (70) with p replaced by r_θ. With the table's $N = 16$ and background it is 2.0807581×10^{-5} versus 1.1559770×10^{-5}. The full source also contains the cross profile, which enters the local dilaton even though its first three moments do not enter the final invariant.

Here the source weight is derived from the state's population in the selected projection; it is not assigned to an external mismatch label. Changing the projection means changing a physical interaction, not renaming an unchanged algebra. The two algebras are abstractly isomorphic but are differently embedded relative to the fixed state and coupling. This supplies a seed–interface–source link. It does not identify that finite-dimensional pointer algebra with a Type III geometric candidate or explain why the coupling selects this seed rather than another. Those are the remaining selection and geometry bridges.

8.5 Response of a given Jackiw–Teitelboim model

Take the JT action [52, 53]

$$I = \frac{1}{16\pi G}\int \sqrt{-g}\,\Phi(R + 2/L^2) + I_{\mathrm{matter}}, \qquad ds^2 = -\frac{4L^2\,du\,dv}{(1+uv)^2}, \tag{18}$$

with the usual boundary terms, $\kappa = 8\pi G$, $D = 1 + uv$ and the vacuum dilaton $\bar\Phi = H_0(1 - uv)/D$, $H_0 > 0$. The field equations are $\nabla_\mu\nabla_\nu\Phi - g_{\mu\nu}\Box\Phi + g_{\mu\nu}\Phi/L^2 = -\kappa T_{\mu\nu}$, and in these coordinates

$$\Phi_{uu} + \tfrac{2v}{D}\Phi_u = -\kappa T_{uu}, \qquad \Phi_{vv} + \tfrac{2u}{D}\Phi_v = -\kappa T_{vv}, \qquad \Phi_{uv} + \tfrac{2}{D^2}\Phi = \kappa T_{uv}. \tag{19}$$

The matter is a conformal theory with matched left and right central charges, and the state-independent anomaly is removed by subtracting the same background. The left-moving input is unchanged, and the state-dependent source is right-moving, $T_{uu} = F(u)$ with $T_{uv} = T_{vv} = 0$.

Proposition 8.4 (Response with fixed initial data)**.** *Let F be integrable with $\int |u|^2|F|du < \infty$. On a connected coordinate domain with $D \neq 0$, write the perturbation as $\Phi - \bar\Phi = (A(u) + vB(u))/D$ and impose the initial coefficient data $A(u), B(u) \to 0$ as $u \to -\infty$. For a source bounded below in support this is equivalent to zero perturbation before the source. The unique solution with these data is*

$$\Phi = \bar\Phi + \frac{A(u) + vB(u)}{D}, \qquad A = -\kappa\int_{-\infty}^{u}(u-z)F(z)dz, \qquad B = -\kappa\int_{-\infty}^{u} z(u-z)F(z)dz. \tag{20}$$

The perturbation is linear in F. With $E_j = \int u^j F\,du$, after the source (or in the asymptotic limit for sources with tails) the invariant $\mathcal{C} = \Phi^2 - L^2(\nabla\Phi)^2$ takes the vacuum value

$$\mathcal{C} = H_0^2 + 2\kappa H_0 E_1 + \kappa^2(E_1^2 - E_0E_2), \tag{21}$$

and inside the support $\partial_u\mathcal{C} = \kappa F\,(2H_0u + uA - B)$.

Proof. Write $\psi = D\,(\Phi - \bar\Phi)$. The vv equation of (19) reads $\psi_{vv} = 0$, so $\psi = A(u) + vB(u)$, and the other two equations reduce to $A'' = -\kappa F$, $B'' = uA''$ and $B' = uA' - A$, which

Eq. (20) satisfies. The absolute first and second moments make both retarded coefficients tend to zero at $-\infty$. Two solutions differ by $\alpha + \beta u + \gamma v - \alpha uv$, whose coefficients are $A_h = \alpha + \beta u$ and $B_h = \gamma - \alpha u$. Their two zero limits force $\alpha = \beta = \gamma = 0$. Every vacuum solution has numerator $\alpha + \beta u + \gamma v - \alpha uv$ and invariant $\alpha^2 + \beta\gamma$; after the source $\alpha = H_0 + \kappa E_1$, $\beta = -\kappa E_0$, $\gamma = \kappa E_2$. The balance law follows by differentiation. □

The coefficient condition cannot be replaced by $\Phi - \bar{\Phi} \to 0$ at each fixed v. The homogeneous solution $\eta v/(1 + uv)$ has that limit for every fixed v along nonsingular points. It even tends to zero uniformly on the subdomain $u < 0$, $v \leq 0$, since its absolute value is at most $|\eta|/|u|$; no uniform limit near $D = 0$ is implied. Adding it leaves the source unchanged but changes the late invariant by $-\kappa E_0 \eta$. It is excluded by $B(-\infty) = 0$. Source comparisons below hold H_0, the coordinate matching and both coefficient boundary data fixed.

The JT curvature stays $-2/L^2$; the dilaton and the invariant respond. A dilaton difference is not by itself a higher-dimensional curvature or a mass, and comparing boundary proper times needs a boundary normalization.

The matching condition matters. Set $A_F, B_F \to 0$ at $+\infty$, or exactly to zero after a compact source, instead of the initial data of Proposition 8.4. The final-state-matched solution differs from the retarded solution by

$$A_F = A_R + \kappa(uE_0 - E_1), \qquad B_F = B_R + \kappa(uE_1 - E_2).$$

For the chiral diffeomorphism states of Consciousness I [25], with a Cut at a to the left of the support, the relative entropy is $D_a = 2\pi \int (u - a)F(u)\, du$. The final-state-matched solution gives $\delta\Phi(a, 0)/(4G) = -D_a$, while the initial-state-matched solution gives $\delta\Phi(a, 0) = 0$ at a Cut before the pulse. The first identity compares an earlier section with a matched final state; it does not say that a pulse removes area before it arrives. The dilaton–modular-energy relation itself is known [58]; what is used here is its type-correct pairing with a specified source and matching condition.

The same equations show that a null-only record of the source is incomplete. A genuine vacuum-density shift $T_{\mu\nu} = -\delta\rho\, g_{\mu\nu}$ has zero null components but gives the particular solution $\delta\Phi = \kappa L^2 \delta\rho$. A probe that reads only null energy cannot certify that a source is absent, which is the source-completeness condition relevant to the vacuum curvature test of Ref. [26].

9 Response that changes the next operation

A source-to-geometry calculation becomes a return only when the geometry changes an actual record and that record changes a later interaction. The readout permission matters. In the minimally coupled JT sector, instruments on the same matter metric with the same inputs have identical adaptive record laws even when the dilaton differs (Proposition G.1). A boundary tied to a fixed dilaton value evades that obstruction. Its echoes reconstruct the three vacuum coefficients, and a source-compensated inverse includes finite probe backreaction (Propositions G.2 and G.3). An exact current contact instead obtains an invariant-dependent bit, including the contact's own source (Propositions G.5 and G.6). The bit selects the next physical projection. Its same-field conditional covariance remains part of the history; detector replacement does not reset the field.

These are finite physical records, not permission to read an exact metric. The worked current contact has total variation only 2.3650×10^{-4} between its two initial geometries. Its fixed-cutoff energy budget also forbids indefinitely many contacts of unchanged positive source charge. Appendix G contains the timing, stress and accumulated-source bounds.

9.1 An acquired record drives a new material source

A record can act without first being converted into a supplied mechanical initial condition. Keep the two classical scalar fields and the material action of Section J.4, and append $i\langle\Psi,\dot\Psi\rangle - \langle\Psi, H_c\Psi\rangle$ to its proper-time Lagrangian. The normalized state Ψ contains a finite history clock, probe P, record R, replacement blank B and an old purification Q. Set

$$H_c(\Phi,y) = H_{\rm hist} + E_{\rm off}\mathbf{1} + \lambda(\Phi-\phi_0)Z_P + \frac{\chi_R}{2}P_R(y-d)^2, \qquad P_R = |1\rangle\langle 1|_R, \quad \chi_R \geq 0. \tag{22}$$

$H_{\rm hist}$ is the positive gate-conjugated clock of Q-RAIF C. Its first nonidentity edge performs the Y_P acquisition C_Y, the next performs a true SWAP of P and B, and the remaining edges are identities. There is no proper-time switch. Initially $R = 0$, so the spring has zero expectation. The actual acquisition opens its force channel. $H_c \geq 0$ while $E_{\rm off} > |\lambda(\Phi - \phi_0)|$. The target d, couplings and initial clock are prepared hardware. The scalar fields, mechanical coordinates and geometry use an Ehrenfest mean-action approximation [61, Section 2]; the finite quantum core and its old reference are retained in full, without sampling or collapsing R.

Here θ is a material angle, y its storage coordinate, $I, J, k, K > 0$ their inertia and elastic parameters, and $U(\Phi)$ the material support potential. Proper-time derivatives obey $I\ddot\theta + k(\theta - y) = -\gamma\dot\theta$, $\gamma = g_+^2 + g_-^2$, with no incoming classical waves. Their emitted powers are $P_\pm = g_\pm^2\dot\theta^2$. The mechanical stored energy is $e = I\dot\theta^2/2 + J\dot y^2/2 + k(\theta-y)^2/2 + Ky^2/2$, and $w = \dot\Phi$ along the wall. The positive wall junction and its two invariant charges are specified by Eqs. (128) and (137); all of these mechanical and gravitational data are supplied.

Let $p = \langle P_R\rangle$, $z = \langle Z_P\rangle$ and $E_c = \langle H_c\rangle$. Variation of this one action gives

$$\begin{aligned} i\dot\Psi = H_c\Psi, \qquad F_y = \chi_R(y-d)p, \qquad F_\Phi = \lambda z, \\ J\ddot y + k(y-\theta) + Ky = -F_y, \qquad \dot e = -P_+ - P_- - F_y\dot y, \qquad \dot E_c = F_y\dot y + F_\Phi w. \end{aligned} \tag{23}$$

The angle equation and $P_\pm = g_\pm^2\dot\theta^2$ are unchanged. Here p is an expectation obtained by variation, not a decoded number fed to a separate control policy. The mass and its tangent balance are

$$M = m_{\rm rest} + U + e + E_c, \qquad \dot M = (U' + F_\Phi)w - P_+ - P_-. \tag{24}$$

Consequently Eq. (137) holds with U' replaced by $U' + F_\Phi$ in its acceleration equation; both invariant and null-clock equations retain their directions. Differentiating the mass junction with these equations gives zero. The separate work balances are

$$\begin{aligned} e(\tau) + Q_{\rm out}(\tau) + \int_0^\tau F_y\dot y\, ds = e(0), \\ E_c(\tau) - E_c(0) = \int_0^\tau (F_y\dot y + F_\Phi w)\, ds. \end{aligned} \tag{25}$$

The clock interaction and conditional spring are included in E_c. The spring breaks the symmetry of Proposition J.6; F_Φ cannot be set to zero in this coupled calculation.

Proposition 9.1 (A nonzero record-mediated source return)**.** *Take two initial geometries with the same $\Phi = \phi_0$, $w = 0$, mass, mechanical data and quantum input, but different $\mathcal{C}_-$; set $\mathcal{C}_+$ by the mass junction. In the positive exterior domain, either $\lambda = 0$ or $\chi_R = 0$ makes their proper emitted powers identical. With both interactions present, there are positive finite-clock data for which the emitted powers differ on a nonempty finite interval. Their record, mechanical response, source and geometry obey the same action and work account.*

Proof. For $\chi_R = 0$, the mechanical subsystem is independent of the reader and geometry. For $\lambda = 0$, the joint mechanical and quantum equations are independent of geometry. Either condition fixes identical mechanical sources for the two initial geometries.

For a nonzero example choose $\theta(0) = \dot\theta(0) = 0$, $y_0(y_0 - d) \neq 0$ and probe state $(I + \upsilon X_P)/2$, $\upsilon \neq 0$, with a blank R and its old purification held. Let $r \geq 1$ be the C_Y edge, $a_j > 0$ the clock edge strengths and $G_r = \prod_{j=1}^{r} a_j$. The initial acceleration difference δa follows from the junction equations, so that $\delta\Phi(t) = \delta a\, t^2/2 + O(t^3)$. The shortest-path Dyson expansion gives

$$\delta p(t) = K_r t^{2r+3} + O(t^{2r+4}), \qquad K_r = \frac{\upsilon G_r^2 \lambda \delta a}{6r!(r-1)!(r+3)}. \tag{26}$$

An insertion of $\lambda\delta a\, t^2 Z_P/2$ after $j < r$ forward edges contributes the time integral $t^{r+3}(j+1)(j+2)/(r+3)!$. Their sum is $t^{r+3}r/[3(r+3)r!]$. Using $C_Y^* P_R C_Y = (I + Y_P)/2$ in the interference with the unperturbed shortest path gives the coefficient above. The insertion after all r edges has Hermitian $P_R Z_P$ in that interference and zero real part. Further graph paths and spring insertions raise the order; the spring annihilates the blank record before acquisition.

Integrating the mechanical equations then yields

$$\delta(P_+ + P_-)(t) = -\frac{2\gamma k^2 y_0 \chi_R (y_0 - d) K_r}{I^2 J (2r+4)(2r+5)(2r+6)} t^{2r+7} + O(t^{2r+8}). \tag{27}$$

The baseline angle velocity is $k y_0 t/I + O(t^2)$. Different initial $\mathcal{C}_-$ give a nonzero δa for generic supported data, since initially $\dot w = -U'\beta_+\beta_-/M - \phi_0/L^2$. The displayed coefficient is nonzero for the stated nonzero parameters. The coupled finite-dimensional equations are analytic on their strict positive exterior domain. Local existence and continuity therefore give a nonempty admissible family, not just a numerically resolved difference. □

The proof isolates the causal mechanism analytically, rather than attributing a small numerical difference to geometry. Blocking either sensing or record actuation removes the source contrast. Keeping the record but blocking actuation preserves a readout difference while removing the later source difference. Its numerical realization and complete mass/work checks are in Appendix J. The two histories here differ in initial gravitational data; they do not purport to generate those data from algebra selection.

For histories with common initial geometric constraints, the distinct spatial construction of Proposition J.7 has a nonzero source–dilaton–sensor path and a later material response. Its finite-width regulator, support and synchronization are physically varied.

The complete material response also contains ordinary clock and force effects; only the separately isolated path is gravitational. A third construction, Proposition J.8, couples the actual full-mode quartet state to paid clock/director fields and constrained JT bulk evolution. It supplies a local Ward identity and a causal prepared bulk domain, not moving-wall gate matching or the full thermal apparatus.

A finite record error alone does not control an unbounded stress. The response comparison uses weighted source bounds (Proposition A.1), and the coupled trajectory–reader bound requires domain-preserving strong solutions and the stated bounded-coupling derivative assumptions (Proposition A.2). Appendix A gives these estimates and the separate full-mode clock-work matching result. None converts fixed-window existence into sustained operation.

10 Maintenance and its gravitational price

10.1 Entropy removal and the source it can constrain

A fixed unitary correction can return its control port coherently. Removing entropy injected by unobserved noise is a different task. To price it, the state whose entropy is removed must include the controller's usable side information. The heat account must also distinguish a thermal reservoir from a supply of pure memories.

HAFF G supplies a record-driven program and executor in which selecting a physical coupling changes its source under one fixed executor [55]. The dephased quartet below adds a specified full renewal debit. It does not identify the two material realizations or inherit a thermal controller from that program loop.

Proposition 10.1 (Maintenance entropy account)**.** *Let $V = SC$ include the maintained system and every retained controller and side-information register. A correction acts unitarily on finite-dimensional VAB, initially in $\rho_V \otimes \alpha_A \otimes \tau_B$, where $\tau_B = e^{-\beta H_B}/Z$, $\beta = (k_B T)^{-1} > 0$. Put*

$$s = S(V) - S(V'), \qquad p = S(A') - S(A), \qquad Q = \operatorname{tr} H_B(\rho'_B - \tau_B).$$

Then, with $I_{V:A:B} = S(V) + S(A) + S(B) - S(VAB)$,

$$\beta Q + p = s + I_{V':A':B'} + D(\rho'_B \| \tau_B) \ \geq s. \tag{28}$$

If the marginal entropy of C is returned, s equals the decrease of $S(S|C)$. In a sequence of such corrections, a bounded total purity consumption and a positive asymptotic entropy-removal rate $\dot{s}$ imply $\liminf Q_{[0,t]}/t \geq k_B T \dot{s}$.

Proof. Unitary invariance of total entropy and the initial product give $\Delta S(V) + \Delta S(A) + \Delta S(B) = I_{V':A':B'}$. Also $\beta Q = \Delta S(B) + D(\rho'_B \| \tau_B)$. Combine the identities and sum over corrections. The rate statement follows when the accumulated p divided by elapsed time tends to zero. □

This is the Landauer entropy balance, with the memory-purity term displayed explicitly; the equality and the role of accessible side information are standard [13, Theorem 3 and Section 5.1]. For initially correlated or nonthermal inputs the same calculation instead gives

$$\beta Q + p = s + (I_{\text{out}} - I_{\text{in}}) + D(\rho'_B \| \tau_B) - D(\rho_B \| \tau_B).$$

These consumed correlations and bath free energy can pay for correction. A classical copy of the error permits a reversible conditional flip with $s = 0$, although the system's marginal entropy decreases. Negative quantum conditional entropy gives another reason to include the side information. A pure-memory swap has $p = s$ and can have $Q = 0$. A retained oscillator's closed unitary loop also has no entropy removal by itself. Thus a count of corrections or a marginal noise entropy is not automatically $\dot{s}$. In a cyclic device $\dot{s}$ must be established from its actual noise and controller dynamics. For nondegenerate systems, heat and supplied work are further separated by the change in internal energy.

Proposition 10.2 (A finite-rate chiral exhaust floor)**.** *Consider a unitary, zero-chemical-potential chiral conformal exhaust with central charge $c_{\mathrm{ex}} > 0$, thermal input at $T \geq 0$, and an extensive entropy-current account. Energy and temperature are defined by the same translation generator and clock. Assume that the outgoing modes export net dimensionless entropy current $x \geq \dot{s} > 0$, where $\dot{s}$ is the actual maintenance entropy-removal rate after all retained side information and sublinear resource consumption are accounted for. The entropy-current maximum at fixed energy is the thermal one. Then the excess heat current over the incoming thermal stream obeys*

$$\dot{Q} \geq k_B T \dot{s} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} \dot{s}^2. \tag{29}$$

Proof. Write $a = \pi c_{\mathrm{ex}} k_B^2/(12\hbar)$. The thermal energy and dimensionless entropy currents are $J_E(T) = aT^2$ and $J_S(T) = 2aT/k_B$. The first is the standard chiral CFT current [11, Eq. (1)]; the second follows from its extensive Gibbs entropy. Entropy maximization at fixed energy is the Gibbs variational principle, underlying the single-channel information-flow bound [10]. Choose $T_* = T + k_B x/(2a)$ so that the thermal output has the required entropy. Any output with at least that entropy has energy current at least aT_*^2. Subtracting aT^2 gives

$$\dot{Q} \geq k_B T x + \frac{k_B^2 x^2}{4a} = k_B T x + \frac{3\hbar x^2}{\pi c_{\mathrm{ex}}}.$$

The expression increases for $x \geq 0$. □

The current account is a physical condition on the exported modes, not the entropy of a Type III local interval. Equation (44) establishes the needed joint entropy rate in a fresh-port account; matching those ports to a conformal current must preserve its capacity and entropy budget. At fixed capacity and positive rate the quadratic term survives even at $T = 0$. Thermal output saturates the channel inequality, but this alone does not construct a correcting apparatus. The currents are long-window, extensive Gibbs limits. For an observation window τ with net exported entropy X_τ, bounded rates and sublinear boundary corrections give

$$Q_\tau \geq k_B T X_\tau + \frac{3\hbar X_\tau^2}{\pi c_{\mathrm{ex}} \tau} + o(\tau).$$

The current formula is not an exact inequality for an arbitrarily short finite packet. Applying the Gibbs bound to the entire output also includes correlations between packets; it does not require each output to be a separate thermal state.

10.2 A conditional gravitational reading

The price becomes a geometric reading only with a physical transport map. The following JT coefficient reads the total routed heat in one matched clock and reference. It does not assign curvature to entropy that remains in a memory or exits through another channel.

Proposition 10.3 (A heat account read through the dilaton)**.** *Compare two JT solutions with the same H_0, boundary coefficient data and coordinate and clock normalization. Suppose their source difference F is the exhaust of the correction in Proposition 10.1, with finite absolute second moment and $\int F du = Q$. This last equality requires that the specified outgoing channel carry all the heat in the same energy normalization. Then its retarded coefficient difference obeys*

$$-\frac{A'(+\infty)}{\kappa} = Q \;\geq\; k_B T(s-p). \tag{30}$$

For a compact exhaust and any $u_2 > u_1$ after it, the left side is exactly $-[A(u_2) - A(u_1)]/[\kappa(u_2-u_1)]$. No pointwise sign of F is required.

Proof. Differentiate Eq. (20): $A'(u) = -\kappa \int_{-\infty}^{u} F(z)dz$. Take the late limit and apply Eq. (28). □

The reference for Q is the incoming thermal reservoir transported through the same exhaust, before its heat uptake; H_B is its u-translation generator. A thermal background has a nonintegrable stationary flux, so the JT solution here is the difference from that same transported thermal background. The identification $E_0 = Q$ is an energy-transport hypothesis: other exhaust channels, energy stored in the device, work beams crossing the observation surface and redshift must be accounted for separately. With a fixed transport factor $E_0 = \zeta Q$, Eq. (30) acquires the same factor ζ. The coefficient slope is read from the dilaton in the declared matched coordinate frame, not an invariant under arbitrary changes of boundary clock. The relation links an actual entropy-removal process to a specified dilaton reading. The JT dynamics and heat-routing map are its geometric inputs.

For a matched running single-chiral source, the marginal-horizon balance is $dS_h/d\eta = 2\pi F\xi^2$. If the charged heat is exactly $F\xi^2$ and the physical input is thermal at $T_H = 1/(2\pi)$ in the same rapidity clock, the long-window bound becomes $\overline{\dot{S}_h} \geq \dot{s} + 6\dot{s}^2/c_{\rm ex}$. The actual thermal-source and reference-clock terms, positive-domain conditions and nonthermal contact excess are retained in Appendices B and H. A reparametrized vacuum is not a prepared heat reservoir.

11 Selection and response over growing windows

11.1 A conjecture on a sustainable selection–response loop

The unresolved question is whether one reused local body can combine selection, renewal, sharp exhaust and informative geometric feedback. Fix two physical preparations on a logical receiver A and its current purifier R:

$$|\Phi_Z\rangle = \sqrt{1-\epsilon}|00\rangle + \sqrt{\epsilon}|11\rangle, \qquad |\Phi_X\rangle = (H_A \otimes I_R)|\Phi_Z\rangle, \quad 0 < \epsilon < 1/2.$$

The logical labels denote Q-RAIF C's neutral pair codes, with their spectator sea retained. Here R is the paid, resettable current purifier in the third and fourth body species, available to joint neutral gates. Old references outside this source quartet are denoted R_{old} and are not acted upon. The quartet is a resettable working source alongside the series' protected unknown content H, not a replacement for it; access to the current R comes from those paid body species and joint neutral gates, not from an old-purifier permission. Each cycle's inaccessible environment applies the same pointer-aligned channel

$$\mathcal{D}_q(\rho) = (1-q)\rho + qZ_A\rho Z_A, \qquad 0 < q < 1/2, \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

No noise error label is available to the apparatus. Renewal returns the complete quartet, including the current purifier.

Proposition 11.1 (Pointer alignment and the complete renewal debit)**.** *For $\rho_p = \mathcal{D}_q(|\Phi_p\rangle\langle\Phi_p|)$, return to its specified pure quartet removes entropy $h_p = S(\rho_p)$, where*

$$h_Z = h\left(\frac{1-\sqrt{1-16q(1-q)\epsilon(1-\epsilon)}}{2}\right) > 0, \qquad h_X = h(q) > h_Z. \tag{31}$$

The receiver-only debits are 0 for Z and $h(\epsilon + q(1-2\epsilon)) - h(\epsilon)$ for X. Approximate quartet return subtracts its residual joint entropy from h_p, with finite-dimensional entropy continuity controlling the error. Program updates, sensors and usable controller registers enter the full maintenance debit in Proposition 10.1 as well.

Proof. The mixture of $|\Phi_p\rangle$ and $Z_A|\Phi_p\rangle$ has nonzero eigenvalues $(1 \pm d_p)/2$, with $d_p^2 = 1 - 4q(1-q)(1 - |\langle\Phi_p|Z_A|\Phi_p\rangle|^2)$. The overlaps are $1-2\epsilon$ and zero. Strict monotonicity of h on $(0, 1/2)$ proves the contrast. Tracing R gives the receiver debits. The reference marginal is unchanged by the noise, so the difference between each complete and marginal debit equals the receiver–purifier mutual information lost to the noise. □

Pointer alignment lowers the complete renewal price without making it zero. For $\epsilon = .1, q = .05$, $h_Z = .0877516288$ and $h_X = .1985152433$ nats per cycle. The marginal Z state is unchanged, but its purifier correlations are not. The loop must establish that its additional charged operations preserve an actual complete-debit contrast $\bar{s}_X > \bar{s}_Z$.

11.2 Three distinct targets

The finite return is target A: an actual acquisition changes a source, its response changes another acquisition, and the retained history remains in the same finite mean-action evolution. Target B is sustained thermal renewal of the body, controls and ports over growing runs while the independent unknown content stays held. Target C combines that renewal with a sharp complete exhaust and nonvanishing causal response gain. A does not prove B, and B alone does not prove C.

Conjecture C.1, stated in full in Appendix C, makes C quantitative. At fixed cycle rate and exhaust capacity, it requires complete-output sharpness, normalized whole-window response contrast, and a positive average gain at *all* declared decision opportunities on the actual-run history measure. Every sensor, program update, delay, buffer, work bank and current purifier enters the account; old inaccessible references are not reset. Resource size may grow with the run, while uncharged purity and correlation consumption have no positive asymptotic rate.

This target has real obstructions. Accurate direct purifier dumping retains cross-species redundancy and cannot make the complete Gibbs deficit vanish. Individually sharp program phases still pay a positive Jensen variance cost when their entropy rates differ at nonzero duty fractions. An exploration scheme whose only response updates occur on a vanishing fraction of opportunities has vanishing average gain, even with infinitely many such updates. These facts do not exclude every joint protocol; they exclude tempting combinations of successful components. The complete statements and resource clocks remain in Appendix C.

The following fixed-source converter supports one part of the target. It does not supply the adaptive source's product structure.

Proposition 11.2 (Finite-carrier sorting of the dephased source)**.** *Q-RAIF C's charge-sector sorter [14] applies to L independently realized copies of either ρ_p under its finite-carrier hypotheses: the source marginal is $\rho_p^{\otimes L}$, initially a product with the complete thermal carrier; their external purifications are retained; joint source operations act on both the receiver and its current R pair, while identity on old references refers to systems outside that quartet; the source is neutral in the original zero-mode quartet sector; the original neutral gates and paid loading, unloading and pipeline stages are available; and weighted energy tails control the implementation limit. It returns the source near its blank with entropy slack $o(L)$ and complete output Gibbs deficit $o(V)$, $V = L/\nu$, at*

$$T_{*,p} = T_H + \frac{6\nu h_p}{\pi c_{\mathrm{ex}}}, \qquad c_{\mathrm{ex}} = 4.$$

The same spectral argument applies to a supplied deterministic word with product marginal $\bigotimes_{j=1}^{L} \rho_{p_j}$ and actual entropy $s_L = \sum_j h_{p_j}$. The finite-bank target uses s_L plus the same subextensive margin. If $s_L/L \to s$, its temperature tends to $T_H + 6\nu s/(\pi c_{\mathrm{ex}})$, with no convergence rate required; otherwise there need not be one limiting temperature. This is a block converter, not an adaptive moving apparatus or a per-cycle heat-reading law.

The spectral proof is in Appendix H.1. It uses a supplied product marginal and controlled energy tails, not a local reset certificate for an adaptive history.

12 Relation to Existing Approaches

AdS/CFT [31] and its area–entropy relation [27] supply the dictionary used in Theorem 5.6; it is not derived here or granted to arbitrary selected algebras. MERA [32, 33] implements coarse-graining through disentanglers and isometries, with network depth resembling a radial AdS direction. Different networks can describe the same boundary data. Relating either construction to physical selection requires the explicit algebra-to-geometry map.

Jacobson's horizon Clausius argument [34] shares the entropy-based approach to gravitational dynamics. Section 8 addresses its source-side input: which selecting processes produce the stress whose response is computed. Loop quantum gravity [29, 35] instead quantizes the metric with background independence. Its relation to a path of selected algebras would require a representation map from spin-network states. These connections motivate the proposal without supplying its missing selection law.

13 Conclusion

The gravitational proposal is that physical histories form a localization structure whose slow evolution admits a common geometric law. Modular stability does not select that structure uniquely. A specified half-sided modular family supports conditional holographic matching; the conformal scalar bridge instead relates one prepared source and its Wald surface coefficient to an all-ball test in the same spacetime. Nonzero-frequency preparations can identify the intrinsic response within its stated class. The consistency condition itself still needs a physical explanation.

The source mechanism is explicit. Changed correction, protection and projection interactions change stress. A finite JT mean-action model retains the acquired quantum record, old reference, mechanics, clock, support and recoil, and proves a nonzero record-mediated source return. The spatial return and full-mode quartet bulk strengthen different interfaces. They are evidence for a common theory, not one already joined higher-dimensional apparatus.

Maintenance can favor one selected structure over another. Pointer alignment lowers the complete renewal debit, and a thermal chiral channel of fixed capacity places a quadratic rate price on its export. Matched horizon transport can read that price, provided the actual source, clock reference and heat routing are supplied. Sharp conversion of a fixed product source does not establish sustained renewal or sharp informative adaptive return.

The remaining mechanism is therefore specific: histories must form the accessible sector, the selecting interactions must carry their reaction stress, and one calibrated source must satisfy the same geometric consistency test across those histories. At second order the geometry, propagation, source and region prescription must vary together. A visibility filter cannot replace this dynamics, and changing an algebra at fixed stress cannot create a second first-order sourced Einstein tensor. Establishing this common selection–source–geometry law would turn the adiabatic algebra proposal from an interpretation of supplied gravity into a mechanism for its effective dynamics.

Certificate guide: HAFF-D technical supplement

The main argument and its core results remain in this paper. The complete technical appendices are in `HAFF_D_supplement.pdf`, *Sources, Return and Apparatus Certificates: Technical Supplement to HAFF-D*. This is supplementary documentation for the present paper, not a separate paper. Appendix letters and all existing result and equation numbers are retained. Cross-document references below link to that file; the supplement links back to the main argument.

Main claim or question	Technical certificate	Scope of support
Conditional modular route	Appendix E; Theorem E.5	Definitions and candidate checks, not a physical selection law.
Source changed by actual interaction	Appendix F; Propositions F.1–F.3	Source identification requires the stated reference permissions.
Finite source–response return	Appendices A, G; Propositions A.2, G.7	A fixed-window mean-action return; every source and clock debit remains.
Complete maintenance price	Appendix B; Proposition B.1	Correlations and the complete output deficit enter the export bill.
Horizon transport and clock	Appendix H; Propositions H.1, H.5	A matched source and running-clock reference are required.
Physical preparation and source	Appendix I; Proposition I.2	Finite preparation and the stated material/source matching.
Support, recoil and later reading	Appendix J; Proposition J.6	Retained support and old reference; not a fresh-body replacement.
Spatial and full-mode extensions	Sections J.7, J.8, J.9	Distinct layouts and domains; not one joined apparatus.
Sustained renewal and sharp return	Appendices C, D	Conjecture C.1 remains open; component certificates do not close targets B and C.

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Measurement as Accessibility: Measurement Permissions, Stable Records and Public Readout

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Abstract

A measurement context supplies probes, environmental dynamics and physical record ports. These determine different objects: implementable effects, persistent content and readable records. We compute their relations in a fresh-memory collision model. For two pure conditional fragment states of overlap r, the exact pointer-effect permission set is $\sqrt{ab} + \sqrt{(1-a)(1-b)} \geq r$; the closest permitted sharp pointer test has error $(1 - \sqrt{1-r^2})/2$. Its operator span does not determine this resolution. Holevo and single-measurement information count different numbers of readable records: one environment has 12 Holevo records but only 8 certified readable ones. Conditional product fragments give explicit public-agreement laws, and pooled one-sided local tests attain the joint pure-state discrimination exponent. Recording does not require a pure memory. We derive a purity-deficit budget and exhibit partially mixed qubits whose exact permissions form a parallelogram, whose local measurement saturates Holevo information, and whose reduced source channel is the same complete dephasing for every memory polarization. At zero polarization this channel produces no record. Reusing a spent port can erase its record and restore coherence, so freshness cannot be inferred from a channel label. For a common unknown reader orientation, agreement alone cannot certify correctness; prepared-pointer references give a finite calibration bound and an explicit residual common-error term. The same actual fragment readout supports Paper A's finite correlation-geometry certificate. The construction accounts for preparation, readout and shared standards without selecting a unique global outcome or deriving the Born rule.

1 Introduction

1.1 Three questions about measurement

Standard quantum mechanics gives rules for outcome probabilities but does not say how definite records arise from states that assign amplitudes to several possibilities. Collapse models change the dynamics, many-worlds accounts deny that one outcome is selected, and decoherence explains the suppression of interference without selecting a result. This

paper does not add to that list. It asks a set of narrower questions that every account has to answer in some form, and it answers them inside a declared physical context.

The first question is one of *permission*. A probe is a physical channel. Which effects on the system can the probe actually implement, given its coupling and initial state? The second concerns *stability*: which effects, or which record values, survive the environmental dynamics over the relevant time window? The third concerns *public readout*: when can several readers, each with access to a different part of the environment, obtain the same value, and how often is a shared value wrong?

The three answers are different kinds of object. Each probe's implementable effects form a convex set inside an operator system; a family of probes gives their union, which is convex only if the permitted family also includes the required randomized choices. Exact fixed points of a channel form an operator system, and under further hypotheses a von Neumann algebra. Records concern a classical label and the information that environmental fragments carry about it. Treating the three as one "accessible algebra" obtained by intersection loses the information that matters, and the paper keeps them apart.

The central claim of the paper is that measurement, in the operational sense used here, is a property of an interaction context: what a declared probe family can implement, what the declared dynamics leaves fixed, and what declared fragments record. We do not explain why one global outcome occurs, and we assume the Born rule.

The formation claim is nevertheless physical. Initially independent memories acquire correlations with a pointer through actual unitary couplings; their conditional distinguishability determines what a later reader can report. Record formation is therefore more than attaching a label to a supplied channel. Its limits are also physical: the same reduced dephasing may be produced with a port that records nothing, and a second contact can undo a record. The apparatus and elementary system–port split remain inputs. Their selection is the separate problem of Papers A and B, not an unexplained premise hidden inside the word *measurement*.

1.2 Results

Section 2 fixes the typed data of a context. Section 3 states the three tests and separates two notions of redundancy, one based on Holevo information and one based on what a measurement can read. Section 4 introduces a record model in which fresh memories collide once with a pointer, and computes:

- the exact set of pointer effects that a fragment can implement (Theorem 4.2) and the distance from a sharp pointer projection to that set (Corollary 4.3);
- a nesting of the three tests: fragment effects lie in the pointer algebra, which equals the fixed algebra of the induced dephasing (Proposition 4.4);
- the gap between Holevo and accessible information, with closed-form fragment lengths for both notions of redundancy (Propositions 4.5 and 4.6);
- a purity-deficit budget and exact nonzero readout from partially mixed memories (Propositions 4.7 and 4.8);

Section 5 turns to several readers: conditional independence, the error exponents of joint, pooled local and majority readout, and a finite calibrated public-readout bound when

the readers share an unknown orientation. Section 6 collects the model's conditions and interfaces. Section 6.2 compares the approach with decoherence theory, quantum Darwinism and other frameworks, and Section 6.3 identifies the actual record interface consumed elsewhere.

Several ingredients are standard: the exact Holevo quantity of a branching state, the Helstrom bound, and the role of redundancy in quantum Darwinism [10, 11, 12]. What the paper adds is the typed separation, the exact permission set of a fragment, the nesting of the tests in one dynamics, the counting difference between Holevo and readable records, the mixed-memory realization and a calibration-sensitive public-readout certificate.

2 Typed accessibility data

2.1 Algebraic background

In the algebraic formulation of quantum theory a system is described by an algebra of observables, and states are positive normalized functionals on it [8, 9]. In algebraic quantum field theory local algebras are attached to regions without a global tensor product, and tensor product structures can be induced by the accessible observables [20, 22]. HAFF builds on this view and asks which operator structures a physical context selects. Paper A treats the reduction of a global state by physical channels and shows that two channels can generate the same algebra while resolving different things [1]. The present paper applies the same caution to measurement.

2.2 The objects of a context

Let $\mathcal{M}_U$ be the ambient von Neumann algebra. A context $\mathbf{c}$ specifies allowed probe channels, an environmental dynamics, a record algebra, time scales and resolution thresholds.

Definition 2.1 (Probe channel and implementable effects)**.** *For each allowed probe α, let*

$$\mathcal{R}^*_{\alpha,\mathbf{c}} : \mathcal{B}(\mathcal{K}_\alpha) \longrightarrow \mathcal{M}_U$$

be its normal unital completely positive Heisenberg map. The set of ***implementable effects*** *of the context is*

$$\mathrm{Eff}(\mathbf{c}) := \{\mathcal{R}^*_{\alpha,\mathbf{c}}(F) : 0 \le F \le \mathbf{1},\ \alpha \text{ allowed}\},$$

and the ***probe-reachable operator system*** *is its ultraweakly closed linear span,*

$$\mathrm{Eff}_{\mathrm{span}}(\mathbf{c}) := \overline{\mathrm{span}}^{\,uw}\,\mathrm{Eff}(\mathbf{c}).$$

The family $\mathfrak{R}_{\mathrm{probe}}(\mathbf{c}) := \{\mathcal{R}_{\alpha,\mathbf{c}}\}_\alpha$ is part of the context, not an observable algebra.

For a single probe Eff is convex, contains 0 and $\mathbf{1}$, and is closed under $E \mapsto \mathbf{1} - E$. The span is the natural object for linear reconstruction of expectation values from many runs. It is not a list of effects that one run of an allowed probe implements: Section 4 gives probes whose span is the whole pointer algebra while the sharp pointer projections are not implementable. The same distinction appears for reductions in Paper A, where the effects a channel reads out, $\Phi^*([0, \mathbf{1}])$, are in general fewer than the positive contractions in the range of Φ^* [1].

Definition 2.2 (Environmental fixed set)**.** *For a normal UCP semigroup $\{\mathcal{E}_t\}_{t\geq 0}$ on $\mathcal{M}_U$, set*

$$\mathrm{Fix}(\mathcal{E}) := \{O \in \mathcal{M}_U : \mathcal{E}_t(O) = O \text{ for every } t \geq 0\}.$$

This is always an ultraweakly closed operator system. It is a von Neumann subalgebra, for example, when the semigroup admits a faithful normal invariant state: equality in the Schwarz inequality then places every fixed element in the multiplicative domain. A discrete-time channel is treated in the same way with $t \in \mathbb{N}$.

Definition 2.3 (Pointer and record algebra)**.** *A **pointer/record algebra** $\mathcal{P}_{\mathrm{env}} \subset \mathcal{M}_U$ is a designated commutative von Neumann algebra generated by a classical record PVM $Z(B)$, or by a finite-resolution coarse graining. For a finite label set, a fragment record channel acts on label states by*

$$\Gamma_{k,*}\big((p_x)_x\big) = \sum_x p_x \rho^x_{E_k}.$$

Its normal UCP Heisenberg dual is

$$\Gamma_k^* : \mathcal{B}(\mathcal{H}_{E_k}) \longrightarrow \mathcal{P}_{\mathrm{env}}, \qquad \Gamma_k^*(F) = \sum_x \mathrm{Tr}(\rho^x_{E_k} F) Z_x.$$

More generally the state map acts on the predual of the commutative record algebra. Approximate pointer observables are treated by their distance to $\mathcal{P}_{\mathrm{env}}$.

There is no default equality among Eff, $\mathrm{Fix}(\mathcal{E})$ and $\mathcal{P}_{\mathrm{env}}$. The declared maps are the probe maps, the dynamics, the inclusions of chosen subalgebras into $\mathcal{M}_U$, and the record channels. A model may establish relations among the objects, for example that every implementable effect lies in $\mathcal{P}_{\mathrm{env}} \cap \mathrm{Fix}(\mathcal{E})$; such relations are results or assumptions of that model.

Definition 2.4 (Measurement-accessibility profile)**.** *The **measurement-accessibility profile** of a context is the typed datum*

$$\mathbf{Acc}(\mathbf{c}) := \big(\mathrm{Eff}(\mathbf{c}),\ \mathrm{Eff}_{\mathrm{span}}(\mathbf{c}),\ \mathfrak{R}_{\mathrm{probe}}(\mathbf{c}),\ \mathcal{P}_{\mathrm{env}}(\mathbf{c}),\ \mathrm{Fix}(\mathcal{E}_{\mathbf{c}});\ \Gamma_{k,*}, T_{\mathbf{c}}, \epsilon, \delta\big). \tag{1}$$

The operator-system and fixed-set statements above hold for general von Neumann algebras under the hypotheses shown. Entropic statements are made for finite alphabets or finite-resolution labels and specified states.

3 Three tests

3.1 Probe reachability

Constraint 3.1 (Probe reachability)**.** *An effect $M(B) \in \mathcal{M}_U$ is implementable in context $\mathbf{c}$ if $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F)$ for some allowed probe α and some probe effect $0 \leq F \leq \mathbf{1}$. A POVM M is implementable if one probe and one probe POVM F give $M(B) = \mathcal{R}^*_{\alpha,\mathbf{c}}(F(B))$ for every outcome set B.*

Implementability is a property of a channel and an effect. A probe Hamiltonian may realize the channel, but the condition $[O, H_{\mathrm{probe}}] \neq 0$ is neither necessary nor closed under sums and products; ideal nondemolition couplings commute with the measured pointer observable. Effects outside $\mathrm{Eff}(\mathbf{c})$ are not ruled out as physical quantities. The context simply supplies no implementation of them, and enlarging the context, for example by allowing more fragments or repeated runs, is a change of context.

3.2 Dynamical stability

Constraint 3.2 (Stability)**.** *An effect O passes the* ***finite-window stability test*** *if*

$$\sup_{0\le t\le T_{\mathbf{c}}} \|\mathcal{E}_t(O) - O\| < \epsilon. \tag{2}$$

Exact stability is the separate statement $O \in \mathrm{Fix}(\mathcal{E})$.

In decoherence theory the pointer observables are those left stable by the system–environment interaction [10]. The threshold ϵ and the window $T_{\mathbf{c}}$ are set by the recording mechanisms and time scales of the context. An implementable effect need not be stable, and an exact fixed point need not be implementable by any allowed probe.

3.3 Environmental redundancy: Holevo and readable records

Let X be the classical label of a pointer PVM, or of a finite coarse graining of a POVM, with distribution (p_x). For each environmental fragment E_k form the classical–quantum state $\rho_{XE_k} = \sum_x p_x |x\rangle\langle x| \otimes \rho^x_{E_k}$. Two quantities measure what E_k carries about X. The Holevo information

$$\chi(X : E_k) = S\Big(\sum_x p_x \rho^x_{E_k}\Big) - \sum_x p_x S(\rho^x_{E_k})$$

bounds what any measurement on E_k can extract. The accessible information

$$I_{\mathrm{acc}}(X : E_k) = \sup_{\text{POVM } N \text{ on } E_k} I(X : Y_N)$$

is what a single measurement on the fragment does extract at best, where Y_N is the outcome. Always $I_{\mathrm{acc}} \le \chi$ [14].

Constraint 3.3 (Redundancy)**.** *Fix $0 < \delta < 1$ and a family of disjoint fragments. The label X has R* ***Holevo records*** *if R disjoint fragments satisfy $\chi(X : E_k) \ge (1-\delta)H(X)$, and R* ***readable records*** *if R disjoint fragments satisfy $I_{\mathrm{acc}}(X : E_k) \ge (1-\delta)H(X)$. For a declared measurement N_k on each fragment one may use $I(X : Y_{N_k})$ in place of I_{acc}.*

Readable records are Holevo records, but not conversely, and the two counts can differ (Proposition 4.6). For continuous spectra X denotes a declared finite-resolution partition. No von Neumann entropy is assigned to an operator, and no expression such as $I(\hat{O} : E_k)$ is used.

Redundancy is what allows several observers to read the same label from different fragments without disturbing one another. It does not by itself say how often their common value is wrong, nor which global outcome is realized. Section 5 computes the former in the model.

4 A record model in which the three tests can be computed

4.1 Fresh memories colliding once with a pointer

The system S is a qubit with pointer projections $P_x = |x\rangle\langle x|$, $x \in \{0, 1\}$. The environment consists of n memory qubits $E_1, \ldots, E_n$, each prepared in $|0\rangle$. The k-th memory collides

once with S through

$$U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k, \qquad R_k = \begin{pmatrix} c_k & -\sqrt{1-c_k^2} \\ \sqrt{1-c_k^2} & c_k \end{pmatrix}, \qquad 0 \le c_k \le 1, \tag{3}$$

and does not interact again. Starting from $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$ with $0 < p < 1$, the state after n collisions is

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_{k=1}^{n} |e_{x,k}\rangle, \tag{4}$$

with $|e_{0,k}\rangle = |0\rangle$ and $|e_{1,k}\rangle = R_k|0\rangle$, so that $\langle e_{0,k}|e_{1,k}\rangle = c_k$. A fragment F is a declared set of memories; write

$$r_F = \prod_{k\in F} c_k, \qquad H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right),$$

where $h(x) = -x\ln x - (1-x)\ln(1-x)$. All entropies are in nats. The label X is the value a pointer measurement on S would return.

The inputs are explicit: the pointer basis, the split into system and memories, the order of collisions, the fragment partition, and the purity of the memories. None of them is derived here. The model is a version of the standard branching model of quantum Darwinism [11, 12]; its role in this paper is to make all components of the profile of Definition 2.4 computable at once.

Proposition 4.1 (Record quantities)**.** *For every fragment F,*

$$\chi(X : E_F) = H_p(r_F), \qquad (\rho_S)_{01} = \sqrt{p(1-p)}\, r_{[n]},$$

$$S(\rho_S) = H_p(r_{[n]}), \qquad I(S : E_{[n]}) = 2H_p(r_{[n]}).$$

Proof. Given $X = x$ the fragment is in the pure state $|e_x^F\rangle = \bigotimes_{k\in F} |e_{x,k}\rangle$, and $\langle e_0^F|e_1^F\rangle = r_F$. The average state $p|e_0^F\rangle\langle e_0^F| + (1-p)|e_1^F\rangle\langle e_1^F|$ has two nonzero eigenvalues with sum 1 and product $p(1-p)(1-r_F^2)$, so its entropy is $H_p(r_F)$, and the conditional entropies vanish. Tracing the memories out of (4) multiplies the off-diagonal element by $\langle e_1|e_0\rangle = r_{[n]}$, and the global state is pure, so $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ and $I(S : E_{[n]}) = 2S(\rho_S)$. □

The reduced dynamics of S is the dephasing channel $\mathcal{D}_k(\rho) = P_0\rho P_0 + P_1\rho P_1 + c_k(P_0\rho P_1 + P_1\rho P_0)$ at each step. Its fixed algebra, for any $c_k < 1$, is the diagonal algebra $\mathcal{P} = \mathrm{span}\{P_0, P_1\}$, which is also the pointer algebra.

4.2 Measurement permissions of a fragment

A reader with access to fragment F implements effects on S through the Heisenberg map

$$\Phi_F^*(A) = \langle e_0^F|A|e_0^F\rangle\, P_0 + \langle e_1^F|A|e_1^F\rangle\, P_1, \qquad A \in \mathcal{B}(\mathcal{H}_F), \tag{5}$$

obtained by coupling S to F through the collisions and tracing out the rest. The span of the image is all of $\mathcal{P}$ as soon as $0 < r_F < 1$, since it contains $\mathbf{1}$ and the operator $\Phi_F^*(|e_0^F\rangle\langle e_0^F|) = P_0 + r_F^2 P_1$. The implementable effects are fewer.

Theorem 4.2 (Permission set of a fragment)**.** *Let* $|e_0\rangle, |e_1\rangle$ *be unit vectors with* $|\langle e_0|e_1\rangle| = r$*. Then*

$$\begin{aligned}&\big\{(\langle e_0|A|e_0\rangle, \langle e_1|A|e_1\rangle) : 0 \le A \le \mathbf{1}\big\}\\ &\qquad = L_r := \big\{(a,b) \in [0,1]^2 :\ \sqrt{ab} + \sqrt{(1-a)(1-b)} \ge r\big\}.\end{aligned}$$

Hence $\mathrm{Eff}_F := \Phi_F^*([0,\mathbf{1}]) = \{aP_0 + bP_1 : (a,b) \in L_{r_F}\}$.

Proof. Inclusion. For an effect A put $a = \langle e_0|A|e_0\rangle$ and $b = \langle e_1|A|e_1\rangle$. By the Cauchy–Schwarz inequality applied to $A^{1/2}e_0$ and $A^{1/2}e_1$, $|\langle e_0|A|e_1\rangle| \le \sqrt{ab}$, and likewise $|\langle e_0|(\mathbf{1}-A)|e_1\rangle| \le \sqrt{(1-a)(1-b)}$. Adding and using the triangle inequality gives $r \le \sqrt{ab} + \sqrt{(1-a)(1-b)}$.

Equality. The function $f(a,b) = \sqrt{ab} + \sqrt{(1-a)(1-b)}$ is a sum of geometric means, hence concave, so L_r is a compact convex set. The set of attained pairs is convex as the image of the convex set of effects under an affine map. At $r = 0$ the states are orthogonal and all pairs in the square are attained by diagonal effects; at $r = 1$ the states are identical up to phase and the attained set is the diagonal. For $0 < r < 1$, a compact convex subset of the plane is the convex hull of its boundary, so it suffices to attain every boundary point of L_r. Choose phases and an orthonormal basis of $\mathrm{span}\{e_0, e_1\}$ with $e_0 = (1,0)$ and $e_1 = (\cos\theta, \sin\theta)$, $\cos\theta = r$, $0 \le \theta \le \pi/2$; effects supported on this span suffice. For $u = (\cos\varphi, \sin\varphi)$ with $\theta - \pi/2 \le \varphi \le 0$, the projection $|u\rangle\langle u|$ gives $a = \cos^2\varphi$ and $b = \cos^2(\theta - \varphi)$, and since $\cos\varphi \ge 0$, $\sin\varphi \le 0$, $\cos(\theta-\varphi) \ge 0$ and $\sin(\theta-\varphi) \ge 0$,

$$f(a,b) = \cos\varphi\cos(\theta-\varphi) - \sin\varphi\sin(\theta-\varphi) = \cos\theta = r.$$

This traces the arc of $\{f = r\}$ from $(1-r^2, 0)$ to $(1, r^2)$. The map $A \mapsto \mathbf{1} - A$ sends it to the arc from $(r^2, 1)$ to $(0, 1-r^2)$. The remaining boundary consists of segments of the square: $b = 0$ with $0 \le a \le 1 - r^2$, attained by $A = t|e_1^\perp\rangle\langle e_1^\perp|$; $a = 1$ with $r^2 \le b \le 1$, attained by $A = |e_0\rangle\langle e_0| + t|e_0^\perp\rangle\langle e_0^\perp|$, $0 \le t \le 1$; and their images under $A \mapsto \mathbf{1} - A$. On each edge of the square f reduces to $\sqrt{1-a}$, $\sqrt{b}$, $\sqrt{1-b}$ or $\sqrt{a}$, so these segments are exactly the parts of the edges in L_r. For $r < 1$, $f(a,a) = 1 > r$, so the level set $\{f = r\}$ inside the square consists of the two arcs. Every boundary point is therefore attained. □

For $0 < r < 1$ the permission set is bounded by two curved arcs and four square-edge segments. The corners $(0,0)$ and $(1,1)$ lie on those segments, not on the arcs $f = r$. It fills the square when $r = 0$ and collapses to the diagonal $\{a = b\}$, the trivial effects $a\mathbf{1}$, when $r = 1$. For $0 < r < 1$ it contains no sharp pointer projection: $(1,0) \notin L_r$ because $f(1,0) = 0$.

Corollary 4.3 (Closest implementable pointer test)**.** *For* $0 \le r \le 1$*,*

$$\min_{E \in \mathrm{Eff}_F} \|E - P_0\| = \varepsilon(r) := \frac{1 - \sqrt{1-r^2}}{2},$$

with $r = r_F$*, attained at* $E = (1-\varepsilon)P_0 + \varepsilon P_1$*. The value* $\varepsilon(r)$ *is the minimal error probability for distinguishing* $|e_0^F\rangle$ *from* $|e_1^F\rangle$ *with equal priors.*

Proof. For $E = aP_0 + bP_1$ one has $\|E - P_0\| = \max(1-a, b)$. If $\max(1-a, b) \le t < 1/2$, then $a \ge 1 - t > 1/2 > t \ge b$. In that region $\partial f/\partial a = \frac{1}{2}\big(\sqrt{b/a} - \sqrt{(1-b)/(1-a)}\big) < 0$

and $\partial f/\partial b = \frac{1}{2}\big(\sqrt{a/b} - \sqrt{(1-a)/(1-b)}\big) > 0$, so $f(a,b) \le f(1-t,t) = 2\sqrt{t(1-t)}$. Since $2\sqrt{\varepsilon(1-\varepsilon)} = r$ and $t \mapsto 2\sqrt{t(1-t)}$ is increasing on $[0,1/2]$, the region contains points of L_r only if $t \ge \varepsilon(r)$, and $(1-\varepsilon,\varepsilon) \in L_r$. The identification with the Helstrom error is the formula $(1-\sqrt{1-4p(1-p)r^2})/2$ at $p = 1/2$ [15]. □

For a fragment of m memories with equal overlaps c, $r_F = c^m$ and $\varepsilon(c^m) = c^{2m}/4 + O(c^{4m})$. Measurement permission for the pointer grows with the number of memories a reader controls, and an η-approximate pointer test becomes available once $c^{2m} \le 4\eta(1-\eta)$. Enlarging F is a change of context, in the sense of Constraint 3.1.

4.3 The three tests in one dynamics

Proposition 4.4 (Nesting)**.** *Let a probe act through a pointer-controlled unitary $U = \sum_x P_x \otimes V_x$ on $\mathcal{H}_S \otimes \mathcal{K}$ with probe state $|\phi\rangle$. Then for every $A \in \mathcal{B}(\mathcal{K})$*

$$\mathcal{R}^*(A) = \sum_x \langle\phi|V_x^* A V_x|\phi\rangle\, P_x \in \mathcal{P}_{\mathrm{env}}.$$

In the record model, for every fragment F with $0 < r_F < 1$ and every $c_k < 1$,

$$\mathrm{Eff}_F \subsetneq \mathrm{Eff}_{\mathrm{span},F} \cap [0,\mathbf{1}] = \mathcal{P} \cap [0,\mathbf{1}] = \mathrm{Fix}(\mathcal{D}_k) \cap [0,\mathbf{1}].$$

Proof. $U^*(\mathbf{1}\otimes A)U = \sum_{x,x'} P_x P_{x'} \otimes V_x^* A V_{x'} = \sum_x P_x \otimes V_x^* A V_x$, and the partial expectation in $|\phi\rangle$ gives the formula. In the model $\mathrm{Eff}_{\mathrm{span},F} = \mathcal{P}$ as noted before Theorem 4.2, the fixed algebra of $\mathcal{D}_k$ is $\mathcal{P}$, and Eff_F is the lens of Theorem 4.2, which omits P_0. □

In this model every effect that the environment can report is a stable pointer effect, and the stable pointer effects are exactly the positive contractions of the span. The tests still differ. The lens measures resolution, the fixed algebra measures persistence, and redundancy, below, counts how many disjoint readers have access. A probe that is not pointer-controlled, for example one that rotates the system before recording, can reach effects outside $\mathrm{Fix}(\mathcal{D}_k)$; those effects are then not stable under the environment that produced the records.

4.4 Holevo records and readable records

Proposition 4.5 (Accessible information of a fragment)**.** *Let*

$$P_e(r) = \frac{1 - \sqrt{1 - 4p(1-p)r^2}}{2}$$

be the Helstrom error for two pure states with overlap r and prior $(p, 1-p)$. For every fragment,

$$I_{\mathrm{acc}}(X : E_F) = h(p) - h(P_e(r_F)) \le \chi(X : E_F) = H_p(r_F).$$

For $p = 1/2$,

$$I_{\mathrm{acc}}(X : E_F) = \ln 2 - h\big(\varepsilon(r_F)\big), \qquad \chi(X : E_F) = h\Big(\frac{1+r_F}{2}\Big),$$

and the two differ whenever $0 < r_F < 1$.

Proof. This binary pure-state optimum is standard [16, Sec. 3.5.4]; the following argument proves it for all POVMs. Write their conditional outcome laws as a_y, b_y, put $q_y = pa_y + (1-p)b_y$, and omit zero-weight outcomes. Their posterior probabilities are $\pi_y = pa_y/q_y$; set $C_y = 2\sqrt{\pi_y(1-\pi_y)}$ and

$$E(C) = h\Big(\frac{1-\sqrt{1-C^2}}{2}\Big).$$

Then $H(X \mid Y) = \sum_y q_y E(C_y)$. For $t = \sqrt{1-C^2}$, $E'(C) = C\,\mathrm{artanh}(t)/t \geq 0$ and $E''(C) = (\mathrm{artanh}(t) - t)/t^3 \geq 0$ for $0 < C < 1$; continuity extends convexity to the endpoints. The outcome-wise Cauchy–Schwarz argument of Theorem 4.2 gives

$$\sum_y q_y C_y = 2\sqrt{p(1-p)} \sum_y \sqrt{a_y b_y} \geq 2\sqrt{p(1-p)}\, r_F.$$

Jensen and monotonicity imply $H(X \mid Y) \geq E(2\sqrt{p(1-p)}r_F) = h(P_e(r_F))$. Diagonalizing $p|e_0^F\rangle\langle e_0^F| - (1-p)|e_1^F\rangle\langle e_1^F|$ shows that its Helstrom measurement has posterior error $P_e(r_F)$ on every nonzero outcome, so it attains the bound. The cases $r_F = 0, 1$ follow directly. The Holevo bound and value are those of Proposition 4.1. At equal priors the displayed forms agree only at $r_F = 0, 1$. □

At $p = 1/2$ and $r_F = 0.8$, $\chi = 0.325083$ while $I_{\mathrm{acc}} = 0.192745$. With $\delta = 0.6$ the threshold $(1-\delta)\ln 2 = 0.277259$ lies between them: the fragment is a Holevo record and not a readable one. The threshold for a Holevo record and the threshold for a measurement-based record are therefore not equivalent, even when the measurement is optimal; the difference concerns one measurement on one fragment, and joint coding over many independent preparations is a different task.

Proposition 4.6 (Fragment lengths)**.** *Let all $c_k = c \in (0,1)$ and $0 < \delta < 1$, and let h^{-1} denote the inverse of h on $[0, 1/2]$. Put*

$$q_\delta = h^{-1}\big((1-\delta)h(p)\big), \qquad a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)},$$

$$e_\delta = h^{-1}\big(\delta h(p)\big), \qquad a_{\mathrm{op}} = \frac{e_\delta(1-e_\delta)}{p(1-p)}.$$

A fragment of m memories is a Holevo record if and only if $c^{2m} \leq a_\chi$, and it is a readable record if and only if $c^{2m} \leq a_{\mathrm{op}}$. The smallest such lengths are

$$m_\chi = \Big\lceil \frac{\ln a_\chi}{2\ln c} \Big\rceil, \qquad m_{\mathrm{op}} = \Big\lceil \frac{\ln a_{\mathrm{op}}}{2\ln c} \Big\rceil,$$

so m_{op} is the smallest readable length for every $0 < p < 1$.

Proof. $H_p(r)$ is a decreasing function of r^2, and $H_p(r) = (1-\delta)h(p)$ exactly when the larger eigenvalue $\lambda = (1 + \sqrt{1 - 4p(1-p)(1-r^2)})/2$ equals $1 - q_\delta$, that is when $r^2 = a_\chi$. For the readable case, Proposition 4.5 gives $I_{\mathrm{acc}} \geq (1-\delta)h(p)$ exactly when $h(P_e(r)) \leq \delta h(p)$, that is when $P_e(r) \leq e_\delta$, which rearranges to $4p(1-p)r^2 \leq 4e_\delta(1-e_\delta)$. Since $e_\delta \leq \min(p, 1-p)$, $a_{\mathrm{op}} \leq 1$. □

When fragments are unequal, the counts depend on the chosen partition. With equal overlaps, m_χ gives the largest Holevo count. m_{op} gives the largest readable count for the same pure binary ensemble at every prior.

p	c	n	δ	m_χ	m_{op}	Holevo records	certified readable
0.5	0.80	60	0.10	5	7	12	8
0.2	0.80	60	0.10	5	7	12	8
0.5	0.95	100	0.05	27	38	3	2

Table 1: Fragment lengths and the number of disjoint equal-length records in n memories. The binary pure-state optimum in Proposition 4.5 makes every listed m_{op} minimal over all POVMs. At $p = 0.2$, I_{acc} is .439321 for $m = 6$ and .458263 for $m = 7$, across the threshold .450362.

4.5 The preparation resource for a record

Proposition 4.7 (Purity-deficit budget)**.** *Replace the memories by systems of dimension d_k in product states $\bigotimes_k \sigma_k$, and let the collisions be pointer-controlled unitaries $P_0 \otimes \mathbf{1} + P_1 \otimes W_k$. Then for every fragment*

$$\chi(X : E_F) = S\Big(p \bigotimes_{k\in F} \sigma_k + (1-p) \bigotimes_{k\in F} W_k \sigma_k W_k^*\Big) - \sum_{k\in F} S(\sigma_k) \le \sum_{k\in F} \big[\ln d_k - S(\sigma_k)\big],$$

while each collision multiplies the off-diagonal element $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. If $\sigma_k = \mathbf{1}/2$ and $W_k = R_k$, no fragment carries any information about X, and the coherence of S is still multiplied by c_k.

Proof. The two conditional states of the fragment are unitarily related products, so both have entropy $\sum_{k\in F} S(\sigma_k)$; the average state has entropy at most $\sum_{k\in F} \ln d_k$. The block $P_0\rho_S P_1 \otimes \sigma_k$ is mapped to $P_0\rho_S P_1 \otimes \sigma_k W_k^*$, so tracing out the memory multiplies $(\rho_S)_{01}$ by $\mathrm{tr}(\sigma_k W_k^*)$. For $\sigma_k = \mathbf{1}/2$ both conditional states equal $\mathbf{1}/2$, and $\mathrm{tr}(R_k^*)/2 = c_k$. □

The initial purity deficits bound the information that these unitary ports can record. They do not require every memory to be pure. Dephasing is not evidence of a readable record: a maximally mixed port can generate the same source channel as a pure port and carry no label information. Paper F uses this resource budget as a quantitative low-record boundary condition [5]. Preparation, correlation and renewal must then be accounted for separately; this entropy bound alone is not a universal work price.

4.6 Partially mixed memories: exact permission and readout

Let each fresh qubit memory start in $\eta_\alpha = (I + \alpha Z)/2$, $0 \le \alpha \le 1$, and use

$$U = P_0 \otimes I + P_1 \otimes X.$$

The conditional states are $\eta_0 = \eta_\alpha$ and $\eta_1 = X\eta_\alpha X = (I - \alpha Z)/2$. Put $\varepsilon_\alpha = (1-\alpha)/2$.

Proposition 4.8 (A mixed-memory record)**.** *For one fresh memory, its exact effect permission set is*

$$\mathrm{conv}\{(0,0), (1,1), (1-\varepsilon_\alpha, \varepsilon_\alpha), (\varepsilon_\alpha, 1-\varepsilon_\alpha)\}. \tag{6}$$

For $\alpha > 0$ the effect span is the full pointer algebra, but the closest permitted P_0 test has distance ε_α. At equal label priors,

$$\chi(X : E) = I_{\mathrm{acc}}(X : E) = \ln 2 - h((1+\alpha)/2) = \ln 2 - S(\eta_\alpha). \tag{7}$$

Measurement of Z_E attains this value and has symmetric error ε_α. For every α, tracing the memory completely dephases the source.

Proof. Only the diagonal entries $u, v \in [0,1]$ of a memory effect matter. The reported pair is $((1-\varepsilon_\alpha)u + \varepsilon_\alpha v, \varepsilon_\alpha u + (1-\varepsilon_\alpha)v)$. Diagonal effects attain every such pair, proving the parallelogram. For any reported pair (a,b), $a - b = \alpha(u-v) \le \alpha$; hence $\max(1-a, b) \ge (1-\alpha)/2$, with equality at $u = 1, v = 0$. The conditional states commute, have entropy $h((1+\alpha)/2)$ and average to $I/2$. Their common eigenbasis measurement attains the Holevo quantity. Finally the source coherence multiplier is $\mathrm{Tr}(\eta_\alpha X) = 0$. □

Records in partially mixed spin environments are already part of quantum Darwinism [12]. For $0 < \alpha < 1$ this construction gives their exact effect permissions and an explicit readout saturating the one-port purity-deficit budget. At $\alpha = 0$ its effects collapse to the trivial diagonal (a, a) and it carries no record, although its source channel is unchanged. At $\alpha = 1$ it records perfectly. Thus neither an effect span nor the complete source channel fixes record quality.

After m fresh collisions, the conditional memory states are $\eta_x^{\otimes m}$. Measuring every Z_E reads the entire commuting ensemble and still saturates Holevo information. The reports are binary independent errors conditional on x. Majority vote has exact error

$$e_m = \Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) > m/2\} + \tfrac{1}{2}\Pr\{\mathrm{Bin}(m, \varepsilon_\alpha) = m/2\} \le e^{-m\alpha^2/2}. \tag{8}$$

The bound is Hoeffding's for the sample mean of signs in $[-1, 1]$. Fano's inequality then gives a sufficient readable-record length: choose m so that the bound is at most $1/2$ and $h(e^{-m\alpha^2/2}) \le \delta \ln 2$. This constructive mixed-state extension needs no classification of all mixed conditional-state permission sets.

4.7 A spent port is not a fresh channel input

The same controlled-X unitary also makes the port distinction exact. Since $U^2 = I$, applying it a second time to the same memory, without intervening measurement, export or dynamics, returns the complete state to its pre-collision value. The first collision dephases the source and writes the α-dependent record; the second removes that record and restores the original source coherence. Two independent fresh collisions instead preserve complete source dephasing and leave two conditional memory copies. This reversal does not erase an external measurement record or a copy already transferred to another port.

The one-step reduced channel cannot be applied again to the source while silently replacing the spent memory by η_α. After the first collision the source and memory are correlated. A second actual interaction acts on that joint state, including its old purification. Paper F studies repeated contact and joint memory updating [5]; the distinction also applies to the retained-port acquisition used by Q-RAIF C [4]. Storage, overwrite, reset and acquisition are different operations even when their isolated channel labels are similar.

5 Public readout

5.1 Agreement, correctness and common error

Proposition 5.1 (Conditional independence of readers)**.** *Let readers $j = 1, \ldots, J$ apply POVMs $N^{(j)}$ to disjoint fragments F_j. Then the joint distribution of the pointer value*

and the outcomes is

$$\Pr(x, y_1, \ldots, y_J) = p_x \prod_{j=1}^{J} \langle e_x^{F_j} | N_{y_j}^{(j)} | e_x^{F_j} \rangle.$$

For two readers who each report a value $\hat{x}_j(y_j) \in \{0, 1\}$ *with error probabilities* $\varepsilon_j(x) = \Pr(\hat{x}_j \neq x \mid x)$*, the agreement rate* A*, the correctness rates* C_j *and the common error rate* $E = \Pr(\hat{x}_1 = \hat{x}_2 \neq x)$ *are*

$$E = \sum_x p_x \varepsilon_1(x) \varepsilon_2(x), \qquad A = \sum_x p_x \big[(1 - \varepsilon_1(x))(1 - \varepsilon_2(x)) + \varepsilon_1(x)\varepsilon_2(x)\big],$$

and $C_1 + C_2 = 1 + A - 2E$*, so that* $\Pr(\textit{correct} \mid \textit{agree}) = 1 - E/A$*.*

Proof. Tracing S out of (4) gives $\rho_E = \sum_x p_x |e_x\rangle\langle e_x|$ with $|e_x\rangle$ a product over memories. A pointer measurement on S commutes with the fragment measurements and returns x with probability p_x, leaving the memories in $|e_x\rangle$. The product form of $|e_x\rangle$ over disjoint fragments gives the factorization. The formulas for E and A follow by summing over the four combinations of right and wrong reports, and the identity $C_1 + C_2 = 1 + A - 2E$ holds event by event: an agreeing correct pair contributes two correct reports, an agreeing wrong pair none, and a disagreeing pair one. □

With equal priors, $c = 0.8$ and two five-memory fragments read with their Helstrom measurements, each reader errs with probability $\varepsilon(0.8^5) = 0.02761$, the readers agree with probability $A = 0.94631$, and an agreeing pair is wrong with probability $E/A = 8.05 \times 10^{-4}$. In this model agreement certifies correctness up to the product of private errors. The identity $C_1 + C_2 = 1 + A - 2E$ and the separation of agreement from correctness are the subject of Foundation II, where a contamination shared by two instruments raises the common error far above the product of private errors [7]. Independent private errors already produce a nonzero common error. An excess beyond the conditional-product formula requires an additional shared influence or a failure of that record law, such as shared instrument contamination or calibration error. T-DOME III makes this precise: under a symmetry acting on all readers at once, their relative sign is detectable from agreement and a common inversion is not, so the error that agreement can never expose is the direction shared by the readers' standards [21].

5.2 Calibrated public readout with a common orientation

Conditional independence is a property of the prepared ports and their actual instruments. It is not implied by agreement or commuting effects. The mixed-memory family gives an explicit calibration problem. Encode the pointer value as $z = (-1)^x$. Suppose all readers share an unknown fixed orientation $C \in \{-1, 1\}$, so each reported elementary sign has conditional mean $C\alpha z$. The physical response $\alpha > 0$ is supplied and stable across reference and target runs. A common inversion leaves every agreement rate unchanged. With equal target priors and no independently anchored reference, the transformations $(C, z) \mapsto (-C, -z)$ give the same observable target-record law, so the absolute orientation is unidentifiable. A known asymmetric source prior would be additional orientation information and a different calibration problem.

The context must supply a preparation of the physical projection P_0 whose label is anchored independently of the unknown reader orientation. Its verification and preparation are resources. If the preparation label is itself inverted by the same unknown C,

the reference cannot remove the ambiguity. Given that permission, prepare $z = +1$ in n_{ref} independent reference trials and measure fresh mixed ports with the same physical C response used for target readout. Estimate C by the sign of their mean, breaking a tie at random. Hoeffding gives

$$\Pr\Big(\widehat{C} \neq C\Big) \leq \delta_{\mathrm{cal}} := \min\{1, e^{-n_{\mathrm{ref}}\alpha^2/2}\}. \tag{9}$$

A change of orientation or response between reference and target runs violates the premise; the bound does not detect arbitrary shared contamination.

Proposition 5.2 (Finite public correctness after calibration)**.** *Let J readers each control m fresh mixed-memory ports of the same target pointer. They measure the memory Z values, take a majority with independent fair tie coins, and multiply their decision sign by $\widehat{C}$. The reference trials are independent of the target ports. With e_m from* (8)*,*

$$A_J := \Pr(\textit{all readers agree}) = (1-e_m)^J + e_m^J, \qquad \Pr(\textit{all agree wrongly}) \leq \delta_{\mathrm{cal}} + e_m^J. \tag{10}$$

Consequently

$$\Pr(\textit{wrong} \mid \textit{all agree}) \leq \min\{1, (\delta_{\mathrm{cal}} + e_m^J)/A_J\}.$$

Proof. Correcting by a common sign does not change agreement. Conditional on the pointer and the true orientation, the fresh ports give independent symmetric majority errors of probability e_m. On the event $\widehat{C} = C$, all reports are wrong with probability e_m^J; on its complement this probability is at most one. Reference–target independence gives the stated bounds. □

For $\alpha = 0.6$, $m = 5$, $J = 2$, the exact private error is $e_5 = 0.05792$ and $A_2 = 0.8908694528$. With $n_{\mathrm{ref}} = 100$ the calibration failure bound is e^{-18}. Even after calibration, the conditional common-error bound is about 0.003766. Increasing the number of readers suppresses their independent mistakes, but a fixed shared calibration error does not disappear through agreement.

The protocol consumes n_{ref} prepared-pointer references and fresh mixed ports, Jm target ports, all corresponding readout bits, access to a common orientation standard, and the communication needed for calibration and agreement. These are prepared resources, not an inference of correctness from consensus. The preparation cost is not identified with the entropy deficit alone. T-DOME III analyzes the remaining common-sign ambiguity [21]; Foundation II distinguishes a shared state or instrument from an independently checked source [7]. Here the reference permission and its finite error are part of one explicit record protocol.

5.3 Many readers: error exponents

Proposition 5.3 (Joint, pooled and majority readout)**.** *Let R disjoint fragments each have overlap $r \in (0, 1)$, and let the prior be $(p, 1-p)$.*

1. *Any measurement on the union of the fragments errs with probability at least $P_e(r^R) = p(1-p)r^{2R} + O(r^{4R})$, and a joint measurement attains this value.*

2. *Let each reader measure $\{|e_0^{F_j}\rangle\langle e_0^{F_j}|,\ \mathbf{1} - |e_0^{F_j}\rangle\langle e_0^{F_j}|\}$ on its own fragment, and decide $\hat{x} = 1$ if at least one reader obtains the second outcome. This pooled local rule errs with probability $(1-p)\, r^{2R}$.*

3. *Let $p = 1/2$, let each reader use its single-fragment Helstrom measurement, and let the decision be the majority of the R reports, with ties broken by a fair coin. The error probability is $r^{R+o(R)}$, and no rule that uses only the R reports has a better exponent.*

Proof. (1) The conditional states of the union are pure with overlap r^R, and the Helstrom bound is attained by the projective measurement onto the positive part of $p|e_0\rangle\langle e_0| - (1-p)|e_1\rangle\langle e_1|$ [15]. (2) Given $x = 0$, every reader obtains the first outcome with certainty. Given $x = 1$, reader j obtains it with probability $|\langle e_0^{F_j}|e_1^{F_j}\rangle|^2 = r^2$, independently by Proposition 5.1, so all R readers obtain it with probability r^{2R}. (3) Each report is wrong with probability $\varepsilon = \varepsilon(r)$ independently. With equal priors and symmetric errors the likelihood ratio of the reports depends only on the number of reports for each value, so the majority rule is optimal among rules that use the reports. Its error is $\Pr(\mathrm{Bin}(R,\varepsilon) > R/2)$ plus half the probability of a tie, which equals $\exp\big[-R\,D(\frac{1}{2}\|\varepsilon) + O(\ln R)\big]$ with $D(\frac{1}{2}\|\varepsilon) = -\ln\big(2\sqrt{\varepsilon(1-\varepsilon)}\big)$ by the standard large-deviation estimate for binomial tails. Finally $4\varepsilon(1-\varepsilon) = 1 - (1-r^2) = r^2$. □

Local readers who only pool their results afterwards thus reach the joint error exponent $2|\ln r|$ per fragment, with a prefactor larger by $1/p$. At $p = 1/2$ and $r = 0.8$, twenty fragments give a joint error of 3.32×10^{-5}, a pooled one-sided error of 6.65×10^{-5}, and a majority error of 1.58×10^{-3}. Readers who are allowed to choose each measurement after learning the earlier outcomes can attain the joint Helstrom error exactly for pure conditional states [17].

The pooled rule requires a matched implementation of the projector onto $|e_0^{F_j}\rangle$. Knowledge of the coupling is one way to supply it; a trusted prepared reference or a previously calibrated fixed reader is another. An incorrect coupling model generally sets up the wrong test. When that relation is unknown, calibration records can change the actual readout permission. T-DOME I computes the value of a readout model in a specified thermodynamic task [6]; the present exponent does not require every successful fixed reader to carry an explicit self-model.

6 Conditions and interfaces

6.1 Preparation and reading permissions

The carrier, pointer basis, collision order, preparation maps and fragment partition enter explicitly. The pure-memory model gives a noncommuting two-state permission lens; the mixed-memory model gives a commuting parallelogram and readable records without complete purity. Both use independent fresh ports. Retained ports must instead be evolved with their actual old correlations, as Section 4.7 demonstrates.

Reader permissions also matter. A one-sided pure-state test requires access to its conditional state vector. The orientation protocol requires an independently anchored pointer preparation and the same stable response in reference and target trials. Neither capability follows from membership of an effect in a generated algebra. Arbitrary shared errors and preparation–reader co-inversions remain outside that calibration certificate.

The global branching state remains a superposition. Born probabilities determine the label distribution used in the readout laws; no unique global outcome is selected by these

calculations. A reader can be any physical device controlling a fragment. The model does not privilege a conscious observer.

Einselection and Paper B's stability analysis address how a given interaction chooses a pointer or protected sector [10, 2]. Paper B's formation constructions add real shared recording ports and Hamiltonian links. Locality and the history of those links must still explain why a particular fragment partition is available. Paper F addresses updating and repeated contact [5]; Q-RAIF C supplies finite acquisition on a carrier retaining quantum history [4]. These are complementary realizations, not permission to replace one model's initial states by another's without a physical bridge.

6.2 Relation to existing approaches

Decoherence and quantum Darwinism. Decoherence theory explains the suppression of interference through entanglement with an environment and identifies pointer observables by their stability [10]. Quantum Darwinism adds that objectivity requires redundant records in many fragments [11], and the exact information content of branching states in spin environments has been analysed in detail [12]. The present paper uses these results and adds the typed distinctions: the permission set of a fragment, the nesting of permission, stability and redundancy in one dynamics, and the difference between Holevo and readable records.

Spectrum broadcast structures. The conditional product form used in Proposition 5.1 is one half of a spectrum broadcast structure, the class of states in which many observers can find the same value independently [13]. The other half, perfect distinguishability of the conditional fragment states, holds here only in the limit $r_F \to 0$. Proposition 5.3 measures how fast readers approach that limit as fragments are added.

QBism and relational quantum mechanics. QBism reads quantum states as an agent's degrees of belief [18], and relational quantum mechanics takes states to be relative to a reference system [19]. Here accessibility is fixed by the interaction structure of a context, and a single objective global state is assumed. The emphasis on relations is shared with relational quantum mechanics; the stable records computed here are candidates for the relata that its relations require.

Algebraic quantum field theory. AQFT treats observable algebras as primary and states as functionals on them [8]. The profile of Definition 2.4 adds typed selection data to that setting. Its entropic statements are finite-dimensional, and no Type III version of the redundancy counts is claimed.

6.3 The actual record interface

Paper A uses these pure fragments as actual readers of its noisy GHZ source [1]. Their visibility $v_i = \sqrt{1 - r_i^2}$ gives the collision-derived law

$$\mathbb{E}(Y_i Y_j) = v_i v_j c_{ij}, \qquad c_{ij} = (1 - 2p_i)(1 - 2p_j).$$

The source sites are correlated; no independence of their answers is assumed. Calibrated finite samples then certify non-isometric shortest-path metrics for two equal-flip-budget noise allocations. This supplies the physical interface a dictionary statement alone lacks.

Paper B's block-constant recorder depends only on sector weights, under its specified commuting noise and recording permissions [2]. Its protected content, fixed algebra and finite channel range differ from this fragment-effect set. The mixed-memory family sharpens the distinction: one unchanged dephasing channel has permissions ranging from trivial to perfect. Paper D and Q-RAIF C add record-controlled physical sources and retained-port acquisition [3, 4]; they grant neither free sharp effects nor fresh inputs in place of spent ports.

The record law supplies two different downstream relations. For Paper F, once-only contact and isolation of spent ports protect earlier records against later operations on the carrier. Redundancy alone supplies no event order: the pointer-controlled gates in Eq. (3) commute, so their full terminal state is unchanged by permuting the contacts. Paper F's order-identification protocol adds actual retained intervention tags and record–swap stations; the tags reveal predecessor relations under its specified one-pass architecture. Neither an order inferred from those relations nor a direction relative to its declared acquisition boundary derives a continuous time parameter [5].

For Foundation II, common content means a common target label and declared record channels, not literal intersection of the readers' separate register algebras. The two fragments can share an accurately readable classical label even when their embedded local algebras intersect only in the scalars. The joint preparation and the response laws must establish this common content; consensus alone cannot fix its orientation or remove a common error [7]. The calibrated protocol above supplies that relation and its error, not a theorem that arbitrary readers agree on a world.

T-DOME I consumes the actual record and its posterior for a task; its action value need not equal the record entropy [6]. Preparation, reading, storage and eventual renewal retain their separate resource conditions. Shared geometry additionally requires source preparation, site labels and a reconstruction rule; the finite orientation certificate here supplies one of those permissions, not all of them.

7 Conclusion

Measurement permissions, stable content and public records are different physical data. In the fresh pure-memory model, the permission lens gives the sharp-test error, while the fixed pointer algebra gives persistence and fragment discrimination gives readable redundancy. Holevo and single-measurement thresholds differ. Pooled local tests attain the joint pure-state error exponent when their readers know the conditional fragment state.

The mixed-memory construction shows why preparation must be accounted for without demanding purity. Its actual readout saturates a purity-deficit budget, its exact permissions form a parallelogram, and its source channel remains complete dephasing even when no record is produced. Reusing the same port reverses its record rather than supplying another independent sample.

Public correctness additionally needs a physical standard. Conditional independence suppresses private common errors, but agreement cannot discover a shared orientation inversion. Prepared-pointer references calibrate that inversion with a finite failure probability; this probability remains in the public error bound and its resources remain in the protocol. Together with Paper A's actual record-to-geometry construction, these results identify a finite operational route from interaction to readable shared structure. They

leave outcome selection, elementary factorization and general spacetime reconstruction as independent explanatory questions.

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Records, Order and Intervention: Conditions for a Temporal Direction

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Abstract

Relations among physical records can identify an event order without a timestamp, but redundancy alone cannot. Pointer-controlled gates on separate fresh ports commute, so every collision permutation gives the same complete endpoint state. We contrast this obstruction with a reversible recording protocol: a carrier visits labelled stations in an unknown order, each station records the arriving classical value and then exchanges it for a fresh random label. Retained copies of those labels and repeated noisy endpoint readings identify the predecessor of every station. For calibrated positive readout bias, a finite-sample correlation certificate recovers the complete directed chain. The local record-before-exchange orientation and one-pass contact architecture are inputs; the global permutation is not.

The direction is relative to a physical preparation boundary. Fresh biased ports and independent label–reference pairs are resources, while inverse evolution starts from the correlated output of the same circuit. A joint purity law bounds the information stored in disjoint fragments, with work and retained side information explicitly counted. Partly mixed ports attain the pointer-controlled bound. Recontact can remove a record and restore source coherence; thermal reset, record transfer and complete history reversal have different targets and prices. Renewing a noisy bank requires removal of its full retained entropy increment, and a specified thermal chiral exhaust adds a finite-rate heat bound. These results separate reconstruction of event order, orientation of a recording process and thermodynamic maintenance. They supply neither a continuous time parameter nor a cosmological initial condition.

1 Introduction

A collection of records says that events left traces. Does it also say which event came first? One can date a record by an external clock, or define influence using interventions at specified times. Neither procedure explains when the relations among the records themselves identify an order. That is the first question of this paper. The second is why a reconstructed chain has a direction relative to a preparation boundary, and what must be supplied to keep making useful records.

The distinction is already sharp in a simple collision model. A pointer controls one gate on each of several isolated memories. The final memories can contain many readable

copies of its value, yet the complete final state is invariant under the order of those gates. No endpoint measurement can recover that order. An intervention can change this conclusion only through the actual interactions it inserts; assigning an influence diagnostic a time argument does not recover an unknown ordering.

We give a contrasting protocol in which endpoint relations do identify the order. A finite carrier visits each labelled station once. At every station a CNOT first writes the arriving classical bit into a biased read port; a SWAP then replaces the carrier by an independent random label whose copy is retained. The public data are the final labels and read ports, indexed by station names, not by their positions in the visit sequence. Their correlation matrix has one nonzero entry per column. It identifies the predecessor chain from the known initial label, including which station was last, without reading the final carrier.

The architecture supplies a directed local operation and a family of possible one-pass histories. It does not supply which of the $n!$ permutations occurred. Conversely, recovering that permutation does not explain the external parameter used to implement the gates, elapsed duration or the origin of the architecture. Reversible gates also admit the inverse history. Its actual input is the forward circuit's output, with its changed correlations, rather than a second independent preparation of fresh ports.

Section 2 proves the endpoint obstruction. Section 3 constructs the relational record and its finite-sample certificate. Section 4 locates its direction and resource boundary. Section 6 then distinguishes this reconstruction from a timed influence test and retains a recontact counterexample. Sections 5–8 supply the record and purity account, and Section 9 states what persistence adds. The appendices retain the acquisition/action, reset and renewal proofs in their original regimes.

2 When the complete endpoint forgets the order

Fix a pointer PVM $\{P_x\}$ on a system S and distinct ports E_i. A collision with port i has $U_i = \sum_x P_x \otimes V_{x,i}$, with each $V_{x,i}$ acting only on E_i. The rotations of Section 5 and the mixed-port CNOTs of Appendix B.1 have this form. No order-dependent drift, intervening operation or retained controller tag is included in the compared composition.

Proposition 2.1 (Endpoint non-identification for commuting record gates)**.** *For one fixed initial joint state of S, all ports and any untouched reference, the complete final state after applying each U_i once is independent of the permutation of the collisions. Thus no endpoint local or joint measurement on these systems, including adaptive measurements, can distinguish the permutations. Redundancy and entropy data cannot do so either.*

Proof. For $i \neq j$, orthogonality of the pointer projections gives $U_iU_j = \sum_x P_x \otimes V_{x,i}V_{x,j} = U_jU_i$. The factors on distinct ports commute. Every ordered product is therefore the same unitary, also after tensoring identity on the reference. Its action on the fixed initial state is identical. Every endpoint instrument consequently has the same outcome law. □

This is stronger than equality of one-port marginals. It holds even for an initially coherent pointer and for a reader controlling all modeled outputs jointly. If the driving controller retains a record of its programmed order, that is an additional order tag, not a counterexample. Intermediate observations or inserted gates are also different data. A high-redundancy record of a fixed pointer value can therefore be completely uninformative about the order in which its copies were made.

3 A chain recorded without timestamps

3.1 The physical preparation and local operation

There are $n \geq 1$ stations, named $1, \ldots, n$, and a qubit carrier S. An unknown permutation $\pi = (\pi_1, \ldots, \pi_n)$ determines a single visit to each station. The station names are identities, not chronological ranks. At station i there is a read port R_i, a replacement label port T_i and a retained label record K_i. A record K_0 accompanies the carrier's initial label.

All bits use the orthogonal computational basis. Prepare independent fair choices $Z_0, \ldots, Z_n$, set $S = K_0 = Z_0$ and $T_i = K_i = Z_i$, and independently prepare $R_i = N_i$ with $\Pr(N_i = 1) = \varepsilon_i < 1/2$. The N_i are independent of one another and of all labels. The correlated pair T_i, K_i is an actual preparation, not knowledge of a secretly chosen value. Only an orthogonal classical encoding is copied. No arbitrary unknown quantum state is cloned.

The known local operation is

$$V_i = \mathrm{SWAP}_{S,T_i}\ \mathrm{CNOT}_{S \to R_i}, \tag{1}$$

with the CNOT acting first. It writes before replacing the carrier. Every exited port remains isolated and retained; there is no erasure or discarded history in the gate sequence. The final readout permits the Z measurements of $K_0, \ldots, K_n$ and $R_1, \ldots, R_n$. Write their outcomes as K_i and Y_j. Neither π nor a timestamp is supplied to the reader, and the final carrier and spent T_i are not measured by this readout. Independent trials repeat the same unknown order and preparation law. The readout pairs all endpoint values from the same trial; unpaired single-port marginals would not supply these relations.

Define $\mathrm{pred}_\pi(\pi_1) = 0$ and $\mathrm{pred}_\pi(\pi_k) = \pi_{k-1}$ for $k > 1$. This is notation for the hidden order, not a readout label.

Theorem 3.1 (Endpoint relations identify the unknown permutation)**.** *For the circuit and preparation above, put $g_j = 1 - 2\varepsilon_j > 0$. The measured endpoint correlation matrix obeys*

$$C_{ij} := \mathbb{E}\,(-1)^{K_i + Y_j} = g_j\, \mathbf{1}\{i = \mathrm{pred}_\pi(j)\}, \qquad i = 0, \ldots, n, \quad j = 1, \ldots, n. \tag{2}$$

Consequently these endpoint relations uniquely identify the directed chain $0 \to \pi_1 \to \cdots \to \pi_n$ among the permitted permutations, without measuring S at the endpoint.

Proof. Immediately before the first visit the carrier is Z_0. At a visit to j, the CNOT sends $R_j \mapsto N_j \oplus S$ and leaves S unchanged. The SWAP then sends $T_j \mapsto S_{\mathrm{old}}$ and $S \mapsto Z_j$. Induction along the actual visits gives

$$Y_j = N_j \oplus Z_{\mathrm{pred}_\pi(j)}, \qquad (T_j)_{\mathrm{out}} = Z_{\mathrm{pred}_\pi(j)}, \qquad S_{\mathrm{out}} = Z_{\pi_n}. \tag{3}$$

The $K_i = Z_i$ never change. For $i = \mathrm{pred}_\pi(j)$ the measured sign is $(-1)^{N_j}$, with expectation g_j. For every other i, the two distinct independent fair labels have zero sign correlation and are independent of N_j, giving zero. This proves Eq. (2). Each column has exactly one positive entry, so it gives one predecessor edge. The assumed single-pass architecture makes these edges one chain starting at the known vertex 0 and visiting every station once. Following that chain recovers the permutation. □

The gates do not merely copy the same pointer repeatedly. The carrier's value is changed at each station, so subsequent records refer to a newly supplied label. The retained K_i reveal which station supplied the value that another station recorded. This is the information missing from the commuting-gate model. The local record-before-exchange direction and the known starting label orient the reconstruction; they do not name the global predecessors in advance.

3.2 A finite-sample certificate

Corollary 3.2 (Calibrated finite-sample order recovery)**.** *Suppose the independent repeated trials have one fixed permutation and a calibrated lower bias* $g_{\min} = \min_j g_j > 0$. *For* N *trials define* $\widehat{C}_{ij} = N^{-1}\sum_{\ell=1}^{N}(-1)^{K_i^{(\ell)}+Y_j^{(\ell)}}$. *Declare an edge* $i \to j$ *when* $\widehat{C}_{ij} > g_{\min}/2$, *and require the declared edges to form one chain from* 0 *through all stations. The probability of failing to recover the correct chain, including rejection, is at most*

$$2n(n+1)\exp\bigl[-Ng_{\min}^2/8\bigr]. \tag{4}$$

For $0 < \zeta < 1$, *a safe integer choice with bound strictly below* ζ *is*

$$N = 1 + \left\lfloor \frac{8}{g_{\min}^2} \log \frac{2n(n+1)}{\zeta} \right\rfloor. \tag{5}$$

Proof. On the strict event $\max_{i,j} |\widehat{C}_{ij} - C_{ij}| < g_{\min}/2$, all false entries fall below threshold and every true entry exceeds it. For one sign variable $A \in [-1, 1]$, the centered log moment-generating function has second derivative equal to a tilted variance, at most one. Its value and first derivative at zero vanish, hence it is at most $\lambda^2/2$. Independence across trials and the Chernoff bound give $\Pr\Bigl(|\widehat{C}_{ij} - C_{ij}| \geq a\Bigr) \leq 2e^{-Na^2/2}$. Take $a = g_{\min}/2$ and a union bound over $n(n+1)$ entries. The added integer in Eq. (5) makes N strictly larger than the real-valued logarithmic bound. □

Equality at the threshold lies outside the sufficient strict event; it is covered by the failure bound, not silently classified as safe. Entries within a trial need not be statistically independent for this union bound. Nor does chain validation certify all preparation and contact assumptions from the same data: those define the tested model.

For $n = 3$, $\varepsilon = (1/10, 1/5, 3/10)$ and the hidden order $(2, 1, 3)$, station-indexed data give

$$C = \begin{pmatrix} 0 & 3/5 & 0 \\ 0 & 0 & 2/5 \\ 4/5 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}. \tag{6}$$

Rows are $0, 1, 2, 3$ and columns $1, 2, 3$, not chronological positions. The matrix recovers $0 \to 2 \to 1 \to 3$. Here $g_{\min} = 2/5$ and $\zeta = .01$ give $N = 390$ and bound .00983364. Appendix A enumerates all six permutations by the actual reversible gates with rational weights.

3.3 What removes identification

Discarding the K_i makes all Y_j independent fair bits in the stated independent-noise model: they use distinct fair predecessor labels. Their complete joint law, not just

their marginals, is then independent of π. If every $\varepsilon_j = 1/2$, even the (K, Y) joint law is independent of π. With two fair columns, ambiguity can already remain: at $n = 3$, $\varepsilon_1 = \varepsilon_2 = 1/2$ and $0 \le \varepsilon_3 < 1/2$, orders $(1, 3, 2)$ and $(2, 1, 3)$ have the same permitted endpoint law, since both give predecessor 1 for station 3 and the other read ports are independent fair bits. A single missing column need not destroy identification; the remaining chain constraints may force its edge. Thus positive $g_{\min}$ is a sufficient condition for this threshold certificate, not a necessary condition for every reconstruction.

These are readout-relative failures. Equation (3) shows that the retained spent T_j still contain the predecessor labels. Granting their measurement recovers the chain even with completely fair R_j. This differs from Proposition 2.1, whose complete modeled endpoint state is permutation invariant.

Independence of the labels is substantive. If every Z_i is one common fair bit, all labels and record laws are the same for every permutation. Reconnection, repeated visits or hidden inputs can likewise destroy the one-pass predecessor interpretation. Merely correlating the N_j with each other while keeping them independent of all Z_i does *not* destroy Eq. (2): its proof uses only the label independence and each port's bias. The sampling bound also survives within-trial shared noise when trials remain independent. Label-dependent noise or additional unobserved couplings require a new identification argument.

4 Which boundary gives the chain a direction?

The reconstruction has an oriented local operation and a fresh preparation boundary. At entry, R_j is independent of the incoming carrier and its noise is biased toward zero. The labels for different stations are independent; each label nevertheless has its own retained copy. The global order is inferred from the relations produced by these operations, rather than selected by increasing record entropy.

The inverse of every V_j exists: it swaps first and then applies the CNOT. In reverse global order these inverses return the entire actual output, including every spent port and label record, to its input. That reverse input is not a fresh preparation of the declared form. Initially $T_j = K_j$ with certainty. At the forward endpoint, $T_j = K_{\mathrm{pred}_\pi(j)}$ and is independent of K_j, while R_j remains correlated with that spent label when $g_j > 0$. The reverse gates use those actual correlations. Microreversibility therefore coexists with a directed reconstruction relative to the acquisition boundary. No probability suppression of inverse histories follows without a specified ensemble and measure.

For the independent preparation above, there are $3n + 2$ modeled qubits per trial: S, K_0 and the triples T_j, K_j, R_j. No separate register secretly stores chronological ranks. With $h(x) = -x \ln x - (1 - x) \ln(1 - x)$, their complete initial entropy is

$$S_{\mathrm{in}} = (n + 1) \ln 2 + \sum_j h(\varepsilon_j),$$

and the reversible circuit preserves it. Relative to the maximally mixed state of these degenerate registers, its entropy deficit is

$$(3n + 2) \ln 2 - S_{\mathrm{in}} = (n + 1) \ln 2 + \sum_j [\ln 2 - h(\varepsilon_j)]. \tag{7}$$

The first term is the stock of label–copy correlations; each constituent label marginal is fair. The second is the initial read-port purity deficit. This is a state-resource account, not a universal attained work price.

Every independent trial uses a new prepared instance of those registers, or an explicitly paid restoration of the complete needed state. Spent T_j and old records are not erased from that account because the declared readout ignores them. With degenerate endpoint Hamiltonians and pulses off at the endpoints, the ideal gate sequence has zero net endpoint work. Preparing labels and biased ports, implementing controls, reading and retaining results, providing timing, and reusing the device remain physical tasks. Sections 7 and 9 price specific aspects of them. The protocol identifies an unknown event permutation; it does not produce the time parameter or the apparatus that realizes the contacts.

5 Records and how to count them

5.1 The record model

The system S is a qubit with pointer projections P_x, $x \in \{0,1\}$, and initial state $\sqrt{p}\,|0\rangle + \sqrt{1-p}\,|1\rangle$. Memory qubits $E_1, \ldots, E_n$ start in $|0\rangle$, and the k-th collides once with S through $U_k = P_0 \otimes \mathbf{1} + P_1 \otimes R_k$, where R_k is a real rotation with $\langle 0|R_k|0\rangle = c_k \in [0,1]$. After n collisions

$$|\Psi_n\rangle = \sqrt{p}\,|0\rangle|e_0\rangle + \sqrt{1-p}\,|1\rangle|e_1\rangle, \qquad |e_x\rangle = \bigotimes_k |e_{x,k}\rangle, \qquad \langle e_{0,k}|e_{1,k}\rangle = c_k. \tag{8}$$

For a fragment F put $r_F = \prod_{k\in F} c_k$ and

$$H_p(r) = h\left(\frac{1+\sqrt{1-4p(1-p)(1-r^2)}}{2}\right), \qquad h(x) = -x\ln x - (1-x)\ln(1-x).$$

Paper E shows that $\chi(X : E_F) = H_p(r_F)$, that the coherence of S is $\sqrt{p(1-p)}\,r_{[n]}$, and that $S(\rho_S) = S(\rho_{E_{[n]}}) = H_p(r_{[n]})$ [3]. Here X is the value a pointer measurement on S returns. Entropies are in nats.

This pure-port model will be compared with the partly mixed controlled-X model below. The inputs are the pointer basis, the split into system and memories, the fragment partition, the order of collisions with its external parameter, and the preparation of the memories. The collisions can be implemented by couplings $g_k(t)P_1 \otimes \sigma_y$ that vanish at the start and end of each pulse and commute with a pointer Hamiltonian $H_S = \varepsilon P_1$; with degenerate memory Hamiltonians the recording unitary has zero net endpoint work. Preparation of the ports, calibrated pulse control and later bank renewal are separate resources. T-DOME I gives the full thermodynamic account of this protocol [5].

5.2 Redundancy and protocol-relative comparisons

Definition 5.1 (Redundancy index)**.** *For a classical label X, a classical–quantum state ρ_{XE} and a declared family of disjoint fragments $\{E_k\}$, define*

$$\mathcal{R}_\delta^{\chi} = \#\big\{k : \chi(X : E_k) \geq (1-\delta)H(X)\big\}, \qquad \mathcal{R}_\delta^{\mathrm{acc}} = \#\big\{k : I_{\mathrm{acc}}(X : E_k) \geq (1-\delta)H(X)\big\}.$$

The first counts Holevo records and the second records that a permitted single measurement can read; $\mathcal{R}_\delta^{\mathrm{acc}} \le \mathcal{R}_\delta^{\chi}$, and Paper E gives closed-form fragment lengths for both in the collision model. With $p = 1/2$, $c_k = 0.8$ and $\delta = 0.1$, blocks of five memories are Holevo records and blocks of seven are readable, so after n collisions $\mathcal{R}_\delta^{\chi} = \lfloor n/5 \rfloor$ and $\mathcal{R}_\delta^{\mathrm{acc}} = \lfloor n/7 \rfloor$. High redundancy supports public readability of the label; it does not select one global outcome. Here the quoted pure-port counts permit arbitrary fragment POVMs, as in Paper E. A restricted readout menu needs its own accessible-information optimization; ideal algebra closure does not grant those measurements.

Definition 5.2 (Protocol-relative comparisons)**.** *Fix a dynamics, a fragment family and a label X, and let α, β be configurations along the resulting history. Write $\alpha \preceq_{\mathcal{R}} \beta$ if $\mathcal{R}_\delta(\rho_\alpha) \le \mathcal{R}_\delta(\rho_\beta)$, and $\alpha \preceq_{\mathrm{inc}} \beta$ if the algebra of memories that have interacted at α is contained in that at β. Each relation is a preorder on its declared domain.*

In the collision model both comparisons follow the collision order. They need not agree in general: decoherence may increase the number of records while the coherence-sensitive effects of the system, whose expectation values scale with $r_{[n]}$, become unreadable on S itself.

5.3 Temporal inputs

The construction uses temporal orientation at several points: the parameter of the dynamics, the order of collisions, the comparison of records at different parameter values, and a boundary condition of fresh memories at one end of the history. These comparisons describe record growth along that supplied history, not an inverse reconstruction of its unknown permutation. Proposition 2.1 shows that their endpoint values may leave the permutation completely unidentified. The purity account below identifies what the acquisition boundary supplies and what its use costs.

6 Intervention and the agent's time order

The unknown-order reconstruction of Theorem 3.1 uses endpoint statistics and an explicitly restricted family of contact histories. The following diagnostic instead compares interventions at already specified parameter values. Its ordering clause is an input and cannot prove the origin of that order.

Definition 6.1 (Protocol-relative causal-influence diagnostic)**.** *Fix a record-forming protocol with external parameter t, labels X_A, X_B, and a family of interventions on X_A. Write $X_A \rightsquigarrow_{\mathsf{P}} X_B$ when (i) both labels have records in the declared fragments, (ii) the X_A records form no later than the X_B records according to t, and (iii) interventions on X_A, with the rest of the protocol held fixed, change the distribution of X_B or its records.*

The diagnostic is not asserted to be transitive, antisymmetric or independent of the intervention family. Redundancy makes an influence readable across fragments; the intervention structure carries the causal content. The next two propositions compute the diagnostic in the collision model.

Proposition 6.2 (Interventions reach only later records)**.** *In the collision model let an arbitrary channel $\mathcal{T}$ act on S between the j-th and $(j+1)$-th collisions.*

1. *The final reduced state of $E_1, \dots, E_j$ is the same as without $\mathcal{T}$.*

2. *If $\mathcal{T}$ is the pointer flip $\rho \mapsto \sigma_x \rho \sigma_x$, the final state is $\sum_x \sqrt{p_x}\,|1-x\rangle|e_x^{\le j}\rangle|e_{1-x}^{>j}\rangle$: the earlier memories record the value x and the later ones record $1-x$.*

3. *If the Kraus operators of $\mathcal{T}$ are diagonal in the pointer basis, the joint distribution of the pointer value and all records is unchanged.*

Proof. (1) After the j-th collision, $\mathcal{T}$ and the later collisions act on $S \otimes E_{j+1} \otimes \cdots \otimes E_n$, and the partial trace over these systems is invariant under any operation on them. (2) σ_x maps the branch $\sqrt{p_x}|x\rangle|e_x^{\le j}\rangle$ to $\sqrt{p_x}|1-x\rangle|e_x^{\le j}\rangle$, and the later collisions then write the value $1-x$. (3) Diagonal Kraus operators commute with the pointer-controlled collisions and with the pointer measurement, and they leave the populations unchanged. □

For a flip after the second of four collisions with overlaps $(0.3, 0.5, 0.2, 0.4)$ and $p = 1/2$, the reduced states of the first two memories and of the last two are each unchanged, while the joint state of the first and third memories changes by 0.437 in its largest matrix element: the flip is visible only in the relation between earlier and later records. In this fresh-port protocol, interventions on the source can alter later records, while detached earlier marginals remain fixed. The flip example makes the change visible in joint relations even when single-fragment marginals do not change.

Proposition 6.3 (Repeated interaction lets a later intervention rewrite an earlier record)**.** *Let a memory collide with S twice through the same rotation with $c = \cos(\pi/4)$, and let $p = 1/2$. After the first collision the memory holds $H_{1/2}(2^{-1/2}) = 0.4165$ nats about the initial pointer value. If nothing happens between the collisions, the second collision makes the record perfect, $\ln 2$ nats. If S is flipped between the collisions, the second collision leaves the memory with no information about the initial pointer value.*

Proof. Write R for the rotation by $\pi/4$, so $R^2|0\rangle = |1\rangle$. Without the flip, the memory ends in $|0\rangle$ if $x = 0$ and in $R^2|0\rangle = |1\rangle$ if $x = 1$. With the flip, it is in $|0\rangle$ after the first collision if $x = 0$ and is then rotated once, ending in $R|0\rangle$; if $x = 1$ it is in $R|0\rangle$ after the first collision and is not rotated again. Both branches end in $R|0\rangle$. □

For the coherent source $|+\rangle$, the same two-qubit protocol also gives the entropy and population ledger:

$$|\Psi_1\rangle = (|0,0\rangle + |1\rangle R|0\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{noflip}}\rangle = (|00\rangle + |11\rangle)/\sqrt{2}, \quad |\Psi_{2,\text{flip}}\rangle = |+\rangle \otimes R|0\rangle.$$

Source populations stay $1/2, 1/2$, and joint entropy stays zero. Put $a = h[(1+2^{-1/2})/2] = 0.4164955$. At first contact, $S(S) = S(E) = a$ and $I(S:E) = 2a$. The no-flip continuation gives $\ln 2, \ln 2, 2\ln 2$. The flip continuation gives $0, 0, 0$: source coherence and both marginal purities return while the earlier pointer information disappears. For a pointer-diagonal source Hamiltonian, energy stays fixed; source free energy first falls by $k_B T a$ and then recovers it on the flip branch. In fact $U(X_S \otimes I)U = X_S \otimes R$, so the echo decouples every source input. Reading or discarding the port after the first contact would change this coherent protocol. The source trace distance of the inputs $|+\rangle, |-\rangle$ falls to $2^{-1/2}$ at first contact and returns to 1 after the echo: this is distinguishability backflow, not merely an entropy change.

The alignment of the intervention order with the record order thus rests on a specific feature of the dynamics: memories that have recorded do not interact with the source

again. Under repeated interaction, whether an early memory ends as a perfect record or as no record can be decided by an intervention made after it first recorded.

An agent can therefore date an intervention by its effect on records within the stated one-contact architecture. It cannot change a detached port by acting only on the source. Acting directly on that port can relabel, transfer or reset it, with the different balances in Appendix C; reconnecting it to the source can also restore coherence. The alignment comes from the actual interaction menu, fresh stock and external order, not from a universal prohibition on changing the past.

7 The purity of the past

Bath balances below use finite entropies, relative entropies and mean energy changes. An infinite reservoir requires a controlled limit of finite reservoirs preserving those balances, not subtraction of divergent global entropies.

Theorem 7.1 (Purity law for records)**.** *Let a system S, memories $E_1, \dots, E_n$ of dimensions d_k with degenerate Hamiltonians, and a bath B with Hamiltonian H_B start in the product state $\rho_S \otimes \bigotimes_k \sigma_k \otimes \tau_B$, where τ_B is the Gibbs state of H_B at inverse temperature β. Let them evolve under a time-dependent Hamiltonian whose interaction terms vanish at the initial and final times and whose bath and memory terms are then H_B and the given degenerate Hamiltonians, let W be the work done on the whole and ΔE_S the change of the energy of S. Let X be the outcome of a final measurement of a PVM on S with rank-one projections, and let $F_1, \dots, F_R$ be disjoint sets of memories. Then*

$$\sum_{j=1}^{R} \chi(X : E_{F_j}) \le \sum_{k=1}^{n} \big[\ln d_k - S(\sigma_k)\big] + \beta\,(W - \Delta E_S) + \big[H(X) - S(\rho_S)\big]. \tag{9}$$

Without a bath the middle term is absent, and the bound holds for an arbitrary joint unitary on S and the memories, with no condition on their Hamiltonians.

Proof. Let ρ be the final state, and $\rho' = \sum_x (P_x \otimes \mathbf{1})\rho(P_x \otimes \mathbf{1})$ its pinching in the pointer basis. Pinching does not decrease entropy and unitary evolution preserves it, so $S(\rho') \ge S(\rho_S) + \sum_k S(\sigma_k) + S(\tau_B)$. Since the P_x have rank one, $S(\rho') = H(X) + S(E_{[n]}B \mid X)$, where the conditional entropy refers to the classical–quantum state of X, the memories and the bath. Write G for the memories outside $\bigcup_j F_j$. Subadditivity of conditional entropy gives

$$S(E_{[n]}B \mid X) \le \sum_j S(E_{F_j} \mid X) + S(G \mid X) + S(B \mid X) \le \sum_j S(E_{F_j} \mid X) + \ln d_G + S(\rho_B),$$

with ρ_B the final state of the bath. Nonnegativity of $D(\rho_B \| \tau_B)$ gives $S(\rho_B) \le S(\tau_B) + \beta \Delta E_B$. Hence

$$\sum_j S(E_{F_j} \mid X) \ge S(\rho_S) + \sum_k S(\sigma_k) - H(X) - \ln d_G - \beta \Delta E_B.$$

Since $\chi(X : E_{F_j}) = S(E_{F_j}) - S(E_{F_j} \mid X) \le \ln d_{F_j} - S(E_{F_j} \mid X)$, summing over j gives (9) with $\beta \Delta E_B$ in the middle. The memory Hamiltonians are degenerate and the interactions vanish at the endpoints, so the first law gives $\Delta E_B = W - \Delta E_S$. Without a bath the same argument applies with B omitted. □

The bound is attained without a bath. If S starts in $|0\rangle$, a single memory starts maximally mixed, and a controlled-NOT with the memory as control and S as target is applied, then the memories have no purity, $H(X) - S(\rho_S) = \ln 2$, and $\chi(X : E_1) = \ln 2$. In that example the label is created by copying the memory's randomness into a pure system, which is why the system can contribute one record's worth. For a bath-free pointer-controlled unitary, $U = \sum_x P_x \otimes V_x$, the system term is absent: the conditional states of the memories are unitary images of their initial state, and the argument of the proof gives $\sum_j \chi(X : E_{F_j}) \leq \sum_k [\ln d_k - S(\sigma_k)]$. If in addition each V_x is a product over memories, as in the collision model, each fragment is bounded separately by its own purity, $\chi(X : E_F) \leq \sum_{k \in F} [\ln d_k - S(\sigma_k)]$ (Paper E's purity-deficit bound). Without the product form the per-fragment bound can fail: with E_1 maximally mixed, E_2 pure, $p = \frac{1}{2}$ and V_1 the swap of E_1 and E_2, the fragment $\{E_1\}$ holds 0.216 nats about X.

Corollary 7.2 (Purity budget of a record history)**.** *If R disjoint fragments are Holevo records at level δ, then*

$$R(1-\delta)H(X) \leq \beta \sum_k \big[F(\sigma_k) - F(\mathbf{1}/d_k)\big] + \beta\,(W - \Delta E_S) + H(X) - S(\rho_S),$$

where F is the nonequilibrium free energy at temperature T. In particular, memories that start in thermal equilibrium can hold R records only if the work not stored in the system satisfies $\beta(W - \Delta E_S) \geq R(1-\delta)H(X) - [H(X) - S(\rho_S)]$. In a pointer-controlled process without a bath, $R(1-\delta)H(X) \leq \sum_k [\ln d_k - S(\sigma_k)]$.

Proof. Each record contributes at least $(1-\delta)H(X)$ to the left side of (9). For a degenerate Hamiltonian the thermal state of a d-dimensional memory is $\mathbf{1}/d$, and $\beta[F(\sigma) - F(\mathbf{1}/d)] = D(\sigma \| \mathbf{1}/d) = \ln d - S(\sigma)$. The pointer-controlled statement is the summed bound stated after Theorem 7.1. □

The acquisition boundary is thus a quantitative resource, not a requirement that every memory be pure. Stable old records can also serve as standards across time, provided their couplings and orientation connect the old and present instruments [15]. The bound counts simultaneously present information in declared disjoint fragments, not all records that could ever be made, read, erased and remade. With product maximally mixed ports, no bath and pointer-controlled recording, it gives zero; an old correlation can change that boundary, as the conditional bound below shows. The theorem specifies what the acquisition history must supply. It does not explain the primitive origin of that stock.

7.1 Acquisition, action and retained references

Partly mixed ports attain the pointer-controlled purity bound. For a fair pointer and a fresh port with error ε, a computational-basis reading attains $\ln 2 - h(\varepsilon)$ nats. The complete acquisition/action ledger, posterior and proof are in Appendix B.1. For one port, reversible feedback moves its initial purity into the source at zero endpoint gate work; it does not complete a thermal cycle.

The label copies in Section 3 illustrate why marginal purity is not the only stock. The conditional budget of Proposition B.2 includes an old readable reference. Two marginally fair copies of one bit allow a new XOR record to be read jointly even though the new port alone is uninformative. Preparation and preservation of that joint reference are part of the history. A quantum conditional mutual information, by itself, still grants no measurement permission.

8 Records and dissipation

Proposition 8.1 (Records bound the coherence that survives)**.** *In the collision model with $0 < p < 1$, let $0 < \delta < 1$, let q_δ satisfy $h(q_\delta) = (1-\delta)h(p)$ with $q_\delta \le 1/2$, and put*

$$a_\chi = \frac{(1-2q_\delta)^2 - (2p-1)^2}{4p(1-p)} \in (0,1).$$

If R disjoint fragments, of any sizes and any collision overlaps c_k, are Holevo records, then

$$r_{[n]}^2 \le a_\chi^R, \qquad |(\rho_S)_{01}| \le \sqrt{p(1-p)}\, a_\chi^{R/2}.$$

If the pointer Hamiltonian is diagonal in the pointer basis, the system's nonequilibrium free energy falls by

$$F_S(0) - F_S(n) = k_B T\, H_p(r_{[n]}) \ge k_B T\, H_p\big(a_\chi^{R/2}\big).$$

If instead S starts in $pP_0 + (1-p)P_1$, every record quantity is unchanged and $F_S(n) = F_S(0)$.

Proof. A fragment is a Holevo record exactly when $r_F^2 \le a_\chi$ (Paper E's Holevo fragment criterion). The overlaps multiply over disjoint fragments, and the memories outside them contribute factors at most 1, which gives the first two bounds. The collisions preserve the pointer populations, so the energy of S is unchanged and $F_S(0) - F_S(n) = k_B T[S(\rho_S(n)) - S(\rho_S(0))] = k_B T\, H_p(r_{[n]})$; H_p decreases in r. For the diagonal initial state the conditional fragment states are the same, so all $\chi(X : E_F)$ are unchanged, and ρ_S does not change at all. □

Records and dissipation in the system therefore coincide only when the system has coherence to lose. As R grows, $H_p(a_\chi^{R/2}) \to h(p)$ and the system loses all the free energy carried by its coherence. A system that is already diagonal acquires the same records at no cost to itself. In the pure-port model, the marginal purity of written memories is transferred into correlations. The complete unitary preserves joint entropy; the mixed-port ledger makes that transfer explicit.

Proposition 8.2 (Records along a CP-divisible dynamics)**.** *The reduced dynamics of S after n collisions is $\mathcal{D}_n \circ \cdots \circ \mathcal{D}_1$, where $\mathcal{D}_k$ keeps the populations and multiplies the coherence by c_k. Every intermediate map $\mathcal{D}_m \circ \cdots \circ \mathcal{D}_{l+1}$ is completely positive and trace preserving, and for equal overlaps $c \in (0,1)$ applied at intervals Δt the family embeds in the dephasing semigroup generated by $\mathcal{L}(\rho) = \frac{\gamma}{2}(Z\rho Z - \rho)$ with $\gamma = -\ln c/\Delta t$. Along this dynamics $\mathcal{R}_\delta^\chi$ and $\mathcal{R}_\delta^{\mathrm{acc}}$ grow without bound in n.*

Proof. Each $\mathcal{D}_k$ is a dephasing channel, and compositions of channels are channels. The semigroup multiplies the coherence by $e^{-\gamma t}$, which equals c^m at $t = m\Delta t$. The record counts are $\lfloor n/m_\chi \rfloor$ and $\lfloor n/m_{\mathrm{op}} \rfloor$ for equal overlaps. Unbounded record counts refer to a family with growing fresh-port supply, not an unbounded yield from a fixed finite bank. □

Record growth thus does not require memory effects in the reduced dynamics of the system. The memories, once written, have a mixture-of-product marginal correlated through X and are independent conditional on a pointer reading; a Markovian system dynamics says nothing about whether the environment's outputs are correlated. The

converse separation also holds. T-DOME I gives an autonomous two-qubit model in which correlation built between a system and an environment qubit returns to the system at a recurrence time while the environment holds no information at all about the pointer, and it shows that the pure-dephasing spin–boson model with a Lorentz–Drude spectral density has no backflow at any temperature [5]. Backflow and records are logically independent.

Proposition 8.3 (Local and global entropy)**.** *For disjoint fragments $F_1, \dots, F_R$ of the collision model,*

$$\sum_{j=1}^{R} S(\rho_{F_j}) = \sum_{j} \chi(X : E_{F_j}), \qquad S\big(\rho_{\cup_j F_j}\big) \le h(p).$$

If the F_j are Holevo records, the total correlation among them, $\sum_j S(\rho_{F_j}) - S(\rho_{\cup_j F_j})$, is at least $[R(1-\delta)-1]h(p)$.

Proof. The conditional states of every fragment are pure, so $S(\rho_{F_j}) = \chi(X : E_{F_j})$. The union of the fragments has conditional states that are pure as well, so $S(\rho_{\cup F_j}) = H_p(r_{\cup F_j}) \le H_p(0) = h(p)$. Subtracting gives the last claim. □

With $p = 1/2$, $c = 0.8$ and $n = 60$ memories in twelve blocks of five, the local entropy is 7.661 nats while the entropy of all sixty memories is 0.693147 nats, equal to $\ln 2$ to nine digits. A reader with access to single fragments sees entropy grow with every record. The global entropy of the environment saturates at the entropy of the label. Which of the two grows without bound is fixed by the access structure, not by the dynamics.

9 What must be restored to keep recording

Order recovery consumes access to a sequence of prepared trials. It does not grant a renewable device. A protected old record can persist without further entropy removal under compatible noise; independent flips create a different maintenance task.

For a bank of conditionally independent copies of one retained label, with initial error ε, an inaccessible independent flip of probability p changes the error to $q = p + (1-2p)\varepsilon$. Returning the complete label–bank–controller state removes $R[h(q) - h(\varepsilon)]$ nats. Proposition D.1 prices that complete target, rather than the number of distinct labels. Proposition D.2 constructs label-controlled thermal swaps that restore it for positive target error and approach the entropy-removal price. Their gates, gaps, thermal preparations and spent bath ports remain charged; they do not prepare or renew the label boundary for free. The full statements and proofs are in Appendix D.

A sole thermal chiral exhaust of fixed capacity adds the long-window quadratic rate term in Eq. (15). Those exhaust assumptions are separate from the thermal-qubit implementation. They connect actual maintenance to HAFF-D's conditional source and horizon readings, not event order to a gravitational law.

Reset is also distinct from undoing the historical relations. A label flip retains information, a SWAP can move it to another port, and a thermal blank reset has a specified entropy decrease. The inverse of the entire recording circuit restores the complete input when all required systems and gates are accessible. Appendix C retains the different reset permissions, finite-bath obstruction, reference-dependent bound and work account. None of these operations supplies a universal probability measure on reversed histories.

10 Which arrows coincide

Table 1 collects the directions studied above in four specified settings: the collision model with a coherent or a diagonal initial system state, the collision model with one repeated collision, and the autonomous two-qubit model of T-DOME I, in which a system qubit and an environment qubit build and return correlation without any pointer record [5].

Direction	collision, coherent S	collision, diagonal S	one repeated collision	two-qubit recurrence
Record count grows	yes	yes	can fall	no records
Local entropy of fragments grows	yes	yes	can fall	constant E
Marginal memory purity	falls	falls	falls, then returns	no fresh bank
System free energy falls	yes	no	falls, then returns	falls, then returns
Interventions reach only later records	yes	yes	no	no records
Local thermal reset bound $\propto R$	yes	yes	no final record	—
Correlation flows back to S	no	no	yes, with flip	yes

Table 1: Specified protocols, not universal arrows. The repeated-contact column uses the coherent, unmeasured flip continuation of Proposition 6.3. Reset prices refer to blank targets, not arbitrary changes of record content.

With pure fresh ports interacting once, record accumulation, local entropy growth and decreasing marginal port purity align; detached marginals also fix which source interventions can still reach them. Local thermal reset prices that bank relative to a blank target. These relations do not make every record update an entropy-producing operation. The mixed-port ledger shows that acquisition and action redistribute finite stock, while the echo reverses the local entropy and correlation changes. Source free-energy loss depends on initial coherence. Reduced-system backflow, record acquisition and bank renewal remain separate questions.

11 Time in physical theory

In Newtonian mechanics and in the Schrödinger equation time is an external parameter, and the equations have no intrinsic arrow. In canonical quantum gravity the Wheeler–DeWitt equation $\hat{H}|\Psi\rangle = 0$ contains no time parameter [11]. The Page–Wootters construction recovers an effective time from correlations between a clock subsystem and the rest of a stationary universal state [12], but it explains an ordering rather than its asymmetry. The thermodynamic arrow supplies the asymmetry through a coarse-graining and a low-entropy boundary condition [9, 10].

The present dynamical models use an external parameter. In Section 3, a known family of one-pass histories leaves the global permutation unknown to the reader, and endpoint relations recover it. This reconstructs a discrete event order under stated physical

permissions. A continuous clock variable and the asymmetry of its acquisition boundary are additional questions.

12 Relations and interfaces

The purity law gives the record boundary of the past hypothesis a quantitative form, while separating record accumulation from a source's loss of free energy [9, 10]. Mixed-environment redundancy [13] is one instance; the theorem also prices stock drawn from work and a bath. Page–Wootters correlations can supply ordering within a stationary state [12]. A direction of recording additionally needs the acquisition boundary and actual contact architecture specified here.

Complete-access reversal is compatible with retrocausal programmes [16]; restricted thermal reset is a different task. The timed causal-influence diagnostic computes supplied interventions in a physical recording process. It does not turn them into a derived causal partial order. The endpoint theorem instead solves a particular inverse problem: station-labelled intervention records identify an unknown global permutation within a known architecture. The two claims have different data and assumptions.

Paper E supplies the recording instruments [3]. The present posterior, intervention order and reference-dependent budgets are the record-side interfaces to T-DOME I's thermodynamic agent [5]; Paper C implements a separate maintained sensing body [1]. Paper D's heat and horizon bounds apply only under their matched source conditions [2]. The purity law supplies a physical input for Paper G's question of how selection permissions form [4]; it does not derive those permissions, clocks or a geometric algebra.

13 Conclusion

Records identify an order only when their relations depend on that order. Repeated pointer copying into distinct ports does not meet this condition: its commuting gates leave the same complete endpoint state for every permutation. Neither redundancy, entropy nor a more powerful endpoint measurement repairs that obstruction.

A record-before-exchange circuit does meet it. Independent classical labels and their retained copies expose each station's predecessor through a single nonzero correlation entry. Calibrated positive readout bias gives a finite-sample certificate for the unknown global chain, without a timestamp or final-carrier measurement. The known local gate direction and one-pass architecture are indispensable inputs. Fair read ports, missing reference access or correlated labels can remove identification; reading retained spent labels changes that permission boundary.

The direction belongs to a prepared history. Inverse gates act on the correlated actual output and can restore the full input. No cosmological arrow or suppression of inverse histories follows from reversibility and a freshness condition alone. The joint purity law and conditional record budget locate the stock in fresh ports, work or usable reference correlations rather than assuming pure marginals everywhere.

Recontact can remove a record and restore coherence. Sustaining a noisy bank instead requires complete-state renewal, realized here by label-controlled thermal swaps. A thermal exhaust adds its rate price. Order recovery is thereby connected to the resources that make records usable, without deriving its time parameter, contact architecture or primitive acquisition boundary.

A Exact checks of the order protocol

For $n = 3$ and $\varepsilon = (1/10, 1/5, 3/10)$, the check enumerates the 2^7 independent label/noise inputs with rational probabilities. It initializes the full eleven-register basis state, applies each actual CNOT and SWAP in turn, then reads only the endpoint K, Y coordinates in station order. All six permutations have distinct permitted endpoint laws and the correlation formula recovers each chain. The order supplied to the generator is used only to check the reconstruction, not appended to its readout data. The check also enumerates all 2^{11} basis inputs per permutation: the gates are bijective and their reverse sequence restores every input.

For the static obstruction, choose the three controlled rotations

$$(c_1, s_1) = (3/5, 4/5), \qquad (c_2, s_2) = (5/13, 12/13), \qquad (c_3, s_3) = (8/17, 15/17).$$

They give exactly equal full 16×16 rational operators for all six collision orders. This operator check covers coherent states and references as well as diagonal preparations; Proposition 2.1 proves the general statement.

Removing all K makes the six Y laws equal. With every read port fair, all six (K, Y) laws are equal, but the spent-T correlations recover all orders. With only one fair read port the six laws remain distinct in this instance. With two fair columns there are three distinct endpoint laws, including the equal-law pair $(1, 3, 2)$ and $(2, 1, 3)$ stated above. A common label-independent error bit across all read ports retains the predecessor formula. These are exact checks of the same circuit and its boundary changes, not noisy fits or a general chronology theorem.

B Partly mixed acquisition and conditional reference budgets

B.1 Partly mixed ports: acquisition, reading and action

Let a fair classical pointer X be held in S, initially $I/2$. Each of R fresh ports starts independently in $\eta_\alpha = (I + \alpha Z)/2$, with $0 \le \alpha \le 1$. Put $\varepsilon = (1-\alpha)/2$ and $i_\alpha = \ln 2 - h(\varepsilon)$. Apply controlled X from S to each port, using the same physical mixed-port coupling as Paper E. The resulting state is diagonal:

$$P(x, \boldsymbol{y}) = \tfrac{1}{2} \prod_{j=1}^{R} (1-\varepsilon)^{\mathbf{1}_{y_j = x}} \varepsilon^{\mathbf{1}_{y_j \neq x}}. \tag{10}$$

An actual Z reading obtains Y_j. The stated calibration fixes which port value labels which pointer; a common unknown reversal would not be removed by collecting more ports.

Proposition B.1 (A saturated acquisition and action ledger)**.** *Every port has Holevo and accessible information i_α, equal to its initial purity deficit. Write $I_R = I(X : \boldsymbol{Y})$. Then*

$$\begin{aligned} S(SE) &= \ln 2 + Rh(\varepsilon), & S(E \mid X) &= Rh(\varepsilon), \\ S(E_j) &= \ln 2, & S(E) &= Rh(\varepsilon) + I_R, \\ \sum_j S(E_j) - S(E) &= Ri_\alpha - I_R, & H(X \mid \boldsymbol{Y}) &= \ln 2 - I_R. \end{aligned} \tag{11}$$

For $R = 1$, feedback controlled by the actual port, $D = X \oplus Y$, leaves the source in η_α and the port in $I/2$, independently. For degenerate source and port Hamiltonians, the source free energy increases by $k_B T i_\alpha$, at zero net endpoint work for the ideal recording and feedback unitaries.

Proof. Conditional ports have entropy $h(\varepsilon)$, while their fair-label averages are $I/2$. Their states commute, so Z reading attains Holevo information. Thus $\sum_j \chi = R i_\alpha$ saturates the pointer-controlled bound for any number of independent ports, including partly mixed ones. Conditional product structure gives $S(SE)$ and $S(E \mid X)$; the definition of I_R gives $S(E)$. For one port, $Y = X \oplus N$, where N is independent with $\Pr(N = 1) = \varepsilon$. Fair X makes Y independent of N; feedback gives $D = N$. Both gates are basis permutations and preserve the complete joint entropy. The source entropy falls from $\ln 2$ to $h(\varepsilon)$, giving the free-energy increment. □

The posterior is available from the actual record, not just from the algebra it generates. For $k = \sum_j y_j$,

$$\Pr(X = 1 \mid \boldsymbol{y}) = \frac{(1-\varepsilon)^k \varepsilon^{R-k}}{(1-\varepsilon)^k \varepsilon^{R-k} + (1-\varepsilon)^{R-k} \varepsilon^k}, \tag{12}$$

with the deterministic limits interpreted on their support. A reversible gate controlled by $\boldsymbol{y}$ can send X to $X \oplus g(\boldsymbol{y})$. Majority is the minimum-error choice here; ties may be resolved by a fixed label. All original ports remain. Computing and retaining an extra classical copy or a decision register would require its own physical stock, so the ideal implementation uses the ports themselves as controls.

The one-port result locates the resource after action: initial memory purity has moved into the source, not disappeared into an unpriced bath. It supplies free energy for a subsequent task, not work already delivered to a work store. T-DOME I implements such a task and accounts for its reset. Neither the source nor the bank has completed a thermodynamic cycle. Paper C's blank-body transfer is another use of finite stock, with different dynamics; its resource account must be retained when the two mechanisms are composed.

B.2 Old references change the acquisition boundary

Proposition B.2 (Conditional budget with a retained reference)**.** *Let X be classical, K a retained reference, and $E = E_1 \otimes \cdots \otimes E_R$, with $\dim E_j = d_j$. The initial state may correlate all of them. Recording applies $\sum_x |x\rangle\langle x| \otimes I_K \otimes V_x$, with V_x acting on E only. Then*

$$\sum_j I(X : E_j \mid K)_{\text{out}} \le \sum_j \ln d_j - S(E \mid XK)_{\text{in}}. \tag{13}$$

For independent fresh product ports the right side is their summed purity deficit. This bounds final conditional information, not only its increment during recording.

Proof. The controlled unitary preserves $S(E \mid XK)$. Strong subadditivity gives $\sum_j S(E_j \mid XK)_{\text{out}} \ge S(E \mid XK)_{\text{out}}$. Also $S(E_j \mid K) \le S(E_j) \le \ln d_j$. Subtracting these inequalities proves the claim. □

Take independent fair bits X, L, an old reference $K = L$ and a new port $E = L$. Each port is marginally maximally mixed. Controlled X writes $Y = L \oplus X$ into E; E alone carries no information about X, but reading K, E recovers it exactly. Here

$S(E \mid XK)_{\text{in}} = 0$ and $I(X : E \mid K)_{\text{out}} = \ln 2$, attaining Eq. (13). The usable stock is the old correlation and permission to read the reference jointly, not a hidden pure marginal. Creating and retaining the two correlated copies of L belongs to the earlier history's account.

For classical readable K, the conditional information is the Holevo quantity averaged over its records, and in this commuting example it is actually attained. A quantum reference need not admit that interpretation: the entropy inequality remains valid, but its conditional mutual information is not by itself a measurement permission or accessible information. This distinction connects acquisition to T-DOME I's retained side information without identifying unlike tasks.

C Undoing records

Proposition C.1 (Local and joint reset)**.** *Let the pure-port model's memories have degenerate Hamiltonians. Each eraser couples the memories it controls to its own independent Gibbs bath at inverse temperature β, with interactions off at endpoints. A reset with final memory entropy tending to zero has:*

1. *for separate fragment erasers,*

$$\liminf W^{\text{local}} \geq k_B T \sum_j S(\rho_{F_j}) = k_B T \sum_j \chi(X : E_{F_j}) \geq R(1-\delta) k_B T h(p)$$

   when those fragments are Holevo records;

2. *for one eraser controlling all memories but not S, reset work infimum $k_B T S(\rho_{E_{[n]}}) = k_B T H_p(r_{[n]})$, approached with vanishing reset error and increasingly large, engineered thermal reservoirs;*

3. *for an agent controlling S and every memory, the inverse $U_n^{-1} \cdots U_1^{-1}$ restores the initial product state and source coherence at zero net endpoint work in the ideal recording model.*

Proof. For one controlled memory M and its initially independent bath, unitarity and subadditivity give $\Delta S_B \geq S(\rho_M) - S(\rho'_M)$. The Gibbs identity gives $\beta W = \beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$, since the degenerate memory energy is unchanged. Hence $\beta W \geq S(\rho_M) - S(\rho'_M)$. Apply this separately, or to the entire bank. Separate baths cannot exploit correlations between fragments; their marginal inputs fix these entropy decreases. The pure-port bank marginal has rank at most two. A unitary acting only on the degenerate bank maps its two eigenvectors to a logical qubit tensored with blank remaining bits. The logical qubit has eigenvalues $(1-z, z)$, $0 \leq z \leq 1/2$, and entropy $h(z) = H_p(r_{[n]})$. For $z > 0$, thermal swaps through decreasing positive error probabilities to $\epsilon > 0$ approach $k_B T[h(z) - h(\epsilon)]$, as in Proposition D.2. Taking $\epsilon \to 0$ gives the joint reset infimum; for $z = 0$ the bank unitary suffices. The same thermal-swap method applies more generally [14]. The inverse recording unitaries have the same unchanged endpoint energy as the forward gates. □

Exact purification of a mixed memory is not attainable with a finite-dimensional, full-rank Gibbs bath at finite energies and temperature. Its input joint rank is $\text{rank}(\rho_M) d_B > d_B$, whereas a pure final marginal would force the joint state to have rank at most d_B.

Unitary evolution preserves rank. Thus the work infimum is not an exact finite-bath reset guarantee; finite reset error uses the entropy decrease in the proof, with nonnegative bath relative-entropy and correlation terms.

For $p = 1/2, c = 0.8, n = 60$, the corresponding asymptotic reset bounds are $19.505\, k_B T$ for sixty separate ports, $7.661\, k_B T$ for twelve five-port fragments, $5.368\, k_B T$ for eight seven-port readable fragments, and $0.693\, k_B T$ for the joint bank. The local price reflects the access structure, not a larger amount of distinct label content.

Lemma C.2 (Reset with an old reference)**.** *Let M, K have zero endpoint Hamiltonians and the bath be thermal and independent of MK. Allow a joint unitary on MKB that leaves the marginal of K unchanged. Then*

$$\beta W = \beta Q \geq S(M \mid K)_{\text{in}} - S(M').$$

In the pure-reset limit the lower bound is $S(M \mid K)_{\text{in}}$.

Proof. The decrease of the retained joint entropy is $S(MK)_{\text{in}} - S(M'K')_{\text{out}}$. Since $S(M'K') \leq S(M') + S(K') = S(M') + S(K)_{\text{in}}$, it is at least $S(M \mid K)_{\text{in}} - S(M')$. The independent-bath argument above applies to the joint retained state. □

The target here preserves the declared reference marginal. If other reference correlations must also be restored, they belong to the retained target and its ledger. Quantum conditional entropy is not a universal attained work price without an allowed protocol. For two classical copies $M = K = L$ of a fair bit, controlled XOR from K resets M exactly at zero endpoint work and leaves K unchanged. Its conditional entropy is zero, although $S(M) = \ln 2$. Permission to use the reference changes the reset task.

Changing a record is not resetting it. A unitary bit flip reverses its labels at zero endpoint work for a degenerate memory and retains all information. Swapping a record into an independent maximally mixed port makes the old memory uninformative at zero endpoint work, but transfers the record into the other port instead of destroying the joint history. Local thermal reset, such transfer, error correction and inversion of the complete source–bank unitary cannot be assigned one erasure bill. Exact history reversal is allowed when all required gates and systems are accessible. Neither its probability nor a suppression of reversed histories follows without an ensemble and measure [16].

D Renewing records instead of making new ones

A written distinction and a renewable distinction are different resources. The purity law prices acquisition from its declared product boundary. It does not make an old record disappear when fresh blank memories run out. For example, a copied Z label remains unchanged under Z dephasing, although each fair-label record has zero marginal purity. Maintaining it costs no entropy removal in that noise model. Independent bit flips give a different, quantitative renewal problem.

Proposition D.1 (The entropy price of a cyclic record bank)**.** *Let a classical label X be retained unchanged, and let R degenerate binary registers C_i be conditionally independent copies of X with error $\varepsilon \in [0, 1/2)$. Between corrections each undergoes an independent flip of probability $p \in (0, 1/2]$, whose noise port is inaccessible. The controller starts independent of XC and is unchanged by this noise. A correction returns the complete*

joint state of X, the bank and its controller to the pre-noise state. Its bath starts thermal and independent, and its interactions vanish at the endpoints. Put

$$q = p + (1-2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0. \tag{14}$$

The entropy removed from the retained state is exactly $Rd_{p,\varepsilon}$ per correction, and its bath heat obeys $\beta Q \geq Rd_{p,\varepsilon}$.

If corrections occur at frequency ν and the sole exhaust has the thermal, zero-charge, extensive-current properties of HAFF-D's chiral exhaust [2], the mean heat current obeys

$$\overline{\dot{Q}} \geq k_B T \nu R d_{p,\varepsilon} + \frac{3\hbar}{\pi c_{\mathrm{ex}}} (\nu R d_{p,\varepsilon})^2. \tag{15}$$

Thus an exhaust-power budget $\overline{\dot{Q}} \leq J$ permits at most

$$R \leq \frac{\sqrt{(k_B T)^2 + 4\alpha J} - k_B T}{2\alpha \nu d_{p,\varepsilon}}, \qquad \alpha = \frac{3\hbar}{\pi c_{\mathrm{ex}}}. \tag{16}$$

Here c_{ex} is dimensionless chiral capacity, and $\nu R d_{p,\varepsilon}$ is the dimensionless entropy removed per unit time, in nats per time. The current bound is a long-window extensive average, not an individual pulse identity. No inference from vanishing budget to vanishing records follows when $d_{p,\varepsilon} = 0$.

Proof. Before noise $S(XC) = H(X) + Rh(\varepsilon)$, and after it $S(XC) = H(X) + Rh(q)$. Cyclicity of the entire retained state prevents a controller or a correlation with the label from absorbing that entropy without charge. For its independent Gibbs bath B, a joint unitary and subadditivity give $\Delta S_B \geq Rd_{p,\varepsilon}$. The Gibbs identity $\beta Q = \Delta S_B + D(\rho'_B \| \tau_B)$ proves the heat bound. The actual net extensive entropy current is at least $\nu R d_{p,\varepsilon}$. Substitution into HAFF-D's increasing finite-rate bound gives Eq. (15); inversion gives Eq. (16). □

Proposition D.2 (Thermal correction with an accessible label)**.** *Under the renewal proposition's conditional-iid error law, assume $0 < \varepsilon < q \leq 1/2$, and allow retained X to control reversible XOR gates on each bank register. Supply an independent Gibbs bath qubit with gap $\Delta_\varepsilon = k_B T \ln[(1-\varepsilon)/\varepsilon]$ for each register. XOR by X, swap with that bath qubit, and XOR by X again restore the complete retained X, C, controller target. Per register the bath heat and mean driven work are*

$$Q = W = k_B T (q - \varepsilon) \ln[(1-\varepsilon)/\varepsilon]. \tag{17}$$

With $q = t_0 > t_1 > \cdots > t_m = \varepsilon$, successive swaps with independent Gibbs qubits of gaps $\Delta_i = k_B T \ln[(1-t_i)/t_i]$ give

$$\beta Q = \beta W = \sum_i (t_{i-1} - t_i) \ln[(1-t_i)/t_i] \longrightarrow h(q) - h(\varepsilon) \tag{18}$$

as the partition mesh tends to zero. All stages have finite dimensions and finite gaps for a fixed positive target error. The finite-step excess is exactly

$$\beta Q - [h(q) - h(\varepsilon)] = \sum_i D(\mathrm{Bern}(t_{i-1}) \| \mathrm{Bern}(t_i))$$

per register; the spent baths end independent of the retained target.

Proof. The first XOR converts $C_i = X \oplus N_i$ into the independent error bit N_i of probability q, without changing X. The thermal swap replaces it by an independent error bit of probability ε; the second XOR restores the intended conditional law. The bath output has probability q, so its energy gain is $\Delta_\varepsilon(q - \varepsilon)$. Controllers remain untouched, and conditional independence is restored. For several swaps, the output of bath qubit i has probability t_{i-1}. Summing its energy gain gives Eq. (18). The right-endpoint Riemann sums tend to $\int_\varepsilon^q \ln[(1-t)/t]\,dt = h(q) - h(\varepsilon)$. The degenerate retained registers have no endpoint energy change, so the first law gives $W = Q$. Each swap's heat equals its error-bit entropy decrease plus $D(\mathrm{Bern}(t_{i-1})\|\mathrm{Bern}(t_i))$. Telescoping gives the stated finite-step excess. All initial error and bath bits are independent after the first XOR, and the swaps merely permute them; hence the spent baths are independent of the final target. □

This is a label-preserving use of the thermal swap construction of Ref. [14]. These swaps are driven gates between degenerate bank bits and nondegenerate bath qubits, not spontaneous energy-conserving collisions. The label's control permission, thermal preparations, clock and switching belong to the implementation. Spent bath qubits retain the exported entropy and are not reset by the construction. It realizes cyclicity of the declared retained bank, not an autonomous battery-complete apparatus or an indefinitely recycled bath. The vanishing-error limit $\varepsilon \to 0$ also requires growing gaps or an appropriate limiting reservoir. This thermal-qubit implementation does not establish the separate sole-chiral-exhaust conditions or sharpness of its finite-rate bound.

For $p = .12$ and $\varepsilon = .05$, one correction removes .2378232484 nat per register. Eight correlated copies of one fair label require 1.9025859874 nats of renewal, despite carrying only one bit of distinct label content. Their independent degradation, not the number of distinct labels, sets this maintenance bill. The cyclic target and the independent noise are substantive inputs. Jointly correlated errors, an accessible noise record, a noncyclic pure controller or a protected bank can change the cost. The heat-current claim additionally needs the actual exhaust assumptions; it is not a Landauer bound on every control pulse.

This gives the record side of a maintained observer. T-DOME III and Brain Anchor let retained history choose which access to repair [17]; Eq. (16) prices one implementation of the retained bank on which such a choice relies. Foundation's exact fixed algebras describe passive protection, not a requirement to export positive entropy indefinitely. Under HAFF-D's further matched horizon-flux conditions, the bank's removal rate can enter the conditional horizon-entropy bound. Those conditions connect a real renewal process to gravity; record existence alone does not.

E Joint-state checks

The mixed-port check constructs the complete diagonal density operator and applies the recording CNOTs, not separately chosen conditional laws. For $\alpha = 0, .2, .8, 1$ and $R = 1, 2, 3, 5$, it verifies Eq. (11), the actual Z-record posterior and one-port feedback. The maximally mixed keyed example is an explicit three-bit permutation. Forty full-density examples with a quantum reference and two initially correlated ports test the conditional bound; the entropy proof establishes it generally.

The coherent echo check applies both collisions and the intermediate flip to the same pure state, verifying $UX_SU = X_S \otimes R$, all marginal and joint entropies, and the recovery of coherence. For renewal, an eight-input XOR–SWAP–XOR permutation verifies the full label–bank target. Thermal chains with $1, 4, 16, 64, 256, 1024$ stages verify the heat

and its relative-entropy excess. At $p = .12, \varepsilon = .05$, one stage costs $0.31799941\, k_B T$ per register; 1024 stages cost $0.23789030\, k_B T$, approaching $0.23782325\, k_B T$. These checks certify the finite-dimensional specified circuits, not the physical chiral exhaust.

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Structural Limits of Unification: Internal Selection, Revisable Permissions and Physical Sources

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Abstract

Can a theory explain the selection rules by which its own physical systems read and act? The operational limit established here is relative to declared permissions, not a universal obstruction to a unified theory. We distinguish an internally executed rule, a history-formed permission and an explanation of the dynamics that makes either possible. An operational equivalence result fixes one boundary: when two dynamics agree on an instrument-preserved observable algebra, no finite adaptive experiment within those permissions can distinguish them. This boundary can move when permissions change; it does not establish a final, unrevisable Cut or prohibit physical self-reference.

The constructive side is a record-driven program with a fixed executor. Controlled couplings select protected algebras, while shared spectra permit coherent choices without central program labels. A qubit loop selects its own protection and a physical field source. Noise limits the duration of history-dependent source contrast. A preceding quantum record can select the two sustained configurations with a maximum contrast equal to its trace distance. Sharp readout selection has a different cost: for the great-circle qubit family, program dimension scales as $\delta^{-1/2}$ under the stated basis-state error criterion, equally for pure and mixed programs. These results support formation of revisable permissions inside a physical architecture. They separate this attainable internal closure from the stronger claim that a model explains every one of its own physical inputs.

1 Introduction

A record can change what a system is able to do next. A damaged body may install a different protective encoding; a reader may move its probe; an internal estimate may lower the authority given to an unreliable response. These changes concern actual operations. To explain them, a theory of selected structure needs a physical account of how a history selects and revises the relations that carry its own activity.

A second question concerns the scope of such an explanation. Does internalizing each selector also explain the laws and physical permissions that implement it? A finite programmer cannot settle this question for every possible theory. Conversely, leaving

an executor in one model's inputs is not evidence that its organization can never be explained. We ask which inputs a construction actually replaces, which distinctions its own experiments can settle, and what a further unification would have to connect. Section 2 makes these three questions separate before constructing the internal selector.

Papers A–F separate parts of this problem. Channels determine which distinctions are read in Paper A; couplings and observation windows determine which survive in Paper B; instruments create records in Paper E; their retention and renewal have resource prices in Paper F [1, 2, 4, 5]. Paper C connects sensing and intervention to a body's viability, while Paper D asks how selected structure and physical sources constrain geometry [25, 3]. Here the question is whether the choice of operation can itself arise from records inside the system.

An operation, a stored description and the device that executes it are different physical objects. We specify an internal program register, a fixed executor and an update instrument. Under these inputs the system selects its couplings, maintains a choice through the records it protects, and changes the matter it emits. The choice need not be a central environmental record. Its origin in a distinguishable history and its effect on a source can be tested separately.

The main construction is a qubit with two record-selected protective configurations. Their history-dependent source weights remain different over a finite noisy window. We then identify the exact information that can pass from a quantum record to this source choice, including a construction that reaches the trace-distance bound. An error bound makes the same finite window robust to imperfect execution. These statements supply a causal interface from history to permission and source; they do not use a reconstructed geometry as a substitute for an actual change of matter.

The cost of selecting operations is separate from the cost of retaining their records. Sharp alternatives require orthogonal program supports. For the one-parameter great-circle qubit family the required register dimension grows as $\delta^{-1/2}$, and mixing cannot improve it under the two basis-state error criterion. Section 7 states this result; Appendix A contains the full proof and numerical bounds. Section 8 places the finite target-register boundary in its proper scope. Section 9 develops the broader formation mechanism and its failure conditions.

Breuer's state self-measurement limit [12] and Wolpert's physical inference limits [13] concern what an embedded system can know. Here we ask what it can choose and execute. The existence of self-reference is not in dispute: self-reproducing automata, recursive programs and fixed-point constructions provide positive examples [14, 21, 22]. What requires explanation is the physical relation between a description, its execution and the history that revises it.

2 Internal closure and explanatory reach

Three claims of completeness require different evidence. *Execution closure* means that the declared operations, records and updates are physical parts of the modeled process. *Formation* means that some usable relation or permission depends on acquired history rather than a preinstalled answer. *Explanatory completeness* would additionally account for the organization supplied to that process. The finite loops below establish the first and particular instances of the second. Their success alone cannot establish the third; nor does their use of specified primitives disprove it.

There is an operational way to locate a model's current boundary. It concerns the entire allowed experiment, not only a snapshot or an internal verbal description.

Proposition 2.1 (An internal experiment can identify only its operational class)**.** *Let $\mathcal{B}$ be a unital observable subalgebra of a finite-dimensional ambient algebra. Two models have initial states ρ_0, ρ_1 with the same restriction to $\mathcal{B}$. For every permitted command a, their Heisenberg evolution channels $\Phi_{0,a}^*, \Phi_{1,a}^*$ preserve $\mathcal{B}$ and agree there. Every outcome map of the permitted instruments also preserves $\mathcal{B}$ and has the same restriction in both models. Use the same adaptive command policy, depending only on the retained outcomes and model-independent private random choices. Then every finite outcome history has the same probability in the two models. With equal prior probabilities, no decision based on those histories identifies the model with success exceeding $1/2$.*

More generally, a property inferred exactly from these experiments must be constant on their operational equivalence classes.

Proof. For a fixed history, pull back the terminal identity through its outcome maps and evolution channels, in reverse composition order. Each intermediate effect remains in $\mathcal{B}$; the two models therefore give the same effect there, and their initial states give it the same expectation. Conditioning on any private random seed and then averaging preserves equality. Adaptive choices do not change the argument because the same realized history selects the same command. Thus the complete finite record laws agree. Any decision function has the same distribution under both hypotheses, giving success $1/2$ at equal priors. If a property differed inside one equivalence class, the same record law could not identify both values exactly. □

A qubit gives a direct example. Compare the identity channel with complete Z dephasing. They agree on $\mathcal{B} = \mathrm{span}\{I, Z\}$, so experiments whose preparation, evolution and outcome maps remain closed on this algebra cannot distinguish them. The statement even holds with the same initial $|+\rangle$ in both models if only those outcome maps are allowed. If an X measurement is admitted after one use, its $+$ probability is 1 versus $1/2$; their record laws now have total variation $1/2$. The boundary was an experimental restriction, not a prohibition imposed on every possible observer. Extra initial correlations or a new transducer must be included in the model before applying the proposition.

This is ordinary operational indistinguishability, used here to separate a closed internal account from a unique account of its underlying dynamics. It provides no theorem that all theories are incomplete: a sufficiently rich instrument family may separate the candidate models. It also provides no theorem that every selection must be written into a central classical record. The coherent program example below denies that additional premise.

A useful unification must replace some independently supplied relation by a joint mechanism with consequences beyond either component alone. HAFF-A and B make stable central sectors depend on a dephasing process, while their spatial interpretation still needs a response dictionary. Paper F treats the distinct identification of event order and its directional boundary. Q-RAIF B and C constrain work and propagation through the same represented generators when their physical alignment is supplied. Paper D's common-source test relates prepared stress, modular data and the geometric response, while modular consistency itself remains a demand on the proposed mechanism. These are explicit, different explanatory tasks. None is discharged by renaming an executor, and none requires deriving every primitive law before an intermediate organization can be explained.

The stronger philosophical proposal is that explanation proceeds through such revisable relations rather than a final internally certified description of everything. The results support revisability and concrete limits on certification. They do not establish that nature has an ontologically necessary last Cut, that an external selector must exist, or that no more unified physical theory is possible. Those conclusions would require premises beyond the operational and programming results.

3 A typed model of internal selection

An element of an observable algebra is an operator, not a description of an algebra, and a formula naming a selection rule is syntax until an encoding maps it into physical degrees of freedom. The following model makes that encoding explicit.

Definition 3.1 (Internal selection model)**.** *A finite internal selection model consists of*

1. *a finite-dimensional system S, on which selected operations act;*
2. *a program register P of dimension d_P and program states $|p\rangle \in P$, $p \in \mathcal{P}$, unit vectors that need not be orthogonal;*
3. *an executor, a fixed physical operation on $P \otimes S$, which for each program state determines an operation on S: a channel $\Phi_p(\rho) = \mathrm{Tr}_P\, \mathcal{G}(|p\rangle\langle p| \otimes \rho)$, a measurement E^p obtained by following $\mathcal{G}$ with a fixed measurement, or a program-controlled coupling to an environment;*
4. *records, the classical outcomes of instruments, and an update rule, a fixed instrument that prepares the next program state from records.*

A selection rule is internalized if the choice of operation is made by program states under a fixed executor and a fixed update rule. A configuration (p, ρ) is self-grounding if ρ is left stable by the operation selected by p and the update rule, applied to the records that ρ produces, returns p.

The space of descriptions is the set of program states, the encoding is the register, the evaluation map is the executor, and the admissible fixed points are the self-grounding configurations. The executor's dispatch map, $p \mapsto \Phi_p$, is physical dynamics; it is not itself a program. Whether it can be made one is the question of Section 8.

4 Selecting one's own accessible algebra

Paper B shows that a system coupled to its environment through commuting operators $A_1, \ldots, A_K$ at positive rates γ_k has, as its stable algebra, the commutant $\{A_k\}'$, and that environmental records can carry only the label of the block of that algebra. We now let the system choose its own couplings. Take an orthonormal program basis and program-controlled couplings

$$A_k = \sum_{p \in \mathcal{P}} |p\rangle\langle p| \otimes A_k^{(p)}, \tag{1}$$

where for each program p the operators $A_1^{(p)}, \ldots, A_K^{(p)}$ on S commute. Let $|p, x\rangle$ be the product of $|p\rangle$ with a joint eigenvector of the $A_k^{(p)}$, with eigenvalues $a_k^{(p)}(x)$.

Proposition 4.1 (Program selection and central records)**.** *For the couplings* (1)*:*

1. *they commute, and the algebra fixed by the generated dephasing semigroup is* $\{A_1, \dots, A_K\}'$*.*

2. *Compressed to the subspace* $|p\rangle \otimes S$ *of a program* p*, this algebra is* $|p\rangle\langle p| \otimes \{A_1^{(p)}, \dots, A_K^{(p)}\}'$*: each program selects the algebra that Paper B assigns to its own couplings.*

3. *The matrix element between* $|p, x\rangle$ *and* $|q, y\rangle$ *is multiplied by* $\exp[-\frac{t}{2}\sum_k \gamma_k(a_k^{(p)}(x) - a_k^{(q)}(y))^2]$*. Coherence between two programs survives exactly between joint eigenvectors with equal eigenvalue tuples. Disjoint joint spectra for* p *and* q *make every operator connecting those two programs decay. The label projector* $|p\rangle\langle p| \otimes \mathbf{1}_S$ *is central in the stable algebra if and only if the joint spectrum of* p *is disjoint from that of every other program. In that case environmental fragments can record its label.*

Proof. The operators are block diagonal in p and commute within each block, so they commute; the basis $|p, x\rangle$ diagonalizes all of them, with eigenvalues $a_k^{(p)}(x)$. Paper B's proposition on commuting couplings then gives the Schur-multiplier form in (3) and the fixed algebra in (1). Its compression to $|p\rangle \otimes S$ consists of the operators there that commute with the $A_k^{(p)}$, which is (2). A program label is central precisely when it is constant on each joint eigenspace, which gives the disjointness condition in (3). The record statement is Paper B's result that fragments carry only block labels, applied to the joint system. □

When the programs have disjoint joint spectra, their labels belong to the centre of the stable algebra and can become redundant environmental records. Shared eigenvalue tuples protect coherence between the corresponding configurations; the spectra need not coincide completely for this exception to occur. With $A = |0\rangle\langle 0| \otimes Z + |1\rangle\langle 1| \otimes X$ on a program qubit and a system qubit, both programs have eigenvalues ± 1, and the superposition $(|0, 0\rangle + |1, +\rangle)/\sqrt{2}$ is an eigenvector of A. Replacing X by $X + 3$ makes the spectra disjoint, and the same superposition decays to an equal mixture of the two selections. Both cases are computed in the accompanying check script.

For more than two programs, disjointness from just one alternative is not enough. With $A = |p\rangle\langle p| \otimes Z + |q\rangle\langle q| \otimes (Z + 3\mathbf{1}) + |r\rangle\langle r| \otimes Z$, the spectra of p and q are disjoint, but $|p, 0\rangle\langle r, 0|$ commutes with A and not with $|p\rangle\langle p| \otimes \mathbf{1}_S$. Thus p is not a central label. The condition in Proposition 4.1 concerns the whole program family.

Internal selection can also change a material source. Use the field normalization of Paper D, in which a coupling through a projector R obeys $U^\dagger j(u)U = j(u) + Rg'(u)/2$. The program controls that projector while the executor, drive and switching schedule stay fixed.

Proposition 4.2 (A program-selected material source)**.** *Let* R_p *be projectors on* S *and put*

$$A = \sum_p |p\rangle\langle p| \otimes R_p, \qquad U = \exp(-i\lambda A \otimes Q), \qquad Q = b + b^\dagger.$$

Take an initial state ρ_{PS} *independent of the field state* σ*. Set* $r = \mathrm{Tr}(\rho_{PS}A)$*. The output field marginal is exactly*

$$\sigma_{\mathrm{out}} = (1 - r)\sigma + r\, e^{-i\lambda Q}\sigma e^{i\lambda Q}.$$

When the displayed current moments exist, its vacuum-subtracted stress is

$$T_{\mathrm{out}}(u) = T_{\mathrm{in}}(u) + r\left[g'(u)\langle j(u)\rangle_\sigma + \frac{g'(u)^2}{4}\right].$$

In a fixed classical program branch p, a non-atomic quadrature spectral law and $\lambda \neq 0$ give the exact multiplicative domain $\{R_p\}'$ of the induced system channel. Thus changing the program can change both the represented algebra and the field source under the same executor.

Proof. Since $A^2 = A$, write $U = (1-A)\otimes 1 + A \otimes e^{-i\lambda Q}$. The off-diagonal terms have zero partial trace over PS, because $\mathrm{Tr}(A\rho_{PS}(1-A)) = 0$. This gives the field mixture even when the program is coherent or correlated with S. Squaring $j + Ag'/2$ gives the stress formula: initial independence supplies the mixed expectation $\langle Aj\rangle = r\langle j\rangle$, and $\langle A^2\rangle = r$. Within branch p, the channel is the two-block dephasing channel of Paper D; a non-atomic spectral law makes its nonzero-gap characteristic value strictly smaller than one, so equality in the Schwarz inequality holds exactly on $\{R_p\}'$. □

For $R_0 = (1+Z)/2$, $R_1 = (1+X)/2$ and $\rho_S = (1+0.8Z)/2$, a program bit prepared in 0 or 1 gives $r = 0.9$ or 0.5, respectively. Their conditional pointer algebras differ, although the two coupling spectra are identical. The field stress therefore need not reveal the program label as a central sector to respond to its choice. With a classical program mixture π, $r = 0.9\pi_0 + 0.5\pi_1$. A record-driven program update can change this weight without an externally assigned emission probability. The global algebra $\{A\}'$ stays fixed; the selected branch algebra on S and the outgoing source vary. This separates internal selection from a change of the executor. Paper D supplies the corresponding JT response for this same wavepacket with matched initial coefficient data; the current statement identifies its physical source, not the origin of the JT action or of a Type III geometric sector. The program, field preparation and switching remain resources of the executor.

5 A self-grounding selection

A selection rule can take as input a record that the selection itself protects. A small model gives two self-grounding fixed points and their history-dependent selection.

A memory qubit M is subject to rotation noise $H = \frac{\omega}{2}Y$. A classical program bit p selects a dephasing coupling of M with rate γ: through Z for $p = 0$ and through X for $p = 1$. Strong dephasing in a basis protects a record in that basis against rotation, by the quantum Zeno mechanism [19, 20]. Every time τ the memory is read in the protected basis, and the update rule sets p equal to the outcome. The rule is internalized in the sense of Definition 3.1, and it is self-applied: the record that sets the selection is a record in the algebra that the selection protects.

After a read the configuration is one of four: $F_0 = (p = 0, |0\rangle)$, $T_1 = (p = 1, |1\rangle)$, $F_1 = (p = 1, |-\rangle)$ and $T_0 = (p = 0, |+\rangle)$. In F_0 and F_1 the protected record agrees with the program it sets; these are the self-grounding configurations. In T_1 and T_0 the record has just switched the program to a basis in which that record is not sharp.

Proposition 5.1 (Self-grounding configurations)**.** *Without rotation noise, F_0 and F_1 are absorbing, and from T_1 and T_0*

$$\Pr(\textit{absorbed in } F_1 \mid T_1) = \tfrac{2}{3}, \qquad \Pr(\textit{absorbed in } F_1 \mid T_0) = \tfrac{1}{3}.$$

From any initial program and memory state the loop reaches F_0 or F_1 with probability one.

Proof. A read in the basis that is already sharp returns the same outcome, so F_0 and F_1 repeat. In T_1 the memory $|1\rangle = (|+\rangle - |-\rangle)/\sqrt{2}$ is read in the X basis, giving F_1 or T_0 with probability $\frac{1}{2}$ each; in T_0 the memory $|+\rangle$ is read in the Z basis, giving F_0 or T_1 with probability $\frac{1}{2}$ each. Writing a and b for the absorption probabilities into F_1 from T_1 and T_0, $a = \frac{1}{2} + \frac{1}{2}b$ and $b = \frac{1}{2}a$, so $a = \frac{2}{3}$, $b = \frac{1}{3}$. Every read from a transient configuration ends in an absorbing one with probability $\frac{1}{2}$, and the first read from an arbitrary initial state produces one of the four configurations. □

Corollary 5.2 (History selects a sustained source)**.** *In the zero-rotation loop, insert before each read a fresh field interaction of Proposition 4.2, using $R_0 = (1 + Z)/2$ and $R_1 = (1 + X)/2$. Each cycle uses a distinct tensor-factor port, in the same state and initially independent of the program, memory and all previous ports. The four-state transition law and its absorption probabilities are unchanged. In F_0 the output field is $e^{-i\lambda Q}\sigma e^{i\lambda Q}$; in F_1 it is σ. Thus the two self-grounding configurations carry different steady field sources under one executor. Starting from T_1 or T_0, the late-cycle mean source weight tends to $1/3$ or $2/3$, respectively.*

Proof. Within either program branch the field coupling is diagonal in the basis of the ensuing read. It changes no outcome probabilities, so the transition matrix of Proposition 5.1 is unchanged. The configurations $F_0 = (0, |0\rangle)$ and $F_1 = (1, |-\rangle)$ are eigenvectors of A with eigenvalues 1 and 0. They remain product states of program and memory, and each pass gives the claimed field output. The source weight therefore converges to the indicator of absorption in F_0. Its expectation is one minus the corresponding probability of absorption in F_1, which gives the stated fractions. □

The read/update history generates the source difference under one executor. Fresh ports and memory operations sustain it. Each cycle supplies a fresh product-state field port, a protected-basis read, the program update and its protecting environment. For a centered field input the coupling's mean switching work is $r_n E_g$, where $E_g = \lambda^2\bar{\omega} = \int g'^2 du/4$ and r_n is the actual source weight. Noncentered inputs add the displayed current cross term. Incoming beam preparation, memory retention and eventual reset have their own charges. The $1/3$ and $2/3$ weights arise from absorption of the two transient histories, not a continuing supply of new source information. Distinct modes of a common bosonic field must be orthogonal and prepared in a product state to realize these ports; mere time translation does not suffice. For Paper D's exact packet, a translation by Δ has overlap modulus $4a^2/(4a^2 + \Delta^2)$. Source accounting for such overlapping pulses requires their joint field state. The full-line packet is a scattering idealization; a record decided at a finite time needs a compact switch scheduled after that decision, as constructed in Paper D. With rotation, the following lifetime uses the stated continuous dephasing generator.

The dephasing plays no role without noise, because the reads are made in the protected basis. With noise it decides whether the self-grounding configurations last.

Proposition 5.3 (Lifetime of a self-grounding selection)**.** *With rotation noise, the probability of leaving F_0 (or F_1) in one cycle is $\varepsilon = \frac{1}{2}(1 - z(\tau))$, where z solves $\ddot{z} + 2\gamma\dot{z} + \omega^2 z = 0$, $z(0) = 1$, $\dot{z}(0) = 0$. For $\gamma > |\omega|$,*

$$z(\tau) = \frac{\lambda_+ e^{\lambda_- \tau} - \lambda_- e^{\lambda_+ \tau}}{\lambda_+ - \lambda_-}, \qquad \lambda_\pm = -\gamma \pm \sqrt{\gamma^2 - \omega^2},$$

so $\varepsilon \approx \omega^2\tau/(4\gamma)$ *when* $\gamma \gg |\omega|$, $\gamma\tau \gg 1$ *and* $\omega^2\tau/\gamma \ll 1$. *Without protection* $\varepsilon = \sin^2(\omega\tau/2)$*; with protection in the wrong basis* ε *approaches* $\frac{1}{2}$ *in the strong-dephasing regime.*

Proof. In F_0 the Bloch vector starts at $z = 1$. The rotation gives $\dot{x} = \omega z$, $\dot{z} = -\omega x$, and dephasing through Z adds $-2\gamma x$ to $\dot{x}$. Eliminating x gives the stated equation; a Z read returns 1 with probability $\frac{1}{2}(1 - z)$. The case F_1 is the same with x and z exchanged, since rotation about y treats both records alike. For $\gamma \gg \omega$, $\lambda_+ \approx -\omega^2/(2\gamma)$ dominates. Without dephasing $z = \cos\omega\tau$. Dephasing in the wrong basis destroys the record at rate 2γ. □

For $\omega = 0.1$ and $\tau = 1$, the escape probability per cycle is 1.42×10^{-3}, 2.37×10^{-4} and 2.49×10^{-5} for $\gamma = 1$, 10 and 100, against 2.50×10^{-3} without protection. Protection in the wrong basis gives 0.433 at $\gamma = 1$ and approaches 0.5 at the larger rates. At $\gamma = 100$ a self-grounding selection lasts on average 4.0×10^4 cycles before the loop passes through a transient configuration and settles again, in F_0 or F_1.

The residence time alone does not determine how long two histories produce different sources. The same four-state process gives that distinction exactly.

Proposition 5.4 (History-dependent source window)**.** *Let* $\gamma > 0$, $\omega \neq 0$ *and* $\tau > 0$. *In each cycle first evolve for time* τ *with the selected rotation–dephasing generator, then apply the fresh field interaction of Corollary 5.2, and then read and update. Put*

$$k = \sqrt{\gamma^2 - \omega^2}, \quad f = e^{-\gamma\tau}\left(\cosh k\tau + \gamma\frac{\sinh k\tau}{k}\right), \quad b = \omega e^{-\gamma\tau}\frac{\sinh k\tau}{k},$$

with analytic continuation when $\gamma < |\omega|$ *and* $\sinh(k\tau)/k = \tau$ *at* $k = 0$. *Set* $\varepsilon = (1 - f)/2$, $q_+ = (1+b)/2$, $q_- = (1-b)/2$. *In the post-read order* (F_0, T_1, F_1, T_0) *the transition matrix and the source-weight vector are*

$$T = \begin{pmatrix} 1-\varepsilon & \varepsilon & 0 & 0 \\ 0 & 0 & q_+ & q_- \\ 0 & 0 & 1-\varepsilon & \varepsilon \\ q_- & q_+ & 0 & 0 \end{pmatrix}, \qquad r = \begin{pmatrix} 1-\varepsilon \\ q_- \\ \varepsilon \\ q_- \end{pmatrix}. \tag{2}$$

For a post-read distribution μ_n*, the next field output has mean stress*

$$T_{\mathrm{in}}(u) + (\mu_n r)\left[g'(u)\langle j(u)\rangle_\sigma + g'(u)^2/4\right].$$

The chain has the unique stationary law

$$\pi = \frac{(q_-, \varepsilon, q_+, \varepsilon)}{1 + 2\varepsilon}, \qquad r_\infty = \pi r = \frac{1}{2} - \frac{b}{2(1 + 2\varepsilon)}.$$

For initial states T_1 *and* T_0*, define*

$$R_1^\circ = \varepsilon + f\frac{q_-^2}{1 - q_+q_-}, \qquad R_0^\circ = \varepsilon + f\frac{q_-}{1 - q_+q_-}.$$

Their actual source weights satisfy, for every integer $n \geq 0$,

$$|\mu_n r - R_i^\circ| \leq n\varepsilon + (q_+q_-)^{\lfloor n/2\rfloor}, \qquad \mu_0 = T_i. \tag{3}$$

Thus a small escape rate preserves different history-selected sources over $1 \ll n \ll \varepsilon^{-1}$*, although both histories ultimately have the same stationary mean.*

Proof. The two Bloch generators on (x, z) are

$$\begin{pmatrix} -2\gamma & \omega \\ -\omega & 0 \end{pmatrix}, \qquad \begin{pmatrix} 0 & \omega \\ -\omega & -2\gamma \end{pmatrix}.$$

Exponentiating them gives (2). The source weight is the population of the selected projector immediately before the read, not its value in the preceding post-read state. The intervening field coupling preserves that population. It equals the probability that the next program is 0. For nonzero rotation and positive dephasing, the Bloch norm strictly decreases from each of the four pure initial states over any positive interval. Hence $0 < \varepsilon < 1$ and $0 < q_{\pm} < 1$. The graph of T is connected and has self-loops, giving uniqueness and convergence. Direct multiplication gives $\pi T = \pi$ and the stated source weight.

For the window bound, replace the two F rows by absorbing rows, retaining $q_{\pm}$ in the transient rows and retaining the same observable r. In this comparison chain the absorption probabilities into F_0 are $q_-^2/(1 - q_+q_-)$ and $q_-/(1 - q_+q_-)$; the corresponding limiting expectations of r are R_1° and R_0°. Transient survival after n steps is at most $(q_+q_-)^{\lfloor n/2 \rfloor}$. Couple the true and comparison chains until their first escape from an F state. That event has probability at most $n\varepsilon$, and $0 \le r \le 1$, proving (3). □

The history window and the lifetime of one configuration are different. The characteristic polynomial is

$$(z - 1)\left[z^3 + (2\varepsilon - 1)z^2 + (\varepsilon^2 - q_+q_-)z + \varepsilon^2 - \varepsilon + q_+q_-\right].$$

For small ε the slow root is $1-\varepsilon/(1-q_+q_-)+O(\varepsilon^2)$, so the late history contrast decays on a scale of about $(1-q_+q_-)/\varepsilon$ cycles. In the strong-protection regime of Proposition 5.3, this is about $3\gamma/(\omega^2\tau)$, whereas a single configuration has mean residence $1/\varepsilon \approx 4\gamma/(\omega^2\tau)$. At $\gamma = 100$, $\omega = 0.1$, $\tau = 1$, the source weights from T_1 and T_0 are respectively 0.33844 and 0.66090 after 1000 cycles, and 0.49370 and 0.50579 after 10^5 cycles. Both approach 0.4997500.

The loop selects a protection because the record it protects says so, and its history selects a material source through the same program-controlled executor. Finite noise makes that selection metastable rather than permanent. The rotation–dephasing stage, field interaction and read are sequential operations here; simultaneous rotation during a finite field pulse would require a different transition law. The protecting environment, fresh packets, read instrument and update map are supplied. The loop generates the history-dependent choice, not these resources.

6 When a history changes the next source

The four-state loop starts with explicit configurations. An earlier history need not already be a classical choice among them. Let R be a retained physical record, and let $\Lambda : R \to P \otimes M$ be a fixed preparation channel acting after that history has formed. It can include a measurement, an outcome record and a controlled preparation. The next fresh field port remains independent of R, P, M and old emitted ports. This time ordering gives a direct criterion for what the history can cause.

Proposition 6.1 (Record-to-source distinguishability)**.** *For two record states ρ_0, ρ_1 and the projector A in Proposition 4.2, the two source weights satisfy*

$$r_0 - r_1 = \mathrm{Tr}[(\rho_0 - \rho_1)\Lambda^*(A)], \qquad |r_0 - r_1| \le D(\rho_0, \rho_1) := \tfrac{1}{2}\|\rho_0 - \rho_1\|_1.$$

For the zero-rotation loop, the largest contrast over fixed preparation channels Λ equals $D(\rho_0, \rho_1)$. A channel attaining it prepares F_0 or F_1 from an actual binary record measurement. Both subsequent histories remain in their chosen configuration, so the same contrast occurs in every later fresh source port.

Proof. The first equality is the adjoint-channel identity. Complete positivity and unitality of Λ^* give $0 \le \Lambda^*(A) \le 1$. Writing the trace-zero Hermitian operator $\rho_0 - \rho_1$ as its positive part minus its negative part bounds the expectation of any effect by D. To attain the bound, let H be the spectral projector onto its positive part. Measure $\{H, 1 - H\}$ and prepare F_0 on outcome H, F_1 otherwise. The resulting source weight is $\mathrm{Tr}(\rho_i H)$ because F_0, F_1 have A-eigenvalues $1, 0$. Their difference is D. Proposition 5.1 preserves these configurations and Corollary 5.2 preserves their source weights. □

The discrimination identity is a standard trace-distance fact; its role here is to determine exactly how much record information can become a difference in protected configuration and actual source. A nonzero source contrast is possible precisely when the two record states differ, provided this binary measurement and the two preparations are available. A specified executor can fail to use that difference: if $\Lambda^*(A)$ is orthogonal to $\rho_0 - \rho_1$ in trace pairing, its outgoing source is the same. Distinguishable histories need not yield distinguishable sources. Conversely, changing a conditional pointer algebra need not change r; a maximally mixed M gives $r = 1/2$ for both the Z and X selections.

This construction does not preserve an arbitrary coherent input record under measurement. For a classical history stored in orthogonal states the preparation can be controlled without deleting it. An independent old content–reference state Ω_{CE} is preserved jointly, even conditioned on the new outcome, when all preparation and loop operations act on R, P, M and leave C, E untouched. If R is already correlated with CE, the old marginal is preserved by the nonselective channel but conditioning can change its state. This is the distinction required when selection is claimed to retain an earlier body or history.

Ideal execution is not required for a finite history effect. Suppose each complete read/update instrument still outputs the four listed configurations, its actual sampled transition in cycle j is $\widetilde{T}_j$, and its actual source weight is $\widetilde{r}_n$, with

$$\max_x \mathrm{TV}(\widetilde{T}_j(x,\cdot), T(x,\cdot)) \le a_j, \qquad \|\widetilde{r}_n - r\|_\infty \le c_n.$$

These are errors of the stated four-state instrument and observable, not an identification of every physical error with a flip.

Corollary 6.2 (A robust finite source window)**.** *From either transient initial configuration T_i, the actual source weight obeys*

$$|\widetilde{\mu}_n \widetilde{r}_n - R_i^\circ| \le n\varepsilon + (q_+ q_-)^{\lfloor n/2 \rfloor} + \sum_{j=1}^{n} a_j + c_n.$$

Consequently the contrast between the two histories is at least

$$\frac{|f| q_+ q_-}{1 - q_+ q_-} - 2n\varepsilon - 2(q_+ q_-)^{\lfloor n/2 \rfloor} - 2\sum_{j=1}^{n} a_j - 2c_n,$$

whenever the right-hand side is positive.

Proof. At each step, couple the true row and its ideal counterpart with mismatch probability at most a_j. After n steps their distributions differ in total variation by at most $\sum_j a_j$. Expectations of $r \in [0,1]$ differ by at most that amount, and changing the source observable costs at most c_n. Apply Eq. (3). Finally, $|R_0^\circ - R_1^\circ| = |f|q_+q_-/(1-q_+q_-)$, and use the triangle inequality for the two source weights. □

For sufficiently small escape and execution errors, this bound certifies a nonempty interval of different physical consequences. Lower escape rates extend that interval only if execution errors and fresh-port correlations remain controlled. Holding a noisy selection indefinitely would need renewed resources or a stronger protection mechanism. The present bound supplies no free infinite stream.

7 The price of selecting sharp readouts

The fixed executor may be followed by a fixed measurement, with a fixed ancilla. A program selects effects E_z^p on a labelled target basis $\{|b_z^p\rangle\}$. We use the worst-case basis-state error

$$\delta_p = \max_z \left(1 - \langle b_z^p | E_z^p | b_z^p \rangle\right).$$

The program dimension counts the selectable register; a fixed ancillary state does not enlarge it. This criterion concerns the target outcome probabilities, and does not compare complete instruments in diamond norm.

Theorem 7.1 (Cost of sharp selection)**.** *1. If two pure program states select two distinct labelled sharp measurements exactly, they are orthogonal. Exact mixed programs for distinct sharp measurements have orthogonal supports. A register of dimension d_P selects at most d_P distinct labelled sharp measurements exactly.*

2. Let S be a qubit, and suppose that for every Bloch angle θ on a great circle some pure program state selects the sharp measurement along θ with error at most δ, where $0 < \delta < 1$. Then

$$d_P \;\geq\; L(\delta) := \max_{M \geq 1} \frac{M}{1 + \frac{1}{3}\left(2\sqrt{\delta} + \delta\right)^2 (M^2 - 1)}, \tag{4}$$

and $L(\delta)\sqrt{\delta} \to \sqrt{3}/4 \approx 0.433$ as $\delta \to 0$. Conversely, $N = \lceil \pi/(2 \arcsin\sqrt{\delta}) \rceil$ orthogonal program states with a controlled rotation achieve error δ, and $N\sqrt{\delta} \to \pi/2$.

3. Part (2) holds unchanged for mixed program states, with $0 < \delta < 1$. For each target a pure program in the same register has exactly the same two basis-state errors. Consequently $d_P \geq L(\delta)$ and $\liminf_{\delta \to 0} d_P \sqrt{\delta} \geq \sqrt{3}/4$; the same grid gives the upper bound. Mixing cannot reduce the required program dimension under this criterion.

The proof and representative numerical bounds are in Appendix A. The constants $\sqrt{3}/4$ and $\pi/2$ are lower and constructive upper bounds; the result establishes the exponent, not an optimal leading constant. Exact orthogonality is the measurement counterpart of the no-programming theorem for unitaries [15]. The equality of the pure and

mixed costs here follows from the convex joint numerical range of two Hermitian operators [18]. Three independent errors need not admit the same pure replacement. The asymptotic exponent applies to this great-circle family and this criterion; universal processors have other resource bounds [17].

A two-axis reader needs only two orthogonal choices [11]. Tracking a continuum of axes to increasing accuracy needs a growing register. Storing and refreshing the chosen programs has a further purity and free-energy account [4, 5]. Coherently returning a program does not, by itself, assign it a reset heat. Conversely, a large register does not supply the entropy outlet needed to renew noisy records.

8 What the executor leaves specified

Rules, programs, their revision and self-grounding selections can be internal. Within Definition 3.1, a full copy of an executor acting on $P \otimes S$ acts on dimension $d_P d_S$, whereas each selected operation acts on S or part of it, of dimension at most $d_S < d_P d_S$ when $d_P \geq 2$. It cannot be that full executor. A meta-register P' can select among executors on $P \otimes S$ under a fixed meta-executor. At the top of each stipulated finite hierarchy the model still supplies an executor as physical dynamics. This is a feature of the architecture, not a general prohibition on a system containing its own control law.

The top dispatch map is specified dynamics in this architecture. A meta-register can make the current executor selectable, at a resource cost set by the family to be selected. For the one-parameter great-circle qubit family, Theorem 7.1 gives order $\delta^{-1/2}$ register dimension; it assigns no common exponent to every continuous family. Stored classical choices also use the record resources described above. The dimension argument excludes a full executor acting on the larger space from being an operation on the smaller target; it permits storing a description of that executor. In infinite dimension the counting argument fails, since a separable Hilbert space is isomorphic to its tensor product with $\mathbb{C}^2$. Kleene and Lawvere fixed points [21, 22], and the order-theoretic results of Scott and Tarski [23, 24], also permit self-reference. Existence of a self-representing physical executor in that setting remains open.

Every stipulated finite hierarchy specifies its top dispatch map. Self-application is different: the loop of Section 5 applies a rule to its own protected records and reaches fixed points without adding levels. A claim that every operation is selected faces a separate resource test. By Theorem 7.1, an executor selecting every sharp readout exactly would require infinitely many orthogonal program states.

The distinction between structure and knowledge has a concrete meaning here. The dispatch map is physical dynamics; program contents are records. When their coupling spectra are disjoint they can be public environmental records, while overlapping eigenvalue tuples can hide some coherence. Proposition 4.2 adds a further distinction: a program need not become a central label to change the field source through the operation it selects.

9 From selectable rules to formed permissions

The loop forms a choice among supplied couplings. A self-frame needs more: history must establish a relation between a task and the body's own reading or action channels. T-DOME II separates an aligned axis from a signed relation that can actually predict [26].

If opposite couplings produce the same retained axis, the axis can be perfectly aligned while its posterior prediction is zero. Outcome-bearing records supply the missing sign. A useful program must encode whichever distinctions its future task consumes.

Self-readout then becomes a problem of attribution. A changed residual can come from the world, from a damaged body, or from a miscalibrated interface. T-DOME III uses identifiable references and paid maintenance to decide among these causes [9]. Its action is based on retained value and body condition; confidence alone need not determine whether repair is worthwhile. Imprint W supplies a complementary mechanism in which the system uses its own reading history to choose a later inquiry [11]. These are mechanisms for revising programs, with their own record laws and budgets.

Brain Anchor locates formed relations in a body: diagnostic outcomes can alter routes and install a different protected encoding [29]. Such relations can be distributed among components. Calling their aggregate a program register does not imply that a single place stores a full description, or that every relation becomes a central environmental label. Proposition 4.1 already gives a finite counterexample to the latter inference. A mathematical encoding is an interface for analysis; a physical implementation must say where its records, actuators and update couplings reside.

Q-RAIF C gives a further, independent source implementation [27]. Its actual quantum history enters a complete positive Hamiltonian and changes material motion and interacting field source. The fresh-port projector mixture used here is simpler and has different inputs. The two constructions support the same causal question, but their clocks, controllers and source objects cannot be exchanged merely because both have an internal program. Returning a field response to the same body requires the actual sensor and finite-resource operations, as studied in Foundation III [28]. A reconstructed geometry, a field stress and the total apparatus stress remain distinct.

The more ambitious generation problem concerns how these operational relations become available in the first place. A fixed meta-register only relocates a supplied dispatch map. A more informative mechanism is adaptation of the body: records change local couplings, routes or encodings, and the new relations alter which later records and acts are reliable. This can generate a permission from primitive interactions without generating the primitive interactions themselves.

Conjecture 9.1 (Formation of revisable permissions)**.** *Consider a body with locally adjustable couplings and a finite family of reversible encodings of the same logical content. Let actual diagnostic instruments act on relational degrees of freedom outside that logical algebra, and distinguish configurations with different task values. Suppose exploratory operations reach each necessary diagnostic relation, the best configuration has a nonzero value gap on each stationary interval, and paid update and maintenance resources suffice over that interval. There exist local record-driven update dynamics for which task-relevant relations become usable permissions from initially uncalibrated couplings, while previously encoded content remains within a stated error budget. When the task or body changes slowly enough, the same retained relations support detection and revision, rather than only repeating a fixed choice.*

The proposed mechanism is concrete: exploration creates diagnostic records; local retained relations select a route or encoding; an actuator installs it; subsequent errors and maintenance outcomes modify those relations. History-selected protection, signed self-frames and local body learning support different parts of this mechanism. Its quantitative test compares formed relations with supplied ones under the same preparation, diagnostic,

installation and content-loss budgets. Formation must lose its advantage when diagnostics alias task-distinct configurations, exploration cannot reach the relevant coupling, or noise destroys old content before learning and migration finish. A distributed learner that changes only labels while all implemented instruments remain fixed also fails this test.

The conjecture asks for a common finite construction with these inputs and outcomes. It is stronger than the supplied two-choice loop, and is not used in its proof. An infinite stationary guarantee would additionally need a sustainable resource supply and control of rare failures. A finite family chosen by the theorist can demonstrate a formation mechanism; it does not explain the origin of every physical interaction or privilege one universal observable algebra.

10 Conclusion

An internal selector can be physical, self-applied and revisable. In the finite construction, records choose their protection and alter the next field source under one executor. Shared coupling spectra allow coherent choices without a central program label. The transition law separates residence in one configuration, a finite history-dependent source window and eventual loss of contrast. Trace distance fixes how much an earlier record can contribute to source selection, and accumulated execution errors bound the useful window.

These are formation and execution results inside declared dynamics. The great-circle sharp-readout price limits a particular programmable family, not every continuum or physical self-reference. Operational equivalence limits which underlying dynamics the declared experiments can certify; enlarging their permissions can remove that limit. The formation conjecture asks how learned body relations can supply such new permissions while old content and resources survive. It is a stronger target than changing a program label.

The limits therefore attach to identifiable physical relations, not to an untyped claim that a system can never contain its own rule. A common account of formed sectors, event order, control and propagation, and source and geometry would explain more than the present selector. Its missing relations remain assigned to those mechanisms. The source realization here uses specified fresh bosonic ports; it is not Paper D's sustained thermal apparatus or a proof of its joint sharp-exhaust and return conjecture. Neither operational closure nor its extension settles the separate question of first-person experience.

A Proofs and constructions for sharp selection

Here we give the full argument for the readout cost. A readout here is a measurement on S implemented by a fixed executor: a unitary on $P \otimes S \otimes A$, with an ancilla A in a fixed state, followed by a fixed measurement. Purifying the fixed ancilla and dilating the measurement gives projections F_z pulled back to the input space. The embedding $|p\rangle \mapsto |p\rangle|a\rangle$ keeps program overlaps and has span dimension d_P; the ancillary dimension is not a selectable program resource. Write G_z for the induced POVM effects on the original $P \otimes S$, after compression to the fixed ancilla state. A pure program $|p\rangle$ selects effects $E_z^p = (\langle p| \otimes \mathbf{1})G_z(|p\rangle \otimes \mathbf{1})$. Its target is a labelled orthonormal basis $\{|b_z^p\rangle\}$ of S, with selection error $\delta_p = \max_z\big(1 - \langle b_z^p|E_z^p|b_z^p\rangle\big)$. For a mixed program σ, use $E_z^\sigma = \mathrm{Tr}_P[(\sigma \otimes \mathbf{1})G_z]$ and the same error on the target basis states. This is a basis-state error criterion, not a diamond-norm distance between instruments.

Lemma A.1 (Near-orthogonality)**.** *For programs p, q and outcomes $z \neq z'$,*

$$|\langle q|p\rangle|\,|\langle b^q_{z'}|b^p_z\rangle| \leq \sqrt{\delta_p} + \sqrt{\delta_q} + \sqrt{\delta_p\delta_q}.$$

Proof. Put $v = |p\rangle|b^p_z\rangle|a\rangle$ and $w = |q\rangle|b^q_{z'}\rangle|a\rangle$. Then $\langle v|F_z|v\rangle = \langle b^p_z|E^p_z|b^p_z\rangle \geq 1 - \delta_p$, so $\|(\mathbf{1} - F_z)v\| \leq \sqrt{\delta_p}$, and likewise $\|(\mathbf{1} - F_{z'})w\| \leq \sqrt{\delta_q}$. Write $v = F_z v + (\mathbf{1} - F_z)v$ and $w = F_{z'}w + (\mathbf{1} - F_{z'})w$. Since $F_{z'}F_z = 0$, the inner product $\langle w|v\rangle = \langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle$ consists of three cross terms bounded by $\sqrt{\delta_p}$, $\sqrt{\delta_q}$ and $\sqrt{\delta_p\delta_q}$. □

The two-expectation fact used below is the two-Hermitian-operator case of joint numerical-range convexity, equivalent to the Toeplitz–Hausdorff theorem [18]. A rank-reduction proof makes clear why the replacement stays in the original register.

Lemma A.2 (Matching two error rates)**.** *For any two Hermitian operators B_0, B_1 on a finite program register P and any density operator σ on P, there is a unit vector $|p\rangle \in P$ such that*

$$\langle p|B_z|p\rangle = \mathrm{Tr}(\sigma B_z), \qquad z = 0, 1.$$

Proof. If σ has rank $r \geq 2$, the Hermitian operators on its support form a real vector space of dimension $r^2 \geq 4$. The three real linear constraints

$$\mathrm{Tr}\, H = \mathrm{Tr}(HB_0) = \mathrm{Tr}(HB_1) = 0$$

therefore have a nonzero Hermitian solution H supported there. Its zero trace implies that it has both positive and negative eigenvalues. Since σ is positive definite on its support, the interval of t for which $\sigma + tH \geq 0$ contains a neighbourhood of zero and has finite endpoints. At either endpoint the rank strictly decreases, while the trace and both expectations remain unchanged. Repeat on the smaller support until the density operator has rank one. □

Theorem 7.1 is proved below with the same fixed ancilla and basis-state error criterion.

Proof. (1) With $\delta_p = \delta_q = 0$ the lemma gives $\langle q|p\rangle\langle b^q_{z'}|b^p_z\rangle = 0$ for all $z \neq z'$. If $\langle q|p\rangle \neq 0$, each $|b^p_z\rangle$ is orthogonal to every $|b^q_{z'}\rangle$ with $z' \neq z$, hence proportional to $|b^q_z\rangle$, and the two labelled measurements coincide. Each mixed program's errors are averages of nonnegative pure-component errors. Zero average forces every vector in its spectral decomposition to select the same measurement exactly. The pure result makes the supports for distinct measurements orthogonal. Their nonzero dimensions sum to at most d_P.

(2) Choose M targets at angles $\theta_j = 2\pi j/M$ with program states $|p_j\rangle$. For $z = 0$, $z' = 1$ and Bloch angle φ between the targets, $|\langle b^q_1|b^p_0\rangle| = |\sin(\varphi/2)|$, so the lemma gives $|\langle p_j|p_l\rangle| \leq (2\sqrt{\delta} + \delta)/\sin(\pi|j-l|/M)$. The Gram matrix Γ of the $|p_j\rangle$ has $\mathrm{Tr}\,\Gamma = M$ and, by the identity $\sum_{k=1}^{M-1} \csc^2(\pi k/M) = (M^2-1)/3$,

$$\mathrm{Tr}\,\Gamma^2 \leq M + M\,\tfrac{1}{3}(2\sqrt{\delta} + \delta)^2(M^2 - 1).$$

Then $d_P \geq \mathrm{rank}\,\Gamma \geq (\mathrm{Tr}\,\Gamma)^2/\,\mathrm{Tr}\,\Gamma^2$, which is (4). With $c = \frac{1}{3}(2\sqrt{\delta} + \delta)^2$ the maximum over M lies near $M = c^{-1/2}$ and equals $\frac{1}{2}c^{-1/2}(1 + O(\sqrt{c}))$, which gives the limit. For the converse, the executor $\sum_j |j\rangle\langle j| \otimes R(\theta_j)^\dagger$ followed by a measurement in the computational basis selects the sharp measurement along $\theta_j = 2\pi j/N$ exactly; using the nearest θ_j for a target θ misaligns the axes by at most π/N, which costs error $\sin^2(\pi/(2N)) \leq \delta$.

(3) Fix a target basis $\{|b_0\rangle, |b_1\rangle\}$ and define positive operators on the original register P by

$$B_z = \mathbf{1}_P - (\mathbf{1}_P \otimes \langle b_z|)G_z(\mathbf{1}_P \otimes |b_z\rangle), \qquad z = 0, 1.$$

A mixed program σ has target errors $\mathrm{Tr}(\sigma B_z)$. Lemma A.2 supplies a pure program in the original P matching both errors exactly. Choose it for each target angle. The fixed executor then has a pure-program family with the same error bound δ and dimension d_P, so part (2) applies. Conversely, pure programs are included among mixed programs. This equivalence concerns the two target errors, not preparation costs or equality of the complete instruments; arbitrary additional error constraints need not admit a rank-one replacement. □

For k positive error operators $B_1, \ldots, B_k$, a mixed program whose errors are all at most δ always has a pure spectral component with every error at most $k\delta$: some component has the sum of its errors no larger than the mixed sum. Exact matching of all errors need not hold when $k \geq 3$: the three positive operators $(I + \sigma_x)/2$, $(I + \sigma_y)/2$, $(I + \sigma_z)/2$ all have expectation $1/2$ in the maximally mixed qubit, while no pure state matches all three. The two-error equality is therefore stronger than this general factor-k bound.

δ	$L(\delta)$	$N(\delta)$	$L\sqrt{\delta}$	$N\sqrt{\delta}$
10^{-2}	4.15	16	0.415	1.600
10^{-4}	43.1	158	0.431	1.580
10^{-6}	432.8	1571	0.433	1.571

Table 1: Common pure- and mixed-program lower bound (4) and the grid construction for selecting every sharp qubit measurement on a great circle to worst-case basis-state error δ.

D'Ariano and Perinotti give finite exact programming for covariant measurements and polynomial approximate programming for general measurements [16]. This does not remove the sharp-selection constraint.

For N grid programs and a uniformly distributed angle the mean error is $\frac{1}{2} - \frac{N}{2\pi} \sin \frac{\pi}{N} \approx \pi^2/(12N^2)$. Doubling the register dimension therefore divides the leading mean error by four.

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On the Closure of Structure: Origins, Persistence and Renewed Participation

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What would it mean for a world to begin or end as an organized structure? The question is not exhausted by the lifetime of an observer. It concerns the formation of distinctions, the relations through which they become accessible, and the possibility that these relations cease or form again. The Holographic Alaya–Field Framework approaches this cosmic question from within physical processes, without assigning an external observer the work of selecting their world.

A trace can remain after its reader is gone, and spacetime can persist after a region no longer supports active readers. Conversely, a continuing reader need not possess a complete description of itself. These distinctions prevent a local model of maintenance from replacing the question of structural origin and closure.

The world line now separates several steps. Interactions can select persistent sectors whose coherence-decay rates define a coupling-dependent Euclidean relation. A response distance requires a separately identified reversible transfer process; neither that process nor its spatial interpretation follows from dephasing alone [1, 2]. Records can identify an otherwise unknown order of interactions in a specified reversible protocol, but they do not thereby create its underlying evolution parameter [6]. Control can form useful permissions and alter physical sources; common source and modular constraints can then restrict the geometric response under their stated field assumptions [3, 4, 7, 5]. A cosmological account must connect these steps within one dynamics, not extrapolate any one finite construction into a history of the universe.

1 Structural closure and cosmic origin

Three readings of closure should be distinguished. Operational closure means that a declared family of states and operations supports its next step. Explanatory closure means that the structures doing this work are accounted for rather than repeatedly supplied by another apparatus. Cosmic closure concerns the global origin, persistence and possible return of those structures. A finite operating loop establishes the first under its inputs; neither of the other two follows by renaming the loop.

One possible reading of a structural beginning is a regime in which durable subsystem distinctions and useful records have not yet formed. A structural ending would be a regime in which no admissible process can continue the distinctions and revisions at issue. They might be equivalent for a specified restricted set of questions without being

the same physical state, the same boundary condition or the same event. This restricted equivalence keeps the original question of origin and terminus meaningful; it does not assert their identity.

In HAFF G, agreement on an invariant accessible algebra and on every permitted instrument implies agreement of all finite adaptive record laws [7]. That result supplies a precise meaning of equivalence for a fixed experimental class. It does not establish a universal final boundary: enlarge the permissions and the two conditions may become distinguishable. A cosmic identification would therefore need a physically justified class of all relevant interventions, rather than a convenient reader that sees neither difference. Whether such a relational equivalence has an additional ontological meaning remains an interpretive question, not a consequence of the record theorem.

2 A trace is not a continuing reader

Consider a fair classical label X copied perfectly into R two-level memories. Its joint state is

$$\rho_{XC} = \frac{1}{2}\sum_{x=0}^{1} |x\rangle\langle x| \otimes (|x\rangle\langle x|)^{\otimes R}.$$

Every memory alone is maximally mixed; the bank has no marginal purity. Nevertheless each copy determines X. Dephasing in the Z basis leaves this joint record unchanged. The product state with the same maximally mixed marginals carries no such record. The difference is in the correlations, not in the local inventories.

An archive of this kind may outlast the machinery that formed it. A later body with appropriate access can still read it. Without that machinery, however, the archive does not acquire a new observation, test an interpretation or correct its own readout. Passive protection and active participation are different physical achievements. Foundation's distinction between held content and return into another operation makes this difference constructive [9]. The continuing operation must actually be available; a comparison state used to bound an error is not a free preparation of another program.

This is also a philosophical distinction. What persists as a trace need not persist as an agent. The past can remain physically relevant through future interventions even after its original reader is gone. Conversely, continuity of an agent need not mean preserving every past datum or every microscopic component. In the framework's agency account it means maintaining an organization that can carry usable relations into later choices, repair its channels and revise what those relations warrant [3, 8]. This is a proposed criterion of causal participation, not a claim that causal organization alone explains experience.

3 Acquisition and renewal have different limits

HAFF F bounds record acquisition at a declared initial resource boundary [6]. Its assumptions include initially independent, unused memories and a specified thermal work account. The initial purity

$$B = \sum_k [\ln d_k - S(\sigma_k)]$$

is then part of what can pay for new correlations, together with supplied resources and the system's own contribution. The example above explains why one cannot impose this

product boundary anew at each time. Old correlations are resources even when $B = 0$. Running out of fresh purity does not erase them.

Renewal asks another question: what must be done when an existing useful relation degrades? In F's conditional-independent bank, $C_i = X \oplus N_i$, with error probability $\varepsilon < 1/2$, independent flips of probability $0 < p \le 1/2$ give

$$q = p + (1 - 2p)\varepsilon, \qquad d_{p,\varepsilon} = h(q) - h(\varepsilon) > 0,$$

where $h(u) = -u \ln u - (1 - u) \ln(1 - u)$. Returning the complete retained label, bank and controller to their joint target removes $Rd_{p,\varepsilon}$ nats. An independent Gibbs bath at temperature T must receive

$$Q \ge k_B T R d_{p,\varepsilon}.$$

Here the flip record is unavailable, the bath is initially independent of the noisy retained state, and the controller cannot absorb the entropy without being restored. For $p = .12$ and $\varepsilon = .05$, eight copies of one fair label require 1.9025859874 nats of correction. The number of distinct labels is one; the independent degradation of eight carriers still has a price.

A sole thermal chiral exhaust with the additional current assumptions in F and D has a quadratic finite-rate price. That result does not apply to every bath or every control pulse. Exact protection gives $d = 0$ in the bank model and defeats a blanket assertion that records require positive upkeep. Accessible noise records, correlated errors and a controller allowed to consume its own purity alter the boundary as well. The distinction matters before any cosmological extrapolation.

The joint-memory argument extends beyond independent copies. Proposition 1 in the appendix shows that an unavailable partial reset of strength λ destroying a retained relation of at least ι nats requires correction heat at least $k_B T \lambda \iota$, under its independent Gibbs-bath and complete-target conditions. If correction is also cyclic in total retained energy, finite supplied work bounds the number of renewals. This is a conditional resource theorem, not a claim that all bodies suffer that noise or that all cosmic evolution has a positive cost per update.

4 An unfinished self can continue

A demand for complete self-description makes participation seem impossible for the wrong reason. A self-model need only support the inquiries and actions on which continuation depends. T-DOME II gives a concrete example: a finite history forms a two-bit reading frame that beats every history-independent direction at one test read, even when the latter receives its exact regression coefficient for free. Retained capacity and total cost are not matched. History must retain the signed relation between the task and the body's channels; an accurately learned but unsigned axis can leave prediction useless [8]. The useful self-frame therefore consists in how a system can reach its task, not in an inventory of everything it is.

Some distinctions are inaccessible to a body's legitimate references. Foundation II separates that failure of identifiability from disagreement, and separates agreement from correctness [9]. A hidden symmetry can make one parameter unknowable while leaving a valuable intervention completely identifiable. In other cases inquiry resolves the ambiguity, but its preparation and maintenance consume resources. A continuing body must decide which error matters and which probe can repair it before the inquiry destroys the

content it was meant to help. Brain Anchor's learning-versus-survival tradeoff supplies one physical version of this problem [12].

Finite self-knowledge is consequently not equivalent to paralysis. A body may track residual errors, revise its references and retain a margin of viable action without ever obtaining a complete account of itself. The stronger philosophical claim is that an imperfect, historically formed organization can be a continuing subject precisely through this revisability. Its identity lies partly in the maintained relation between history and possible action, not in an immutable perfect image.

A rival account could identify continuation with persistence of the stored pattern. It then misses a relevant difference between two copies of the same archive: one has working probes and can revise its routing after an error, while the other's channels have permanently failed. Their stored descriptions agree; their possible next histories do not. The active criterion explains that difference without demanding exact microscopic identity. Two functioning copies may both inherit a causal history, so this criterion does not by itself select a unique first-person successor. That further identity question, like experience, cannot be decided by the survival of a data pattern alone.

That claim still needs an account of first-person experience. Consciousness III's explanatory gap cannot be crossed by renaming calibration as awareness [10]. A candidate bridge must explain why these particular internal relations matter to experience, distinguish alternatives that make the same reports, and expose a relevant intervention. Neither Proposition 1 nor a finite self-frame proves that a body feels anything. The thermodynamic argument addresses the support for continuation; the experience argument addresses what, if anything, that continuation is like.

5 A cosmological participation-window conjecture

The universe's duration and a body's active history need not have the same limit. Dyson's cooling and scaling scenario explicitly allows indefinitely continued life and communication from finite energy under its assumed laws [13]. Finite resources alone therefore do not establish a finite active window. In asymptotically de Sitter models a detector can encounter cosmological-horizon thermal radiation [14]. Krauss and Starkman examine how horizon restrictions, recoverable energy and memory requirements change the prospect of indefinite activity [15]. These are competing conditional mechanisms, not observational proof of the far future.

The framework suggests a more specific conjecture.

Conjecture 1 (Resource-limited participation)**.** *In a late-time causal domain with finite accessible usable work, a positive environmental temperature floor and unavoidable loss of task-relevant relations, any fixed-capacity body that continues at a fixed nonzero task quality can perform only finitely many successful inquiry–revision–action updates, even while passive traces and spacetime structure persist.*

The proposed mechanism is loss of correlations on which usable permissions depend. A retained history supports diagnosis; diagnosis changes routing or protection; the revised body supports another action and its physical source. Mixing damages the relations required to repeat that loop. Restoration exports entropy or consumes a fresh resource. A horizon may restrict replenishment and a thermal floor may prevent the cost per restoration from tending to zero. Exhaustion then ends this participation without erasing every trace.

The bank and partial-reset propositions establish this mechanism in two specified memory models. G supplies finite history-dependent source differences, while Imprint and Brain connect retained relations to subsequent inquiry and bodily change [7, 11, 12]. The conjecture reaches further: an interacting adaptive body can compress memories, migrate codes and replace its task. To prove it for a proposed body class one must obtain a uniform positive resource cost per successful update after optimizing those operations. Neither a count of bit erasures nor a total marginal purity inventory supplies that result automatically. The temperature floor, accessible work bound and irreducible relation loss must also be established for the actual cosmological domain. They are not consequences of selecting an algebra.

The conjecture is weakened or falsified by a body at the stipulated quality with an indefinitely usable protected relation, by vanishing renewal cost after admissible compression or migration, by a cooling law permitting a summable sequence of positive costs, or by unbounded replenishment. An accessible noise record can break the proposed loss mechanism. Discovery of one such regime would identify which part of the mechanism fails, rather than merely extending a numerical lifetime. A static surviving archive is not a counterexample to this active claim, but a continuing reader with the stated resources would be.

The sustained selection–response conjecture in D has different resource quantifiers [4]: it asks for families of longer runs whose installed energy, size and operating domain may grow, while one body is reused within each run. The participation conjecture here fixes capacity and limits the total accessible work. Neither quantifier refutes the other. Foundation III's finite semiclassical mean-action source–response construction [5] concerns a specified operating window; it does not establish indefinitely renewed thermal operation.

6 Origin, return and what closure would mean

An unused record port and an exhausted one can be locally indistinguishable and have different futures. The former may be independent of the source and backed by a usable gradient. The latter can carry correlations with an old history or lack the resources needed for another operation. Allowed future interventions can distinguish them. Equal local states therefore do not identify origin with terminus.

The stronger origin proposal in G is physical formation of permissions: a body with locally adjustable relations encounters diagnostic tasks, retains their results and uses them to install a useful coupling [7]. Exploration, a distinguishable value difference and enough survival time must be supplied by the primitive dynamics. This moves the problem beyond an external program name, without proving that every supplied executor arises in this way. Enlarging an executor introduces another joint interaction to explain; it does not prove a universal final boundary or forbid a finite system from carrying a useful description of itself.

A cosmological return would require more still. One concrete mechanism candidate is that contraction or another geometric transition creates newly accessible nonequilibrium gradients; interactions then allocate part of that work to fresh record ports and bodies, while old correlations and exported entropy remain accounted for in other degrees of freedom. A repeated active phase would be a renewal of resources and formation, not necessarily recurrence of the global state. The physical transition must have its own dynamics, show that the gradients can be used, and explain why old entropy does not

obstruct the new phase. Local thermal swaps or recurrence of a finite register do not establish any of these cosmological steps. A bounce that restores density but no usable work would fail the mechanism; so would a transition requiring an uncharged deletion of the old history. No such return is forced by the participation-window conjecture.

Conjecture 2 (Formation and return of relational structure)**.** *There exist cosmological dynamics in which effective subsystem distinctions, stable record relations and a spatial response description form in one regime, lose their ability to support continuing participation in another, and become physically available again through a later resource-producing transition. The later regime need not repeat the global state or erase the earlier history.*

This is a joint formation conjecture, not a periodic-universe prediction. Its existential scope is a research target: the failure tests below exclude proposed witnesses, not all possible cosmological dynamics. A testable realization must first specify its dynamical family and admissible transitions, for example a particular bounce or phase-transition model with a closed conservation and entropy account. Its witnesses would have to derive the effective decomposition and response dictionary from the same evolving interactions, identify record order without supplying it as a decoded label, and account for usable gradients, clocks, discarded correlations and entropy across the transition. The geometric law and the transition itself require their own dynamical argument. Existing sector-selection, order-reading and sourced-response results establish components under distinct inputs; they do not yet give that cosmological witness.

The discriminating question is whether renewed activity is accompanied by reformation of the relations that make a world accessible, rather than only by newly charged batteries in a permanently stipulated architecture. A proposed model fails this stronger test if its decomposition, spatial dictionary or record standards are simply reinstalled by hand at each epoch. It also fails if the required transition violates its own conservation or entropy account. By contrast, a model demonstrating those structures and their budgets would turn a metaphor of cosmic return into a physical mechanism. The present record and renewal bounds supply tests for such a model, not a substitute for constructing it.

Closure thus has several meanings that must not be confused. A record can become unreadable to one body; a body can lose the capacity to revise itself; a cosmic domain can cease to support new active readers. None entails the nonexistence of its remaining physical structures. Nor must an imperfect self finish knowing itself before it can live. The framework's most consequential prospect is that history changes what a body can do next, and that this relation can be maintained and revised for a time without being complete.

A history can remain after the means of continuing it have gone. A continuing self need not wait to become a finished account of itself.

A Finite-body renewal bound

The bank is not the only setting in which a retained relation has a renewal cost. The following result treats an entire finite memory as one object, allows correlated and quantum content, and makes the noise assumption explicit. Entropies and mutual informations are in nats.

Proposition 1 (Renewing a useful relation under partial reset)**.** *Let θ_{XMC} be a target state of a retained reference X, a memory M of dimension d, and a controller C. Suppose $I_\theta(M:X) \geq \iota > 0$. During each degradation interval the unavailable noise acts as*

$$\mathcal{D}_\lambda(\theta) = (1-\lambda)\theta + \lambda\, \theta_{XC} \otimes \tau_M, \qquad \tau_M = I_M/d, \quad 0 < \lambda \leq 1.$$

A correction restores the complete target θ_{XMC}, using a bath initially independent of the noisy retained state and Gibbs at temperature T. If the correction is a possibly driven unitary on the retained system and bath, with no other entropy sink and vanishing endpoint interaction energy, then

$$Q \geq k_B T \lambda \iota. \tag{1}$$

If the total retained-system energy, including internal couplings, is equal immediately before and after the correction, and driven work and the counted bath heat are the only energy flows, then $W = Q$. A degenerate memory with additive, unchanged apparatus energies is one sufficient realization. For N renewals with $T_j \geq T_* > 0$, $\lambda_j \geq \lambda_* > 0$, and $I_{\theta_j}(M_j : X_j) \geq \iota_* > 0$, a finite total supplied work $\sum_j W_j \leq E_{\mathrm{av}}$ implies

$$N \leq \frac{E_{\mathrm{av}}}{k_B T_* \lambda_* \iota_*}. \tag{2}$$

For the last statement each interval must satisfy the preceding cyclic correction and energy conditions, even when its target differs.

Proof. Write $Z = XMC$ and $\eta = \mathcal{D}_\lambda(\theta)$. Entropy concavity gives

$$\begin{aligned} S(\eta) - S(\theta) &\geq \lambda[S(\theta_{XC}) + \ln d - S(\theta)] \\ &= \lambda[\ln d - S_\theta(M|XC)] \\ &= \lambda[\ln d - S_\theta(M) + I_\theta(M:XC)] \\ &\geq \lambda I_\theta(M:X) \geq \lambda\iota. \end{aligned}$$

The last line uses $S(M) \leq \ln d$ and monotonicity of mutual information under discarding C. The proof is valid for quantum X and for a controller correlated with the target.

Initially Z and the Gibbs bath B are independent. Entropy conservation under the correction and subadditivity at its end give $\Delta S_B \geq S(\eta) - S(\theta)$. For $\tau_B = e^{-\beta H_B} / \operatorname{Tr} e^{-\beta H_B}$,

$$\beta Q = \Delta S_B + D(\rho'_B \| \tau_B), \qquad \beta = (k_B T)^{-1}.$$

This proves (1). With vanishing endpoint bath coupling, the first law reads $W = \Delta E_{XMC} + Q$. The total-energy and energy-flow assumptions give $\Delta E_{XMC} = 0$ and hence $W = Q$; summing the lower bounds proves (2). □

The reset mixture models a lost alternative, not an accessible branch flag the controller can undo for free. It acts on the whole designated memory, so no protected subsystem within that memory escapes the stated channel. The cost is paid for restoring the joint relation, not for restoring a local marginal. A fair classical copy has a maximally mixed memory marginal and still satisfies the bound. An initially independent maximally mixed memory has $\iota = 0$ and does not.

This extends the renewal argument beyond independent copies. It does not establish that every physical body experiences this noise, that every update needs a fixed amount

of information, or that every target can be restored under a body's allowed access. The bound is necessary where such a correction exists. A consumed controller, an accumulating waste register or an accessible auxiliary noise record must be included among the resources. If the noise environment is retained, its correlations cannot be discarded when defining the target. Calling only the memory cyclic would hide the expenditure. The entropy proof permits an externally scheduled unitary; it is not a proof of autonomous apparatus closure. A physical battery that takes part in the entropy account must be included or separately shown not to supply an additional entropy sink.

Equation (2) bounds successful renewals, not necessarily calendar time. A system can postpone a last update arbitrarily far. A finite upper bound on an uninterrupted active interval additionally requires a positive minimum update rate and a bound on gaps. Hibernation, cooling toward zero temperature, decreasing task quality and improving protection are substantive alternatives, not semantic exceptions.

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